

LINE: Queueing Analysis Algorithms

User manual for C++

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Chapter 1

Introduction

1.1 What is LINE?

LINE is an open-source software package to analyze queueing models via analytical methods and simulation. The tool aims at simplifying the computation of performance and reliability metrics in models of systems such as software applications, business processes, computer networks, and others. LINE decomposes a high-level system model into one or more stochastic models, typically extended queueing networks, that are subsequently analyzed using either numerical algorithms or simulation.

LINE supports 170+ modeling features: node types (queues, delays, caches, fork-join, routers), 39 scheduling disciplines (FCFS, LCFS, PS, priority variants, polling), distributions (Erlang, hyperexponential, MAP, MMPP, phase-type, matrix-exponential), cache replacement, finite capacity, impatience (balking, reneging), retrial queues, batch arrivals, dependencies (load, class, joint), and layered queueing network constructs.

LDES, the discrete-event simulation engine of LINE, is built to span the entire feature space of the package, so that every approximation LINE publishes can be validated against a sample path of the same model, expressed once. Several analytical solvers are included alongside it to accelerate the evaluation of specific classes of models, and their solutions can in turn warm-start an LDES run, placing it near steady state already at the start of the simulation (§8.1.3).

A key feature of LINE is that it decouples the model description from the solvers used for its solution. That is, LINE implements model-to-model transformations that automatically translate the model specification into the input format (or data structure) accepted by its solutions algorithms. LINE sends models for evaluation either to native algorithms of external solvers that include Java Modelling Tools (JMT; <http://jmt.sf.net>) and LQNS (<http://www.sce.carleton.ca/rads/lqns/>). Native model solvers are instead based on formalisms and algorithms based on:

- Continuous-time Markov chains (CTMC)
- Discrete-event simulation (LDES)
- Fluid/mean-field ordinary differential equations (FLD)

- Matrix analytic methods (MAM)
- Normalizing constant analysis and generating-function asymptotics (NC)
- Mean-value analysis (MVA)
- Stochastic Simulation Algorithms (SSA)

Each solver encodes a general solution paradigm and can implement both exact and approximate analysis methods. For example, the MVA solver implements both exact mean value analysis (MVA) and approximate mean value analysis (AMVA). NC is named for the normalizing constant, but its remit is wider: it is also LINE's default solver for *asymptotic expansions of generating functions*, and that is where such methods belong. Most of them expand the generating function of a normalizing constant, as `pana`, `le`, `ble`, `kt`, `bkt`, `lekt`, `bk`, `gleint` and the Norlund-Rice family do; `morrison` instead expands the generating function of the balance equations themselves, of a model that has no product form and hence no normalizing constant to expand. A method is hosted by NC because it is a transform expansion, not because it computes a constant. The offered methods typically differ for accuracy, computational cost, and the subset of model features they support. A special solver, called `AUTO`, is supplied that provides an automated selection of the solver to use for a given model.

LINE also includes two meta-solvers that can leverage other solvers to analyze composite models such as layered queueing networks or queueing networks in random environments. The native meta-solvers include:

- Random environment solver (ENV)
- Layered network meta-solver (LN).

Figure 1.1 summarizes how these components are layered internally. A high-level system model is first handled by a meta-solver, when composite, and then dispatched to one of the network solvers. Each solver exposes an analyzer that selects a concrete solution method, which in turn is built on top of a set of reusable numerical APIs.

LINE can be applied to models specified in the following formats:

- *LINE modeling language*. This is a domain-specific object-oriented language designed to resemble the abstractions available in JMT's queueing network simulator (JSIM). This requires the model direct coding of the system model.
- *Layered queueing network models (LQNS XML format)*. LINE is able to solve a sub-class of layered queueing network models, either specified using the LINE modeling language or according to the XML metamodel of the LQNS solver.
- *JMT simulation models (JSIMg, JSIMw formats)*. LINE is able to import and solve queueing network models specified using JSIMgraph and JSIMwiz. LINE models can be exported to, and visualized with, JSIMgraph and JSIMwiz.
- *JSON model serialization*. LINE also accepts models supplied in a portable JSON format that conforms to the `line-model.schema.json` specification, enabling model interchange across codebases and with external tools, as described in Section D.2.

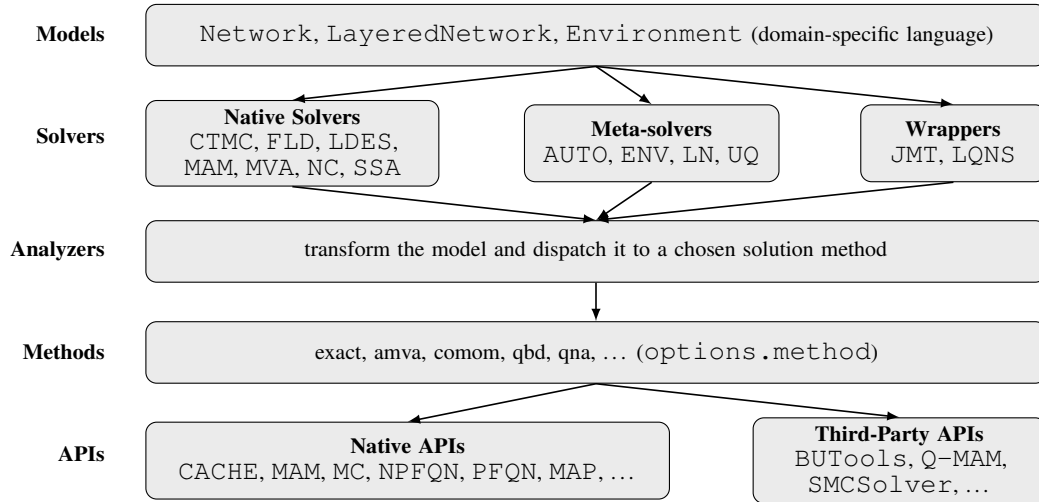


Figure 1.1: Layered internal architecture of LINE. A model is evaluated by a native solver, by a meta-solver that coordinates other solvers, or handed to an external tool through a solver wrapper; each solver runs an analyzer that selects a solution method, and methods are implemented on top of shared numerical APIs.

1.2 Network solvers

Solvers analyze objects of class `to` to return average, transient, distributions, or state probability metrics. A solver can implement one or more *methods*, which although featuring a similar overall solution strategy, they can differ significantly from each other in the way this strategy is actually implemented and on whether the final solution is exact or approximate.

A ‘method’ flag can be passed upon invoking a solver to specify the solution method that should be used. For example, the following invocations are identical:

```

MVA(model, "exact").avg_table();
SolverOptions opt;           // as MVA.defaultOptions
opt.method = "exact";
MVA(model, opt).avg_table();
  
```

In what follows, we describe the general characteristics and supported model features for each solver available in LINE and their methods.

Available solvers

The following solvers are available within LINE 3.0.x:

- **AUTO:** This solver uses an algorithm to select the best solution method for the model under consideration, among those offered by the other solvers. Analytical solvers are always preferred to simulation-based

solvers. This solver is implemented by the `AUTO` class.

- **CTMC:** This is a solver that returns the exact values of the performance metrics by explicit generation of the continuous-time Markov chain (CTMC) underpinning the model. As the CTMC typically incurs state-space explosion, this solver can successfully analyze only small models. The CTMC solver is the only method offered within LINE that can return an exact solution on all Markovian models, all other solvers are either approximate or are simulators. This solver is implemented by the `CTMC` class.
- **FLD:** This solver analyzes the model by means of an approximate fluid model, leveraging a representation of the queueing network as a system of ordinary differential equations (ODEs). The fluid model is approximate, but if the servers are all PS or INF, it can be shown to become exact in the limit where the number of users and the number of servers in each node grow to infinity [125]. This solver is implemented by the `FLD` class.
- **JMT:** This is a solver that uses a model-to-model transformation to export the LINE representation into a JMT simulation (JSIM) or analytical (JMVA) models [10]. The JSIM simulation solver can analyze also non-Markovian models, in particular those involving deterministic or Pareto distributions, or empirical traces. This solver is implemented by the `JMT` class.
- **LDES:** This is LINE’s own discrete-event simulator, a native simulation solver built on the SSJ library. Like JMT, it analyzes the model by discrete-event simulation and supports non-Markovian distributions and a broad range of scheduling, routing, and blocking features, but it runs in-process (or through a standalone native binary) without requiring an external tool. This solver is implemented by the `LDES` class.
- **LQNS:** This is a wrapper for the `lqns` analytical solver and the `lqsim` simulator distributed with the LQNS suite, which solve layered queueing networks. The model is exported to the LQNS XML format, the back-end tool is run as an external process, and its output is parsed back into the LINE result tables; both commands must therefore be available on the system path. On a `LayeredNetwork` it is the external counterpart of LINE’s own layered solver, `LN`. It also accepts a `flat`, which it solves with the `qnsolver` utility of the same suite, so that product-form models can be evaluated with the MVA algorithms implemented within LQNS. On a `flat` the methods specify the multiserver handling algorithm: Conway’s multiserver approximation (`qns.conway`) [44], which extends the multinomial all-servers-busy approximation of de Souza e Silva and Muntz [51]; Rolia’s multiserver approximation (`qns.rolia`) [136, 137], in the per-class form of Franks [64] that LQNS implements; load-dependent MVA by Reiser-Lavenberg (`qns.reiser`) [129, 131]; and Zhou’s arrival-theorem binomial approximation (`qns.zhou`) [172, 173]. This solver is implemented by the `LQNS` class.
- **MAM:** This is a matrix-analytic method solver, which relies on quasi-birth death (QBD) processes to analyze open queueing systems. This solver is implemented by the `MAM` class.
- **MVA:** This is a solver based on approximate and exact mean-value analysis. This solver is typically the fastest and offers very good accuracy in a number of situations, in particular models where stations have a single-server. Besides mean-value analysis, the MVA solver also hosts LINE’s closed-form analytical decomposition methods for open networks, namely the Queueing Network Analyzer (`qna`) [165] and its

index-of-dispersion counterpart, the Robust Queueing Network Analyzer (`rqna`) [166], which characterizes each flow by its full index of dispersion for counts to capture bursty, non-renewal traffic and bounds the mean workload at each queue by robust queueing. In the near-immediate feedback elimination of `rqna`, our implementation keeps the strictly higher-utilization stations as explicit queues and collapses only the equal- or lower-utilization ones, so that the network traffic rates and the variability injected by upstream bottlenecks are preserved. This solver is implemented by the `MVA` class.

- **NC:** This solver uses a combination of methods based on the normalizing constant of state probability to solve a model. The underpinning algorithm are particularly useful to compute marginal and joint state probabilities in queueing network models. This solver is implemented by the `NC` class.
- **SSA:** This is a Stochastic Simulation Algorithms based on the CTMC representation of the model. Contrary to the `JMT` simulator, which has online estimators for all the performance metrics, `SSA` estimates only the probability distribution of the system states, indirectly deriving the metrics after the simulation is completed. Moreover, the `SSA` execution can more efficiently parallelized on multi-core machines. Moreover, it is possible to retrieve the evolution over time of each node state, including quantities that are not loggable in `JMT`, e.g., the active phase of a service or arrival distribution. This solver is implemented by the `SSA` class.

1.3 Automatic solver selection

The `AUTO` class (aliases: `SolverAuto`, `SolverAUTO`, `LINE`) provides interfaces to the core solution functions (e.g., `avg`, ...) that dynamically bind to one of the other solvers implemented in `LINE` (`CTMC`, `NC`, ...). It is often difficult to identify the best solver without some performance results on the model, for example to determine if it operates in light, moderate, or heavy-load regime.

The selection is therefore driven by structural features extracted from `sn`: the proportions of FCFS, processor-sharing and delay stations, the presence of class-switching nodes, the mean number of servers per queue, the number of jobs per chain, and the service process types. These features are normalized and combined into ratios that a decision rule maps to a solver.

When the selected solver does not implement the operation requested (for instance `solver_ctmc_transient_analyzer`, which not every solver provides), `AUTO` falls back along the list of feasible solvers ordered by increasing average runtime. Since the list ends with the simulators, a solution strategy always exists. Direct instantiation of a solver remains preferable once the appropriate one is known, as it avoids the feature-extraction step.

Selection methods. The constructor of `AUTO` () accepts an `options.method` argument that fixes how the back-end solver is chosen. Categorical options trigger a single back-end consistent with the requested intent:

- `options.method=default` or `heur`: heuristic auto-selection. `LINE` instantiates a list of feasible solvers ordered by typical execution time and dispatches the requested function to the first one that

supports the model (and the operation, e.g. `solver_ctmc_transient_analyzer`). For `Network` models the candidate list is, in increasing average runtime, MVA, NC, MAM, FLD, SSA, CTMC, LDES; JMT is deliberately not among the candidates, because LDES subsumes its feature set and automatic selection therefore has nothing to gain from an external simulation process (JMT remains reachable through the explicit per-solver delegation below); for `LayeredNetwork` models it is LQNS followed by LN wrappers around NC, MVA, MAM, FLD; for `Environment` models it is ENV wrapping MVA, NC, or FLD.

- `options.method=sim`: best simulation back-end. Routes to LDES (`Network`, with SSA taking precedence for event-level sampling), LQNS (`LayeredNetwork`), or ENV wrapping MVA (`Environment`).
- `options.method=exact`: best exact analytical back-end. Routes to NC for closed `Network` models, to CTMC for non-closed ones, to LN wrapping NC for layered models, and to ENV wrapping NC for random environments.
- `options.method=fast`: fast approximate back-end (MVA or its LN/ENV wrapping).
- `options.method=accurate`: accurate approximate back-end (FLD or its LN/ENV wrapping).
- Per-solver delegations: passing the lowercase solver name forces the corresponding back-end with its default method. Recognised values are `mam`, `mva`, `nc`, `fluid`, `jmt`, `ssa`, `ctmc`, `ldes`, `env`, `ln`, `lqns`.

1.4 Choosing the codebase that solves the model

LINE exists as four codebases – MATLAB, Java, Python and C++ – that implement the same algorithms against the same internal representation. Two of them can also hand a model to another: the model is written once and the `lang` option decides which codebase actually solves it. This is what cross-checks a result without rewriting the model, and what makes a codebase’s faster or more precise engine reachable from the one the model was built in.

The C++ port is the codebase other implementations delegate TO rather than FROM: it has no `lang` option of its own, and its front end is the `line-cli` binary. A model written in Python is solved by it with `lang='cpp'`, which serializes the model to `model.json` and runs `line-cli` on it; that route, the binary lookup and the solvers it serves are documented in the Python manual (`LINE-user-python.pdf`). The corresponding delegation to the JAR, `lang='java'`, is documented in the MATLAB manual (`LINE-user-matlab.pdf`) and the Python one. From MATLAB the C++ port is reached by exporting the model with `linemodel_save` and running `line-cli` on the file.

1.5 Obtaining the latest release

This document contains the user manual for the latest version of LINE, which can be obtained from:

<http://line-solver.sf.net/>

LINE 3.0.x has been tested using

Alternatively, a development version of the codebase can be cloned from Github at <https://github.com/imperial-qore/line-solver>.

1.6 References

To cite the LINE solver, we recommend to reference:

- G. Casale. “Integrated Performance Evaluation of Extended Queueing Network Models with LINE”, in *Proc. of WSC 2020*, ACM Press, Dec 2020.

The following are references for the individual tools used by LINE. Please cite them if your use of LINE focuses on features developed in these tools:

- *BuTools* [78]: G. Horváth and M. Telek. “BuTools 2: A Rich Toolbox for Markovian Performance Evaluation”, in *Proc. of VALUETOOLS*, 2017.
- *SMCSolver* [14]: D. Bini, B. Meini, S. Steffé, J. F. Pérez, and B. Van Houdt. “SMCSolver and Q-MAM: Tools for Matrix-Analytic Methods”, in *SIGMETRICS Performance Evaluation Review*, 39(4), 2012.
- *MAMSolver* [132]: A. Riska and E. Smirni. “MAMSolver: A Matrix Analytic Methods Tool”, in *Proc. of the 12th Int. Conf. on Modelling Techniques and Tools (TOOLS)*, LNCS 2324:205–211, 2002.
- *KPC-Toolbox* [35]: G. Casale, E. Z. Zhang, and E. Smirni. “KPC-Toolbox: Best Recipes for Automatic Trace Fitting Using Markovian Arrival Processes”, in *Performance Evaluation*, 67(9):873–896, 2010.
- *JMT* [10]: M. Bertoli, G. Casale, and G. Serazzi. “The JMT Simulator for Performance Evaluation of Non-Product-Form Queueing Networks”, in *Proc. of the 40th Annual Simulation Symposium (ANSS)*, 2007.
- *LQNS* [63]: G. Franks, P. Maly, M. Woodside, D. C. Petriu, A. Hubbard, and M. Mroz. “Layered Queueing Network Solver and Simulator User Manual”, Carleton University, 2013.
- *Q-MAM* [14]: D. Bini, B. Meini, S. Steffé, J. F. Pérez, and B. Van Houdt. “SMCSolver and Q-MAM: Tools for Matrix-Analytic Methods”, in *SIGMETRICS Performance Evaluation Review*, 39(4), 2012.

1.7 Contact and credits

LINE is the collective work of many students and researchers hosted at the QORE Lab (<https://qore.doc.ic.ac.uk/>) at Imperial College London. Please refer to the AUTHORS files in the codebase for detailed credits to individual project contributors.

Project coordinator: Giuliano Casale, Department of Computing, Imperial College London, 180 Queen’s Gate, SW7 2AZ, London, United Kingdom. Web: <http://wp.doc.ic.ac.uk/gcasale/>

1.8 Copyright and license

Copyright Imperial College London (2012-Present). LINE is freeware and open-source, released under the 3-clause BSD license.

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1.10 Use of AI assistance

During the development of LINE the authors used Anthropic’s Claude models, namely Opus 4.0, Opus 4.5, Opus 4.8, Opus 5, Sonnet 4.5, Sonnet 5 and Fable 5, as well as OpenAI’s Codex (GPT-5.1-Codex), to assist with implementation, porting across the MATLAB, Java, Python and C++ codebases, test authoring, refactoring and documentation drafting. The tool is provided as-is, without warranty of any kind, under the terms of its license. Algorithms taken from the scientific literature have been checked against the numerical examples reported in the corresponding papers, in addition to being cross-checked against simulation and against the reference implementation. The methods themselves, and their published references, remain the work of their respective authors; model assistance concerns the realization of those methods in code, not their derivation. No such model is an author or a copyright holder of any part of LINE. The last stable LINE release developed without any such assistance is 2.0.39.

Chapter 2

Getting started

2.1 Installation and support

Quick Start: You can get started with LINE as follows:

1. Obtain the latest release

- Stable release (zip file): <https://sf.net/projects/line-solver/files/latest/download>

Ensure that the files are decompressed in the installation folder.

2. Configure and build the library and its command-line front end with CMake:

```
cmake -S cpp -B cpp/build -DCMAKE_BUILD_TYPE=Release
cmake --build cpp/build -j4
```

This writes the `line-cli` executable under `cpp/build/`. The solver itself is header-only, so a program uses it by putting `cpp/include` on its include path; no library needs to be linked.

The command above builds every target of the project, which includes the regression suite, the benchmarks and the example gallery alongside the front end. That is a long compilation: several of the dispatch translation units instantiate the solver templates over each arithmetic backend and take minutes apiece on their own, so the full build runs into the tens of minutes. To obtain just the command-line front end, name it as the target,

```
cmake --build cpp/build --target line-cli -j4
```

or turn the other targets off at configure time with `-DLINE_MP_BUILD_TESTS=OFF`, `-DLINE_MP_BUILD_BENCH=OFF` and `-DLINE_MP_BUILD_EXAMPLES=OFF`.

3. LINE is now ready to use. For example, solve the model file shipped in the root of the distribution with

```
cpp/build/line-cli -f example.json -s mva -a avg
```

`example.json` and `example_lqn.json` are a queueing network and a layered network in LINE's own JSON format (Appendix D.2), which is also what `io::write_network_json` writes.

4. To use LINE inside a C++ program, include the model builder, the node and distribution types, and the solver of interest, e.g.

```
#include "line/lang/qn/network_builder.h"    // Network<T>
#include "line/lang/qn/nodes.h"              // Source, Queue, Sink, OpenClass, ...
#include "line/lang/distributions.h"         // Exp, Erlang, HyperExp, ...
#include "line/solvers/solver.h"             // Network, SolverMVA, AvgTable, ...
using namespace line;
using namespace lang; // model enumerations: SchedStrategy, DropStrategy, ...
```

All code listings in this manual assume these two using-directives, so model enumerations and distributions appear unqualified (`SchedStrategy::FCFS` rather than the fully qualified `line::lang::SchedStrategy::FCFS`); solver namespaces such as `qn`, `mva` or `ctmc` are kept explicit when a specialized analysis below the public solver facade is required.

The first four headers are the baseline every model needs: the builder declares the `Network<T>` template, `nodes.h` the station and class types, `distributions.h` the service and arrival laws, and `solver.h` the `Network` alias of `qn::Network<double>` that the listings use unqualified. The facade header declares the shared `SolverOptions`, `AvgTable` and `SolverX` classes. A listing needs a lower-level solver header only for an analysis that the facade does not expose.

2.1.1 Software requirements

Certain features of LINE depend on external tools and libraries. The recommended dependencies are:

- A C++17 compiler (GCC 9 or later, or Clang 10 or later).
- CMake version 3.16 or later.
- The Boost.Multiprecision headers, header-only and permissively licensed, which supply the exact (`cpp_rational`) and high-precision (`cpp_bin_float`) arithmetics.
- Optionally GMP and MPFR, selected with `-DLINE_MP_USE_GMP=ON`; both are LGPL, so they are not used unless requested.
- LAPACK (`-DLINE_MP_USE_LAPACK`, on by default), which supplies the eigenvalues, singular values and Schur decompositions of the `double` backend.
- Optionally HiGHS (`-DLINE_MP_USE_HIGHS`, on by default), the MIT-licensed sparse linear-programming backend behind the QRF bounds and NLP entry points.

LAPACK is nominally optional but required in practice: the flag only asks CMake to look, and when `find_library` does not find `liblapack` the build still succeeds while `eig_values`, `svd_values`,

`svd_full`, `schur_decomposition`, `schur_reorder` and `sylvester_schur` all refuse by name at run time. Everything reached through them refuses with them, the fluid second-moment machinery above all: `fluid_lyapunov`, the stationary covariance of the linear-noise approximation, and the refined mean-field expansion both reduce to the reachable subspace through a full SVD, so the regression suite fails as well. Configure reports `line_mp: LAPACK enabled when it is present` and `line_mp: LAPACK not found`; `eig/svd` entry points will refuse by name when it is not; that line is the one to check before reading any result that depends on it.

Without HiGHS the dense tableau of `util/simplex.h` caps the QRF bounds at a few hundred columns and the QRF NLP entry points refuse by name. It is also required to run the regression suite, whose `tests/test_lp_highs.cpp` asserts that the backend is compiled in and whose `example_bas_small.m` case (3894 columns) does not terminate on the dense tableau.

The compiler, CMake and the library dependencies are installed as follows. On Debian and Ubuntu, with `apt`:

```
sudo apt install build-essential cmake libboost-dev liblapack-dev libblas-dev
sudo apt install libgmp-dev libmpfr-dev      # only for -DLINE_MP_USE_GMP=ON
```

With `dnf`:

```
sudo dnf install gcc-c++ make cmake boost-devel lapack-devel blas-devel
sudo dnf install gmp-devel mpfr-devel      # only for -DLINE_MP_USE_GMP=ON
```

On macOS, with Homebrew and the Xcode command-line tools (`xcode-select --install`) for the compiler:

```
brew install cmake boost lapack
brew install gmp mpfr      # only for -DLINE_MP_USE_GMP=ON
```

Homebrew keeps `lapack` keg-only, because macOS ships its own LAPACK inside `Accelerate.framework` and the two would collide on the default library path. LINE looks for a plain `liblapack`, so export the keg before configuring, otherwise configure reports LAPACK as not found and the entry points above refuse by name:

```
export CMAKE_PREFIX_PATH="$(brew --prefix lapack):$CMAKE_PREFIX_PATH"
```

HiGHS is packaged as `libhighs-dev` on Debian and on Ubuntu 25.04 or later, and as `highs` in Homebrew:

```
sudo apt install libhighs-dev      # Debian, Ubuntu 25.04 and later
brew install highs                 # macOS
```

Older Ubuntu releases, including the 22.04 and 24.04 LTS, have no package for it, and neither does `dnf`; on those it is built from source:

```
git clone --depth 1 https://github.com/ERGO-Code/HiGHS.git
cmake -S HiGHS -B HiGHS/build -DCMAKE_BUILD_TYPE=Release \
```

```
-DBUILD_SHARED_LIBS=ON -DBUILD_TESTING=OFF \
-DCMAKE_INSTALL_PREFIX=/usr/local
cmake --build HiGHS/build -j4
sudo cmake --install HiGHS/build && sudo ldconfig
```

The LINE CMake project then finds it with no further configuration and reports `line_mp: HiGHS enabled at configure time`; when it is absent, configure says so and continues with the dense tableau. To install HiGHS without root, give it a writable `-DCMAKE_INSTALL_PREFIX` and export that prefix in `CMAKE_PREFIX_PATH`, and its `lib/` in `LD_LIBRARY_PATH`, before configuring LINE.

- KPC-Toolbox (<https://github.com/kpctoolboxteam/kpc-toolbox>): version 0.3.4 or later.
- MAMSolver (<https://www.cs.wm.edu/MAMSolver/>): tools for matrix-analytic methods.
- M3A (<https://github.com/imperial-qore/M3A>): version 1.0.0.
- BuTools (<https://github.com/ghorvath78/butools>): version 2.0 or later.
- Q-MAM/SMC (<https://win.uantwerpen.be/~vanhoudt/tools/QBDfiles.zip>).

One optional dependency is resolved at run time rather than at configure time:

- Symbolic analysis backend (<https://hub.docker.com/r/imperialqore/line-sage-rest>): a SageMath computer algebra system behind a small REST service, used for symbolic steady-state analysis, exact parametric sensitivities and fluid Jacobians (§9.1). It requires Docker and is obtained with

```
docker pull imperialqore/line-sage-rest:latest
```

after which LINE starts and stops the service on its own. The image is not fetched automatically on first use. The C++ port carries no computer algebra system of its own, so this backend is the only way to run symbolic analysis from it: `--arith exact` is exact rational arithmetic, which is a different thing.

2.1.2 Checking the installation

Every distribution ships an environment check that reports which of the above dependencies are reachable. It is run as

```
cpp/build/line-cli --version
cpp/build/line-cli --list-api
```

The first prints the version string, the second the list of API functions the port exposes. The C++ port has no separate environment check: the optional dependencies are resolved once, at configure time, and CMake reports each one on its own `line_mp: line`, so rerunning `cmake -S cpp -B cpp/build` and reading those lines is the check. The regression suite is built by the same CMake project and is run with

```
cmake --build cpp/build --target line_mp_tests -j4 && cpp/build/line_mp_tests
```

2.1.3 Getting help

For discussions, bug reports, new feature requests, please create a thread on the Sourceforge forums:

- General discussion: <https://sf.net/p/line-solver/discussion/help/>
- Bugs and issues: <https://sf.net/p/line-solver/tickets/>
- Feature requests: <https://sf.net/p/line-solver/feature-requests/>

2.2 Getting started examples

2.2.1 Tutorials

The sixteen step-by-step tutorials that introduce the modelling language, from a single $M/M/1$ queue to layered, cached, random-environment, and Petri-net models, are collected in the LINE primer for this language (, chapter *Tutorials*). The present manual documents each construct they use in full detail, and the LINE internals manual (`LINE-internals-cpp.pdf`) documents the solution algorithms.

2.2.2 Model gallery

LINE includes a collection of classic, commonly occurring, queueing models, reached . They include single queueing systems (e.g., $M/M/1$, $M/H_2/1$, $D/M/1$, ...), tandem queueing systems, and basic queueing networks. For example, to instantiate and estimate the mean response time for a tandem network of $M/M/1$ queues we may run The C++ port has no model gallery: its front end reads models rather than generating them, so the equivalent of a gallery entry is a JSON model file solved with

```
cpp/build/line-cli -f example.json -s mva -a avg
```

which prints a station-class table of the same shape as the ones above, one row per (station, class) pair, for whichever model the file holds. The distribution ships `example.json` and `example_lqn.json` in its root; any other model is written by hand in the format of Appendix D.2, exported from one of the other three codebases, or produced by `io::write_network_json`. The examples in the gallery may also be used as templates to accelerate the definition of basic models. Tutorial 10 of the primer shows a gallery instantiation of a $M/E_2/1$ queue.

Table 2.1 lists the gallery models grouped by category. Each entry is a parameter-free factory available in MATLAB (`gallery_*.m`), in the Java JAR (`Gallery.gallery_*`()), and in Python (`from line_solver import gallery_*`). Layered models return a `LayeredNetwork`, the random-environment model returns an `Environment`, and all others return a `Network`.

Table 2.1: Model gallery entries by category.

Category	Gallery functions
Single queues	gallery_aphm1, gallery_coxm1, gallery_detm1, gallery_dm1, gallery_erlm1, gallery_gamm1, gallery_hypm1, gallery_mapm1, gallery_merl1, gallery_mhyp1, gallery_mml, gallery_mmap1, gallery_mpar1, gallery_parm1, gallery_replayerm1, gallery_um1
Processor sharing	gallery_erlm1_ps, gallery_erlmips, gallery_mml_ps
Multiserver	gallery_erldk, gallery_hyperlk, gallery_mapmk, gallery_mdk, gallery_merlk, gallery_mhypk, gallery_mmapk, gallery_mmk
Multiclass / priority	gallery_mml_multiclass, gallery_mml_prio, gallery_mml_ps_multiclass, gallery_mmap1_multiclass
Tandem / linear	gallery_hyphyp1_linear, gallery_hyphyp1_tandem, gallery_merl1_linear, gallery_merl1_tandem, gallery_mhyp1_linear, gallery_mhyp1_tandem, gallery_mml_linear, gallery_mml_tandem, gallery_mml_tandem_multiclass
Feedback	gallery_mml_feedback, gallery_mml_ps_feedback
Reentrant lines	gallery_erlerl1, gallery_erlerl1_reentrant, gallery_erlm1_reentrant, gallery_hyperl1_feedback, gallery_hyperl1_reentrant, gallery_hyphyp1_reentrant, gallery_hypm1_reentrant, gallery_lukumar_reentrant, gallery_merl1_reentrant, gallery_mhyp1_reentrant, gallery_mml_ps_reentrant, gallery_mml_reentrant
Closed networks	gallery_cqn, gallery_cqn_multiclass, gallery_repairmen
Fork-join	gallery_fj_closed, gallery_fj_open
Caches	gallery_cache_lru, gallery_cache_routing
Finite capacity	gallery_fcr, gallery_mmlk
Layered (LQN)	gallery_lqn_basic, gallery_lqn_workflows, gallery_multitier, gallery_multitier_storage
Random environment	gallery_renv_breakdown
Random generators	gallery_qn_random, gallery_lqn_random

The `gallery_qn_random` and `gallery_lqn_random` entries wrap the `NetworkGenerator` and `LayeredNetworkGenerator` random model generators. Both accept an optional seed argument (default fixed) so that repeated calls return the same model; `gallery_qn_random` uses a deterministic (cyclic) topology to keep the generated network reproducible under the seed.

2.3 Controlling verbosity

Solver verbosity may be configured at program start using, e.g.: Verbosity is a property of the front end rather than of a global object: the `line-cli` flag `--verbosity (-v)` takes `silent`, `standard` or `debug`,

```
cpp/build/line-cli -f example.json -s mva -a avg -v debug
```

while a program that calls the solver headers directly prints nothing at all. The three available verbosity levels are:

- `VerboseLevel.SILENT`: Suppresses all solver output messages
- `VerboseLevel.STD`: Shows standard solver output messages (default)

- `VerboseLevel.DEBUG`: Shows detailed solver output including debug information

At STD verbosity each solve emits a single banner line before its result table, in the form

```
<solver> analysis [method: <requested>/<resolved>; type: <accuracy>, <randomness>; lang: ...  
<language>; env: <version>] completed in <t>s.
```

The `lang` and `env` fields name the codebase that actually solved the model and its runtime version. They are worth reading rather than skipping: they are not fixed by the edition of this manual, since MATLAB solves through the JAR under `options.lang='java'` and through native Python under `options.lang='python'`, and a banner is the quickest way to tell which one answered. Example banner lines from this edition are: The C++ port emits one such banner, from the LDES front end, and it carries no `env` field:

```
LDES analysis [method: default; type: approximate, randomized; lang: cpp] completed in ...  
1.204813s. Iterations: 10000.
```

The other solvers print no banner: as noted above, `line-cli` reports through `-v debug` and the solver headers print nothing when called directly.

Chapter 3

Basic queueing networks

Throughout this chapter, we discuss the specification of `Network` models. The chapter covers what a queueing network needs in order to be specified and solved: its nodes, its job classes, the routes the jobs take, and the distributions of their service and inter-arrival times. Kept together, these are exactly the ingredients of a product-form model, class switching included. The features that take a model out of product form are collected in Chapter 5, and the formalisms that are not queueing networks in Chapter 6. LINE currently supports open, closed, and mixed networks with non-exponential service and arrivals, and state-dependent routing. All solvers support the computation of basic performance metrics, while some more advanced features are available only in specific solvers. Each `Network` model requires in input a description of the nodes, the network topology, and the characteristics of the jobs that circulate within the network. In output, LINE returns performance and reliability metrics. Technical background on these models can be found in books such as [17, 96] or in tutorials such as [5, 95].

Figure 3.1 shows a small queueing network of the kind LINE analyzes, and names on the diagram the metrics it returns. A fixed population of N users sits at terminals: each user thinks for Z seconds, then submits a request. The request queues at the CPU, is then served by one of two disks, and on completion returns to its user, who begins thinking again. In the vocabulary used throughout this chapter, the users, the CPU and the disks are *nodes*; the CPU and the disks are *stations*, since a request spends time there waiting and in service, and the terminals are a *delay* station, where a user is never queued behind another. The requests circulating in the network are the *jobs*, and all jobs that follow the same route and service requirements form a *class*. The loop drawn by the return path makes the model *closed*: no job enters or leaves, so exactly N jobs are present at all times. Had the requests instead been generated by an external `Source` and left through a `Sink`, the model would be *open*.

3.1 Default performance metrics

The default metrics supported by all solvers are as follows:

- *Mean queue-length* (`QLen`). This is the mean number of jobs residing at a node when this is observed at a

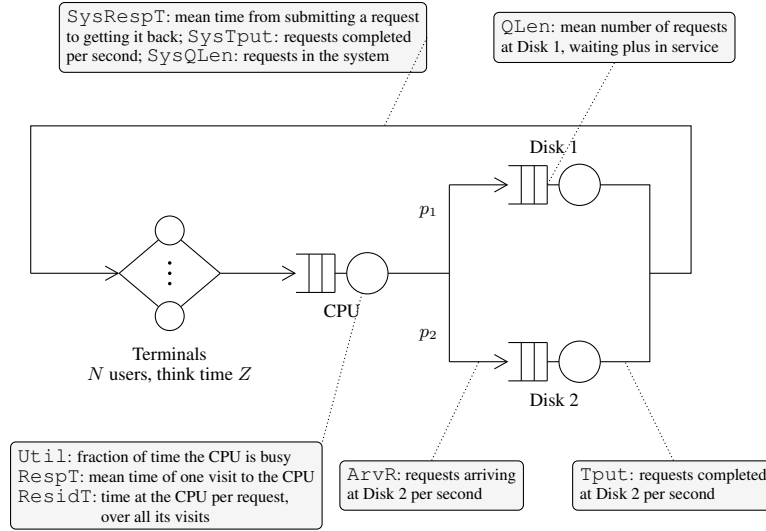


Figure 3.1: An extended queueing network: N users at terminals submit requests that are served by a CPU and by one of two disks before returning to their user. The annotations name the metrics LINE reports, per station and class for $QLen$, $Util$, $RespT$, $ResidT$, $ArvR$ and $Tput$, and per chain for the system metrics.

random instant of time.

- **Mean utilization ($Util$).** For nodes that serve jobs, this is the mean fraction of time the node is busy processing jobs. In both single-server and multi-server nodes, this is a number normalized between 0 and 1, corresponding to 0% and 100%. In infinite-server nodes, the utilization is set by convention equal to the mean queue-length, therefore taking the interpretation of the mean number of jobs in execution at the station.
- **Mean response time ($RespT$).** This is the mean time a job spends traversing a node within a network. If the node is visited multiple times, the response time is the time spent for a single visit to the node.
- **Mean residence time ($ResidT$).** This is the total time a job accumulates, on average, to traverse a node within a network. If the node is visited multiple times, the residence time is the time accumulated overall visits to the node prior to returning to the reference station or arriving to a sink.
- **Mean throughput ($Tput$).** This is the mean departure rate of jobs completed at a resource per time unit. Typically, this matches the mean arrival rate, unless the node switches the class of the jobs in which case the arrival rate of a class may not match its departure rate.
- **Mean arrival rate ($ArvR$).** This is the actual rate at which jobs of a given class arrive at a station, measured in jobs per unit time. The arrival rate differs from throughput primarily in networks with class-switching mechanisms, where jobs may arrive at a node in one class and depart in a different class. In such cases, the arrival rate for a class reflects the incoming flow before any class transformation occurs, while the

throughput represents the outgoing flow after transformation. For nodes without class-switching, the arrival rate and throughput are identical under steady-state conditions due to flow balance.

The metrics above characterize individual nodes. Response times, throughputs and queue lengths are also reported end-to-end, in which case they are called *system* metrics and are indexed by chain rather than by station-class pair:

- *System response time* (SysRespT): time a job spends in the network, measured from its arrival at the Source for open chains, or from its departure from the reference station for closed chains, until it reaches the Sink or returns to the reference station.
- *System throughput* (SysTput): rate at which such end-to-end visits complete.
- *System queue-length* (SysQLen): total number of jobs present in the network, summed over all stations and classes.

These are obtained with the `AvgResult::CN` method, which returns a table, or with `AvgResult::CN`, which returns the same quantities as arrays.

3.2 Network object definition

3.2.1 Creating a network and its nodes

A queueing network can be described in LINE using the `Network` class constructor with a unique string identifying the model name:

```
Network model("myModel");
```

The returned object of the `Network` class offers functions to instantiate and manage resource *nodes* (stations, delays, caches, ...) visited by jobs of several types (*classes*).

A node is a resource in the network that can be visited by a job. A node must have a unique name and can either be *stateful* or *stateless*, the latter meaning that the node does not require state variables to determine its state or actions. If jobs visiting a stateful node can be required to spend time in it, the node is also said to be a *station*. A list of nodes available in `Network` models is given in Table 3.1.

We now provide more details on each of the nodes available in `Network` models.

Queue node. A `Queue` specifies a queueing station from its name and scheduling strategy, e.g.

```
Queue queue(model, "Queue1", SchedStrategy::FCFS);
```

specifies a first-come, first-served queue. It is alternatively possible to instantiate a queue using the `QueueingStation` constructor, which is merely an alias for `Queue`.

Queueing stations have by default a single server. The `set_number_of_servers` method can be used to instantiate multi-server stations. For heterogeneous servers with different capabilities, use `ServerType` to define server types and `HeteroSchedPolicy` to control which server types handle which job classes.

Table 3.1: Nodes available in `Network` models.

Node	Description
Cache	A node to switch job classes based on hits/misses in its cache
ClassSwitch	A node to switch job classes based on a static probability matrix
Delay	A station where jobs spend time without queueing
Fork	A node that forks jobs into tasks
Join	A node that joins sibling tasks into the original job
Logger	A node that logs passage of jobs
Place	A node that holds tokens in a stochastic Petri net
Queue	A node where jobs queue and receive service
Router	A node that routes jobs to other nodes
Sink	Exit point for jobs in open classes
Source	Entry point for jobs in open classes
Transition	A timed or immediate node that fires tokens between places in a stochastic Petri net

Valid scheduling strategies are specified within the `SchedStrategy` static class and include: If a strategy requires class weights, these can be specified directly as an argument to the `set_service` function or using the `set_sched_param` function, see later the description of DPS scheduling for an example.

Delay node. Delay stations, also called infinite server stations, may be instantiated either as objects of `Queue` class, with the `SchedStrategy.INF` scheduling strategy, or using the following specialized constructor

```
Delay delay(model, "ThinkTime");
```

As for queues, for readability it is possible to instantiate delay nodes using the `DelayStation` class which is an alias for the `Delay` class.

Source and Sink nodes. As seen in the M/M/1 getting started example, these nodes are mandatory elements for the specification of open classes. Their constructor only requires a specification of the unique name associated with the nodes:

```
Source source(model, "Source");
Sink sink(model, "Sink");
```

The `Source` node supports state-dependent arrival rates via the load-dependent scaling mechanism. Instead of specifying a fixed inter-arrival distribution, the effective arrival rate can vary as a function of the total number of jobs in the system. This is achieved by assigning a load-dependent rate function to the `Source` station using the `lldscaling` mechanism, in the same way load-dependent service rates are specified for queue nodes (see Section 5.2.1). When a load-dependent arrival rate is defined, the `CTMC` and `SSA` solvers automatically account for state-dependent rates during analysis, enabling accurate modeling of systems where arrival rates diminish as congestion increases (e.g., finite source populations or flow-controlled networks).

ClassSwitch node. This is a stateless node to change the class of a transiting job based on a static probabilistic policy. For example, it is possible to specify that all jobs belonging to class 1 should become of class 2 with probability 1.0, or that a transiting job of class 2 should become of class 1 with probability 0.3. This example is instantiated as follows

```
Matrix<double> C(2, 2);
C(0, 0) = 0.0; C(0, 1) = 1.0;
C(1, 0) = 0.3; C(1, 1) = 0.7;
const std::size_t cs = model.add_class_switch("ClassSwitchPoint", C);
```

The matrix is indexed by class, so the classes must be declared before the node that switches them.

Router node. This node is able to route jobs according to a specified `RoutingStrategy`, which can either be probabilistic or not (e.g., round-robin). Upon entering a `Router`, a job neither waits nor receives service; it is instead directly forwarded to the next node according to the specified routing strategy. A `Router` can be instantiated as follows:

```
Router router(model, "RouterNode");
```

An example of use of this node is given in the round-robin load-balancing tutorial of the LINE primer. Routing strategies need to be specified for each class using the `set_routing` method and valid choices are as follows

- Random routing (`RoutingStrategy.RAND`)
- Round robin (`RoutingStrategy.RROBIN`)
- Weighted round robin (`RoutingStrategy.WRROBIN`)
- Probabilistic routing (`RoutingStrategy.PROB`)
- Join-the-shortest-queue (`RoutingStrategy.JSQ`)
- Shortest queue of d (`RoutingStrategy.SQ`)

The state-dependent strategies (`RROBIN`, `WRROBIN`, `JSQ`, `SQ`) are supported by both `SSA` and `LDES`. `SQ` samples d candidate destinations with replacement and dispatches to the shortest among them. For `WRROBIN`, weights are supplied via additional arguments to `set_routing`; the dispatcher then cycles through a list in which each destination appears as many times as its integer weight. For example, assume that `oclass` is a class of jobs. In order to route jobs in this class with equal probabilities to every outgoing link we set

```
model.set_routing(router, oclass, RoutingStrategy::RAND);
```

It should be noted that `set_routing` is also available for all other nodes such as queueing stations, delays, etc. Therefore, the added value of the `Router` node is the ability to represent certain system elements that centralize the routing logic, such as load balancers.

3.2.2 Scheduling parameters

Upon setting service distributions at a station, one may also specify scheduling parameters such as weights as additional arguments to the `set_service` function. For example, if the node implements discriminatory processor sharing (`SchedStrategy.DPS`), the command

```
const mam::Map<double> aph = mam::aph_fit_mean_scv(0.2, 10.0);
queue.set_service(class2,
    Distrib<double>::map_dist(aph.D0, aph.D1, ProcessType::APH));
model.set_sched_param(queue, class2, 5.0);
```

The weight is a separate call: `set_service` installs the distribution only. `assigns` a weight 5.0 to jobs in class 2. The default weight of a class is 1.0.

3.2.3 Job classes

Jobs travel within the network placing service demands at the stations. The demand placed by a job at a station depends on the class of the job. Jobs in *open classes* arrive from the external world and, upon completing the visit, leave the network. Jobs in *closed classes* start within the network and are forbidden to ever leave it, perpetually cycling among the nodes.

Open classes

The constructor for an open class only requires the class name and the creation of special nodes called `Source` and `Sink`

```
Source source(model, "Source");
Sink sink(model, "Sink");
```

Sources are special stations holding an infinite pool of jobs and representing the external world. Sinks are nodes that route a departing job back into this infinite pool, i.e., into the source. Note that a network can include at most a single `Source` and a single `Sink`.

Once source and sink are instantiated in the model, it is possible to instantiate open classes using

```
OpenClass class1(model, "Class1");
```

LINE does not require explicitly associating source and sink with the open classes in their constructors, as this is done automatically. However, the LINE language requires explicitly creating these nodes since the routing topology needs to indicate the arrival and departure points of jobs in open classes. However, if the network does not include open classes, the user will not need to instantiate a `Source` and a `Sink`.

Closed classes

To create a closed class, we need instead to indicate the number of jobs that start in that class (e.g., 5 jobs) and the *reference station* for that class (e.g., `queue`), i.e.:

```
ClosedClass class2(model, "Class2", 5, queue);
```

The reference station indicates a point in the network used to calculate certain performance indexes, called *system performance indexes*. The end-to-end response time for a job in an open class to traverse the system is an example of a system performance index (system response time). The reference station of an open class is always automatically set by LINE to be the . Conversely, the reference station needs to be indicated explicitly in the constructor for closed classes since the point at which a class job completes execution depends on the semantics of the model.

LINE also supports a special class of jobs, called *self-looping jobs*, which perpetually loop at the reference station, remaining in their class. The following example shows the syntax to specify a self-looping job, which is identical to closed classes but there is no need later to specify routing information.

```
Network model("model");
// Block 1: nodes
Delay delay(model, "Delay");
Queue queue(model, "Queue1", SchedStrategy::FCFS);
// Block 2: classes
ClosedClass cclass(model, "Class1", 10, delay, 0);
delay.set_service(cclass, Exp(1.0));
queue.set_service(cclass, Exp(1.5));
// Block 3: topology
Routing P;
P.set(cclass, cclass, delay, delay, 0.7);
P.set(cclass, cclass, delay, queue, 0.3);
P.set(cclass, cclass, queue, delay, 1.0);
model.link(P);
```

The self-looping class of the example above is omitted here for brevity; the C++ builder declares it as `SelfLoopingClass`, alongside `OpenClass` and `ClosedClass`. Note that any routing information specified for the self-looping class will be ignored.

Mixed models

LINE also accepts models where a user has instantiated both open and closed classes. The only requirement is that, if two classes communicate by means of a class-switching mechanism, then the two classes must either be all closed or all open. In other words, classes in the same chain must either be both closed or open. Furthermore, for all closed classes in the same chain, it is required for the reference station to be the same.

The `FCFSRPrio` (FCFS preemptive-resume with priority) strategy combines priority-based scheduling with preemptive-resume semantics. When a higher-priority job arrives at a queue using this strategy, it preempts the currently running lower-priority job. The preempted job retains its accumulated service progress and resumes from where it was interrupted once all higher-priority jobs have been served. This is suitable for modeling systems such as interrupt-driven processors or prioritized packet queues. The `FCFSRPrio` strategy is supported by the CTMC, LDES, JMT, and SSA solvers.

The `SRPT` (Shortest Remaining Processing Time) strategy is a preemptive, non-priority discipline that always serves the job with the least remaining service requirement. When a new job arrives with a shorter

remaining time than the job currently in service, the server preempts the current job and begins serving the new arrival. SRPT is known to minimize mean response time in single-server queues. In LINE, SRPT is supported by the LDES and JMT solvers. The priority variant `SRPTprio` applies SRPT within priority groups: jobs are first partitioned by priority level, and within each level the SRPT discipline is used to select the next job.

```
Queue queue_pr(model, "PreemptQueue", SchedStrategy::FCFSPPRIO);
Queue queue_srpt(model, "SRPTQueue", SchedStrategy::SRPT);
```

The FSP (Fair Sojourn Protocol) strategy [65] is a preemptive single-server discipline that ranks jobs by their *virtual processor-sharing finish time*: the time at which each job would complete under a hypothetical processor-sharing shadow running over the same set of jobs. The server serves the job whose virtual finish time is smallest.

```
Queue queue_fsp(model, "FSPQueue", SchedStrategy::FSP);
```

Class switching

In LINE, jobs can switch their class while they travel between nodes (including self-loops on the same node). For example, this feature can be used to model queueing properties such as re-entrant lines in which a job visiting a station a second time may require a different average service demand than at its first visit.

A chain defines the set of reachable classes for a job that starts in the given class r and over time changes class. Since class switching in LINE does not allow a closed class to become open, and vice-versa, chains can themselves be classified into *open chains* and *closed chains*, depending on the classes that compose them.

Jobs in open classes can only switch to another open class. Similarly, jobs in closed classes can only switch to a closed class. Thus, class switching from open to closed classes (or vice-versa) is forbidden. More details about class-switching are given in Section 3.2.5.

Reference station

Before we have shown that the specification of classes requires choosing a reference station. In LINE, reference stations are properties of chains, thus if two closed classes belong to the same chain they must have the same reference station. This avoids ambiguities in the definition of the completion point for jobs in a chain.

For example, the system throughput for a chain is defined as the sum of the arrival rates at the reference station for all classes in that chain. That is, the solver counts a return to the reference station as a completion of the visit to the system. In the case of open chains, the reference station is always the `Source` and the system throughput corresponds to the rate at which jobs arrive at the sink `Sink`, which may be seen as the arrival rate seen by the infinite pool of jobs in the external world. If there is no class switching, each chain contains a single class, thus per-chain and per-class performance indexes will be identical.

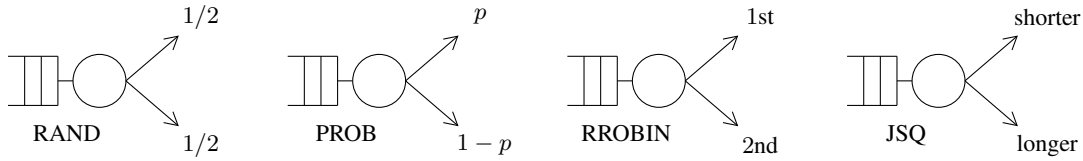


Figure 3.2: Routing strategies at a node with two outgoing links.

Reference class

Occasionally, it is possible to encounter situations where a job needs to change class while remaining inside the same station. In this case, **LINE** modifies the network automatically to introduce a class-switching node for the job to route out of the station and immediately return to it in the new class.

One complication of the approach is that, by departing the node and returning to it, the job visits the station one additional time, affecting the visit count to the station and therefore performance metrics such as the residence time. To cope with this issue, **LINE** offers a method for the class objects, called `visit`, that allows users to specify whether the visit of that class to the reference station should be considered upon computing the residence times across the network for the chain to which the class belongs. By default, all classes traversing the reference station are used in the visit count calculation.

3.2.4 Routing strategies

Figure 3.2 compares the strategies on a node with two outgoing links.

Probabilistic routing

Jobs travel between nodes according to the network topology and a routing strategy. Typically a queueing network will use a probabilistic routing strategy (`RoutingStrategy.PROB`), which requires specifying routing probabilities among the nodes. The simplest way to specify a large routing topology is to define the routing probability matrix for each class, followed by a call to the `link` function. This function will automatically add certain nodes to the network to ensure the correct class switching for jobs moving between stations (`ClassSwitch` elements).

In the running case, we may instantiate a routing topology as follows:

```
Routing P;
P.set(class1, class1, source, queue, 1.0);
P.set(class1, class1, queue, queue, 0.3); // self-loop with probability 0.3
P.set(class1, class1, queue, delay, 0.7); // transition with probability 0.7
P.set(class1, class1, delay, sink, 1.0);
P.set(class2, class2, delay, queue, 1.0); // closed class jobs start at the delay
P.set(class2, class2, queue, delay, 1.0);
model.link(P);
```

When used as arguments to a cell array or matrix, class, and node objects will be replaced by a corresponding numerical index. Normally, the indexing of classes and nodes matches the order in which they are instantiated in the model and one can therefore specify the routing matrices using this property. In this case, we would have

```
Routing P;
// class1: source -> queue, queue -> {queue, delay}, delay -> sink
P.set(class1, class1, source, queue, 1.0);
P.set(class1, class1, queue, queue, 0.3);
P.set(class1, class1, queue, delay, 0.7);
P.set(class1, class1, delay, sink, 1.0);
// class2: delay <-> queue
P.set(class2, class2, delay, queue, 1.0);
P.set(class2, class2, queue, delay, 1.0);
model.link(P);
```

Entries that are not set are zero, so the C++ builder fills the matrix pair by pair rather than row by row. Where needed, the functions return the numerical index associated with a node name, for example . Class and node names in a network *must be unique*. The list of names already assigned to nodes in the network can be obtained with the functions of the `Network` class.

It is also important to note that the routing matrix in the last example is specified between *nodes*, instead of between just stations or stateful nodes, which means that for example elements such as the `Sink` need to be explicitly considered in the routing matrix. The only exception is that `ClassSwitch` elements do not need to be explicitly instantiated in the routing matrix, provided that one uses the `link` function to instantiate the topology. Note that the routing matrix assigned to a model can be printed on the screen in a human-readable format using the function, e.g., `The port has no routing-matrix printer`. The expanded routing that `link` produces is the struct field `sn.rt`, indexed by (stateful node, class) pairs, and a caller prints whatever part of it is of interest:

```
const qn::NetworkStruct<double>& sn = model.get_struct();
const Matrix<double>& rt = sn.rt; // routing over (stateful node, class) pairs
```

Routing probabilities for Source and Sink nodes

In the presence of open classes, and in mixed models with both open and closed classes, one needs only to specify the routing probabilities *out* of the source. The probabilities out of the sink can all be set to zero for all classes and destinations (including self-loops). The solver will take care of adjusting these inputs to create a valid routing table.

Simplified definition of tandem and cyclic topologies

Tandem networks are open queueing networks with a serial topology. LINE provides functions that ease the definition of tandem networks of stations with exponential service times. For example, Tutorial 1 of the primer, on the M/M/1 queue, illustrates a simplified way to specify a serial routing topology, i.e., The C++ builder has

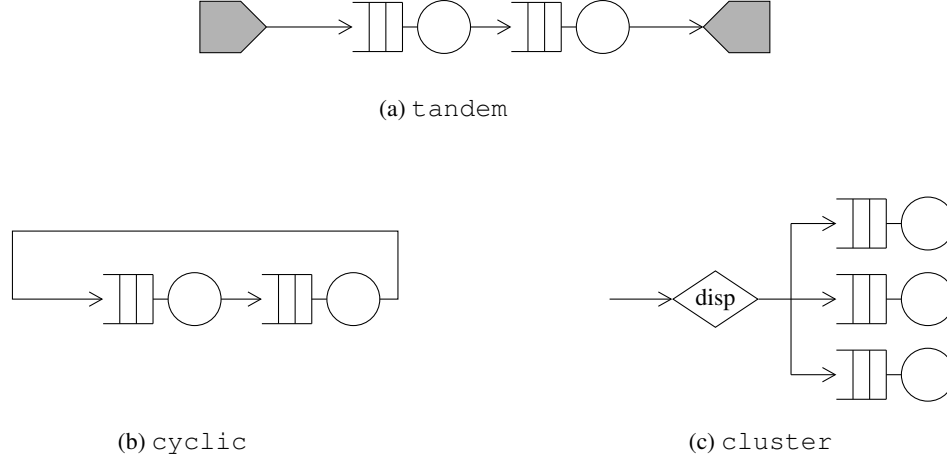


Figure 3.3: The topologies produced by the simplified constructors.

no `serialRouting` helper: a serial topology is written as the two entries `P.set(source, queue, 1.0)` and `P.set(queue, sink, 1.0)`. In a similar fashion, we can also rapidly instantiate a tandem network consisting of stations with PS and INF scheduling as follows. The parameterized topology generators are not part of the C++ port: a tandem network is built node by node with `add_queue / add_delay` and linked with an explicit routing matrix, or read from a `model.json` file produced by one of the other implementations. The above snippet instantiates an open network with two queueing stations (PS), two delay stations (INF), and exponential distributions with the given inter-arrival rates and mean service times. The functions provide static constructors for networks with other combinations of scheduling policies, namely only PS, only FCFS, or FCFS and INF.

These functions only require replacing the arrival rate vector A by a vector N specifying the job populations for each of the closed classes, e.g., As for the tandem generators, `cyclicPs` and its variants have no C++ counterpart; the cyclic topology is the routing matrix `P.set(i, i+1, 1.0)` closed by `P.set(M, 1, 1.0)`.

Figure 3.3 shows what each constructor builds.

Simplified definition of cluster topologies

A cluster is a network in which jobs flow from an arrival point through a dispatcher (modelled by a `Router`) that distributes them across M parallel servers, after which they leave the system. `LINE` provides static constructors that build the open topology `Source → Dispatcher → Server[1..M] → Sink` and its closed counterpart `Think(Delay) → Dispatcher → Server[1..M] → Think`, as well as a mixed topology in which both families share the dispatcher and the servers. Service times at the servers are exponential and the dispatcher routes jobs according to `RoutingStrategy.RAND` by default. For example, an open PS cluster and a closed FCFS cluster can be created as follows: The

cluster generators are not part of the C++ port either. A cluster is built explicitly: one `add_queue` per server, `set_number_of_servers` for its multiplicity, and a routing matrix that spreads the arrivals over the servers. For richer configuration, LINE also exposes a chainable builder with setters `setDispatching`, `setScheduling`, `setStationServers`, `setClosed`, and `setMixed`, together with helpers `compareDispatching`, `compareScheduling`, `sweepArrivalRate`, and `sweepNumStations` for parametric studies. Worked examples are available under (e.g. `cl_basic`, `cl_closed`, `cl_compare`, `cl_multiclass`, `cl_sweep`).

3.2.5 Class switching

Depending on the specified probabilities, a job will be able to switch its class only among a subset of the available classes. Each subset is called a *chain*. Chains are computed in LINE as the weakly connected components of the routing probability matrix of the network when this is seen as an undirected graph. The function `produceChains` produces the list of chains for the model, inclusive of a list of their composing classes.

The definition of class switching in a model is integrated in the specification of the routing between stations as described in the next subsection.

Probabilistic class switching

In models with class switching and probabilistic routing at all nodes, a routing matrix is required for each possible pair of source and target classes. For instance, suppose that in the previous example the job in the closed class `class2` switches into a new closed class (`class3`) while visiting the queue node. We can specify this routing strategy as follows:

```
ClosedClass class3(model, "Class3", 0, queue, 0);

Routing P;
P.set(class1, class1, source, queue, 1.0);
P.set(class1, class1, queue, queue, 0.3); // self-loop
P.set(class1, class1, queue, delay, 0.7);
P.set(class1, class1, delay, sink, 1.0);
P.set(class2, class3, delay, queue, 1.0); // class switching on the edge
P.set(class3, class2, queue, delay, 1.0);
model.link(P);
```

Importantly, LINE assumes that a job switches class an instant *after* leaving a station, thus the performance metrics of a class at the node refer to the class that jobs had upon arrival to that node.

3.2.6 Service and inter-arrival time processes

A number of statistical distributions are available to specify job service times at the stations and inter-arrival times from the station. The class `Markovian` offers distributions that are analytically tractable, which are defined using absorbing Markov chains consisting of one or more states (*phases*) and called phase-type distributions. They include as special cases the distributions shown in Table 3.4.

Marked and time-inhomogeneous families: what accepts them. MPH is stationary and renewal and lowers to the same M3A cell as `MarkedMAP`, so every solver that accepts a marked arrival process accepts it too: CTMC, MAM, MVA, SSA and LDES. The schedule-bearing families are narrower. `MAPt` and `Pht` are declared only by FLD under method `kp` and by LDES, and `MMAPt`, `MPht` and `BMMAPt` by LDES alone. A fluid answer for a marked schedule is obtained instead by declaring the K per-mark marginal `MAPt` streams, which is exact in the mean limit.

In all of the marked families a mark routes *arrivals* only. At a `Source` the mark selects the class of the arriving job, bound through `setMarkedArrival`; as a service process a marked law is sampled for its duration and the mark is discarded, so no class switch occurs on completion. A `MAPt`, `Pht`, `MMAPt` or `MPht` service is moreover walked forward from the service start epoch, which is exact only while service is delivered continuously at unit rate. LDES therefore admits such a service at INF and at the non-preemptive disciplines, and refuses processor sharing, preemption, load dependence and heterogeneous server pools by name rather than approximating them.

A `BMMAPt` carries a third label, and it is the one that changes how many jobs move. At a `Source` an epoch releases a whole *batch* of jobs, all of the class its mark names, so the job rate is $\sum_c \sum_b b \theta D_{cb} e$ and exceeds the epoch rate whenever a batch larger than one has mass; the `Source` row of the throughput table reports the job rate, so that it balances against the stations downstream. As a *service* process it is a batch Markovian service process, that is a bulk server: the station has no per-job service time at all, only a station-level firing clock armed when it becomes busy, and one firing completes $\min(b, n)$ of the n jobs present, with the mark discarded as for any marked service. A bulk server is a single server by construction, so LDES admits it only at a single-server single-class FCFS queue and refuses multiserver, multiclass, preemptive, polling, pass-and-swap, setup, load-dependent, pooled, immediate-feedback and blocking variants by name. The same holds of a `BMAP` service, which is the stationary case of exactly this. A batch process and `setArrivalBatch` are mutually exclusive on one stream: the process already carries its own sizes, so a second batch law bolted on top would multiply the two.

Fitting a distribution

The `function` is available for all distributions that inherit from the `Markovian` class. This function provides exact or approximate matching of the first two moments, depending on the theoretical constraints imposed by the distribution. For example, an Erlang distribution with $SCV=0.75$ does not exist, because in a n -phase Erlang it must be $SCV=1/n$. In a case like this, `function` will return the closest approximation, e.g., a 2-phase Erlang ($SCV=0.5$) with unit mean. The Erlang distribution also offers a function (μ, n) , which instantiates a n -phase Erlang with given mean μ .

In distributions that are uniquely determined by more than two moments, `function` chooses a particular assignment of the residual degrees of freedom other than mean and SCV. For example, `HyperExp` depends on three parameters, therefore it is insufficient to specify mean and SCV to identify the distribution. Thus, `function` automatically chooses to return a probability of selecting phase 1 equal to 0.99. Compared to other choices, this particular assignment corresponds to a higher probability mass in the tail of the distribution. `function` instead assigns p in a two-phase hyper-exponential distribution so that $p/\mu_1 = (1 - p)/\mu_2$.

Some distributions admit no closed-form two-moment inverse and are fitted by an approximation, in which case the mean is matched exactly and the SCV only approximately. This is the case for `Weibull`, whose shape parameter is obtained from the Justus et al. (1976) relation $k = c^{-1.086}$ with $c = \sqrt{\text{SCV}}$, the scale then being chosen so that the mean is exact. The relation was published for $1 \leq k \leq 10$, i.e. $\text{SCV} \leq 1$, where the realized SCV is within about 3% of the request; outside that range the error grows quickly, reaching 12% at $\text{SCV} = 2$ and 48% at $\text{SCV} = 4$. Use a phase-type distribution instead when the second moment must be matched exactly at high variability.

Three-moment fitting for APH and PH. The acyclic phase-type (APH) and general phase-type (PH) classes additionally expose static factory methods that fit a distribution from the first three moments, providing finer control over higher-order behaviour such as burstiness and skewness of the service time. For both APH and PH, three equivalent entry points are available:

- fits the distribution from the first three standard moments (mean, squared coefficient of variation and skewness).
- accepts the first three central moments (mean, variance and skewness).
- accepts the first three raw moments $E[X^k]$ for $k = 1, 2, 3$.

Each variant returns an APH (resp. PH) instance whose parameters reproduce the requested moments, falling back to an exponential or immediate distribution when the requested mean is below `GlobalConstants.FineTol`. For example,

```
// APH fitted to the first three RAW moments E[X], E[X^2], E[X^3]
const mam::Map<double> aph = mam::aph_fit(1.0, 5.0, 45.0).aph;
queue.set_service(class1,
                  Distrib<double>::map_dist(aph.D0, aph.D1, ProcessType::APH));
```

`aph_fit` takes raw moments; `mam::aph_fit_mean_scv(mean, scv)` is the two-moment fit, and `Distrib<double>::erlang_fit(mean, scv)` the Erlang one. The Coxian class supports the same three-moment family of methods through its inherited APH interface. Both APH and PH also offer for two-moment fitting.

Inspecting and sampling a distribution

To verify that the fitted distribution has the expected mean and SCV it is possible to use the `Distrib::mean` and `Distrib::scv` functions, e.g.,

```
const Distrib<double> dist = Exp(1.0);
std::printf("%g\n", dist.mean);
std::printf("%g\n", dist.scv);
```

The mean and the SCV are fields of the distribution, computed by its factory; `dist_moment(dist, k)` returns the k -th raw moment. Moreover, the `sample` function can be used to generate values from the obtained distribution, e.g. we can generate 3 samples as The C++ port does not draw pseudo-random variates

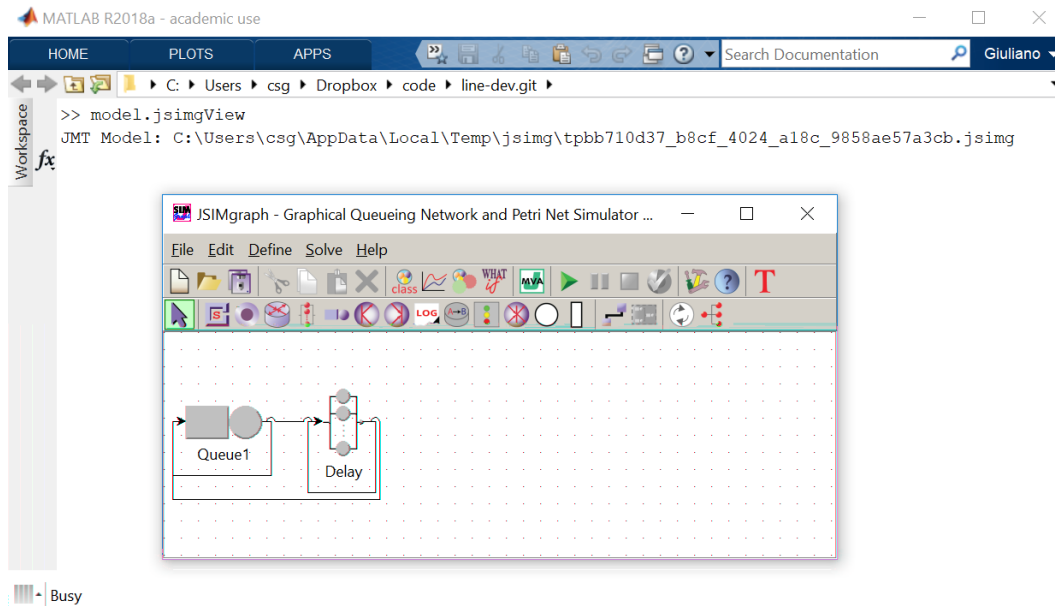


Figure 3.4: jsimView function

from a distribution object: sampling exists only inside the simulators (`ssa::solver_ssa`) and in the CTMC sample-path getter (`ctmc::solver_ctmc_sample`). The `dist_cdf` and `dist_cdf` functions return the cumulative distribution function at the specified point or within a range, e.g.,

```
std::printf("%g\n", dist_cdf(dist, 5.0) - dist_cdf(dist, 2.0));
```

There is one entry point, `dist_cdf`; the interval probability is the difference of its values. For more advanced uses, the distributions of the `Markovian` class also offer the possibility to obtain the standard (D_0, D_1) representation used in the theory of Markovian arrival processes by means of the `dist_to_map` function [17].

3.3 Debugging and visualization

JSIMgraph is the graphical simulation environment of the JMT suite. LINE can export models to this environment for visualization purposes using the command `The JSIM viewers belong to the JMT bridge, which the C++ port does not implement: it neither writes JSIM XML nor launches an external tool. An example is shown in Figure 3.4. Using a related function, , it is also possible to export the model to the JSIMwiz environment, which offers a wizard-based interface.`

The port has no graph export. The topology it works on is the struct itself: `sn.nodes`, `sn.stations` and the routing table `sn.rt` returned by `model.get_struct()`.

3.3.1 TikZ export

The network topology can also be exported as $\text{\LaTeX}$ source, so that a model diagram can be included directly in a paper or a report. The TikZ exporter is provided by the JAR and is not part of the C++ port.

Table 3.2: Scheduling strategies available in LINE

Strategy	Code	Description
Discriminatory processor-sharing	DPS	Processor sharing with class-dependent weights
Earliest due date	EDD	Job with earliest deadline is served first (non-preemptive)
Earliest deadline first	EDF	Preemptive variant of EDD: an arriving job with an earlier deadline preempts the in-service job
Foreground-background (Least attained service)	FB / LAS	Preemptive; serves the job with least attained service. FB and LAS are synonyms
First-come, first-served	FCFS	Jobs are served in order of arrival (non-preemptive)
FCFS preemptive-independent	FCFSPI	FCFS where a higher-priority arrival preempts the in-service job and the preempted job restarts service from scratch
FCFS preemptive-resume	FCFSPR	FCFS where a higher-priority arrival preempts the in-service job and the preempted job resumes from where it stopped
Fair sojourn protocol	FSP	Preemptive; serves the job with smallest virtual processor-sharing finish time [65]. Supported by LDES
Generalized processor-sharing	GPS	Processor sharing with configurable weights per class
Infinite-server	INF	All jobs receive service simultaneously (delay station)
Last-come first-serve	LCFS	Most recently arrived job is served first (non-preemptive)
LCFS preemptive-independent	LCFSPI	LCFS where an arriving job preempts the in-service job and the preempted job restarts service from scratch
LCFS preemptive-resume	LCFSPR	LCFS where an arriving job preempts the in-service job and the preempted job resumes from where it stopped
Longest expected processing time	LEPT	Job with longest expected processing time is served first
Longest job first	LJF	Job with longest service requirement is served first
Limited processor sharing	LPS	Processor sharing with limit on concurrent jobs
Longest remaining processing time	LRPT	Preemptive scheduling that always serves the job with the largest remaining service requirement; the dual of SRPT
Polling	POLLING	Server cycles through queues in round-robin fashion
Processor-sharing	PS	Server capacity is shared equally among all present jobs
Preemptive shortest job first	PSJF	Preemptive scheduling that prioritises jobs by their original service requirement (not the residual one), avoiding the constant churn of SRPT but still favouring short jobs
Shortest expected processing time	SEPT	Job with shortest expected processing time is served first
Shortest elapsed time first	SETF	Non-preemptive variant of FB/LAS: at each completion the next job in service is the one with least attained service, but it is not preempted by later-arriving jobs
Service in random order	SIRO	Jobs are selected randomly for service
Shortest job first	SJF	Job with shortest service requirement is served first

Table 3.3: Priority scheduling strategies available in LINE

Strategy	Code	Description
FCFS with priority	FCFSPRIO	FCFS within priority classes, non-preemptive. Also named HOL (head-of-line), the two being aliases
FCFS preemptive-resume with priority	FCFSRPRIO	FCFS with preemptive-resume priority
FCFS preemptive-independent with priority	FCFSPIPRIO	FCFS with preemptive-independent priority
LCFS with priority	LCFSPRIO	LCFS within priority classes, non-preemptive
LCFS preemptive-resume with priority	LCFSRPRIO	LCFS with preemptive-resume priority
LCFS preemptive-independent with priority	LCFSPIPRIO	LCFS with preemptive-independent priority
Processor-sharing with priority	PSPRIO	PS within priority classes
Discriminatory PS with priority	DPSPRIO	DPS within priority classes
Generalized PS with priority	GPSPRIO	GPS within priority classes
SRPT with priority	SRPTPRIO	Shortest remaining processing time within priority classes

Table 3.4: Phase-type distributions available in LINE

Distribution	Description
Exponential	$\text{Exp}(\lambda)$, where λ is the rate of the exponential
n -phase Erlang	$\text{Erlang}(\alpha, n)$, where α is the rate of each of the n exponential phases
2-phase hyper-exponential	$\text{HyperExp}(p, \lambda_1, \lambda_2)$, that returns an exponential with rate λ_1 with probability p , and an exponential with rate λ_2 otherwise
n -phase hyper-exponential	$\text{HyperExp}(p, \lambda)$, that builds a n -phase hyper-exponential from a rate vector $\lambda = [\lambda_1, \dots, \lambda_n]$ and phase selection probabilities $p = [p_1, \dots, p_n]$
2-phase Coxian	$\text{Coxian}(\mu_1, \mu_2, \phi_1)$, which assigns phases μ_1 and μ_2 to the two rates, and completion probability from phase 1 equal to ϕ_1 (the probability from phase 2 is $\phi_2 = 1.0$)
n -phase Coxian	$\text{Coxian}(\mu, \phi)$, which builds an arbitrary Coxian distribution from a vector $\mu = [\mu_1, \dots, \mu_n]$ of n rates and a completion probability vector $\phi = [\phi_1, \dots, \phi_n]$ with $\phi_n = 1.0$
n -phase acyclic phase-type	$\text{APH}(\alpha, T)$, which defines an acyclic phase-type distribution with initial probability vector $\alpha = [\alpha_1, \dots, \alpha_n]$ and transient generator T
n -phase matrix exponential	$\text{ME}(\alpha, A)$, which defines a matrix exponential distribution with initial vector $\alpha = [\alpha_1, \dots, \alpha_n]$ (possibly non-normalized) and matrix parameter A . This generalizes phase-type distributions by allowing non-stochastic initial vectors [16]
n -phase concentrated matrix exponential	$\text{CME}(m, n)$, a subclass of ME defining the matrix exponential distribution of odd order $n = 2k + 1$ with mean m whose squared coefficient of variation is minimal for that order [79]. Its SCV decays as $O(1/k^2)$, hence well below the bound $1/n$ attainable by a phase-type distribution of the same order, which makes it an inexpensive representation of a near-deterministic timing. Available orders are returned by <code>CME.getSupportedOrders()</code> , and <code>CME.fitMeanAndSCV(m, c^2)</code> returns the lowest order meeting a target SCV
Rational arrival process	$\text{RAP}(H_0, H_1)$, which defines a correlated arrival process with hidden transition matrix H_0 and visible transition matrix H_1 , where $H_0 + H_1$ forms a generator matrix. This generalizes the Markovian arrival process (MAP) [3]
Markovian arrival process	$\text{MAP}(D_0, D_1)$, which defines a point process with phase-dependent arrival rates via matrices D_0 (hidden transitions) and D_1 (arrivals)
Markov-modulated Poisson process	$\text{MMPP2}(\lambda_1, \lambda_2, \sigma_1, \sigma_2)$, a two-phase MMPP with arrival rates λ_i and switching rates σ_i
Batch Markovian arrival process	$\text{BMAP}(D)$, which generalizes MAP to allow batch arrivals via a set of arrival matrices $D = \{D_0, D_1, \dots, D_k\}$
General phase-type	$\text{PH}(\alpha, T)$, which defines a phase-type distribution with initial probability vector α and transient generator T , allowing cyclic phase transitions
Markov-modulated deterministic process	$\text{MMDP}(Q, R)$, which defines a fluid process with deterministic flow rates modulated by a background CTMC with generator Q and diagonal rate matrix R . This is the deterministic analogue of MMPP and is supported by the Markovian fluid queue solver (FLD with method <code>mfq</code>)
2-state Markov-modulated deterministic process	$\text{MMDP2}(r_0, r_1, \sigma_0, \sigma_1)$, a two-state MMDP with deterministic rates r_0 and r_1 in the two phases, and switching rates σ_0 (from state 0 to 1) and σ_1 (from state 1 to 0)
Marked Markovian arrival process	$\text{MarkedMAP}(D, K)$, which defines a Markovian arrival process with K marking (color) types represented via the M3A format $D = \{D_0, D_1, D_{11}, \dots, D_{1K}\}$. Supports multi-class arrival flows with correlated inter-arrival times and class assignment
Marked Markov-modulated Poisson process	$\text{MarkedMMPP}(D, K)$, a restriction of MarkedMAP to diagonal D_{1k} matrices, i.e., a MMPP with K arrival classes. Represents a Poisson arrival stream partitioned into K distinguishable job types modulated by a shared background Markov chain
Marked phase-type	$\text{MPH}(\alpha, S, \{s_1, \dots, s_K\})$, a phase-type distribution whose absorption is labelled: the chain runs on the transient phases of (α, S) and leaves through one of K exit vectors, so a completion carries both a duration and a mark. The partition identity $\sum_k s_k = -Se$ makes the K exits account for exactly the absorption that S leaves. It is the renewal special case of MarkedMAP , since $D_0 = S$ and $D_{1k} = s_k \alpha$ [75]
Discrete-time Markovian arrival process	$\text{DMAP}(D_0, D_1)$, the discrete-time counterpart of MAP. Matrices D_0 (no-arrival transitions) and D_1 (arrival transitions) are sub-stochastic with $D_0 + D_1$ having row sums equal to 1. Useful for modelling slotted or frame-based arrival processes
Time-inhomogeneous Markovian arrival process	$\text{MAPt}(\{t_j\}, \{D_0^{(j)}\}, \{D_1^{(j)}\}, \text{cyclic})$, a MAP whose matrices are piecewise constant on a schedule of breakpoints $t_1 < \dots < t_n$, realising elementwise the locally integrable $D_0(t), D_1(t)$ of [90]. When <i>cyclic</i> is set the schedule repeats with period t_n . With one phase it is an NHPP, with one segment a MAP. All segments must share one sparsity pattern
Time-inhomogeneous phase-type	$\text{PHt}(\{t_j\}, \{\alpha^{(j)}\}, \{S^{(j)}\}, \text{cyclic})$, the phase-type counterpart of MAPt , with entry law and sub-generator piecewise constant on a schedule of the same form
Time-inhomogeneous marked Markovian arrival process	$\text{MMAPt}(\{t_j\}, \{D_0^{(j)}\}, \{D_{1k}^{(j)}\}, \text{cyclic})$, MAPt and MarkedMAP crossed: every segment carries K mark blocks, with $D_0^{(j)} + \sum_k D_{1k}^{(j)}$ a generator. The mark of an arrival is drawn from the segment in force at that epoch rather than from a time average, so the class mix may vary over the schedule even at constant aggregate rate. With $K = 1$ it reproduces the MAPt of the same matrices
Time-inhomogeneous	$\text{MPHt}(\{t_i\}, \{\alpha^{(j)}\}, \{S^{(j)}\}, \{s_k^{(j)}\}, \text{cyclic})$, an MPH whose parameters follow a schedule. It is stored lowered to

Chapter 4

Basic analysis methods

The metrics defined here are the ones a queueing model is usually built to obtain: the mean number of jobs at a station, the fraction of time its servers are busy, the time a job spends there, and the rate at which work flows through it. Every solver of Section 1.2 returns them through the same handles, so the calls below are independent of the solver in use. The analyses that go beyond averages, and the ones that need a state or a sample path to be named, are deferred to Chapter 7.

4.1 Performance metrics

As discussed earlier, LINE supports a set of steady-state and transient performance metrics. Table 4.1 summarizes the definition of the associated random variables. For each metric, one or more analysis types may be available, which are extensively discussed in the next sections.

4.2 Steady-state analysis

4.2.1 Station average performance

LINE decouples network specification from its solution, allowing to evaluate the same model with multiple solvers. Model analysis is carried out in LINE according to the following general steps:

Step 1: Definition of the model. This proceeds as explained in the previous chapters.

Step 2: Instantiation of the solver(s). A solver is an instance of the `Solver` class. LINE offers multiple solvers, which can be configured through a set of common and individual solver options. For example, the C++ port implements the native solvers only. The available entry points are `mva::solver_mva_run`, `nc::solver_nc_run`, `ctmc::solver_ctmc_run`, `mam::solver_mam_run`, `fluid::solver_fluid_run`, `ssa::solver_ssa`, `ba::solver_ba_run`, and the meta-solvers `env::solver_env`,

Table 4.1: Performance metrics

Metric	Acronym	Description
Queue-length	QLen	Number of jobs of class r (or chain- c) residing at a node i
Utilization	Util	Utilization of class- r (or chain- c) jobs at node i , scaled in $[0,1]$ for multi-server nodes, equal to QLen at infinite server nodes
Response time	RespT	Time that a class- r (or chain- c) jobs spends for a single visit at node i
Residence time	ResidT	Cumulative time that a class- r (or chain- c) jobs spends across all visits at node i
Arrival rate	ArvR	Arrival rate of class- r (or chain- c) jobs at node i
Throughput	Tput	Throughput of class- r (or chain- c) jobs at node i
System Response time	SysRespT	For an open chain c , this is the time from leaving the source to arriving at the sink for <i>any</i> class in the chain. For a closed chain c , this is the interval of time between two successive visits to the reference station in any two <i>completing classes</i> within the chain.
System Throughput	SysTput	For an open chain c , this is the departure rate towards the sink for <i>any</i> class in the chain. For a closed chain c , this is the rate of arrival of <i>completing classes</i> in the chain at the reference station.
System queue length	SysQLen	Total mean number of jobs in chain c across all stations. For a closed chain, equal to the total number of jobs in the chain. For an open chain, related to SysRespT and SysTput through Little's law: $\text{SysQLen}(c) = \text{SysTput}(c) \cdot \text{SysRespT}(c)$.
Fork-join queue length	FJQLen	Number of jobs present within a fork-join scope, counting from the moment they are dispatched at the fork until synchronization is completed at the join. Measured per fork-join pair.
Fork-join response time	FJRespT	Time from fork dispatch to completion of synchronization at the join, measuring the end-to-end latency of the parallel section across all spawned branches.
Marginal queue-length distribution	ProbMarg	Probability mass function $P(n_{i,r} = k)$ of the number of class- r jobs at station i , marginalized over all other classes and stations. Returned by <code>solver_mva_get_prob_marg</code> .
Joint total queue-length law	ProbSysMarg	Joint probability $P(n_1 = k_1, \dots, n_M = k_M)$ that every station holds a given TOTAL number of jobs, all classes summed out. Exact for closed product-form networks with load-independent single-server queues and infinite servers, and returned by <code>solver_nc_getprob_sys_marg</code> under the NC solver only.

`ln::solver_ln` and `uq::solver_uq_run`. The external tools are reached through `jmt::` (JMT) and `lqns::` (`lqns/lqsim` for a layered model, `qnsolver` for a flat one, via `lqns::solve_network_run_analyzer`). `lqns::solve_network_run_analyzer` returns a handle to a simulation-based solver based on JMT, configured with default options.

Step 3: Solution. Finally, this step solves the network and retrieves the concrete values for the performance indexes of interest. This may be done as follows, e.g.,

```
mva::MvaOptions opt;
const mva::AvgResult<double> res =
    mva::solver_mva_run_analyzer(model.get_struct(), opt, Matrix<double>());
const Matrix<double>& QN = res.QN; // queue length, (station, class)
const Matrix<double>& UN = res.UN; // utilization
```

```
const Matrix<double>& RN = res.RN; // response time, per visit
const Matrix<double>& TN = res.TN; // throughput
const Matrix<double>& AN = res.AN; // arrival rate
const Matrix<double>& WN = res.WN; // residence time, summed on visits
```

A C++ solve returns every mean measure in one `AvgResult` rather than through one getter per metric. Alternatively, all the above metrics may be obtained in a single method call as

```
const mva::AvgResult<double> res =
    mva::solver_mva_run_analyzer(model.get_struct(), opt, Matrix<double>());
// res.QN, res.UN, res.RN, res.TN, res.AN, res.WN, res.CN, res.XN
```

In the methods above, LINE assigns station and class indexes (e.g., i , r) in order of creation in order of creation of the corresponding station and class objects. However, large models may be easier to debug by checking results using class and station names, as opposed to indexes. This can be done either by requesting LINE to build a table with the result

```
SolverMVA solver(model);
AvgTable avg_table = solver.avg_table();
avg_table.print();
AvgTable queue_class = avg_table(queue, cclass);
double qlen = avg_table.get("QLen", "MyQueue", "MyClass");
```

The C++ table supports object-based filtering, named-cell access and stream output. which however tends to be a rather slow data structure to use in case of repeated invocations of the solver, or by indexing the matrices returned by `avg` using the model objects. That is, if the first instantiated node is `queue` with name `MyQueue` and the second instantiated class is `cclass` with name `MyClass`, then the following commands are equivalent

```
res.QN(0, 1); // 0-based indexing
res.QN(model.station_index(queue) - 1, cclass - 1);
```

Node and class handles are 1-based indices returned by the `add_*` calls, whereas the result matrices are 0-based, so a handle is converted by subtracting one; `station_index` maps a node index to its station index. Similar methods are defined to obtain aggregate performance metrics at chain level at each station, namely `AvgResult::QN` for queue-lengths, `AvgResult::UN` for utilizations, `AvgResult::RN` for response times, `AvgResult::TN` for throughputs, and the `sn_deaggregate_chain_results` method to obtain all the previous metrics.

Tabular result variants. For ease of inspection, the average results can also be retrieved as labelled tables that report station, node, class, and chain identifiers alongside the numerical values. Seven tabular variants are provided, each tailored to a different aggregation level:

- `avg_table`: one row per (*station*, *class*) pair, listing queue-length, utilization, response time, residence time, arrival rate, and throughput. This is the default view and matches the output of `AvgTable`.

- `sn_deaggregate_chain_results`: one row per (*station*, *chain*) pair, with the class-level entries of a chain aggregated into the chain-level metric. Use it when the chain decomposition is more meaningful than the per-class breakdown.
- `node_metrics`: one row per (*node*, *class*) pair. Unlike the station-level table it also covers non-station nodes such as `Source`, `Sink`, `Router`, `ClassSwitch`, `Fork` and `Join`.
- `AvgResult::CN`: one row per chain, reporting the system response time `SysRespT` and system throughput `SysTput` at the reference station of the chain. It is the natural complement of `AvgResult::CN`.
- `cache_metrics_of`: one row per (*node*, *class*) of each `Cache` node, reporting the true-hit, delayed-hit and miss probabilities, which sum to one, plus per-list rows wherever the algorithm resolves the per-level occupancy, which the exact cache algorithms and the fluid `FLD` analyzer both do. The `ListCost` column reports the mean storage cost held by the list, `NaN` unless item sizes were set. A delayed hit is a request served by a fetch that is already in flight; `CTMC` splits it exactly, while the simulation solvers fold it into the hit class. The `Item` column names the item a class reads in a cache network, where each item is carried by a dedicated class, and is zero otherwise (see 6.2.4).
- `LdesResult::avgOrbitSize`: one row per (*station*, *class*) pair of each retrial station, reporting the mean orbit length in the `Orbit` column. It separates the orbiting population from `QLen`, which also counts the jobs in service, and is empty for models without retrial stations (see 5.3.5).
- `CacheNodeMetrics::itemprob`: one row per (*node*, *item*, *list*) of each `Cache` node, reporting the list capacity `ListCap`, the item storage cost `Size`, the steady-state probability `Prob` that the item resides in that list, and the expected cost `Cost=Size·Prob` it contributes there, which summed over items reproduces the list's `ListCost`. `Size` and `Cost` are `NaN` unless item sizes were set. Two further columns give the delayed-hit backlog, `DelayedHitQLen` and `DelayedHitQLenFull`, which `CTMC` computes exactly and other solvers leave as `NaN`.

Method aliases. MATLAB and Java expose progressively shorter aliases for the methods used most often during interactive analysis, all forwarding to the same canonical implementation. Table 4.2 lists the most common ones. The same short-alias convention (`get`-prefix dropped, then `Table` shortened to `T`, then reduced to initials) extends to the per-chain and per-node-chain variants of Section 4 (e.g., `→ AvgResult::QN`) and to the ensemble-solver methods used by `LN`, `UQ`, and `ENV` (e.g., `get_ensemble_avg → get_ensemble_avg`).

4.2.2 The utilization convention

Throughout `LINE`, `Util` is the mean fraction of a station's service capacity held by jobs of the given class, so that a station's total utilization lies in $[0, 1]$ irrespective of how many servers it owns. Three regimes must be distinguished, and they must not be mixed.

Table 4.2: Common method aliases

Canonical method	Short alias(es)	Description
avg_table	avg_table	Average metrics table per (station, class); default view, matches AvgTable Average metrics table per (node, class); includes non-station nodes Average metrics table per (station, chain) Average metrics table per (node, chain)
sn_deaggregate_chain_results	sn_deaggregate_chain_results	System-level response time and throughput per chain
AvgResult::CN	AvgResult::CN	QLen, Util, RespT, Tput, ArvR, ResidT matrices per (station, class)
solver_mva_run	AvgResult	Chain-aggregated counterpart of AvgResult
sn_deaggregate_chain_results	sn_deaggregate_chain_results	System-level counterpart of AvgResult
AvgResult::CN	AvgResult::CN	Response time cumulative distribution function
solver_ctmc_cdf_respt	solver_ctmc_cdf_respt	Transient (time-dependent) average trajectories
solver_ctmc_transient_analyzer	solver_ctmc_transient_analyzer	Steady-state marginal state probability
solver_mva_get_prob_marg	solver_mva_get_prob_marg	Queue-length (and FCFS response-time) moments per (station, class) Queue-length moments per (station, chain) Total-queue-length moments per station Derivatives of mean measures w.r.t. service rate per (station, class) Cache hit/miss summary per (node, class) Per-item cache hit probabilities Retrial-orbit occupancy per (node, class) Loss rate and loss ratio per (station, class) Loss rate and loss ratio per finite-capacity region
sens_visit_mask	solver_sensitivity_table	

Load-independent stations. The Utilization Law applies,

$$U_{ir} = X_r D_{ir} / c_i = T_{ir} / (\mu_{ir} c_i), \quad (4.1)$$

with c_i the number of servers at station i , D_{ir} the service demand and X_r the class- r throughput. The division by c_i is what keeps a c -server station comparable to a single-server one: an M/M/3 offered $\lambda = 1.2$ with unit mean service reports $U = 0.4$, not 1.2. Infinite-server (delay) stations are the exception and report $U = \text{QLen}$, the mean number of jobs in service, which may exceed one.

Load-dependent and class-dependent stations. When a station's rate varies with its occupancy (setLoadDependence), the capacity that normalizes the utilization is the effective one, $c_i^{eff} = \max(c_i, \max_n \alpha_i(n))$, since a load-dependent station emulates multiserver capacity while c_i remains one. A model built both ways – as a true c -server station and as a load-dependent station with $\alpha(n) = \min(n, c)$ – therefore reports the same utilization. For class-dependent rates the utilization is normalized the same way,

as $U_{ir} = T_{ir}S_{ir}/p_{ir}$, where p_{ir} is the *peak rate scaling* per class that the user declares explicitly when the station is created (`setClassDependence(beta, peakRatePerClass)`), stored in the `NetworkStruct` field `cdscalingpeak`, see Section 5.2.5; a scalar peak is broadcast to every class. This makes the class-dependent utilization follow the same $T_{ir}S_{ir}/c$ convention as an ordinary multiserver station.

Order-independent and pass-and-swap stations. Here $U_{ir} = E[s_{ir}]/c_i$, the mean number of class- r jobs receiving a strictly positive rank rate. Note that s_{ir} counts *jobs*, not servers, so a job served concurrently by several compatible servers counts once; this coincides with the offered load only when a job engages a single server. CTMC and LDES observe s_{ir} directly on the ordered state, while MVA and NC recover its mean from a balanced-fairness recursion, agreeing with CTMC to machine precision. A multi-server station promoted internally to the OI representation for solution purposes (as MVA does for product-form multiserver queues) is still reported with the load-independent formula above; the OI convention applies only to stations the user declares as OI or PAS.

4.2.3 Station response time distribution

Several solvers (CTMC, FLD, JMT, NC, and MAM) support the computation of response time distributions for individual classes through the `solver_ctmc_cdf_respt` function. The function returns the response time distribution for every station and class. The response-time distribution is reached through the front end, `line-cli -f model.json -s fluid -a cdf` (also available for `-s ctmc` and `-s nc`). In-process, the fluid passage time is computed by `fluid::fluid_passage_time`, which takes the steady-state fluid state as its input.

For the isolated single-server queue under egalitarian processor sharing, the exact sojourn time distribution [123, 169] is available separately at the API level as .

4.2.4 Station queue-length and response-time moments

Queue-length and response-time moments beyond the mean are not part of the C++ port: `AvgResult` carries first moments only.

4.2.5 Loss rates and loss ratios

Loss rates and loss ratios are not tabulated by the C++ port; the finite-capacity model is solved, but the dropped-traffic table has no counterpart.

4.2.6 System average performance

LINE also allows users to analyze models for end-to-end performance indexes such a system throughput or system response time. However, in models with class switching the notion of system-wide metrics can be ambiguous. For example, consider a job that enters the network in one class and departs the network

in another class. In this situation one may attribute system response time to either the arriving class or the departing one, or attempt to partition it proportionally to the time spent by the job within each class. In general, the right semantics depends on the aim of the study.

LINE tackles this issue by supporting only the computation of system performance indexes *by chain*, rather than by class. Since a job that switches classes remains in the same chain by definition, the system metrics can be attributed to the chain without ambiguity. The solver functions `AvgResult::CN` and `AvgResult::CN` return system response time and system throughput per chain as observed: (i) upon arrival to the sink, for open classes; (ii) upon arrival to the reference station, for closed classes. The system queue length (`SysQLen`) is related to these two quantities through Little's law and can be obtained as their product; for closed chains, `SysQLen` is simply equal to the fixed number of jobs in the chain.

In some cases, it is possible that a chain visits multiple times the reference station before the job completes. This also affects the definition of the system averages, since one may want to avoid counting each visit as a completion of the visit to the system. In such cases, LINE allows users to specify which classes of the chain can complete at the reference station. For example, in the code below we require that a job visits reference station 1 twice, in classes 1 and 2, but completes at the reference station only when arriving in class 2. Therefore, the system response time will be counted between successive passages in class 2.

```
ClosedClass class1(model, "ClosedClass1", 1, queue, 0);
ClosedClass class2(model, "ClosedClass2", 0, queue, 0);
// a class that does not complete its cycle at the reference station
model.raw_struct().classes[class1 - 1].completes = false;

Routing P;
P.set(class1, class1, delay, queue, 1.0); // within class 1
P.set(class1, class2, queue, delay, 1.0); // class 1 switches into class 2
P.set(class2, class1, queue, delay, 1.0);
P.set(class2, class2, delay, queue, 1.0);
model.link(P);
```

`completes` is a field of the class rather than a setter: `raw_struct` hands out the struct under construction. Note that the `completes` property of a class always refers to the reference station for the chain.

4.3 Controlling the analysis

The options below bound the effort an analysis may spend and fix what makes a run reproducible. They are set on the solver, either as name/value pairs in the constructor or on the `options` structure it exposes, and are shared by every solver of Chapter 1.2; the complete per-solver table is Appendix C.

4.3.1 Accuracy and cost

`options.iter_tol` sets the relative tolerance at which an iterative analysis is declared converged, and `options.iter_max` caps the number of iterations it may take; a run that reaches the cap without meeting the tolerance reports the result together with a warning, rather than failing. `options.tol` plays the same

role for the numerical routines underneath. `options.timeout` bounds the wall-clock time of a single analysis and `options.cutoff` bounds the state space that an exact analysis is allowed to enumerate, by truncating the per-station queue length at the given value; raising it tightens the answer and enlarges the state space, and the analysis is refused when the resulting space does not fit the memory budget.

4.3.2 Simulation controls

When the answer is obtained by simulation, its statistical accuracy is under the analyst's control.

- `options.samples` is the sample budget of the run, measured in simulated events. The estimates converge as its square root, so a tenfold reduction of the confidence interval costs a hundredfold increase of the budget.
- `options.seed` fixes the seed of the pseudo-random number generators, so that repeated invocations return identical results. Two runs of the same model with the same seed and the same sample budget agree exactly.
- `options.confint` requests confidence intervals for the estimates, at the given confidence level, reported alongside the means.
- `options.tranfilter` selects the warm-up removal rule applied before the estimators are accumulated, so that the initial transient does not bias the steady-state estimates.
- `options.timespan` restricts the observation window, and is what turns a steady-state run into a transient one.

Two further controls change the nature of the sample path rather than its length. `options.slotted`, with `options.slotlength`, runs the model on a discrete time scale: every sampled interarrival and service time must fall on the slot lattice, a sample that does not is an error rather than being rounded, and intra-slot events are ordered by a fixed phase. Batch arrivals, declared on the source with `batch`, release a batch of jobs at each arrival epoch and are honoured on the sample path.

4.3.3 Warm start

An analysis may be seeded with the solution of another one, which is useful when a model is solved repeatedly along a parameter sweep, and when a cheap analysis is used to place a costly one near its fixed point. Passing a solver to `seeds` the initial state from that solver's steady state: an exact solver contributes the mode of its stationary distribution, an approximate one its rounded mean queue lengths, with the closed populations conserved. The same information may be supplied directly as `options.init_sol`, a station-major vector of queue lengths. A warm-started run disables warm-up removal, since its initial state is already representative. Which solver may seed which, what an exact and an approximate auxiliary solution each contribute, and what a warm start buys a simulation run are covered in §8.1.3.

4.3.4 Reproducibility and diagnostics

`options.verbose` controls how much the solver reports while it runs, from silent to a full trace of the analysis (Section 2.3). `options.cache` keeps the internal representation of the model between successive calls on the same solver, and `options.keep` retains the temporary files produced by an analysis, which is what makes a run reproducible outside LINE. `options.force` overrides the safety gates that refuse an analysis judged too large; it is the analyst's statement that the cost is acceptable, and it does not make the analysis cheaper.

4.3.5 Citations

The method that answers a request carries a bibliographic record with it. The method `returns` the references for the algorithms used in the last analysis, so that a result can be attributed also when the method was selected automatically. It is the recommended way to cite LINE results in a publication.

The C++ port carries no citation registry: a run reports the algorithm it used in `AvgResult::actualmethod`, and the reference for it is looked up in another edition.

The records themselves live in one table per codebase, keyed by method name: `matlab/src/io/line_citations.m`, `jline.io.LineCitations` in the JAR, and `line_solver.api.io.citations` in Python. Each entry carries the bibliography key as used in `doc/latex/biblio.bib`, a short author-title-venue-year reference, and a one-line `covers` hint naming the part of the solution process the paper accounts for. A method that is added or renamed gets its entry in the same change, so that `citations` stays complete.

Chapter 5

Extended queueing networks

The features in this chapter take a model out of product form: they are the reason LINE carries approximate and simulative algorithms next to the exact ones. Each of them is declared on the model exactly as the basic ones are, and each solver declares which of them it supports (Section 1.2); what changes is the cost of the solution, and whether it is exact.

5.1 Stations and buffers

5.1.1 Finite buffers

The functions `set_capacity` and `set_class_capacity` of the `Station` class are used to place constraints on the number of jobs, total or for each chain, that can reside within a station. The capacity follows standard Kendall notation where K represents the *total system capacity* (queue + in-service jobs). For an M/M/c/K system with c servers, the buffer holds $K - c$ waiting jobs plus c jobs in service. A single class is capped with `set_class_capacity(node, class, k)`, the station-level twin of `set_chain_capacity`. Unlike the latter, which caps every class in one call and can therefore total them into the station capacity, capping one class says nothing about the others, so the station-wide value is left untouched and the tighter of the two limits applies at analysis time. The capacity must be positive, `inf` standing for an unbounded class: a zero is how LINE internally marks a class the station does not serve at all, and so cannot also mean a buffer of size zero.

For example,

```
delay.set_capacity(2);           // total buffer at the station
model.set_class_capacity(delay, cclass, 1); // per-class buffer
const qn::NetworkStruct<double>& sn = model.get_struct(); // re-derives sn.cap, sn.classcap
```

The C++ builder constrains a station per class rather than per chain, and there is no separate refresh call: `get_struct` re-runs the whole derivation, capacities included, on every call. `qn_threechain_hyper1.m` creates an example model with two chains and three classes (specified in `qn_threechain_hyper1.m`) and requires the second station to

accept a maximum of one job in each chain. Note that if we were to ask for a higher capacity, such as , which exceeds the total job population in chain 2, LINE would have automatically reduced the value 7 to the chain 2 job population (2). This automatic correction ensures that functions that analyze the state space of the model do not generate unreachable states.

The `refresh_capacity` function updates the buffer parameterizations, performing appropriate sanity checks. Since `cqn_threeclass_hyper1` has already invoked a solver prior to our changes, the requested modifications are materially applied by LINE to the network only after calling an appropriate `refresh_struct` function, see the sensitivity analysis section. If the buffer capacity changes were made before the first solver invocation on the model, then there would not be the need for a `refresh_capacity` call, since the internal representation of the `Network` object used by the solvers is still to be created.

5.1.2 Heterogeneous servers

Queue nodes can be configured with heterogeneous servers that have distinct service rates and job class compatibilities. This feature enables modeling of systems where servers have different capabilities or speeds, such as heterogeneous CPU pools, mixed-skill service desks, or tiered processing resources. The `ServerType` class allows defining server types within a queue, each with its own characteristics. To add a server type, first create a `ServerType` object with a name and the number of servers of that type, then pass it to the `add_server_type` method. After creating server types, configure their service rates per job class using the same methods as for homogeneous queues, but targeting specific server types. Optionally, the scheduling policy for heterogeneous servers can be controlled using `HeteroSchedPolicy`, which determines how jobs are assigned to available server types. This allows specifying whether jobs can only be served by compatible server types or whether more sophisticated assignment policies are used. The following example demonstrates a queue with two server types, fast and slow servers, where each type has different service rates. Heterogeneous server types are not part of the C++ port: a station has one service distribution per class and a server count set with `set_number_of_servers`. This capability is particularly useful for capacity planning studies where resource pools contain servers with varying performance characteristics, or for modeling systems where specialization or skill matching affects service delivery.

When a job is compatible with multiple server types, the `HeteroSchedPolicy` class determines the assignment strategy. The available policies are listed in Table 5.1. The policy is set via `set_policy`. When the model is exported to JSIM XML for use with the JMT solver, the policy is serialised using JMT's descriptive string form (e.g. "ALIS (Assign Longest Idle Server)") via `HeteroSchedPolicy.toJMTText`, so the JMT engine recognises it.

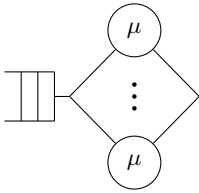
Figure 5.1 contrasts homogeneous and heterogeneous servers with the load-dependent and joint-dependent rate mechanisms discussed in §5.2.4.

5.1.3 Cyclic polling

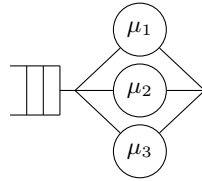
In the polling scheduling policy, the server cyclically visits the input buffer of each job class, processing jobs from that queue for a finite amount of time before switching to the next buffer. This scheduling policy further

Table 5.1: Heterogeneous server assignment policies (HeteroSchedPolicy)

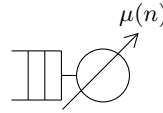
Policy	Description
ORDER	Assign to the first compatible server type in definition order (default)
ALIS	Assign Longest Idle Server: routes to the server type with the longest accumulated idle time (round-robin with busy servers placed at the back of the priority list)
ALFS	Assign Least Flexible Server: among compatible server types prefers the one that can serve the fewest classes (least cross-class flexibility), saving more flexible servers for classes that need them
FAIRNESS	Round-robin with fairness sorting: when a server type is used it is moved to the back of the priority list, so all compatible types are exercised before any one repeats
FSF	Fastest Server First: assigns the job to the compatible server type with the minimum expected service time
RAIS	Random Available Idle Server: selects uniformly at random among compatible server types that have at least one idle server



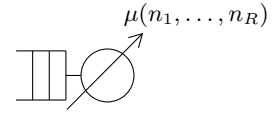
(a) homogeneous servers



(b) heterogeneous servers



(c) load-dependent rate



(d) joint-dependent rate

Figure 5.1: (a) A station with m homogeneous servers, all with the same service rate. (b) A station with heterogeneous servers, each with its own service rate. (c) A load-dependent station, whose service rate is a function of the total number of jobs present. (d) A joint-dependent station, whose service rate is a function of the per-class populations.

requires to specify the polling type at the station, chosen among:

- Exhaustive (`PollingType.EXHAUSTIVE`), where the server moves to the next input buffer only after finishing all jobs in the current buffer, including new arrivals during the service period;
- Gated (`PollingType.GATED`), where the number of served jobs for a buffer equals the jobs therein at the instant where the server switched to processing that buffer. Thus, newly arrived jobs during the service period will be processed at the next period.
- K -Limited (`PollingType.KLIMITED`), where the number of jobs processed is a constant K specified by the user. If the queue empties before k jobs have been served, the server will move on and attend the next buffer.

To specify a polling type at a queue, we may write for instance

```
model.set_polling_type(queue, PollingType::EXHAUSTIVE);
// or for K-Limited with K=2
model.set_polling_type(queue, PollingType::KLIMITED, 2);
```

A polling station cannot at the same time be the blocking station of a true blocking-after-service (BAS)

relationship: the two features both require the same per-station state slot, so a model that declares BAS blocking at a polling station is rejected with an explicit error rather than solved. Blocking-after-service at ordinary (non-polling) stations is fully supported.

5.1.4 Switchover times

In multiclass models, queueing stations alternate over time the processing of jobs of different classes. In some real-world situations, overheads may arise when a server needs to be reconfigured to process a different class of service. Such overheads are referred to as *switchover times*.

LINE supports switchover times in queueing stations. For example, to configure the overhead to begin processing *jobclass2* jobs after *jobclass1* we may write

```
model.set_switchover(queue, jobclass1, Exp(1.0));
```

The C++ port carries one switchover distribution per class, so a pairwise (from, to) switchover is not expressible. A special case arises when the station uses (deterministic) polling scheduling. In this situation, the switchover time specifies the time for moving from the input buffer of class i to the input buffer of class $i + 1 \bmod R$, where R is the total number of input classes to the queue. Using this fact, we need only to specify the switchover time after each class, e.g., the switchover time to begin processing *jobclass2* jobs after *jobclass1* is set as

```
model.set_switchover(queue, jobclass1, Exp(1.0));
```

5.2 Service processes

5.2.1 Load-dependent service

A queueing station i is called *load-dependent* whenever its service rate is a function of the number n_i of resident jobs at the station, summed across the ones in service and the ones in the waiting buffer. For example, a multi-server station with c identical servers, each with processing rate μ , may be shown to behave similarly to a single-server load-dependent station where the service rate is $\mu(n_i) = \mu\alpha(n_i) = \mu \min(n_i, c)$.

LINE presently supports *limited load-dependence* [28], meaning that it is possible to specify the form of the load-dependent service up to a finite range of n_i . As such, the support is currently limited to closed models, which are guaranteed to have a finite population at all times.

To specify a load-dependence service for a queueing station over the range $n_i \in [1, N]$ it is sufficient to call the `set_load_dependence` method, passing a vector of size N in its input with the scaling factor values for each n_i . For example, to instantiate a c -server node we write

```
std::vector<double> alpha(N);
for (std::size_t n = 0; n < N; ++n) alpha[n] = std::min(double(n + 1), double(c));
model.set_load_dependence(queue, alpha);
```

where the i -th element of the vector argument of `set_load_dependence` is the scaling factor $\alpha(n_i)$. It is assumed by default that $\alpha(0) = 1$. The mechanism is admitted at PS, FCFS and DPS stations: $\alpha(n_i)$ scales the total capacity the station clears and the scheduling discipline then divides that capacity among the classes, so the two compose without interfering.

Load-dependent closed networks may be analysed through the NC solver, either exactly with `exact` and `comomld`, or approximately with the saddle-point methods `nr.logit(nrl)`, `nr.probit(nrp)`, `nre` and `rd`, see §8.2.8. The probability metrics `solver_mva_get_prob_marg`, `solver_mva_get_prob_aggr`, and `solver_mva_get_prob_norm_const_aggr` listed in Table 4.1 remain available in the load-dependent setting and are computed from the same normalising constant produced by these methods.

5.2.2 MAP and BMAP analysis

The `MAP` class exposes a number of inspection methods to assess the temporal characteristics of a Markovian arrival process. Given a `MAP` instance, the following methods are commonly used:

- `map_lambda` returns the mean arrival rate λ .
- `map_acf` returns the autocorrelation function of the inter-arrival times at the requested lags.
- returns the asymptotic (and optionally interpolated) decay rate of the autocorrelation function.
- `map_idc` returns the asymptotic index of dispersion for counts; passing a timescale t returns the finite-horizon counterpart.
- returns the time-reversed MAP, useful in departure-process and reversed-time arguments.
- produces a random MAP of the specified phase order.

For example, to inspect the burstiness of a MAP one may write

```
mam::Map<double> m;
m.D0 = D0; m.D1 = D1; // a MAP is the pair of matrices
const double lambda = mam::map_lambda(m);
std::vector<unsigned> lags;
for (unsigned k = 1; k <= 10; ++k) lags.push_back(k);
const std::vector<double> acf = mam::map_acf(m, lags);
const double idc = mam::map_idc(m);
```

There is no random-MAP generator in the C++ port; a MAP is supplied as its two matrices, or built by `mam::map_erlang`, `map_hyperexp` and `map_exponential`.

The counting process. The quantities above summarise the counting process $N(t)$ through its rate and its dispersion. Its distribution is available too: `mc::ctmc_saddlepoint` approximates $\Pr\{N(t) = k\}$, the probability that a MAP records exactly k events in $(0, t]$, by the saddlepoint method [49, 84]. The counting generating function of a MAP is $\sum_{k \geq 0} P(k, t) z^k = e^{t(D_0 + zD_1)}$, so its cumulant generating function is the Perron root $\eta(\theta)$ of $D_0 + e^\theta D_1$, and the contour integral inverting it is evaluated at the saddle θ^* solving $\eta'(\theta^*) = k/t$. Three variants are offered through the method argument: `daniels2` (the default) carries the

amplitude's own curvature along the contour, `daniels` is the first-order form with the Perron amplitude, and `plain` is the bare first-order form.

The expansion parameter is the *variance* of the count, $\kappa_2 = t\eta''(\theta^*)$, and not its mean or the horizon: measured across the process family the relative error follows $0.083/\kappa_2$ for `daniels` and $0.017/\kappa_2^2$ for `daniels2`. Low variability shrinks κ_2 rather than breaking the method, so a renewal Erlang- r stream needs r times the horizon a Poisson one does; below $\kappa_2 = 5$ the call warns, and the per-point value is returned so that the caller can tell how far the answer can be trusted. The cost is one eigenproblem per Newton step and is independent of both t and k , and the logarithm is returned without ever forming the probability, so the accuracy holds well below the smallest positive double – on a three-phase MAP at $t = 500$, $k = 165$, where $\Pr\{N(t) = k\} \approx 1.3 \cdot 10^{-44}$, the second-order form is accurate to seven digits. This is an asymptotic method aimed at rare-event and large-deviation coefficients; for the bulk of the distribution at a moderate horizon, uniformization (`ctmc_uniformization`, `ctmc_foxglyn`) is both exact and faster.

```
const double p = mc::ctmc_saddlepoint(D0, D1, 200.0, 66L);
const mc::SaddlepointResult<double> res =
    mc::ctmc_saddlepoint(D0, D1, ts, ks, mc::SADDLEPOINT_DANIELS2);
```

The Batch Markovian Arrival Process `BMAP` extends `MAP` to allow batch arrivals via the cell array $D = \{D_0, D_1, \dots, D_K\}$, where D_k is the rate matrix for transitions producing batches of size k . The most useful inspection methods are:

- returns the largest batch size K supported.
- returns the average batch size $E[B]$ weighted by the batch arrival rates.
- returns a vector of per-batch-size arrival rates.
- draws n batches, returning inter-batch times X and batch sizes B .
- generates a random `BMAP` of given phase order and maximum batch size.

A typical inspection of a `BMAP` combines its scalar summaries with sampling, e.g., Batch Markovian arrival processes are carried on the wire and by the QBD solver (`mam::qbd_bmapbmap1`), but the `BMAP` constructors and their statistics have no C++ counterpart. A `BMAP` can also be constructed deterministically from an underlying `MAP` and a batch-size probability mass function via .

5.2.3 Non-Markovian distributions and moment fitting

A service or arrival process need not be given as an explicit Markovian representation: a distribution is more often specified by its moments, or drawn from a family with no phase structure at all. The distribution classes therefore expose fitting constructors that match a target mean, or a mean together with a squared coefficient of variation. For example, given mean $\mu = 0.2$ and squared coefficient of variation $\text{SCV}=10$, where $\text{SCV}=\text{variance}/\mu^2$, we can assign to a node a 2-phase Coxian service time distribution with these moments as

```
queue.set_service(class2, Cox2(mu1, mu2, phi1));
```

Table 5.2: Non-Markovian distributions available in LINE

Distribution	Description
Deterministic	<code>Det(μ)</code> assigns probability 1.0 to the value μ
Uniform	<code>Uniform(a, b)</code> assigns uniform probability $1/(b - a)$ to the interval $[a, b]$
Gamma	<code>Gamma(α, k)</code> assigns a gamma density with shape α and scale k
Pareto	<code>Pareto(α, k)</code> assigns a Pareto density with shape α and scale k
Weibull	<code>Weibull(r, α)</code> assigns a Weibull density with shape r and scale α
Lognormal	<code>Lognormal(μ, σ)</code> assigns a lognormal density with parameters μ and σ
Zipf	<code>Zipf(s, N)</code> assigns power-law probabilities $p(k) \propto k^{-s}$ for $k = 1, \dots, N$
Trace-based distributions (replay empirical data from files)	
Replayer	<code>Replayer(filename)</code> reads interarrival or service times cyclically from a CSV file
Trace	<code>Trace(data)</code> wraps an empirical dataset as a distribution
EmpiricalCDF	<code>EmpiricalCDF(x,F)</code> defines a distribution from CDF values F at points x

Table 5.3: Non-Markovian distribution parameter formats in `proc{ist}{r}`

Distribution	Format
Gamma	{shape (α), scale (β)}
Weibull	{shape (r), scale (α)}
Lognormal	{ μ, σ }
Pareto	{shape (α), scale (k)}
Uniform	{min (a), max (b)}
Det	{ t }

The C++ Coxian constructor takes its phase rates directly; a two-moment fit is `mam::aph_fit_mean_scv(0.2, 10.0)`, whose (D0, D1) pair is installed with `Distrib<double>::map_dist`. where `Cox2` is a static class to fit 2-phase Coxian distributions. Inter-arrival time distributions can be instantiated in a similar way, using `set_arrival` instead of `set_service` on the `Source` node. For example, if the `Source` is node 3 we may assign the inter-arrival times of class 2 to be exponential with mean 0.1 as follows

```
source.set_arrival(class2, Exp(0.1));
```

Is it also possible to plot the structure of a phase-type distribution using the `plot` instance method of the Markovian class.

Non-Markovian distributions are also available, but typically they can restrict the available solvers to the JMT simulator. They include the distributions shown in Table 5.2.

When non-Markovian distributions are used with the CTMC, SSA, or MAM solvers, they are automatically converted to phase-type (PH) distributions using a Bernstein polynomial approximation. A warning message is issued when this conversion occurs. The number of phases used in the approximation can be controlled via `options.config.bernstein` (default: 20 phases).

Table 5.3 shows the internal parameter storage format for each non-Markovian distribution. These parameters are stored in the `proc{ist}{r}` field of the network structure.

Lastly, we discuss two special distributions. The `Disabled` distribution can be used to explicitly forbid a

class to receive service at a station. This may be useful to declare in models with sparse routing matrices to debug the model specification. Performance metrics for disabled classes will be set to .

Conversely, the `Immediate` class can be used to specify instantaneous service (zero service time). Normally, solvers will replace zero service times with small positive values ($\varepsilon =$).

5.2.4 Class- and joint-dependent service

A generalization of load-dependent service lets the service rate depend on the per-class populations $n_i = [n_{i,1}, \dots, n_{i,R}]$ at station i , where $n_{i,r}$ is the current number of class- r jobs. LINE exposes this through function handles supplied to the station, and distinguishes two mechanisms according to whether the resulting demands remain in product form.

Class dependence (`setClassDependence`, stored as `cdscaling`) is the *product-form* case of the QD-AMVA model [31]: the demand factorizes as $D_{i,r}(n_i) = \theta_{i,r} \beta_{i,r}(n_{i,r})$, where the scaling $\beta_{i,r}$ read by class r depends only on its own marginal count $n_{i,r}$ (or, equivalently, on the total n_i , which also preserves product form). The handle returns the per-class vector $[\beta_{i,1}(n_i), \dots, \beta_{i,R}(n_i)]$, of which the solver selects class r ; a scalar return is broadcast to all classes. For example

```
// Product-form class dependence: each class reads its own marginal
model.set_class_dependence(queue,
    [c](const std::vector<double>& ni) {
        return std::vector<double>(std::min(ni[0], double(c)), 1.0);
    },
    std::vector<double>(double(c), 1.0));
```

gives class-1 jobs a multiserver-type scaling in their own count while class-2 jobs are always single-server.

Joint dependence (`setJointDependence`, stored as `jdscaling`) is the *non-product-form* case: the handle $\eta_i(n_i)$ may read the joint population vector arbitrarily – for instance a scalar rate driven by a foreign class marginal, $\min(n_{i,1}, c)$ shared across all classes – which violates the QD-AMVA product-form condition. Such models are solved by the same solvers as an approximation, with no exactness or uniqueness guarantee:

```
// Non-product-form joint dependence
model.set_joint_dependence(queue,
    [c](const std::vector<double>& ni) {
        return std::vector<double>(std::min(ni[0], double(c)));
    },
    std::vector<double>(double(c)));
```

The peak rate is mandatory, as in the reference: the utilization of a dependent station is reported against it. In both cases the second argument is the required peak rate scaling per class (a scalar is broadcast to all classes) that normalizes utilization as $U = TS/\text{peak}$. The solver implicitly assumes the handles are smooth and defined also for fractional values of $n_{i,r}$.

Global dependence (`setGlobalDependence`, stored as `gdscaling`) differs from the three mechanisms above in what the handle is allowed to read. The first three are *station-local*: each receives only the population held at the station it belongs to. A global dependence instead receives the *entire* network state, the $(M \times R)$ matrix $n = (n_{i,r})$, and is declared on the model rather than on a node:

```

model.set_global_dependence(
    [] (const std::vector<double>& n) { return phi(n); },
    std::vector<double>{peak});

```

The handle returns a scalar broadcast to every station and class, an $(M \times 1)$ column read per station, or an $(M \times R)$ matrix; the service rate of class r at station i is its declared rate times $\phi_{i,r}(n)$, composing multiplicatively with any load, class or joint dependence already present. The peak argument is required and normalizes utilization exactly as above, the peaks multiplying when a station carries several of these mechanisms.

Since the Whittle balance property [167] is a property of the handle and not of the model text, it is worth checking rather than assuming: `sn_gd_balance` returns the worst relative violation over a bounded lattice, zero to rounding when ϕ is balanced. Note that any $\phi_s(n) = f(n_s)g(|n|)$ satisfies it trivially, so a probe outside that family is needed to exercise the check.

`SolverCTMC`, `SolverSSA` and `SolverLDES` declare the `GlobalDependence` feature; every other solver refuses such a model rather than solving it unscaled, and the combination with a finite capacity region is refused as well. The first two exploit the same observation: within one state $\phi(n)$ is a *constant*, so it need be evaluated only once per state and then multiplies the rate of every station service event there. The generator tabulates it over the enumerated state space; the simulator, which holds one state at a time, collapses that table to a single row. Within `SolverSSA` the next-reaction engine cannot carry it, because its propensity closures receive one station's population slice and never the whole matrix ϕ reads, so a model declaring a global dependence is routed to the serial engine. `SolverLDES` re-times the completions already in flight whenever ϕ changes, the infinite servers among them, and reports utilization as $T E[S]/\text{peak}$ at every station; a bulk (BMSP) server and a pass-and-swap station carry no per-job residual work to rescale and are refused by name.

A handle cannot cross a language boundary, so the JSON model interchange carries ϕ as a *table*, under the model-level key `globalDependence`. The lattice is that of the whole network state, restricted to the (station, class) slots a class can actually occupy: a source holds no jobs, and a class with zero per-class capacity at a station never appears there, so those entries carry no coordinate. The restriction is lossless, since no service event ever fires at a slot the class cannot occupy. A closed class is tabulated up to its own population; an open one up to the truncation given as the third argument of the setter, which should match the cutoff the model is solved at. A lattice above 2×10^5 points is refused by name rather than truncated, since a silently shortened table would make one language disagree with another on the same model.

Table 5.4 summarizes the queue-dependent service mechanisms by what the handle reads and returns, the field that stores them, and whether the resulting demands are product form.

5.2.5 Per-class dependence and tabulated scalings

The class-dependence function of §5.2.4 may also be made *per class*: instead of returning a single scaling shared by every class, $\beta_i(n_i)$ may return the vector $[\beta_{i,1}(n_i), \dots, \beta_{i,R}(n_i)]$, so that each class r receives its own scaling at the same joint population $n_i = [n_{i,1}, \dots, n_{i,R}]$. This is the chain-dependent service rate

Argument read	Return	Setter	Field	Prod. form
total $n_i = \sum_r n_{i,r}$	scalar (shared)	setLoadDependence	lldscaling	yes
own marginal $n_{i,r}$	per-class vector	setClassDependence	cdscaling	yes
total n_i	per-class vector	setClassDependence	cdscaling	yes
joint $[n_{i,1}, \dots, n_{i,R}]$	scalar (shared)	setJointDependence	jdscaling	if OI
joint $[n_{i,1}, \dots, n_{i,R}]$	per-class vector	setJointDependence	jdscaling	if OI
full state $(n_{i,r})$	scalar, $M \times 1$ or $M \times R$	setGlobalDependence	gdscaling	if balanced

Table 5.4: Queue-dependent service mechanisms. `setLoadDependence` takes a numeric vector $\alpha(n_i)$; `setClassDependence` and `setJointDependence` take a function handle of the joint population and a required per-class peak rate. The product-form column indicates whether the BCMP recurrence of [31] is preserved; a joint dependence preserves it only when the station is order independent (OI). A scalar handle that reads only the total n_i is equivalent to `lldscaling` and should be declared through `setLoadDependence`. The last row is declared on the MODEL rather than on a station, since its argument is the whole network state; it preserves a product form exactly when it satisfies the Whittle balance property [167], and only `SolverCTMC` and `SolverSSA` support it.

$\mu_{r,i}(n_i)$ of [142], and it is what allows a station to serve different classes at genuinely different rates in the same state. It is useful for systems whose server efficiency depends on the mix of job classes present, such as heterogeneous workloads competing for a shared resource.

A dependence obtained from measurements, or defined only over a bounded population range, is expressed by having the handle index a table and clamp the population to the range it was tabulated on. Earlier releases exposed this through separate limited joint dependence (LJD) and limited joint class dependence (LJCD) lookup tables, which the handle forms now subsume. A per-class tabulated scaling whose entry r reads only its own marginal is product form and belongs in `cdscaling`; a table that reads foreign marginals is joint and belongs in `jdscaling`.

Per-class dependence is used by the MVA solver during the mean value analysis iteration, and by the NC solver, which solves such models exactly by the multichain convolution of §5.6.3. Utilization at such a station is normalized by the per-class peak rate declared as the second argument of `setClassDependence`, following the convention of §4.2.2.

5.2.6 Discrete distributions

In addition to continuous service- and inter-arrival-time distributions, LINE provides a family of discrete distributions through the `DiscreteDistribution` class hierarchy. These distributions take values on a (possibly finite) integer support and are primarily used as count or popularity models, e.g. to specify the number of items requested from a cache, the number of arrivals per slot, or batch sizes. Table 5.5 summarizes the discrete distributions currently available.

All discrete distributions inherit the same inspection interface as the continuous ones: `Distrib::mean`, `Distrib::scv` and `return` the first two moments and draw n pseudo-random values. In addition, `returns` the probability mass at the integer k , and `returns` the cumulative probability up to k .

A typical use case for `DiscreteSampler` is to specify the popularity profile of items in a `CacheTask`

Table 5.5: Discrete distributions available in LINE

Distribution	Description
Bernoulli	$\text{Bernoulli}(p)$, single trial with success probability $p \in [0, 1]$. Mean = p , support $\{0, 1\}$
Binomial	$\text{Binomial}(n, p)$, number of successes in n independent Bernoulli trials. Mean = np , support $\{0, \dots, n\}$
Poisson	$\text{Poisson}(\lambda)$, count of events occurring in a fixed interval at average rate λ . Mean = Var = λ , support $\{0, 1, 2, \dots\}$
DiscreteUniform	$\text{DiscreteUniform}(a, b)$, uniform probability $1/(b - a + 1)$ on the integers $\{a, a + 1, \dots, b\}$
Geometric	$\text{Geometric}(p)$, number of Bernoulli trials needed to obtain the first success. Mean = $1/p$, support $\{1, 2, \dots\}$
DiscreteSampler	$\text{DiscreteSampler}(p, x)$, fully user-specified discrete distribution. Probability vector p assigns mass p_i to value x_i . If x is omitted it defaults to $1:n$ where n is the length of p

or `ItemEntry`. The example below assigns a non-uniform popularity to four items so that item 1 is requested with probability 0.5 and items 2–4 share the remaining mass: The discrete sampler is a sampling utility of the simulators and is not exposed by the C++ port; a cache popularity is given to `add_cache` as the probability vector itself. The same approach is used by `Zipf( $s, N$ )` to assign power-law popularity to a finite catalogue of N items with exponent s , while `Bernoulli` and `Geometric` are convenient when modelling cache hit/miss processes or first-success counts.

Discrete-time modelling with DMAP. LINE supports discrete-time modelling through the `DMAP` (Discrete-time Markovian Arrival Process) class. Unlike the continuous-time `MAP`, where $D_0 + D_1$ is an infinitesimal generator, in a `DMAP` the matrix $D_0 + D_1$ is a (row-)stochastic transition matrix. A `DMAP` is constructed as follows: The C++ port carries `DMAP` and consumes it on the slotted `MAM` path of Section 5.3.8; the slotted `LDSE` engine is reached through the subprocess client rather than built in. A random two-phase `DMAP` can be generated with `ctmc_rand`. The `DMAP` class inherits statistical accessors from its continuous-time counterpart `()`.

5.2.7 Temporal dependent processes

It is sometimes useful to specify the statistical properties of a *time series* of service or inter-arrival times, as in the case of systems with short- and long-range dependent workloads. When the model is stochastic, we refer to these as situations where one specifies a *process*, as opposed to only specifying the *distribution* of the service or inter-arrival times. In LINE processes include the 2-state Markov-modulated Poisson process (`MMPP2`) and empirical traces read from files (`Replayer`).

For the latter, LINE assumes that empirical traces are supplied as text files (ASCII), formatted as a column of numbers. Once specified, the `Replayer` object can be used as any other distribution. This means that it is possible to run a simulation of the model with the specified trace. However, analytical solvers will require tractable distributions from the `Markovian` class.

5.3 Job behaviour

5.3.1 Class priorities

If a class has a priority, **with 0 representing the highest priority**, this can be specified as an additional argument to both `OpenClass` and `ClosedClass`, e.g.,

```
ClosedClass class2(model, "Class2", 5, queue, 0);
```

The last argument is the priority, as in the reference: a smaller integer is a higher priority. **Important:** Class priorities are only effective when a station uses a priority scheduling policy (see Table 3.3). If no priority scheduling policy is explicitly assigned to a station, class priorities are ignored. In *Network* models, priorities are intended as hard priorities. Weight-based policies such as DPS and GPS may be used, as an alternative, to prevent starvation of jobs in low-priority classes.

5.3.2 Drop strategies

When a station has finite capacity, the `DropStrategy` associated with each class determines what happens to a job that arrives at, or finishes service in, a full station. The drop rule is configured per class via `set_drop_rule` on a station node. The complete enumeration is summarised in Table 5.6; only `Queue` and `Drop` apply to per-station capacity in all four codebases, while `BAS/BBS/ReServiceOnRejection` are blocking semantics typically used in conjunction with finite buffers and routing to downstream finite-capacity destinations. The `Retrial` and `RetrialWithLimit` values are normally not set explicitly: they are assigned automatically when `setRetrial` is called on a queue node (see 5.3.5).

The next listing assigns different drop rules to two classes sharing the same finite-capacity queue: jobs of the first class are rejected on overflow, while jobs of the second class are admitted to the system-level waiting queue.

```
model.set_drop_rule(queue, class1, DropStrategy::DROP);
model.set_drop_rule(queue, class2, DropStrategy::WAITQ);
```

The enumerators are `DROP`, `WAITQ`, `BAS`, `BBS`, `RSRD` and `RETRIAL`.

The same `setDropRule` API is exposed by the `Region` class for finite capacity regions (see 5.5); on `Region` the boolean overload `setDropRule(jobclass, true)/setDropRule(jobclass, false)` is also accepted, with `true` mapping to `DropStrategy.Drop` and `false` to `DropStrategy.Queue`. Solver support for the more advanced strategies depends on the chosen backend: simulators (JMT, LDES, SSA) handle the full range of blocking semantics, while analytical solvers typically require `Drop` or `Queue`.

5.3.3 Impatience (customer abandonment)

A job may leave a queue before being served. This models a caller who hangs up, a user who navigates away from a slow page, or a request that exceeds its deadline. On arrival the job draws a *patience* time from a distribution; if it has not entered service by the time that patience expires, it reneges, leaving the queue

Table 5.6: Drop strategies available in LINE

Strategy	Description
<code>DropStrategy.Queue</code>	Default policy. Jobs that find the station full wait in an external (logical) waiting queue until capacity becomes available. Selected with <code>DropStrategy.Queue</code> or by passing <code>false</code> to the boolean overload of <code>setDropRule</code> on <code>Region</code> .
<code>DropStrategy.Drop</code>	Jobs that find the station full are rejected and leave the system permanently. Selected with the MATLAB constant <code>DropStrategy.DROP</code> or by passing <code>true</code> to the boolean overload of <code>setDropRule</code> on <code>Region</code> .
<code>DropStrategy.BlockingAfterService (BAS)</code>	Blocking-after-service. A job completing service waits at the server until the downstream destination has free capacity, blocking that server in the meantime. Selected via <code>DropStrategy.BAS</code> (MATLAB constant).
<code>DropStrategy.BlockingBeforeService (BBS)</code>	Blocking-before-service. A job is held back from entering service while the downstream destination is full, so that no service capacity is consumed for jobs that cannot complete. Selected via <code>DropStrategy.BBS</code> (MATLAB constant).
<code>DropStrategy.ReServiceOnRejection</code>	Re-service on rejection. A job rejected by a full downstream station is returned to the originating server and served again. Selected via <code>DropStrategy.RSRD</code> (MATLAB constant).
<code>DropStrategy.Retry</code>	Job moves to a virtual orbit and retries after a random delay; the number of retry attempts is unlimited. Set automatically by <code>setRetry(jobclass, delay)</code> .
<code>DropStrategy.RetryWithLimit</code>	As <code>Retry</code> , but with a finite cap on the number of attempts; once the cap is exhausted the job is dropped. Set automatically by <code>setRetry(jobclass, delay, maxAttempts)</code> .

and being routed onward to the next node or to a sink. The patience distribution may be any standard one, exponential for memoryless impatience or deterministic for a fixed timeout, but not a modulated process (MAP, BMAP, MMPP2).

Patience is declared either on a queue, for one job class, or on the class itself, in which case it holds at every queue the class visits. When both are given, the queue-level declaration prevails. Impatience is not part of the C++ port: neither reneging (a patience distribution) nor balking is carried by `qn::Network`. LINE honours the feature both by tracking individual patience times on the sample path and, in exact analyses, by encoding reneging as an additional state transition of the Markov chain. The resulting abandonment rate is what separates offered from carried throughput, and is therefore the quantity of interest whenever a service level agreement is at stake.

5.3.4 Balking

An arriving job that balks inspects the state of the queue and leaves without joining, unlike reneging (see 5.3.3), where it joins and abandons later. A balking policy is configured per class on a queue node with `balk`, passing a `BalkingStrategy` constant and a list of thresholds (`minJobs`, `maxJobs`, `probability`): the first range that contains the queue length at the arrival instant fixes the balking probability, and an unbounded upper limit is written as ∞ (MATLAB), `Integer.MAX_VALUE` (Java) or `float('inf')` (Python).

Three strategies are supported, listed in Table 5.7.

The following example configures a single class to balk with probability 0.3 when the queue holds 5 to 10 jobs, and with probability 1 (certain refusal) when 11 or more jobs are present.

Balking has no C++ counterpart; a station refuses work only through its capacity and its drop rule.

Table 5.7: Balking strategies available in LINE

Strategy	Description
BalkingStrategy.QUEUE_LENGTH	Balking decision is taken from the current queue length. The threshold list assigns a balking probability to each range of queue lengths.
BalkingStrategy.EXPECTED_WAIT	Balking decision is based on the expected waiting time, estimated from the current queue length and the per-class service rate. The customer balks when the estimated wait exceeds a tolerance threshold.
BalkingStrategy.COMBINED	Both queue-length and expected-wait conditions are evaluated; the customer balks when either condition triggers (logical OR). Useful when modelling customers that are sensitive to both visible queue length and perceived waiting time.

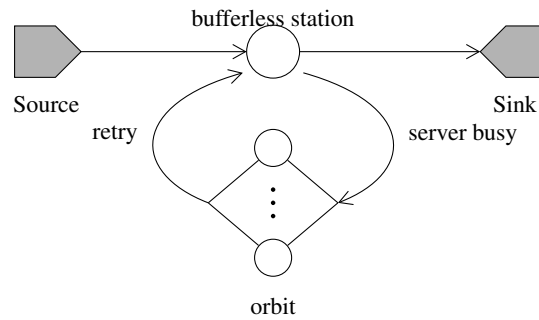


Figure 5.2: A retrial queue. The station has no waiting room, so a job that finds the server busy joins the orbit and retries after a random delay.

A simplified single-threshold form, `setBalking(jobClass, strategy, minJobs, probability)`, is available in Java and configures a single open-ended threshold. Balking combines with the impatience and retrial mechanisms above, as illustrated by the example `json_balking_retrial`.

5.3.5 Retrial queues

In a retrial queue a customer that finds all servers busy joins an orbit and reattempts service after a random delay, a pattern common in call centres, telecommunication systems and cloud platforms. The station is declared with `set_retrial`, giving the retrial delay distribution and optionally a maximum number of attempts (-1 , the default, allows unlimited retries), and makes orbiting customers abandon after a random time. The retrial delay must not be a modulated process, and MAM, CTMC, SSA and LDES declare support for the feature.

```
// retrial delay process, orbit retry rate, and the attempt cap (0 = unbounded)
model.set_retrial(queue, jobclass, Exp(0.5), 0.5, 0);
model.set_retrial(queue, jobclass, Erlang(0.3, 2), 0.3, 3);
```

The retry rate is passed explicitly beside the process, and orbit abandonment has no counterpart.

Figure 5.2 shows the resulting flow.

First-class orbit declaration. The method `declareOrbit` declares a retrial station in a single call, installing the retrial delay and recording the retrial policy and the orbit capacity. Its signature is `setOrbit(class, retrialDistribution, policy, maxOrbit)`, where `policy` is `RetrialPolicy.LINEAR` (default), under which each orbiting job retries at its own rate ν so that an orbit of n jobs retries at rate $n\nu$ (the classical Falin-Templeton queue), or `RetrialPolicy.CONSTANT`, under which the orbit retries as a whole at rate ν while non-empty. The argument `maxOrbit` is the orbit capacity, -1 (default) leaving it unbounded and a non-negative value making it finite, so that a job finding it full is lost; the station capacity is set accordingly, to infinity or to the server count plus `maxOrbit`, since the state-space solvers hold the orbit in the station buffer and a capacity pinned to the server count would leave them nowhere to put an orbiting job. The configuration is read back with `getOrbit`. There is one retrial entry point, `set_retrial`, which installs the orbit; the retrial policy and the finite-orbit bound of the reference are not carried by the C++ port. Note that the finite orbit declared by `maxOrbit` is a property of the model, not a truncation of the solution: it changes the loss probability seen by the arriving jobs, whereas the QBD truncation level discussed above is an accuracy control that leaves the model unchanged.

Orbit metric. At a retrial station `QLen` reports the whole station population, that is the jobs in service together with the jobs orbiting. The mean orbit length alone is returned by `getOrbitLength`, an $(n_{\text{stations}} \times n_{\text{classes}})$ matrix that is zero at stations that are not retrial queues, and by its labelled variant `getOrbitLengthLabel`. Orbit occupancy is not reported separately: the MAM retrial solution returns the queue length as orbiting jobs plus jobs in service, in `AvgResult::QN`.

5.3.6 Batch arrivals

By default a `Source` releases one job per arrival epoch. A batch arrival stream is obtained by attaching a discrete batch-size distribution to a class with `batchSize`: at each epoch generated by the inter-arrival distribution, the source releases a number of jobs drawn from the batch distribution instead of a single job. The batch size must be at least one, so distributions that can return zero are rejected rather than clamped. Batch arrivals are not part of the C++ port: a `Source` carries one inter-arrival process per class and releases one job per epoch. The batch distribution is stored in the `arrivalbatch` field of the network structure and is advertised through the `BatchArrival` language feature. Batch arrivals are simulated by the `LDES` solver. Combining a geometric inter-arrival distribution with a geometric batch size under slotted simulation (Section 8.2.5.1) yields the Geo^X arrival stream, whose analytical reference is `qsys_geoxgeol`.

The dual construction applies at a station. When a `BMAP` is assigned as the `service` distribution, the `LDES` solver interprets it as a batch service process (BMSP): at a service epoch the server removes and completes a batch of jobs whose size is drawn from the `BMAP` batch-size distribution, truncated by the number of jobs present, so that the station behaves as an `M/BMSP/1` queue. The same `BMAP` placed at a `Source` produces batch arrivals instead.

5.3.7 Server breakdowns

A station whose server is unreliable is declared with `on` on a queue node: the server alternates between UP and DOWN, failing after the failure distribution and being restored after the repair distribution, with the solver expanding the joint (queue, server status) chain rather than the ensemble of Section 6.6. The failure clock runs whenever the server is up, so an idle station can fail; arrivals keep queueing while it is down, and a job in service is not lost but resumes on repair. An optional third argument gives the service distribution used while down, one distribution or a per-class list, and omitting it means no service at all while down. The configuration is read back with `.`. Station breakdowns are not declared on the station in the C++ port. A failing server is modelled at the environment level instead, as a node failure of an `Environment` model solved by `env::solver_env`. Only exponential failure and repair distributions are expanded into the joint chain; any other process is rejected with an explicit error, and the random-environment ensemble remains available for it. Breakdowns are registered as the `Breakdown` language feature, so a solver that does not build the joint chain rejects the model instead of silently solving it breakdown-free. The CTMC solver supports both its steady-state and its transient solution.

5.3.8 Slotted models and the discrete-time solver

A model whose interarrival and service laws all live on a slot lattice is a discrete-time model, and `SolverMAM` recognizes it from the distributions alone before any phase-type conversion takes place. The lattice families are `Geometric` (support $\{1, 2, \dots\}$ slots), `DMAP`, `DiscreteUniform` with integral bounds, and `Det` on an integral number of slots; a model that mixes these with continuous laws is solved on the continuous time scale as before. The decision can be forced with `options.config.timescale`, which takes `auto` (the default), `discrete` or `continuous`, and the slot is set with `options.config.slotlength`. Requesting `discrete` on a model that mixes lattice and non-lattice laws raises an error rather than answering on the wrong time scale. A `DMAP` combined with continuous laws is always an error: its (D_0, D_1) are probability matrices, so the continuous machinery would form $(-D_0)^{-1}$ where the law needs $(I - D_0)^{-1}$.

The convention is the late arrival system with delayed access. Within a slot the service completion resolves first, the arrival is appended at the end of the slot and cannot enter service before the next one, and every metric is read after both events. This is the convention of the `Q_DT_*` queues, of the analytical references `dqsys_geogeol` and `dqsys_geoxgeol`, and of the slotted mode of the `LDES` simulator, so the four are directly comparable.

A single queueing station is solved exactly, under the method name `dt.qmam: Q_DT_PH_PH_1` when both the arrival and the service process are renewal discrete phase-type, `Q_DT_MAP_MAP_1` otherwise. Several stations are solved by a discrete-time parametric decomposition under the name `dt.dec`, which is an approximation: superposing slotted streams produces batches, since two streams fire in the same slot with positive probability, so each station is an `M/G/1`-type chain rather than a quasi-birth-death process, and the departure process handed downstream is truncated at a finite level and compressed back to the phase budget of `options.config.space_max`. A tandem of geometric-server queues is nonetheless exact, because the stationary departure stream of a `Geo/Geo/1` queue is Bernoulli.

The discrete-time path covers open, single-class models whose queueing stations are single-server FCFS. Multiserver stations, infinite servers, closed populations and several classes are refused by name rather than silently solved on the continuous scale. The slotted path is reached by the ordinary solver entry point, and the time scale is selected on the command line with `--timescale auto|discrete|continuous`, the slot itself coming from the shared `--slotlength`.

5.4 Routing and synchronization

5.4.1 Other routing strategies

The above routing specification style is only for models with probabilistic routing strategies between every pair of nodes. A different style should be used for scheduling policies that do not require to explicit routing probabilities, as in the case of state-dependent routing. Currently supported strategies include:

- Round robin (`RoutingStrategy.RROBIN`). This is a deterministic strategy that sends jobs to outgoing links in a cyclic order.
- Random routing (`RoutingStrategy.RAND`). This is equivalent to a standard probabilistic strategy that for each class assigns identical values to the routing probabilities of all outgoing links. When a target is invalid its probability is kept to zero, e.g., random routing will not send a job in a closed class to a sink.
- Join-the-Shortest-Queue (`RoutingStrategy.JSQ`). This is a non-probabilistic strategy that sends jobs to the destination with the smallest total number of jobs in it (either queueing or receiving service). If multiple stations have the same total number of jobs, then the destination is chosen at random with equal probability.
- Weighted round robin (`RoutingStrategy.WRROBIN`). Routes jobs cyclically through a list in which each destination appears as many times as its integer weight, so destinations with higher weights are visited proportionally more often. Configure each weight with a per-destination call ; with all weights equal to one this collapses to ordinary round-robin.
- Shortest queue of d , $SQ(d)$ (`RoutingStrategy.SQ`, formerly `KCHOICES`). Samples d candidate destinations uniformly at random *with replacement* from the set of available outgoing links and dispatches the job to the one with the shortest queue, breaking ties by first occurrence in the candidate list [117, 161]. The parameter d defaults to 2 if not specified. Dispatcher memory is not supported. To set a specific value of d use .
- Product-form state-dependent routing (`RoutingStrategy.SDR`), see §5.4.2.
- Event-based routing (`RoutingStrategy.FIRING`). This strategy is used internally by Transition nodes in stochastic Petri net models and routes tokens to output places according to the firing outcomes configured for each mode of the transition. It is set automatically when specifying Petri net firing outcomes and is not intended for direct assignment by the user.
- Disabled routing (`RoutingStrategy.DISABLED`). This pseudo-strategy explicitly marks a (node, class) pair as inactive and prevents jobs of that class from being dispatched out of the node. It is used internally

by the model linker to suppress routing for classes that should not visit a given node, and is not normally set by user code.

For the above policies, the function `addLink` should be first used to specify pairs of connected nodes

```
Routing P;
P.set(queue, queue, 1.0); // self-loop
P.set(queue, delay, 1.0);
model.link(P);
```

There is no `addLink`: an arc is an entry of the routing matrix. Then an appropriate routing strategy should be selected at every node, e.g.,

```
model.set_routing(queue, class1, RoutingStrategy::RROBIN);
```

assigns round robin among all outgoing links from the `queue` node.

A model could also include both classes with probabilistic routing strategies and classes that use round-robin or other non-probabilistic strategies. To instantiate routing probabilities in such situations one should then use, e.g.,

```
model.set_routing(queue, class1, RoutingStrategy::PROB);
Routing P;
P.set(class1, class1, queue, queue, 0.7);
P.set(class1, class1, queue, delay, 0.3);
model.link(P);
```

Probabilistic routing is given by the routing matrix rather than by per-destination setters. `set_routing` assigns the routing probabilities to the two links.

5.4.2 Product-form state-dependent routing

Adaptive strategies such as JSQ and SQ take a model out of product form, so a network using them must be simulated or solved on its state space. The state-dependent routing of Krzesinski [93], the multiclass generalization of the rule of [156], is the exception: its routing probabilities depend on the branch populations and yet the stationary distribution keeps a product form, so `SolverNC` solves such a model exactly and `SolverCTMC` and `SolverSSA` accept it as well.

A node is declared the *entry center* e of a subnetwork $Q(V, V)$ whose branches it feeds:

```
model.set_state_dep_routing(node1, node1, branches, level, C, d);
```

The arguments follow the paper's own indexing. The second names the departure center of $Q(V, V)$, which is the entry center itself in a central-server model. The branch list has an empty first entry, because branch index 1 denotes the complement $M - V$, and lists from index 2 onwards the nodes of each branch, entry center first and departure center last. The level vector gives, for each branch, the index t of the subnetwork holding it, its first entry being unused; C is the vector of coefficients C_t and d the matrix of coefficients d_{tb} . Negative C_t and positive d_{tb} make the routing prefer the least congested branches and impose the population

bounds $m_b \leq d_{tb}/(-C_t)$ and $v_t \leq D_{tt}/(-C_t)$. With one level, four single-center branches, $C_1 = -1$ and $d_{1i} = d$, the entry center routes as $P_{1,i}(N) = (d - n_i)/(4d - V)$ with $V = n_2 + n_3 + n_4 + n_5$, which is adaptive routing with balanced loading; the state-independent counterpart is $P_{1,i} = 1/4$.

The default NC dispatch recognises the declaration and solves the model with it. The method can also be named explicitly: `sdr` evaluates the product form of the paper's eq. (16) exactly by state enumeration and accepts any branch topology, while `sdr.mva` runs instead the Section 4 mean value analysis and convolution, which costs polynomially in the populations rather than the size of the state space but is stated for single-center branches only and requires every C_t to be negative. The examples `sdroute_krzesinski`, which reproduces the paper's Section 5.1.1, and `sdroute_multibranch` illustrate both.

5.4.3 Fork and Join nodes

The fork and join nodes are supported by the JMT and LDES simulators, by the CTMC and SSA solvers through a native fork-join state representation (closed and multiclass models, including), and, for mean performance metrics, by the MVA and MAM analytical solvers (and by LN for layered models). The `Fork` splits an incoming job into a set of sibling tasks, sending out one task for each outgoing link. These tasks inherit the class of the original job and are served as normal jobs until they are reassembled at a `Join` station.

Their specification of Fork and Join nodes only requires the name of the node

```
const std::size_t fork = model.add_fork("Fork");
const std::size_t join = model.add_join("Join", fork);
```

The number of tasks sent by a `Fork` on each output link can be set using the `method` of the `fork` object. To enable effective analytical approximations, presently LINE requires that every join node is bound to a specific fork node, although specific solvers will ignore this information (e.g., JMT).

Quorum (partial) joins. A `Join` waits by default for every sibling of the forked job. It can instead be told to fire on the first k of n siblings, discarding the ones that arrive later, which is the early-reply pattern of replicated storage and quorum protocols:

```
// the quorum rides on the strategy: fire on the second of three siblings
builder.set_join_strategy(join, lang::JoinStrategy::PARTIAL, 2.0);
```

The simulators JMT and LDES reproduce the discipline on the sample path. The analytical solvers MVA and NC charge the k -th order statistic $E[X_{(k)}]$ of the branch completion times at the join instead of $E[\max]$ (`Fj_ordstat_exp`), and the request tail is read from the k -of- n approximation `Fj_tail_ordstat` behind $-a$ percentile. The feature is declared as `JoinPartial` in the feature sets of MVA, NC, JMT and LDES; CTMC, SSA and MAM refuse it by name rather than answering the question of a standard join. A quorum join is also the one exception to the `ArvR - Tput` identity of §4.2.5: the discarded siblings make the join loss rate $\max(0, \text{ArvR} - k \text{ Tput})$. The example `gallery_fj_quorum` is a closed network with a 2-of-3 quorum.

Variable forking levels. The forking level need not be one number shared by every link and every job. Three overrides sit beside the scalar and may be given in any order with respect to the links themselves:

- `set_fork_tasks_per_link(fork, class, n, dest)` gives one class, and optionally one destination, its own number of tasks, leaving the other classes and links on the node-wide value.
- `set_fork_tasks_per_link_dist(fork, class, dist, dest)` makes the number of tasks a draw from `DiscreteSampler`, redrawn independently for every link and every forked job. This is the variable forking level of JMT’s `JobsPerLinkDis`.
- `set_fork_branch_probability(fork, class, dest, p)` fires the branch towards `dest` only with probability p , and emits nothing otherwise. The branches are activated independently, so the number of siblings is random even when the tasks per link are deterministic. The matched `Join` must be told what to wait for, since a job that skipped a branch would block a standard join forever: a branch probability below one requires `JoinStrategy.PARTIAL` or a quorum (§5.4.3).

The scalar remains what every existing consumer reads and is now the *mean*, with the branch probability folded into it. The overrides are exact under JMT and LDES, which draw the degree at the fork epoch, and are seen as the expected degree by the analytical solvers, whose MMT correction is linear in it. The exact CTMC and SSA path serves an integer per-link count together with `JoinStrategy.PARTIAL`, but refuses a random degree and a branch probability below one by name, rather than answering with the certain-fork numbers, because its tag construction fixes the sibling count and the sibling set when the state space is built. The example `fj_variable_fanout` builds the same network in each of the four modes.

Also note that the routing probabilities out of the `Fork` node need to be set to 1.0 towards every other node connected to the `Fork`. For example, a `Fork` sending jobs in class 1 to nodes *A*, *B* and *C*, cannot send jobs in class 2 only to *A* and *B*: it must send them to all three connected nodes *A*, *B* and *C*. A new fork node visited only by class-2 jobs needs to be created in order to send that class of jobs only to *A* and *B*.

Fork nodes also support probabilistic (state-dependent) branching via `RoutingStrategy.PROB`, where different output links carry non-uniform routing probabilities. This is set using `set_fork_routing_strategy` on the `Fork` node and is available when using JMT. This allows a `Fork` node to direct tasks to different parallel branches with configurable proportions, enabling asymmetric fork-join models where not all branches are equally loaded.

After splitting a job into tasks, LINE takes the convention that visit counts refer to the average number of passages at the target resources for the original job, scaled by the number of tasks. For example, if a job is split into two tasks at a fork node, each visiting respectively nodes *A* and *B*, the average visit count at *A* and *B* will be 0.5.

For Fork-Join response time percentiles, MAM automatically detects valid FJ topologies (Source $\rightarrow$ Fork $\rightarrow$ K Queues $\rightarrow$ Join $\rightarrow$ Sink) and applies percentile computation as follows. Response-time percentiles of a fork-join network are not reported by the C++ port; the fork-join solution returns mean measures only.

Figure 5.3 shows the resulting topology.

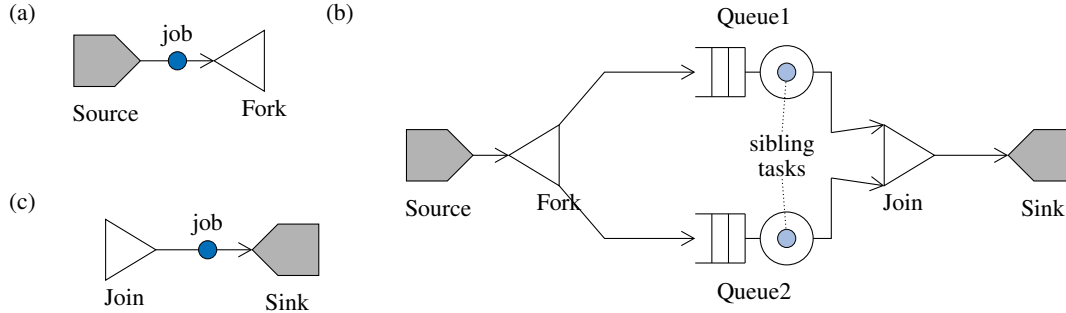


Figure 5.3: Fork and join. (a) A job travelling from the source to the fork. (b) The fork replaces that job with one sibling task per outgoing link, drawn in the lighter shade; the tasks are served as ordinary jobs and are reassembled at the join. (c) The job restored by the join, travelling on to the sink.

5.4.4 Class switching with non-probabilistic routing strategies

In the presence of non-probabilistic routing strategies, such as round-robin or join-the-shortest-queue, one may need to manually specify the details of the class switching mechanism. This can be done through addition to the network topology of `ClassSwitch` nodes.

The constructor of the `ClassSwitch` node requires a probability matrix C such that the element in row r and column s is the probability that a job of class r arriving into the node switches to class s during the visit. For example, in a 2-class model, the following node will switch all visiting jobs into class 2

```
// Block 1: nodes
Source source(model, "Source");
// ...
// Block 2: classes, declared BEFORE the ClassSwitch node
OpenClass jobclass1(model, "Class1", 0);
OpenClass jobclass2(model, "Class2", 0);
// Block 3: the switching matrix and the node that carries it
Matrix<double> C(2, 2);
C(jobclass1 - 1, jobclass2 - 1) = 1.0;
C(jobclass2 - 1, jobclass2 - 1) = 1.0;
const std::size_t csnode = model.add_class_switch("ClassSwitch 1", C);
```

The matrix is passed to the constructor, so the classes it is indexed by must exist first. Note that for a network with M stations, up to M^2 `ClassSwitch` elements may be required to implement class-switching across all possible links, including self-loops.

5.4.5 Logger node

A logger node is a node that closely resembles the logger node available in the JSIMgraph simulator within JMT. Models that include this element can be solved using the JMT and LDES simulators.

A `Logger` node records information about passing jobs in a `csv` file, such as the timestamp of passage and general information about the jobs. The node can be instantiated as follows. The C++ builder has no `Logger` node: the node type exists in the struct, but a model built in C++ writes no trace file.

The file is written in CSV format, one row per job passage, with a column for each enabled field. The `LogTunnel` section that implements the logging is transparent: a job crosses the logger instantaneously and no queueing delay is introduced, so a logger may be placed anywhere in the topology without perturbing the metrics being measured. Placing several loggers yields a trace of job trajectories from which empirical inter-passage distributions can be estimated offline.

Each method can be called either with a single argument, to enable or disable the recording of the corresponding information, e.g. Nothing is logged by the C++ port, so the per-field logger switches have no counterpart. The routing behavior of jobs can be set up as explained for regular nodes such as queues or delay stations.

5.5 Finite capacity regions

A Finite Capacity Region (FCR) provides a mechanism to constrain the total number of jobs that can simultaneously reside within a designated group of stations in a queueing network. Unlike per-station buffer limits, which restrict occupancy at individual queues, an FCR imposes a collective constraint across multiple stations, limiting the aggregate number of jobs across all stations in the region. When the region reaches its capacity, incoming jobs are either dropped or blocked depending on the configured drop strategy. This modeling capability is particularly useful for representing systems with admission control policies, finite buffers spanning multiple processing stages, resource constraints that affect groups of servers, or loss networks where jobs are rejected when the system lacks sufficient capacity.

An FCR is created using the `add_region` method of the `Network` class, which takes as input a cell array or list of stations belonging to the region. The maximum occupancy across all stations in the region is set via `cap`, which specifies the total number of jobs allowed in the region. The drop behavior for each class is configured using `set_drop_rule`, which determines whether arriving jobs are dropped when the region is at capacity.

Figure 5.4 illustrates the mechanism.

```
// nodes of the region, per-class job caps, then the region-wide cap
const std::vector<std::size_t> members{queue1, queue2};
const std::vector<double> classmax{-1.0, -1.0};           // -1 = unbounded
const std::vector<DropStrategy> rule{DropStrategy::DROP};
model.add_region(members, classmax, double(K), rule);
```

A region is declared in one call: its members, the per-class caps, the global cap, and the per-class rule that says whether an excess job is dropped (`DROP`) or waits (`WAITQ`).

Two primary drop strategies are available. The `DropStrategy.Drop` strategy causes jobs to be permanently lost when they arrive at a full region, modeling systems where arrivals are rejected or abandoned. The `DropStrategy.Queue` strategy causes jobs to wait in a virtual queue until capacity becomes available,

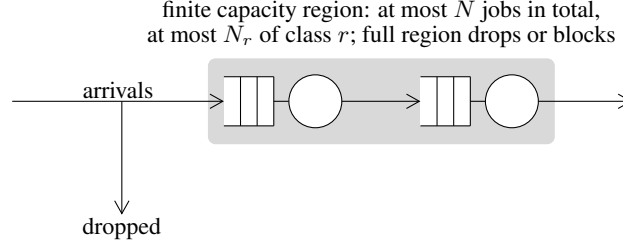


Figure 5.4: A finite capacity region. The constraint applies to the total occupancy of the member stations rather than to any single buffer, and an arrival that would violate it is dropped or blocked according to the drop rule.

modeling systems with admission control or backpressure mechanisms. In addition to global capacity constraints, per-class capacity limits can be specified within a region to enforce class-specific occupancy restrictions.

Beyond the global job cap, a region may declare several further admission constraints, and they compose. Per-class occupancy limits are set with `.`. A pooled resource whose classes consume unequal shares is expressed by giving each class a footprint with `.` and a budget with `.`, which admits a job only while the weighted occupancy stays within the budget. Arbitrary linear admission rules are supplied directly with `.`, where $\mathbf{A}$ has one column per class and one row per constraint.

```
const std::vector<std::size_t> members{delay};
const std::vector<double> classmax{-1.0, -1.0, -1.0};
const std::vector<DropStrategy> rule; // defaults to WAITQ
const std::vector<double> classmem; // per-class memory caps
const std::vector<double> classsize{1.0, 2.0, 3.0}; // weights n_1, n_2, n_3
model.add_region(members, classmax, 9.0, rule, classmem, classsize, 14.0);
```

The class sizes and the region-wide memory budget are arguments of the same call; general linear constraints beyond these two are not supported.

Taken together these declarations define the admission rule $\mathbf{A}\mathbf{n} \leq \mathbf{C}$ on the vector $\mathbf{n}$ of per-class occupancies of the region, an arriving job of class r being admitted only when $\mathbf{A}(\mathbf{n} + \mathbf{e}_r) \leq \mathbf{C}$. Every row is a function of the per-class occupancy of the region alone, so no constraint can distinguish the individual stations inside it: a job moving from one station of the region to another leaves every left-hand side unchanged. The admissible set is coordinate convex, which is what allows the truncated model to retain a product form under the `Drop` policy.

The simulators (JMT, LDES, SSA) and `SolverCTMC` honour the full rule for any region contents. Among the analytical solvers, `SolverNC` recognizes the open single-Delay region under `Drop` as a loss network and solves it from the exact normalizing constant of the product form, assembling $\mathbf{A}$ and $\mathbf{C}$ from all four declarations above. Four methods are offered there through `options.method`: `exact`, the default on an integral admission rule, which evaluates the normalizing constant exactly by contour integration and residues [108]; `rec`, which evaluates the same constant exactly as the sum of the product form over the

admissible set $\{n \geq 0 : \mathbf{A}n \leq \mathbf{C}\}$, by one memoised walk of a multi-valued decision diagram holding that set [6], and which is therefore the default when the region declares fractional class sizes or capacities, where the residue argument cannot count whole units; `erlangfp`, the Erlang fixed-point (reduced-load) approximation [87]; and `mci`, importance-sampling Monte Carlo summation with confidence intervals [138], for regions whose capacities make the exact evaluation too large. The loss-network tutorial in the LINE primer works through a complete example. `SolverFLD` carries a region under its `dae` method (§8.2.4.2), where the cap becomes a linear algebraic constraint solved beside the fluid drift and the blocked jobs are held in an explicit waiting room; that route accepts the waiting-queue rule only, and refuses per-class admission weights and the blocking and retrial rules, which change the event set rather than constrain the drift. The remaining analytical solvers, and every other fluid method, reject models with a finite capacity region rather than returning the unconstrained answer.

A special case of interest is an FCR containing a single queue with dropping enabled. This configuration behaves identically to an M/M/1/K queue, where K is the maximum capacity including both jobs in service and jobs waiting in the buffer. This equivalence provides a convenient way to model loss systems and finite-buffer queues. The examples `fcr_mmlkdrop`, `fcr_mmlwaitq`, `fcr_constraints`, and `fcr_oqndrop` demonstrate various applications of finite capacity regions, including single-queue finite buffers, multi-station regions with different drop policies, and open queueing networks with job loss.

Linear admission constraints. Besides the global and per-class job caps, a region admits an arbitrary set of linear inequalities $\mathbf{A}n \leq \mathbf{b}$ on the vector $\mathbf{n} = (n_1, \dots, n_R)$ of per-class job counts inside it. A job is admitted only while every inequality stays satisfied. The pair is set with `and` and read back with `;`; the example `fcr_lincon` illustrates it.

General linear constraints on a region are not part of the C++ port: a region carries the per-class caps, the global job cap and the memory budget shown above.

Two further per-region metrics are available with the `JMT` and `LDES` solvers, and are useful when classes consume unequal shares of a pooled resource, as in cache or storage models. `FCRWeight` (`MetricType.FCRWeight`) is the time-average total weight of the in-region jobs, summing the per-class weight set with `;` with the default weight 1 it coincides with the number of customers. `FCRMemOcc` (`MetricType.FCRMemOcc`) is the time-average memory occupation, summing the per-class footprint set with `.`

In `JMT` the two map to the native region measures `FCR Capacity` and `FCR Memory`; `LDES` accumulates the same time-weighted quantities directly. They are returned per region and per class as `WeightNfcr` and `MemOccNfcr`, whose row sums are the region aggregates.

5.6 Model adaptation

The `ModelAdapter` class provides static methods for transforming and adapting `Network` models. These methods are useful for model reduction, response time analysis, and preprocessing operations.

5.6.1 Chain aggregation

The function transforms a multi-class model with K classes organized into C chains into a model with C classes, preserving chain population, arrival rates, service demands, and routing while reducing state space when $C \ll K$. The function returns the aggregated model, an aggregation factors matrix α satisfying $\sum_{r \in c} \alpha(i, r) = 1$, and deaggregation information for transforming chain-level results back to class-level metrics. Chain aggregation is internal to the C++ solvers: `mva::sn_chain` aggregates the struct on the way into an algorithm and `mva::sn_deaggregate_chain_results` splits the answer back per class. There is no model-to-model adapter a caller drives. The same transformation is reachable from the model itself, which is the recommended form: As above, chain aggregation is not exposed as a model transformation in C++. Class switching is eliminated by the transformation, since it is internal to a chain. The aggregation is exact for product-form models and approximate otherwise, because a single service process fitted on the chain mean and squared coefficient of variation replaces the per-class ones weighted by α . Chain-level metrics are mapped back to class level with `sn_deaggregate_chain_results`, passing the chain queue lengths and utilizations as empty: splitting the chain queue length by α alone ignores the per-class service times, whereas the response-time branch rescales them by S_{ir}/S_{ic} . The CTMC solver can drive the whole round trip itself, through `options.config.chain_aggregation` (§8.2.3.1).

5.6.2 Flow-equivalent server aggregation

The function replaces a chosen subset of the stations of a closed product-form network with a single flow-equivalent server, returning the reduced model, the aggregated station and the information needed to recover the metrics of the original stations. The reduced model is smaller and cheaper to solve, and its throughput and utilization metrics are exact. The CTMC solver applies the transformation and recovers the collapsed stations' own metrics in a single solve when `options.config.fes_stations` names them (§8.2.3.1). The transformation is shown in Figure 5.5.

The flow-equivalent server aggregation is available at the API level only (`fes::fes_build_isolated`, `fes::fes_compute_throughputs`, `fes::fes_beta_handle`), not as a model-to-model transformation.

5.6.3 Convolution-based FES analysis

The FES station produced by `aggregateFES` stores its throughput lookup table as a *per-class dependence* function (§5.2.5). For each aggregated station the per-class throughput $X_r(n)$ is tabulated over all population vectors up to a per-class cutoff, so that the exact dependence of throughput on the joint class population is preserved; the table is then wrapped as the handle $\beta_{i,r}(n) = X_r(n)$, i.e. Sauer's chain-dependent service rate $\mu_{r,i}(n)$ [142], and stored in the `NetworkStruct` field `cdscaling{ist}`. The population is clamped to the cutoffs the table was built on, so the rate saturates beyond the tabulated range.

When a model contains one or more class-dependent stations, the NC solver uses the multichain convolution algorithm (`pfqn_conv`) to compute exact normalizing constants and performance metrics, building each station factor $X_m(n)$ by the recurrence $X_m(n) = (u_{km}/\mu_{km}(n)) X_m(n - e_k)$ [142]. The convolution

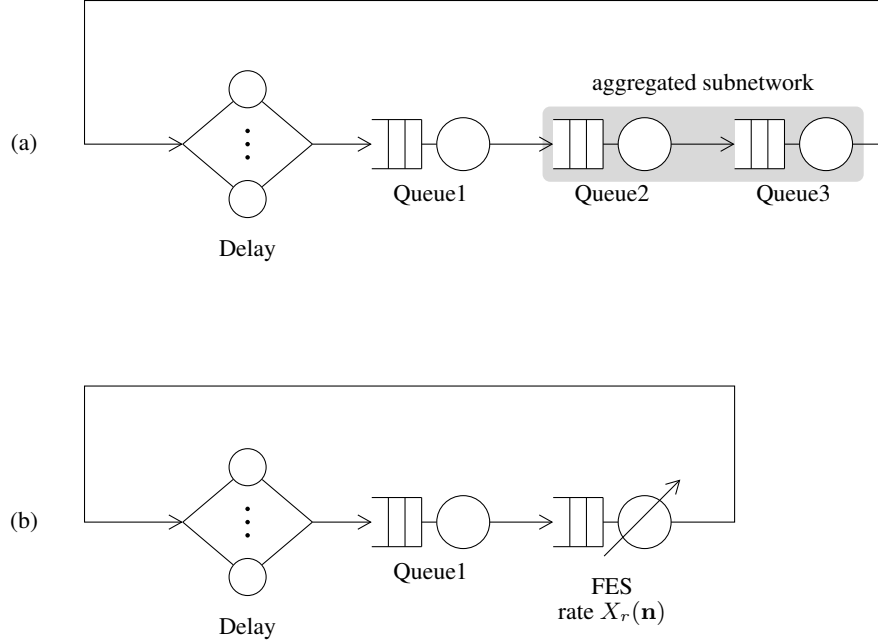


Figure 5.5: Flow-equivalent server aggregation. (a) The closed network before the transformation, with the aggregated subnetwork shaded. (b) The reduced network, in which the two aggregated stations are replaced by a single load-dependent station whose class- r rate is the throughput $X_r(\mathbf{n})$ of the isolated subnetwork.

is invoked automatically by the NC solver when a class-dependence function is detected and `method=exact` is requested; no explicit user action is required beyond creating the FES model via `aggregateFES` with appropriate cutoffs.

Burstiness-preserving FES analysis. The flow-equivalent server of §5.6.2 keeps the mean throughput of the aggregated subnetwork and discards every other property of its output. That is exactly right for a product-form subnetwork, by Norton’s theorem [36], and it is the reason the reduced model of Figure 5.5 is exact. It is also why the transformation cannot represent a *bottleneck switch*, in which the congested resource alternates over time among the aggregated stations: the alternation is a property of the departure stream that a single mean rate cannot carry. LINE therefore also provides the load-dependent MAP flow-equivalent server of [29], in which the composite station serves according to a Markovian arrival process that changes with the population it holds.

The burstiness of a stream is measured by its asymptotic index of dispersion, which combines the squared coefficient of variation of the inter-departure times with their autocorrelation at every lag. It equals one for a Poisson process and grows as samples of similar size aggregate into bursts. A subnetwork made of one station with MAP service and a flow-equivalent server emits inter-departure times that are themselves a MAP,

block bidiagonal over the population held by the flow-equivalent server. Its first three moments and its index of dispersion are obtained by a vector recursion that never forms a dense inverse, and a second-order MAP is then fitted to them: a MAP(2) has a geometrically decaying autocorrelation, so the index of dispersion inverts in closed form into a decay rate, which is handed to the explicit characterization in [76]. Stations are folded one at a time, so a subnetwork of any size reduces to a single load-dependent MAP at a cost independent of the number of stations.

When the aggregate is not overdispersed, that is when its inter-departure times are no more variable than an exponential or their index of dispersion falls below their squared coefficient of variation, the fit returns an exponential. The rule is not merely a feasibility guard and should not be relaxed: an exponential load-dependent server is exact for a product-form subnetwork, whereas a MAP(2) fitted to the marginal inter-departure statistics is not, because the departure stream of a subnetwork is not independent of the rest of the model. With the rule in place the construction degrades gracefully to the classic flow-equivalent server, and reproduces exact mean value analysis to machine precision on an exponential subnetwork.

The API is available in all four languages, as `fes_map_interdeparture`, `fes_map_moments`, `fes_map_aggregate` for the aggregation, and `fes_map_solve`, `fes_map_deaggregate` for the reduced model. The latter reads the block-bidiagonal inter-departure MAP as the generator of the closed model made of the think times and the aggregate, its level being the number of jobs held by the flow-equivalent server; the think time may itself be a MAP, which is how a bounded flash crowd is modelled. Per-station metrics follow by conditioning on the population of the aggregate, which is the decomposition step of the hierarchical analysis [36] and is exact for a product-form subnetwork.

```
auto agg = line::fes::fes_map_aggregate(maps, {1, 1}, N);
auto sol = line::fes::fes_map_solve(agg.fes, ...
line::mam::map_exponential_mean(Z), N);
```

On a closed model with two bursty MAP stations and a delay, the reduced model built this way predicts the throughput within 2.8–5.0% of the exact Markov chain, where the product-form solution of the same model is in error by 21.7–47.0%; when the subnetwork holds a single station, so that no aggregation approximation is involved, the reduced model reproduces the exact chain to machine precision.

5.6.4 Tagged job models

The `tag_chain` function creates a tagged job model for response time distribution analysis. It isolates a single job from a specified chain by creating duplicate tagged classes, enabling the computation of per-job response time distributions using methods such as passage time analysis.

```
const qn::TaggedChain<double> tag =
    qn::tag_chain(model.get_struct(), chain, jobclass);
```

The tagged model is a struct transformation rather than a model object, and is what the response-time distribution getters consume.

5.6.5 Class removal

The `removeClass` function removes a specified job class from a model, updating all station configurations, routing matrices, and class-dependent parameters accordingly. This is useful for analyzing submodels or simplifying complex multi-class networks. The removal is delegated to the nodes, each of which drops its own per-class configuration (service and arrival processes, class capacities, scheduling parameters, routing strategies, and the class-switching matrix of a `ClassSwitch`). Removing the last remaining class, or any class of a model containing a `Cache`, is rejected: the cache item state is indexed by class and cannot be re-indexed on the node alone.

Two variants are available. The `removeClass` method removes the class in place, whereas `ModelAdapter::removeClass` returns a copy and leaves the original model solvable, which is the form to use for ablation studies or a per-class decomposition: Classes cannot be removed from a built model in C++: `qn::Network` grows monotonically, and a model without a class is built as such.

Chapter 6

Other stochastic models

Most formalisms in this chapter are described with the same `Network` class as a queueing network, and are solved through the same solver, but they model something else: an item cache, a stochastic Petri net, and a network in which jobs may be created or destroyed by signals. The chapter opens instead with the two classes that carry a Markov chain written down directly, without a network around it. They are collected here so that the queueing-network chapters stay about queues.

6.1 Markov chains and processes

A model need not be a network. `LINE` accepts a chain written down by the user, namely a `MarkovProcess`, holding an infinitesimal generator Q , or a `MarkovChain`, holding a transition matrix P . No state space is generated in this mode, the given matrix being solved as it stands; a discrete-time chain is carried as its image $P - I$, which has the same stationary distribution. The methods that describe the chain remain available – for the stationary distribution, `getGenerator`, `getStateSpace`, `getProb` for a single state, `getTranProbSys` for the transient distribution and `sampleSys` for a sample path – whereas the average performance metrics are refused, a chain having neither stations nor classes. A state is identified by its row in the state space attached to the chain with `setStateSpace`, or by its index when the chain carries none. In transient analysis the initial distribution is `options.init_sol`, or the uniform distribution when none is given.

```
Matrix<double> Q(2, 2);
Q(0,0) = -0.5; Q(0,1) = 0.5; Q(1,0) = 0.2; Q(1,1) = -0.2;
const std::vector<double> pi = mc::ctmc_solve(Q);

Matrix<double> P(2, 2);
P(0,0) = 0.9; P(0,1) = 0.1; P(1,0) = 0.4; P(1,1) = 0.6;
const std::vector<double> piDtmc = mc::dtmc_solve(P);

const std::vector<double> piT = mc::ctmc_transient(Q, pi0, 10.0);
```

A chain given directly is solved by the API layer: there is no solver object to wrap it in. The two classes also expose the Markov-chain algorithms of the API directly, so that a chain can be analysed without going through a solver at all. On a `MarkovProcess`:

- `solve` and `solveRelative`: the stationary distribution, normalized or relative to a reference state.
- `transient` (`transientProb` in Java, `transient` being a keyword there): the distribution at a finite horizon, its method argument selecting Jensen uniformization [83] or the Fox-Glynn weights [60].
- `timeAverage`: the time-averaged distribution over a horizon, read off the same uniformization series [83] that gives the endpoint distribution.
- `sens`: the sensitivity of the stationary distribution to the generator entries, from the differentiated balance equations [158].
- `aggregate`: an approximate distribution obtained by aggregation-disaggregation over a macrostate partition, by the Courtois [46], Koury-McAllister-Stewart [91], Takahashi [154] or multi-level [151] method, returned together with the nearly-complete-decomposability index of the partition.

On a `MarkovChain`:

- `solve`: the stationary distribution.
- `transient`: the distribution after a number of steps, and `transientUnif` over a continuous horizon.
- `hittingTime`: the expected time to reach a set of states.

On both classes: `stochComp` and `stochCompFull` (the stochastic complement of a subset of states, the latter returning also the blocks it was built from), `isFeasible`, `toTimeReversed` and `sample`. A continuous-time chain converts to discrete time in two distinct ways: `toDTMC` uniformizes it, preserving the stationary distribution, whereas `toEmbedded` returns the embedded jump chain, which records the sequence of states visited and drops the holding times.

6.2 Caches

6.2.1 Cache node

This is a stateful node to store one or more items in a cache of finite size, for which it is possible to specify a replacement policy. The `cache` constructor requires the total cache capacity and the number of items that can be referenced by the jobs in transit, e.g.,

```
qn::CacheParam<double> par;
par.nitems = nitems;
par.itemcap.assign(1, capacity); // one list of this size
par.replacestrat = ReplacementStrategy::LRU;
const std::size_t cacheNode = model.add_cache("Cache1", par);
```

The cache is described by a parameter struct, whose `itemcap` holds one entry per list of the replacement policy.

If the capacity is a scalar integer (e.g., 15), then it represents the total number of items that can be cached and the value cannot be greater than the number of items. Conversely, if it is a vector of integers (e.g.,) then the node is a list-based cache, where the vector entries specify the capacity of each list. We point to [66] for more details on list-based caches and their replacement policies.

Available replacement policies are specified within the `ReplacementStrategy` static class and include:

- First-in first-out (`ReplacementStrategy.FIFO`)
- Random replacement (`ReplacementStrategy.RR`)
- Least-recently used (`ReplacementStrategy.LRU`)
- Strict first-in first-out (`ReplacementStrategy.SFIFO`)

Upon cache hit or cache miss, a job in transit is switched to a user-specified class. More details are given later in Section 3.2.5.

Beyond basic capacity constraints, cache nodes in LINE support item sizes, where each cached item consumes a configurable amount of storage rather than occupying a single slot; this is described in Section 6.2.2. State-dependent routing exploits detailed cache state information to direct jobs along different paths depending on whether their requested item is present in the cache. When a job requests a cached item (a hit), it is classified into a hit class and routed to a delay node representing fast cache access, while jobs experiencing misses are switched to a miss class and directed to a queue representing slower backend retrieval. Solvers such as CTMC and NC employ stochastic complementation techniques to efficiently analyze cache models by eliminating intermediate Router states and computing effective transition rates between observable cache states.

6.2.2 Item sizes and storage cost caps

By default every item occupies one slot of a list, so a list holds as many items as its capacity, whatever they are. When the cached objects have heterogeneous sizes, each item may instead be given a storage cost, a positive integer, and each list a cap on the total cost of the items resident in it:

```
par.itemsize = std::vector<int>{1, 1, 1, 2, 2, 2}; // storage cost of each item
par.costcap  = std::vector<int>{2, 1};           // cap on the cost of each list
```

A single value passed to `CacheParam::costcap` declares one cap for the whole cache. The policy is extended as in [26]: a request whose promotion, or whose insertion after a miss, would push a list above its cap is served, but leaves the cache state unchanged. This applies to the miss list as well, so an item too large for the list it would enter is served without being cached.

The state space is a restriction of a reversible chain, so the product form survives. The constrained normalizing constant obeys a recursion on an arbitrary item, which is either absent from the cache or resident in one of the lists; each residency deducts that item's storage cost from the remaining cap of the list it sits in, and a term whose residual cap has gone negative contributes nothing. The probability that an item resides in a given list is the term it contributes to that recursion, normalized by the constant itself, and the mean storage

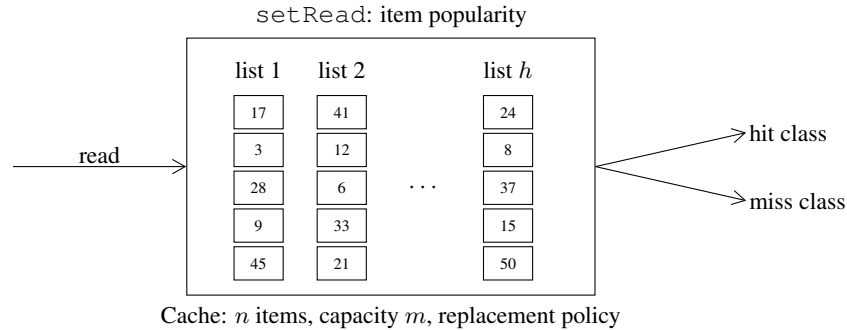


Figure 6.1: The `Cache` node. A read draws an item from the popularity distribution declared with `setRead`; the job then leaves the node in the hit class or in the miss class, and the item lists are updated by the replacement policy. The numbers in the slots are the identifiers of the items currently memorized, all distinct across the h lists.

cost a list holds is the cost-weighted sum of those probabilities over the catalogue, reported as the `ListCost` column of the cache table.

NC evaluates these expressions with `method='exact'`, or by Monte Carlo summation with `method='sampling'` on larger catalogues; the approximate methods have no cost-capped counterpart and are switched automatically, with a warning. `LDES` simulates the rule directly and is the reference when the caps make a list unreachable: the recursion sums over every size-feasible state, which is a strict superset of the states the cache can visit whenever a cap on an intermediate list blocks a promotion path, and `LINE` warns when it detects that case.

Figure 6.1 summarizes the node.

6.2.3 Cache-based class-switching

An advanced feature of `LINE` available for example within the `Cache` node, is that the class-switching decision can dynamically depend on the state of the node (e.g., cache hit/cache miss). However, in order to statically determine chains, `LINE` requires that every class-switching node declares the pair of classes that can potentially communicate with each other via a switch. This is called the *class-switching mask* and it is automatically computed. The boolean matrix returned by the function provides this mask, which has an entry in row r and column s set to true only if jobs in class r can switch into class s at some node in the network.

Upon cache hit or cache miss, a job in transit is switched to a user-specified class, as specified by the `CacheParam::pread` function allows the user to specify a discrete distribution (e.g., `Zipf`, `DiscreteSampler`) for the frequency at which an item is requested. For example,

```
// Zipf(0.5) popularity over nitems, as the read probabilities of initClass
std::vector<double> pop(nitems);
double norm = 0.0;
```

```

for (std::size_t i = 0; i < nitems; ++i) { pop[i] = 1.0 / std::pow(double(i + 1), 0.5); ...
norm += pop[i]; }
for (std::size_t i = 0; i < nitems; ++i) pop[i] /= norm;
par.pread.assign(model.get_struct().nclasses, std::vector<double>());
par.pread[initClass - 1] = pop;
par.hitclass.assign(model.get_struct().nclasses, 0);
par.missclass.assign(model.get_struct().nclasses, 0);
par.hitclass[initClass - 1] = hitClass;
par.missclass[initClass - 1] = missClass;

```

The popularity is given as the read probability vector itself: the C++ port has no Zipf object, and the hit and miss classes are per-read-class entries of the cache parameters. Here can be either open or closed instantiated as usual with the `OpenClass` or `ClosedClass` constructors.

6.2.4 Cache networks

A model may hold several `Cache` nodes arranged in a hierarchy, so that a request that misses at one level is served by the next. The difficulty such a model raises is not the number of nodes but the identity of the item requested. A cache node draws the item referenced by an arriving job from the popularity distribution declared for its class, so a miss routed into a second cache would make that cache sample an item independently of the one that was just missed. Only the aggregate miss rate would then cross the boundary between the two levels, whereas the item-level correlation of the miss stream is precisely the phenomenon a cache hierarchy exists to exhibit.

LINE resolves this by carrying item identity in the job class. Each item is given a dedicated class, whose popularity is a point mass on that item, and the miss class of level l for item f is declared to be the read class of level $l + 1$ for the same item f . Item identity is then preserved by ordinary routing, no class switching takes place on the arc between two caches, and the number of classes grows as nL rather than combinatorially in the number of levels L and items n . Throughout we use f as the item index and l as the level index. Item popularity is expressed by the per-class request rates, or by the per-class populations in a closed model, rather than by a distribution the cache draws from, which is also how the analysis of cache networks is usually parameterized in the literature.

Four functions on the `Cache` node build such a model. The cache the exogenous requests enter declares the user's own per-item classes with `set_item_read_classes`. Each subsequent level is attached with `set_miss_cache`, which mints the per-item classes of the next cache, binds the miss class of each item to the corresponding read class there, and registers the cache-to-cache routing arc. That arc is injected when the model is linked, in the same way as the arcs of a retrieval system, since the classes it connects do not yet exist when the user builds the routing matrix. Lastly, the root of the hierarchy is terminated with `set_item_miss_class`, whose misses leave the network towards the origin server. A cache that is fed by another cache but reached by some other construction may mint its classes directly with `set_item_read_classes`, which `set_miss_cache` itself calls and which is idempotent. All the caches of a network are required to share one item set, and a mismatch in the number of items is refused naming both counts, since padding the shorter catalogue would silently misalign the read probabilities and the item bitmaps.

A tandem of two caches over $n = 3$ items is built as follows.

```

qn::CacheParam<double> c1p, c2p;
c1p.nitems = n; c1p.itemcap = std::vector<int>{1}; c1p.replacestrat = ...
ReplacementStrategy::LRU;
c2p.nitems = n; c2p.itemcap = std::vector<int>{2}; c2p.replacestrat = ...
ReplacementStrategy::LRU;
const std::size_t cachel = net.add_cache("Cache1", c1p);
const std::size_t cache2 = net.add_cache("Cache2", c2p);
std::vector<std::size_t> read(n), hit1(n), hit2(n), miss(n);
for (std::size_t f = 0; f < n; ++f) {
    read[f] = net.add_closed_class("Read" + std::to_string(f+1), 1.0, think);
    hit1[f] = net.add_closed_class("Hit1_" + std::to_string(f+1), 0.0, think);
    hit2[f] = net.add_closed_class("Hit2_" + std::to_string(f+1), 0.0, think);
    miss[f] = net.add_closed_class("Miss_" + std::to_string(f+1), 0.0, think);
    net.set_service(think, read[f], Exp(pAccess[f]));
}
net.set_item_read_classes(cachel, read, hit1);
net.set_miss_cache(cachel, cache2, hit2);
net.set_item_miss_class(cache2, miss);

```

The builder addresses nodes and classes by index, so the terminating call names the cache whose misses leave the network rather than the class that reads it.

The user then draws only the arcs of the intended topology, namely the request arc into the first cache and the return arcs of the hit and miss classes, one per item. The miss hop between the two caches belongs to the construction and is added automatically. The hit and miss classes may be given either one per item, as above, or as a single class shared by every item when the per-item split is not of interest.

Each cache reports one row per item in the table returned by `net.get_row(f)`, the item being named in the `Item` column that this construction adds. The corresponding structural field is `nodeparam[i].classitem`, which records the item each class reads and is zero for a class that reads none, and which is exchanged as the `itemClass` key of the JSON model interchange. Inferring the same information by inspecting the read probabilities would be ambiguous, since a one-hot popularity is also a legitimate way of describing a single-item catalogue.

Two properties of the construction are worth stating. First, the per-item hit probabilities of the first level reproduce, under the independent reference model, the classical result that an LRU cache of unit capacity holds item f with probability p_f , which offers an exact closed-form check of any solver on the model. Second, an LRU cache of unit capacity at both levels makes the second level unreachable, in the sense that it can never register a hit. A miss at the first level inserts the requested item there, so the next miss is necessarily for a different item and consecutive arrivals at the second level are always distinct. A hierarchy of unit caches is therefore exclusive by construction, and the second level must be given room for more than one item before it can exhibit any hit at all.

Cache networks are solved by `CTMC`, which is exact on them, and by the simulation solvers `SSA` and `LDES`, the latter treating hits and misses as exact sample-path events under every replacement policy it supports. They may not be solved by `JMT`. Its cache section admits only a parametric popularity, either Zipf or uniform, and has no empirical discrete form, so a point-mass popularity has no faithful representation there. Rather than export such a model as a uniform draw over the whole catalogue, and thereby solve a different model without saying so, `LINE` refuses it naming the cache and the class concerned.

Table 6.1: Per-mode setters of a Transition node

Setter	Effect on the mode
set_distribution	Firing time distribution
set_enabling_conditions	Token multiplicities required on the input arcs
set_inhibiting_conditions	Token-level inhibition arcs
set_firing_priorities	Integer priority breaking ties among enabled modes
set_firing_weights	Weight randomizing the choice among modes of equal highest priority
set_number_of_servers	Per-mode server count; <code>GlobalConstants::MaxInt</code> gives an infinite-server transition
set_timing_strategy	With <code>TimingStrategy::IMMEDIATE</code> , marks the mode immediate and replaces its firing distribution by the <code>Immediate</code> singleton

6.3 Stochastic Petri nets

6.3.1 Place and Transition nodes

LINE supports stochastic Petri net modeling via `Place` and `Transition` nodes. A `Place` holds tokens representing jobs or resources, while a `Transition` fires when its enabling conditions are met, consuming tokens from input places and producing tokens at output places. Transitions support multiple firing modes, each with its own enabling/inhibiting conditions, firing priorities, and firing rate distributions. Modes allow a single transition to represent different behaviors depending on system state. For example:

```
Place place(model, "Buffer");
Transition trans(model, "Process");
const std::size_t fastMode = trans.add_mode("fast"); // returns a 1-based mode index
const std::size_t slowMode = trans.add_mode("slow");
trans.set_distribution(fastMode, Exp(2.0)); // fast processing mode
trans.set_distribution(slowMode, Exp(0.5)); // slow processing mode
trans.set_enabling_conditions(fastMode, jobclass, place, 1); // requires 1 token
trans.set_enabling_conditions(slowMode, jobclass, place, 1);
```

A mode is added by name and then configured through the setters of Table 6.1, as in the other languages. The arcs are recorded sparsely and the $(\text{node}, \text{class})$ matrices behind them are sized when `get_struct()` is first called, so a place or a class declared *after* the transition still receives a correctly shaped row. Each mode of a `Transition` is configured through the per-mode setters of Table 6.1. Their full use is illustrated in the dedicated Petri-net section (§6.3.3); worked models are `spn_basic_open`, `spn_twomodes` and related files in the `examples/` folder.

Figure 6.2 shows the graphical conventions used for these elements.

6.3.2 Queueing places

By default a `Place` is an *ordinary place*: a token deposited by an input transition is immediately available to the output transitions. LINE also supports *queueing places* in the sense of queueing Petri nets (Bause, 1993), where a scheduling station is embedded inside the place. A place becomes a queueing place as soon as a

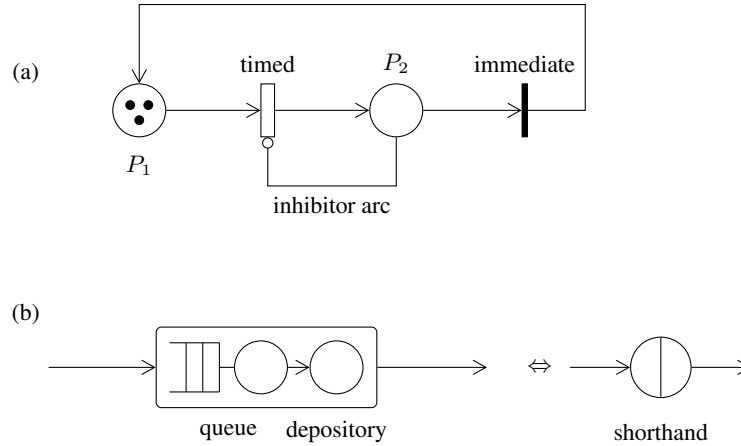


Figure 6.2: Petri-net elements. (a) A timed transition (hollow bar) and an immediate transition (solid bar), with an inhibitor arc that disables the timed transition while P_2 is marked. (b) A queueing place: an arriving token is first served by the embedded queue and only then moves to the depository, where it becomes available to the output transitions; the shorthand symbol on the right is the circle of an ordinary place bisected by a vertical bar.

service process is assigned to a token colour with `set_service`; an arriving token then enters an embedded queue served under the place's scheduling strategy and, upon service completion, moves to a *depository* from which it becomes available to the output transitions. The scheduling strategy is chosen through the optional third constructor argument, and the number of embedded servers through `set_number_of_servers`. For example, a processor-sharing CPU place and an infinite-server think place:

```
Place cpu(model, "CPU", SchedStrategy::FCFS);
cpu.set_service(jobclass, Exp(2.0)); // embedded FCFS queue, rate 2
cpu.set_number_of_servers(2);       // two embedded servers (M/M/2)
```

The three-argument `Place` constructor is `add_place(name, sched)`; the two-argument one still builds the ordinary, infinite-server token container. As elsewhere, it is `set_service` that makes the place a queueing one, which `NetworkStruct::is_queueing_place` reports. The order in which depository tokens become available to the output transitions is governed by the departure discipline (`DepartureDiscipline.NORMAL`, the default, or `DepartureDiscipline.FIFO`), set with `.`. Queueing places are simulated by the `LDSE` solver, which supports the FCFS, LCFS, SIRO and INF embedded scheduling strategies (single- or multi-server) with exponential, deterministic, immediate and renewal phase-type service. The result reported for the place (its token count) is the total marking, i.e. the sum of the tokens waiting, in service and in the depository. Because a queueing place embeds a service station inside a Petri-net place, it cannot be represented as a single node in `JMT` (whose engine keeps token storage and job service in separate nodes); a queueing-place model is therefore rejected by `SolverJMT` and must be expanded into a place plus an adjacent queueing station to be simulated there.

Table 6.2: Internal sections of `Place` and `Transition` nodes

Node	Section	Role
Place	Storage (input)	Token buffer. Accepts tokens deposited by upstream firings and applies the per-class capacity set with <code>set_class_capacity</code> (unbounded by default). The policy is non-preemptive, since tokens in an ordinary place are not served
	ServiceTunnel (server)	Pass-through with no service delay, which is what makes an ordinary place a passive marker store rather than a station
	Linkage (output)	Dispatcher towards the downstream transitions. Each class defaults to <code>RoutingStrategy.RAND</code> , overwritten by the routing matrix installed with <code>model.link(P)</code>
Transition	Enabling (input)	Evaluates the per-mode enabling and inhibiting conditions
	Timing (server)	Draws the firing delay from the per-mode distribution
	Firing (output)	Applies the per-mode firing outcome to the connected places

6.3.3 Petri net specification

LINE supports the specification of stochastic Petri net (SPN) models through dedicated `Place` and `Transition` node types, providing an alternative modeling paradigm to queueing networks. In an SPN, the system state is represented by the distribution of tokens across places, and the system dynamics are governed by transition firings that consume tokens from input places and produce tokens in output places. This formalism is particularly well-suited for modeling systems with synchronization, resource sharing, and complex control flow that may be difficult to express naturally in the queueing network framework.

A `Place` is a stateful node that holds tokens, which correspond to jobs in the queueing network interpretation. Places can have capacity limits and initial token populations. A `Transition` is a node that defines one or more firing modes, each representing a distinct way the transition can fire. Each mode has its own timing distribution, which governs the delay between enabling and firing, its own enabling condition, which specifies the number of tokens required in each input place, and its own firing outcome, which specifies the number of tokens deposited in each output place when the transition fires. The ability to define multiple modes on a single transition allows modeling of complex state-dependent behaviors and routing decisions.

Internal sections of `Place` and `Transition` nodes. Like every node in LINE, a `Place` and a `Transition` are decomposed into three internal sections, listed in Table 6.2, that determine how tokens are accepted, processed and dispatched. The sections are instantiated by the constructors and are not manipulated directly; they are described here because they name the entries appearing in the `nodeparam` tables of §3.2.2.

The following example demonstrates the creation of a simple SPN model. A place `P1` and a transition `T1` are created, and a firing mode is added to the transition. The mode is configured with an exponential firing delay, an enabling condition requiring one token in `P1`, and a firing outcome producing one token in a sink node.

```
Place P1(model, "P1");
Transition T1(model, "T1");
const std::size_t mode = T1.add_mode("Mode1");
T1.set_number_of_servers(mode, GlobalConstants::MaxInt);
```

```
T1.set_distribution(mode, Exp(4));
T1.set_enabling_conditions(mode, jobclass1, P1, 1); // requires 1 token in P1
T1.set_firing_outcome(mode, jobclass1, sink, 1);    // produces 1 token in sink
```

An arc names the class and the node it touches, not an index, so the compiler refuses a place where a class was meant. An arc left undeclared is no arc: enabling and firing default to zero tokens and inhibition to infinity.

In addition to standard enabling conditions, SPN models support inhibiting arcs, which prevent a transition from firing when a place contains too many tokens. This mechanism is useful for modeling resource constraints or conditions where an event should not occur when a system is in a particular state. The `set_inhibiting_conditions` method specifies that a transition cannot fire when a designated place holds more than a specified threshold number of tokens.

```
T1.set_inhibiting_conditions(mode, jobclass1, P2, 2); // inhibit if P2 has >2 tokens
```

An undeclared inhibiting arc materializes as an infinite entry, so a place never named here never inhibits the mode.

SPN models in LINE are analyzed by default with the JMT solver, which translates the Petri net specification into its native simulation engine and performs discrete-event simulation to estimate performance metrics; the CTMC, SSA, and LDES solvers can also analyze models containing Place and Transition nodes. This allows SPN models to benefit from JMT's efficient simulation algorithms while retaining the expressive power of the Petri net formalism. The examples `spn_basic_open`, `spn_twomodes`, `spn_inhibiting`, and `spn_closed_fourplaces` demonstrate various applications of stochastic Petri nets, including open and closed systems, multi-mode transitions for state-dependent routing, inhibiting arcs for resource constraints, and complex token flows.

6.3.4 PNML import and export

A Petri net built in LINE can be written to, and read back from, a PNML document (ISO/IEC 15909-2) in the place/transition grammar <http://www.pnml.org/version-2009/grammar/ptnet>, which is the interchange format of the tools built around that corpus, among them GreatSPN, TINA, PIPE and the Model Checking Contest harnesses.

```
line::io::pnml_save(model.get_struct(), "net.pnml");
auto model = line::io::pnml_load<double>("net.pnml"); // first net
auto model = line::io::pnml_load<double>("net.pnml", "n1"); // by net id
```

The place/transition grammar is *uncoloured*, so it carries less than LINE can express, and the difference is refused rather than approximated. Writing fails, naming the construct, on more than one job class, whose tokens would make the net coloured; on an open class, a Source or a Sink, for which the grammar has no unbounded token source; on a queueing place, which is a station rather than a place; on a firing-rate dependence, which no PNML element can carry; and on a distribution outside the scalar-parameter families, such as a phase-type law or a Markovian arrival process given by matrices.

Table 6.3: Signal types available in `Network` models.

SignalType	Description
NEGATIVE	Removes a random number of jobs from the queue according to a removal distribution (default: 1 job)
CATASTROPHE	Removes all jobs from the queue upon arrival
REPLY	Triggers a reply action for synchronous call semantics

Timing rides in a `toolspecific` block, which is where the grammar puts what it does not define. Each LINE firing mode becomes one PNML transition, so that the arcs of a mode are the arcs of a transition as the grammar requires, and the block records which LINE transition and mode it came from, so that the reader regroups the modes it split. A reader that ignores the block still sees a correct untimed net. In the other direction the grammar says nothing about how long a transition takes to fire, so a file written by another tool is read with every transition timed and exponential at rate 1, the convention of the stochastic Petri net literature and GreatSPN's own default, and its tokens become a single `ClosedClass` holding the initial marking. A file written by LINE reads back with the distributions, servers, priorities and weights it was written with. Inhibitor arcs are read from the `<type value="inhibitor"/>` extension that GreatSPN, TINA and PIPE all write, the grammar itself having no inhibitor arc.

A PNML document is also accepted directly by the command-line front ends, as `-i pnml` or, where the format is inferred from the path, a bare `.pnml` file. Only the solvers whose feature set declares `Transition` can answer for it, namely CTMC, SSA, JMT and LDES:

```
cpp/build/line-cli -f net.pnml -i pnml -s ctmc -a avg
```

6.4 Networks with signals

6.4.1 Signal classes

Networks with signals are supported by the LDES and SSA simulators, by the CTMC solver, which encodes signal arrivals and the induced job removals directly in the state-transition rates, and by the AG solver (§8.2.1), whose agent-based analyzer is the only analytical one that reads the signal marking. No MAM method reads it: each would answer with every signal turned into an ordinary customer, so MAM refuses such a model by name and points at AG instead. Table 6.3 summarizes the signal types available in LINE.

All signal classes are created using the `Signal` class with a required `SignalType` parameter. A `Signal` class can be configured with:

- **Removal distribution:** A discrete distribution specifying how many jobs to remove (default: 1 job). An alternative to the default is a `Geometric` distribution for batch removals.
- **Removal policy:** Determines which jobs are selected for removal when multiple jobs are present:
 - `RemovalPolicy.RANDOM` – Select jobs uniformly at random (default)
 - `RemovalPolicy.FCFS` – Remove the oldest job in the station first

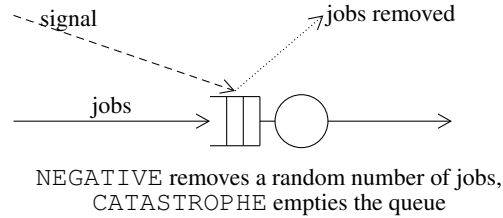


Figure 6.3: Signal classes. A signal arriving at a station does not queue: it removes jobs already present, a random number of them for `NEGATIVE` and the whole queue for `CATASTROPHE`.

- `RemovalPolicy.LCFS` – Remove the newest job in the station first

Figure 6.3 illustrates the effect of a signal arrival.

The following example shows how to create a negative signal class:

```
OpenClass negSignal(model, "NegativeCustomer");
model.set_signal(negSignal, SignalType::NEGATIVE,
                 RemovalPolicy::LCFS); // remove the newest job first
source.set_arrival(negSignal, Exp(0.1));
```

A signal is an ordinary class marked as one; the removal policy is an argument of `set_signal`, whose optional last argument is the batch-removal distribution as a probability vector.

A catastrophe signal removes all jobs from a queue. Use `SignalType.CATASTROPHE`:

```
OpenClass disaster(model, "Disaster");
model.set_signal(disaster, SignalType::CATASTROPHE);
source.set_arrival(disaster, Exp(0.01));
```

Synchronous calls. A signal of type `SignalType.REPLY` carries the return leg of a synchronous call, that is a call whose caller stops and waits for the callee to finish. The signal is bound to the calling class with `,` which names the class whose server is released when the reply arrives. The canonical topology, shown in Figure 6.4, routes the caller from the client station to the callee and switches it to the reply class on departure from the callee, so that the reply travels back to the client. The client keeps a server busy for the whole call, which is what makes the call synchronous, while the callee is not blocked, since its own departure is what creates the reply. The holding station is identified structurally as the station the reply class is routed into, and is recorded in the fields `syncreply` and `replyblock` of the network structure. Because a held server is encoded as a per-class counter, which is exact only where servers are interchangeable or unlimited, the holding station must use `FCFS` or `INF` scheduling. Synchronous calls are also how `LQN2QN` (Section 6.5) flattens the calls of a layered network into an ordinary `Network`.

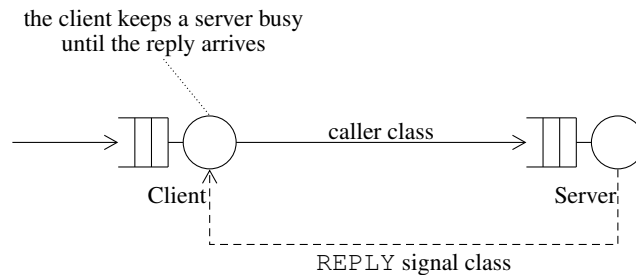


Figure 6.4: A synchronous call. The caller class is routed from the client to the server station and is switched to the `REPLY` signal class on departure from it. The reply releases the server the caller holds at the client, which resumes from where it stopped.

6.4.2 Signal semantics

LINE supports the modeling of networks with signals through the `Signal` class. Signals extend traditional queueing networks by introducing negative customers, which have the distinctive property of removing positive customers from queues upon arrival rather than joining them. This mechanism provides a natural way to model phenomena such as job cancellations, cache invalidations, service interrupts, or resets in computing systems. The interaction between positive and negative customers creates rich dynamics that capture feedback control, load regulation, and catastrophic events in networked systems.

A signal is defined using the `Signal` constructor, which takes a model reference, a descriptive name, and a signal type that specifies the behavior of the signal class. The `SignalType.NEGATIVE` type creates a negative customer class with the semantics that each arrival at a queue removes one positive customer from that queue. If the queue is empty when a negative customer arrives, the negative customer has no effect and is simply absorbed.

```
OpenClass negClass(model, "Negative");
model.set_signal(negClass, SignalType::NEGATIVE);
source.set_arrival(negClass, Exp(lambda_neg));
```

In this example, a negative customer class is created and assigned an arrival process at a source node. When negative customers arrive at queues in the network, they reduce the queue populations by removing positive customers. The routing of negative customers is specified using the same mechanisms as for ordinary job classes, allowing complex patterns of cancellation and control.

```
ag::AgOptions opt;
opt.method = "inap";
const ag::AgResult<double> res = ag::solver_ag(model.get_struct(), opt);
// res.QN, res.UN, res.RN, res.TN
```

The example `ag_gnetwork` demonstrates the construction and analysis of a network with signals featuring both positive customers and negative arrivals, illustrating how signal classes interact with ordinary job classes to produce non-trivial equilibrium behavior. Networks with signals are particularly useful for

modeling systems with dynamic load control, self-regulation, or reset mechanisms, where the conventional queueing network paradigm does not naturally capture the feedback effects of cancellations and interruptions.

6.5 Layered network models

In this section, we present the definition of the `LayeredNetwork` class, which encodes the support in LINE for a class of generalized layered stochastic networks. In their basic form, these models are called layered queueing networks (LQNs) and differ from regular queueing networks as servers, in order to process jobs, can issue synchronous and asynchronous calls among each others. We point to [61] and to the LQNS user manual for an introduction [62]. Contrary to the original LQNs, layered networks in LINE can also include non-queueing servers, such as caches, hence they may be conceptualized as more general layered stochastic networks.

The topology of call dependencies in a layered network makes it possible to partition the model into a set of layers, each consisting of a subset of the servers. Each of these layers is then solved in isolation, updating with an iterative procedure its parameters and performance metrics until the layers solutions jointly converge to a consistent solution.

6.5.1 Basics about layered networks

Layered network models describe a collection of resources called *tasks*, each representing for example a software server, that run on resources called *host processors*. Classes of service exposed by a task are called *entries*. Each entry is an endpoint at which a task can be invoked; for example, if a task represents a web server then its web pages may be described as different entries.

A special task, called the *reference task* is used to represent a group of system users. In this case, the host processor for a reference task can either be real, as in the case of users that are themselves software systems, or fictitious, as in the case of human users.

Figure 6.5 shows a small layered network.

Each entry can be specified by a workflow of operations called *activities*, typically organized as a directed acyclic graph. The time demand that each activity places at the underpinning host processor is called its *host demand* and it is a random variable with a user-specified distribution.

Figure 6.6 shows the precedences an activity graph can express.

Activity graphs may include *calls* to entries exposed by other tasks. This is an abstraction of the calls that distributed system components have among themselves. Calls can be *synchronous*, *asynchronous*, or *forwarding*. Synchronous calls are requests that block the sender until a reply is received, while asynchronous calls are non-blocking and the sender execution can continue after issuing the call. Calls can either be repeated either *deterministic* or *stochastic*, meaning in the latter case that the number of calls issued is a random variable, e.g. geometrically distributed.

Forwarding calls provide a mechanism for delegation chains in layered networks. When a task receives a synchronous request, instead of replying directly to the caller, it can forward the request to another entry

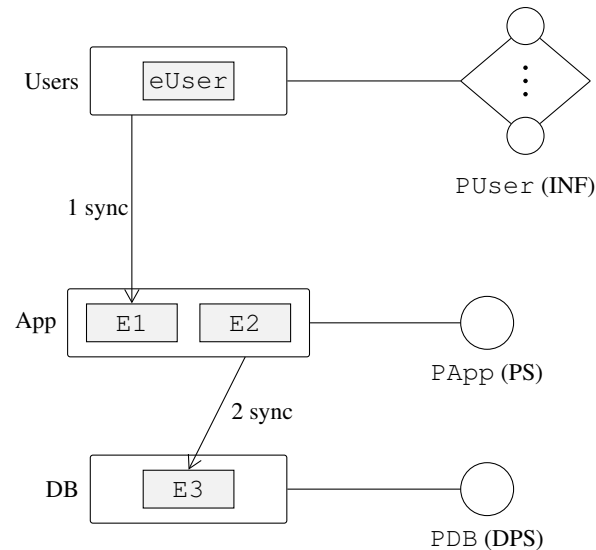


Figure 6.5: A layered network with three tasks. Each task exposes one or more entries and runs on a host processor, drawn to its right; an arc between two tasks is a synchronous call from an entry of the caller to an entry of the callee, labelled by its mean multiplicity.

with a given probability. The forwarding task is released immediately (it does not block waiting for the downstream reply), while the original caller remains blocked until the final entry in the forwarding chain sends the reply. This semantics is useful for modeling multi-tier systems where a front-end server dispatches requests to a back-end without itself waiting for the result—the client blocks end-to-end, but intermediate servers are freed. Forwarding is specified at the entry level using the `forward` method:

```
lqn.forward("E1", "E2", 0.8); // forward 80% of requests from E1 to E2
```

Multiple forwarding destinations can be specified for the same entry, provided that the sum of forwarding probabilities does not exceed one. The remaining probability mass corresponds to the entry replying directly to its caller. Forwarding calls are serialized in the LQNX format using the `<forwarding>` XML element. They are supported by the `LQNS` and `LQSIM` wrappers and by the native `LN` solver, which represents a forwarding call as a caller-side pseudo rendezvous: the forwarding chain is rebuilt so that the original client remains blocked until the last entry in the chain replies, and phase-2 and multiple-destination forwarding are handled by the same construction.

Contrary to ordinary layered queueing networks, a layered network in `LINE` can also feature *cache tasks*, *item entries*, and *cache-access* precedence relations.

- Cache tasks have the basic properties of tasks, but add three specific properties for caching: the total number of items, the cache capacity and the cache replacement policy. Cached items can be either contents or services. Cache capacity indicates the storage constraints of the cache.

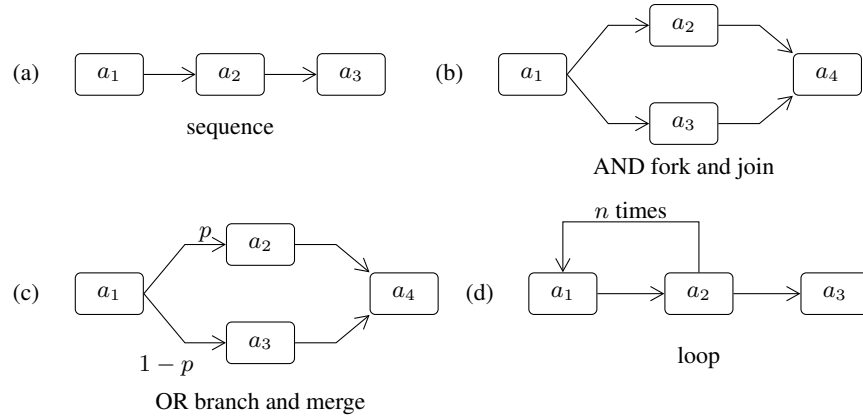


Figure 6.6: Activity graph precedences. (a) A sequence, each activity enabling the next. (b) An AND fork, whose branches run concurrently and are synchronized by an AND join. (c) An OR branch, taking one successor with probability p , merged by an OR join. (d) A loop, repeating an activity a mean number n of times before continuing.

- An item-entry provides instead access to a group of entries of a cache, Item-entries have the basic properties of entries, but add the property of the popularity of the items they give access to.
- A precedence relationship called *cache-access* is defined for the cache hit and miss activities under each item-entry. That is, it is possible to proceed to a different activity depending on whether the cache access produced a cache hit or cache miss. For example, a cache miss can produce a call to a remote entry to retrieve the missing content.

Note that the above extensions are not queueing-based and this explains why these models are referred to in LINE as layered networks and not as layered queueing networks. Similar to the latter, the analysis of a layered networks uses a decomposition of the model into a set of submodels, each being a object, which are then iterative analyzed using different solution methods.

6.5.2 LayeredNetwork object definition

Creating a layered network topology

A layered queueing network consists of four types of elements: processors, tasks, entries and activities. An entry is a class of service specified through a finite sequence of activities, and hosted by a task running on a (physical) processor. A task is typically a software queue that models access to the capacity of the underpinning processor. Activities model either demands required at the underpinning processor, or calls to entries exposed by some remote tasks.

In the `LayeredNetwork` class, the terms *host* and *processor* are entirely interchangeable.

To create our first layered network, we instantiate a new model as

```
lqn::LqnBuilder<double> lqn;
```

The layered model is built by `LqnBuilder`, whose elements are named by string rather than held as objects; `lqn.build()` returns the `LqnStruct` the solver consumes. We now proceed to instantiate the static topology of processors, tasks and entries:

```
lqn.processor("P1", 1, SchedStrategy::PS);
lqn.processor("P2", 1, SchedStrategy::PS);
lqn.task("T1", 5, SchedStrategy::REF, "P1");
lqn.task("T2", std::numeric_limits<double>::infinity(), SchedStrategy::INF, "P2");
lqn.entry("E1", "T1");
lqn.entry("E2", "T2");
```

The processor a task runs on is an argument of `task`, and the task an entry belongs to an argument of `entry`, in place of the `.on()` chaining of the other implementations. An equivalent way to specify the above example is to use the `Host` class rather than the `Processor` class, with identical parameters.

In the above code, the `on` method specifies the associations between the elements, e.g., task `T1` runs on processor `P1`, and accepts calls to entry `E1`. Furthermore, the multiplicity of `T1` is 5, meaning that up to 5 calls can be simultaneously served by this element (i.e., 5 is the multiplicity of servers in the underpinning queueing system for `T1`).

Both processors and tasks can be associated with the standard LINE scheduling strategies. For instance, `T2` will process incoming requests in parallel according as an infinite server node, since we selected the `SchedStrategy::INF` scheduling policy. An exception is that `SchedStrategy::REF` should be used to denote the reference task (e.g. a node representing the clients of the models), which has a similar meaning to the reference node in the object.

Tasks with setup and delay-off times

A task whose servers are switched off while idle does not begin service as soon as a request arrives: the server must first be reactivated, and it is switched off again only after it has been idle for some time. LINE represents this behaviour on any `Task` through two timing properties:

- Setup time: the activation delay paid by a server that resumes from the off state before it can serve the request that woke it up.
- Delay-off time: the idle period a server remains available after emptying its queue, before it powers off. A further arrival within this window is served without paying the setup time again.

These are the setup and close-down times of a server with vacations, and they arise wherever activation is not instantaneous: virtual machines and containers started on demand, power-managed servers under a timeout policy, machines requiring a tool change or a warm-up, and serverless (function-as-a-service) platforms, where the setup time is the cold-start delay and the delay-off time is the keep-alive window. Such a task is instantiated as follows:

```
lqn.task("F1", 4, SchedStrategy::FCFS, "P1");
```

```
lqn.setup_time("F1", Exp(1.0), // activation time
              Exp(0.5)); // delay-off time
```

Setup and delay-off are set together, on the task, by one call. Both `setSetupTime` and `setDelayOffTime` accept either a numeric value (which creates an exponential distribution with the given mean) or a `Distribution` object. Both default to `Immediate`, i.e., a task that sets neither is an ordinary always-on task. Since the two properties belong to `Task`, the `SetupTask` subclass, used above, is only a naming convenience that makes the intent explicit; a plain `Task` carrying either time behaves identically. The former name of this subclass, `FunctionTask`, is retained as a deprecated alias so that existing serverless models keep working.

The same setup and delay-off semantics are available at the level of a flat `Network`, without a layered model, through `setSetupTime` and `setDelayOffTime`, which attaches a per-class setup distribution and delay-off distribution to a queueing station. The two distributions can be read back with `getSetupTime` and `getDelayOffTime`. A model that uses them declares the `SetupDelayOff` language feature and is simulated by the `LDES` solver. A complete example is provided in `lqn_setup` in the `examples/` folder.

Cache tasks

The `CacheTask` class extends `Task` to model in-memory caches embedded in a layered network. A cache task stores up to a finite number of items chosen out of a catalogue, evicting them according to a configurable replacement policy. Cached items are exposed to the rest of the model as `ItemEntry` objects, which behave like ordinary entries but are additionally annotated with a discrete popularity distribution describing how frequently each item is requested.

The `CacheTask` constructor takes the following arguments:

- `model`: the parent `LayeredNetwork`;
- `name`: a unique task name;
- `nitems`: the catalogue size, i.e., the total number of distinct items that may be requested;
- `itemLevelCap`: cache capacity. A scalar value specifies the size of a single-level cache, while a row vector specifies per-level capacities for a multi-level cache hierarchy (e.g., `[2, 4]` for an L1 of size 2 and an L2 of size 4);
- `replStrat`: a `ReplacementStrategy` enum value. The supported policies are `ReplacementStrategy.RR` (random replacement), `ReplacementStrategy.FIFO`, `ReplacementStrategy.SFIFO` (strict FIFO), and `ReplacementStrategy.LRU`. Note that this is narrower than the policy set accepted by a flat `Cache` node, which additionally supports `HLRU`, `CLIMB` and `QLRU`;
- `delayed-hit retrieval`, enabled with `set_retrieval_system`. When it is set, concurrent misses for the same item that arrive while a fetch is already in flight (the fetch being the miss-branch activity together with its backend calls) are parked and released together as delayed hits when that fetch completes, instead of each triggering an independent fetch. Without it, every miss issues its own fetch. Layers that declare

retrieval are solved by a dedicated analytic sublayer, and support for them is currently experimental;

- multiplicity (optional, default 1): number of concurrent threads that may serve cache accesses;
- scheduling (optional, default `SchedStrategy.FCFS`): scheduling discipline used when multiple requests compete for the cache.

An `ItemEntry` extends `Entry` with a popularity descriptor and is constructed as `ItemEntry(model, name, cardinality, popularityDistribution)`, where `cardinality` is the number of items reachable through the entry and `popularityDistribution` is a discrete `Distribution` object (e.g., `DiscreteSampler` or `Zipf`). The hit/miss outcome of a cache access is then routed in the activity graph using the `ActivityPrecedence.CacheAccess` construct (see Section 6.5.2).

The following example defines a single-host LQN where a client task `T1` issues calls to a cache task `C2` hosting four items with capacity two and a round-robin eviction policy:

```
lqn::LqnBuilder<double> lqn;
lqn.processor("P1", 1, SchedStrategy::PS);
lqn.task("T1", 1, SchedStrategy::REF, "P1");
lqn.entry("E1", "T1");

const std::size_t totalitems = 4;
const std::vector<int> itemcap{2};
lqn.processor("PC", 1, SchedStrategy::PS);
lqn.cache_task("C2", 1, SchedStrategy::PS, "PC", totalitems, itemcap,
    ReplacementStrategy::RR);
std::vector<double> pAccess(totalitems, 1.0 / double(totalitems));
lqn.item_entry("I2", "C2", totalitems, pAccess);

lqn.activity("A1", Immediate(), "T1");
lqn.bound_to("A1", "E1");
lqn.sync_call("A1", "I2", 1.0);
lqn.activity("AC2", Immediate(), "C2");
lqn.bound_to("AC2", "I2");
lqn.activity("AC2h", Exp(1.0), "C2");
lqn.replies_to("AC2h", "I2");
lqn.activity("AC2m", Exp(0.5), "C2");
lqn.replies_to("AC2m", "I2");
lqn.cache_access("AC2", "AC2h", "AC2m");
```

`cache_access` is the precedence that splits the read into its hit and miss branches. The two post-cache activities passed to `CacheAccess` are interpreted in order: the first is executed on a cache hit, the second on a cache miss. A complete worked example is given in `lcq_singlehost` in the `examples/advanced/layeredCQ/` folder; `lcq_threehosts` illustrates a hierarchical, multi-tier cache configuration.

Replication and fan-out

Layered networks support replication of processors and tasks, together with explicit fan-in and fan-out annotations on synchronous and asynchronous calls. Replication multiplies a single declared element into a number of identical copies that share the same activity graph but provide independent service capacity, and

is the recommended way to model horizontally scaled microservices, clusters, or any tier with a number of equivalent instances.

The `method` on `Processor` and `Task` sets the replica count $r \geq 1$. The `method` on `Task` declares that each replica of the calling task issues calls to `value` replicas of the called task `dest`; symmetrically, declares that each replica of the called task receives calls from `value` replicas of the source task. The product of fan-in and fan-out values must be consistent with the replica counts of the two tasks involved in each call. The `lqn_sockshop` example in the `examples/basic/layeredModel/` folder demonstrates a microservice-style topology in which an edge task `T1` is replicated $2\times$ and front-end / back-end tasks declare fan-in and fan-out values around it:

```
lqn.processor("P2_1", 1, SchedStrategy::PS, 2.0); // the last argument is the replication
lqn.task("T1", 24, SchedStrategy::FCFS, "P2_1");
```

Replication is an argument of `processor` and `task`; per-task fan-in and fan-out are not carried by the C++ builder. Fan-out destinations and fan-in sources are referenced by task name, so the corresponding tasks must already exist (or be created later in the model). Multiple fan-out destinations may be specified by repeated calls to `setFanOut`; each call appends a new (destination, value) pair.

Servers with queue-dependent service rates

When `LN` decomposes a layered model, every processor and every non-reference task becomes a queueing station in the layer where it acts as a server. The three queue-dependent service mechanisms of §5.2.1–§5.2.4, which for a model are attributes of a `Queue`, may be declared directly on that server, so that a host can slow down as its tasks pile up, or a task can serve one of its entries at a rate that depends on how many requests of each kind it currently holds. The setters carry the same names as on a queue and are available on `Processor/Host` and on `Task`:

- takes the numeric vector $\alpha(n)$ that scales the rate of the layer station when it holds n jobs in total, exactly as for a queue. The scaling multiplies the station rate on top of the multiplicity, so a host of multiplicity m applies $\min(n, m) \alpha(n)$.
- takes the product-form handle $\beta(n)$, and the non-product-form handle $\eta(n)$. Their argument is the per-operand population vector of the server: $n(j)$ counts the jobs the layer station holds on behalf of operand j , which is task j of a processor or entry j of a task, in declaration order. The handle returns a scalar shared by every operand or a per-operand vector, and the peak rate is required, as it normalizes utilization as $U = T S/\text{peak}$.

Declarations are admitted only on `PS` and `FCFS` servers, the disciplines whose layer station admits a rate scaling; any other scheduling strategy is rejected at declaration time. For example, a processor that gains capacity as its tasks queue up, and a task whose first entry is served by two internal workers:

```
lqn.processor("P1", 1, SchedStrategy::PS);
lqn.set_load_dependence("P1", {1.0, 1.6, 1.9, 2.0, 2.0}); // alpha(n), n = 1..5
lqn.task("T1", 5, SchedStrategy::FCFS, "P1");
lqn.entry("E1", "T1");
```

```
lqn.entry("E2", "T1");
lqn.set_class_dependence("T1",
    [](const std::vector<double>& n) { return std::vector<double>{std::min(n[0], 2.0), ...
1.0}; },
    {2.0, 1.0});
```

LqnBuilder carries the three setters under their snake-case names, and the handles take and return a `std::vector<T>` over the operands rather than a matrix. The declarations are stored in the LayeredNetworkStruct fields `lldscaling`, `cdscaling` and `jdsaling` and are applied by LN while it builds the layers, so any layer solver that supports the mechanism can then be used: MVA through its queue-dependent arms and NC through the multichain convolution. A dependence declared over operands is lifted onto the classes of the layer, since a station speaks classes and an operand may occupy the station through several of them. Where each operand maps to exactly one class the lift is faithful and a class dependence remains product form; otherwise LN emits the same numbers as a joint dependence, which is not accompanied by an exactness guarantee. A scalar handle applies to every class of every operand, while classes that belong to no operand keep a unit scaling.

Two limitations follow from the interchange formats. The LQN schema has no element for a rate scaling, so `read_lqnx` warns and omits the declaration, and a model exported to `.lqnx` (hence also one solved by the LQNS wrapper) is analysed without it. Under `options.lang=java` the load-dependence vector is marshalled to the JAR, whereas the class- and joint-dependence handles are not transferable across the language boundary and are rejected with an explicit error; solve those models natively instead.

Servers with class compatibilities

A processor or a task is by default a *homogeneous* pool: any of its servers may run any operand. When only some are eligible – a core pool in which only some cores carry an accelerator – the multiplicity is partitioned into *pools*, each naming the operands it serves, through the `ServerType` class of §5.1.2:

```
lqn.processor("P1", 3, SchedStrategy::PS);
lqn.task("T2", 6, SchedStrategy::FCFS, "P1");
lqn.task("T3", 6, SchedStrategy::FCFS, "P1");
lqn.add_server_type("P1", "S1", 1, {"T2"}, 1.0);
lqn.add_server_type("P1", "S2", 1, {"T2", "T3"}, 1.0);
lqn.add_server_type("P1", "S3", 1, {"T3"}, 1.0);
```

The pools partition the servers, so their counts must sum to the multiplicity and every operand must be reachable; both are checked when the struct is built. A pool's relative speed, one by default, is per pool and not per (pool, operand), since a server free to take several operands does not announce which it picked.

The station clears the *activated-server* rate

$$\mu(n) = \sum_t k_t r_t \min\left(1, \sum_{j: \text{compat}(t,j) \neq 0} n_j\right), \quad (6.1)$$

where pool t holds k_t servers of rate r_t : a pool contributes its full rate as soon as a compatible operand is present. At integer states this depends on n only through its support, the order-independent condition; the

$\min(1, \cdot)$ matters only at the fractional argument a mean-value solver supplies, without which a barely-busy operand would activate every pool it touches.

LN lowers it to a joint dependence normalised by the peak $\sum_t k_t r_t$ of (6.1), so a fully compatible pool reproduces the neutral scaling. That makes the declaration an *approximation* within a layer and confines it to the class-switching layerings `srvn.cs` and `flat.cs`; `srvn.ph` refuses it. The LQN schema cannot express a compatibility graph either, so an `.lqnx` export warns and reads back homogeneous, and LQNS refuses the model.

Describing host demands of entries

The demands placed by an entry on the underpinning host (also called in layered queueing networks the *host demand*) is described in terms of execution of one or more activities. Although in tools such as LQNS activities can be associated with either entries or tasks, LINE supports only the more general of the two options, i.e., the definition of activities at the level of tasks. In this case:

- Every task defines a collection of activities.
- Every entry needs to specify an initial activity where the execution of the entry starts (the activity is said to be “bound to the entry”) and a replying activity, which upon completion terminates the execution of the entry.

For example, in our running example, we may now associate an activity to each entry as follows:

```
lqn.activity("A1", Exp(1.0), "T1");
lqn.bound_to("A1", "E1");
lqn.sync_call("A1", "E2", 3.5);
lqn.activity("A2", Exp(2.0), "T2");
lqn.bound_to("A2", "E2");
lqn.replies_to("A2", "E2");
```

Each property of an activity is one call, in place of the chained builders of the other implementations; `async_call` is the asynchronous variant of `sync_call`. Here, A1 is a task activity for T1, acts as initial activity for E1, consumes an exponentially distributed time on the processor underpinning T1, and requires on average 3.5 synchronous calls to E2 to complete. Each call to entry E2 is served by the activity A2, with a demand on the processor hosting T2 given by an exponential distribution with rate $\lambda = 2.0$.

Activity graphs Often, it is useful to structure the sequence of activities carried out by an entry in a graph. Activity graphs can be characterized by precedence relationships of the following kinds:

- *sequence*: two activities are executed sequentially, one after each other. This is implemented through the construct.
- *loop*: an activity is repeated a number of times. This is implemented in .
- *and-fork*: a serial execution is forked into concurrent activities. This can be materialized using the construct.

- *or-fork*: the server chooses probabilistically which activity to execute next among a set of alternatives. This is implemented in .
- *and-join*: concurrent activities are joined into a single serial execution. This is implemented in .
- *or-join*: merge point for alternative activities that may execute in parallel after a *or-fork*. This is implemented in .
- *cache-access*: split point for cache hit/cache miss results in an activity graph. This is implemented in . For usage examples, see `cache_replc_lru` and `cache_compare_replc` in the `examples/` folder.

A composite example showing fork/join precedences and loops is given in `lqn_workflows` in the `examples/` folder.

An AND-fork precedence spawns several execution paths that proceed independently until they converge at the matching AND-join, which blocks until every spawned branch has completed and then resumes a single thread of control. A branch may contain any sequence of activities, including processor demands, synchronous calls to lower-layer entries, and nested fork-join structures. The construct differs from the fork-join of ordinary queueing networks in that activities are bound to tasks and processors, so contention arises at both levels; and because branches may call entries in different layers, the response time of the forking task is the maximum over branch completion times, each of which depends in turn on lower-layer response times. An entry is therefore complete only when all activities in its activity graph, including every branch of an AND-fork, have synchronized.

Solvers evaluate these structures by treating each fork-to-join branch as a subnetwork, estimating its completion-time distribution, and deriving the distribution of the maximum, which is the synchronization delay at the join. LN approximates that maximum by moment matching on matrix-geometric branch representations, whereas LQNS decomposes the model layer by layer and refines the parameterization iteratively. The `lqn_workflows` example exercises several AND-fork/AND-join pairs spanning multiple layers.

For instance, we may replace in the running example the specification of the activities underpinning a call to E2 as

```
lqn.activity("A20", Exp(1.0), "T2");
lqn.bound_to("A20", "E2");
lqn.activity("A21", Erlang(2.0, 2), "T2");
lqn.activity("A22", Exp(1.0), "T2");
lqn.replies_to("A22", "E2");
lqn.serial("A20", "A21");
lqn.serial("A21", "A22");
```

A serial precedence is written one arc at a time. such that a call to E2 serially executes A20, A21, and A22 prior to replying. Here, A21 is chosen to be an Erlang distribution with given mean (1.0) and number of phases (2).

OrFork with branch probabilities The `OrFork` construct (also available as `Xor`) selects probabilistically between alternative successors of a single activity. The third argument is a row vector of branch probabilities whose entries sum to one and are listed in the same order as the destination activities:

```
lqn.or_fork("A20", std::vector<std::string>{"A21", "A22"},
           std::vector<double>{0.3, 0.7});
```

Here, after completing A20 the task executes A21 with probability 0.3 and A22 with probability 0.7.

Loop precedence The `Loop` construct repeats a body activity a configurable number of times before continuing with a successor. In its three-argument form `Loop(preAct, {loopBody, endAct}, count)`, the activity `preAct` is executed once, then `loopBody` is executed `count` times, and finally `endAct` runs once before exiting the loop:

```
lqn.loop("A20", std::vector<std::string>{"A21"}, "A22", 3.0);
```

A four-argument form `Loop(preAct, loopBody, endAct, counts)` is also accepted as a more explicit equivalent.

AndJoin with quorum By default, `AndJoin` blocks until all parallel branches spawned by the matching `AndFork` have completed. The optional third argument `quorum` relaxes this synchronization so that the join fires as soon as any `quorum` of its predecessors have reported completion, modeling early-reply patterns and partial joins:

```
lqn.and_join(std::vector<std::string>{"A21", "A22", "A23"}, "A24", 2);
```

The last argument is the join quorum. The example releases A24 as soon as any two of A21, A22, A23 have completed.

CacheAccess precedence The `CacheAccess` construct routes the control flow downstream of a cache lookup based on the lookup outcome. The post-cache activity list is interpreted positionally: the first element is the hit branch, the second the miss branch:

```
lqn.cache_access("AC2", "AC2h", "AC2m");
```

This precedence is meaningful only when `preAct` is bound to an `ItemEntry` of a `CacheTask`; see Section 6.5.2 for the cache modeling primitives.

Activity configuration

Beyond the host demand specified at construction time, individual activities expose a number of methods that influence how they participate in the activity graph and in the analysis.

Two-phase activities (setPhase). The two-phase model allows part of an activity's execution to be carried out after a synchronous reply has been sent to the caller. Activities are by default in phase 1 (executed before the reply) and can be moved to phase 2 (executed after the reply, while the caller proceeds in parallel)

using `.`. Only phase values of 1 or 2 are accepted. Phase-2 activities show up in the layered network struct under the `actthink_*` fields.

Per-activity think time (`setThinkTime`). An activity can carry an additional think-time component, separate from the host demand and from the task-level think time, modeling for example a deferred operation. The method `setThinkTime` accepts either a numeric mean (which produces an exponential distribution) or any `Distribution` object. A zero or negligible mean is automatically replaced by an `Immediate` distribution.

Asynchronous calls (`asynchCall`). In addition to `synchCall`, an activity can issue non-blocking calls to entries of other tasks via `asynchCall`. The destination `E` can be passed either as an `Entry` object or by name; the second argument `m` is the mean number of calls per activity execution and defaults to 1.0. Multiple distinct destinations can be added by repeated calls, but a single destination cannot be registered twice – to model multiple calls to the same entry, increase `m` accordingly.

Call groups (`synchCallRoundRobin`, `synchCallJSQ`). An activity’s synchronous calls can also be dispatched to a *group* of target entries rather than to one fixed destination, by `synchCallRoundRobin` or `synchCallJSQ`, where `dests` is a list of two or more target entries and `m` is the total mean number of calls per activity execution, defaulting to 1.0. Round robin sends successive calls to the targets in cyclic order, so each receives $m/\text{numel}(\text{dests})$ of them deterministically; JSQ instead sends every call to whichever target task’s station currently holds the fewest jobs, breaking ties uniformly at random. Both differ from an equal-probability `synchCall` split with the same per-target means only in how the individual calls are correlated over time, not in the long-run rate each target receives.

A call group is representable only under the squashed (`'flat'`) layering, since under `'srvn'` the grouped targets never share a submodel; the solver must therefore be asked for that layering, either as `options.method=flat` or through `setLayering`. Honoring the dispatch rule itself further requires a layer solver capable of state-dependent routing, which only CTMC and SSA provide (SSA alone in the C++ port); `SolverLN` rejects a model containing a call group before it reaches an MVA, NC or FLD layer solver, rather than silently returning a probabilistic split mislabeled as round-robin or JSQ. The examples `lqn_rrobin` and `lqn_jsq` illustrate both constructs. A group survives an export to `.lqnx` as a `<call-group>` element naming its strategy and its target entries, a LINE extension of the schema: the member calls stay ordinary `<synch-call>` elements carrying the same per-target means, so a reader that ignores the extension — `lqns` and `lqsim`, which have no dispatcher — analyses the ungrouped twin instead of a model with the calls dropped.

Open arrivals at entries (`Entry.setArrival`). An entry can be turned into an open arrival point by attaching an arrival distribution. The method `setArrival` accepts any `Distribution` object, e.g., `Exp(2.5)` for Poisson arrivals at rate 2.5 or `Erlang(2, 0.4)` for an Erlang-2 stream. Open-arrival entries surface in the layered network struct through the `arrival_*` fields and let the entry serve external workload alongside (or instead of) calls received from a reference task.

Error Type	Error Type
Activity in REF task replies	Entry called both synchronously and asynchronously
Entry on task calls itself	Repeated definition of parent task
Entry on task calls entry on the same task	Invalid .on() argument for an activity
Cycle in activity graph	Invalid .on() argument for a task
Unsupported	Repeated synch calls
Activity with specification	Repeated asynch calls

Debugging and visualization

The structure of a `LayeredNetwork` object can be graphically visualized as follows. The C++ port has no viewer: a layered model is inspected through the struct `lqn.build()` returns, whose tasks, entries, activities and calls are plain vectors.

The `get_ensemble_avg` and `get_struc` methods can be used to visualize in JMT each layer. This can be done by first calling the `SolverLN::get_ensemble_avg` method to obtain a `LayeredNetwork` consisting of the `LayeredNetwork` objects, each one corresponding to a layer, and then invoking the `get_struc` and `get_ensemble_avg` methods on the desired layer. This is discussed in more details in the next section.

Lastly, we note a number of specification issues that trigger errors in the LQN definition:

The internal representation of a `LayeredNetwork`, returned by `get_struc`, and the decomposition of a layered model into layers are documented field by field in the LINE internals manual (`LINE-internals-cpp.pdf`, appendix *The LayeredNetworkStruct reference*).

6.5.3 Solvers

LINE offers two solvers for the solution of a `LayeredNetwork` model consisting in its own native solver (LN) and a wrapper (LQNS) to the LQNS solver [62]. The latter requires a distribution of LQNS to be available on the operating system command line.

LQNS is an external tool developed by the Real-time and Distributed Systems Group at Carleton University. The wrapper writes an XML file conforming to the LQN metamodel, runs the `lqns` (or `lqsim`) executable as a subprocess, and parses the XML output back into LINE metrics. Its solution methods, and the convergence tolerance and iteration cap that govern them, are selected through command-line flags set from the solver options.

The two layered solvers differ in where the decomposition happens. LN converts the layered model into a set of ordinary queueing networks in which cross-layer synchronization appears as artificial delay stations, then iterates a network solver until the delays are mutually consistent; this makes the whole LINE solver portfolio, and features such as sensitivity analysis and optimization, available to layered models. LQNS instead applies layered algorithms directly to the hierarchy without that intermediate conversion, which converges faster on deeply nested models and supports constructs, notably quorum joins, that the conversion does not yet cover. On standard layered models the two agree; divergences are confined to combinations of scheduling disciplines, replication and activity precedences.

The solution methods available for `LayeredNetwork` models are similar to those for `Network` objects. For example, the `AvgResult` can be used to obtain a full set of mean performance indexes for the

model, e.g., In the C++ port the same solver is `line-cli model.lqnx -s lqns -a avg`, or in-process `lqns::SolverLQNS`; the native meta-solver is `line-cli model.lqnx -s ln -a avg`, or `ln::SolverLN`. Note that in the above table, some performance indexes are marked as NaN because they are not defined in a layered queueing network. Further, compared to the `avgTable` method in `objects`, `LayeredNetwork` do not have an explicit differentiation between stations and classes, since in a layer a task may either act as a server station or a client class.

The main challenge in solving layered queueing networks through analytical methods is that the parameterization of the artificial delays depends on the steady-state performance of the other layers, thus causing a cyclic dependence between input parameters and solutions across the layers. Depending on the solver in use, such issue can be addressed in a different way, but in general a decomposition into layers will remain parametric on a set of response times, throughputs and utilizations.

This issue can be resolved through solvers that, starting from an initial guess, cyclically analyze the layers and update their artificial delays on the basis of the results of these analyses. Both `LN` and `LQNS` implement this solution method. Normally, after a number of iterations the model converges to a steady-state solution, where the parameterization of the artificial delays does not change after additional iterations.

LQNS

The `LQNS` wrapper operates by first transforming the specification into a valid `LQNS` XML file. Subsequently, `LQNS` calls the solver and parses the results from disks in order to present them to the user in the appropriate `LINE` tables or vectors. The `options.method` can be used to configure the `LQNS` execution as follows:

- `options.method=default` or `lqns`: `LQNS` analytical solver with default settings.
- `options.method=exactmva`: the solver will execute the standard `LQNS` analytical solver with the exact MVA method.
- `options.method=srvn`: `LQNS` analytical solver with `SRVN` layering.
- `options.method=srvn.exactmva`: the solver will execute the standard `LQNS` analytical solver with `SRVN` layering and the exact MVA method.
- `options.method=sim` or `lqsim`: `LQSIM` simulator, with simulation length specified via the `samples` field (i.e., with parameter `-A options.samples`).
- `options.method=lqns.default`: alias of `default` retained for backward compatibility with earlier scripts.

These methods apply to a `LayeredNetwork`. The same solver also accepts a flat , which it solves through `qnsolver` with the `qns` methods described in §8.4.2.2. Upon invocation, the `lqns` or `lqsim` commands will be searched for in the system path, and the solver is unavailable if they are not found there. `LINE` ships no `LQNS` binary, names no container image holding one, and never pulls one: `LQNS` is distributed by its authors under an evaluation agreement that forbids providing the software to third parties, so the binary must be one the user has obtained and is licensed for. Obtain it from <http://www.sce.carleton.ca/rads/lqns/>. To exercise a containerised `LQNS` build against the test suite, `run_tests.sh --lqns-docker <image>` places `lqns`,

`lqsim` and `qnsolver` shims on the `PATH` that run the named local image; the solvers themselves still see ordinary local binaries. The same applies to `qnsolver`, which is part of the LQNS distribution and backs the `qns` methods of LQNS on a . Local execution is distinct from `options.config.remote`, which dispatches instead to a remote `lqns-rest` service.

The pragma string passed to `lqns` can be customised through the `options.config.multiserver` entry, which forwards a `-Pmultiserver=...` pragma selecting the LQNS multi-server approximation ('conway' [44], 'reiser' [131], 'rolia' [64, 136], 'zhou' [172, 173], 'suri' [152], or 'schmidt' [143]; the 'default' value maps to 'rolia'). The value is passed verbatim to the back-end and therefore accepts any multi-server pragma supported by the installed LQNS version.

Remote execution As an alternative to local installation of the LQNS tools, LINE supports remote execution via a Docker container running the `lqns-rest` API. This is useful when LQNS binaries are not available locally or when running on systems where compilation is difficult. To enable remote execution, configure the following options:

- `options.config.remote = :` Enable remote execution via REST API.
- `options.config.remote_url = http://localhost:8080:` URL of the `lqns-rest` server (default port is 8080).

The service at that URL is one the user runs; LINE publishes no LQNS image and names none here, for the licensing reason given above. When remote execution is enabled, the model is serialized to LQNX format and sent via HTTP POST to the remote server, which executes the solver and returns the results. This approach supports all solver methods (`lqns`, `lqsim`, etc.) and pragmas available in the local execution mode.

LN

The native LN solver iteratively applies the layer updates until convergence of the steady-state measures; the layering, the parameter updates, and the convergence control are described in the LINE internals manual (LINE-internals-cpp.pdf, Chapter LN). Since updates are parametric on the solution of each layer, LN can apply any of the solvers described in the solvers chapter to the analysis of individual layers, as illustrated in the following example for the MVA solver

```
ln::LnOptions opt;                                // LN defaults
opt.layer_solver = "mva";                          // the solver run in each layer
opt.layer.method = "default";                      // options handed to that layer solver
ln::SolverLN<double> solver(lqn.build(), opt);
const ln::LnSolution<double> res = solver.get_ensemble_avg();
```

A C++ template cannot take a solver factory, so the layer solver is named by `layer_solver` (`mva`, `nc`, `fluid` or `ssa`) and its options are carried in the matching `layer*` field. Options parameters may also be omitted. The LN method converges when the maximum relative change of mean response times across layers from the last iteration is less than `options.iter_tol`.

A method name of the LN solver carries *two* decisions, and its two-part form says so: the part before the dot is the **layering**, which fixes what a submodel is, and the part after it is the **encoding**, which fixes how an activity graph is written into that submodel.

Methods supported by the LN solver include:

- `options.method=srvn.cs`: the SRVN layering with the activity graph encoded as **class switching**. There is one submodel per served element – a host layer for each processor and a task layer for each called task – and inside it the graph becomes routing: one class per task, entry, activity and call, plus the Fork, Join and Router nodes that an AND precedence needs and the class switch that an off-diagonal $P_{r,s}$ mints. This is the encoding of every earlier release and it serves every layered feature the solver implements.
- `options.method=srvn.ph`: the same layering with the activity graph composed instead into one **phase-type** service law per entry, by the exact series-parallel reduction of [141] (sequence by convolution, OR by mixture, loop by the geometric compound, AND by the serialized host law). A layer is then a two-station cycle – a client delay and the server – carrying one closed class per calling task instead of one per graph element, which makes it markedly smaller and faster. It cannot represent second phases, forwarding calls, cache tasks, quorum AND joins, routed call groups or per-entry admission constraints, and it refuses such a model by name rather than degrading it silently.
- `options.method=srvn`: the **alias**, and the default. It resolves to `srvn.ph` where that encoding can serve the model and to `srvn.cs` where it cannot. The decision is made once, at layer-build time, by a non-destructive feasibility probe (the feature gate, then the composition of every entry workflow), and the outcome is reported in `solver.state_lnmethod()`. Naming `srvn.ph` explicitly is therefore *not* the same as taking the default: the alias falls back, the explicit name raises.
- `options.method=flat.cs`: the **squashed** layering, in which every processor and every called task is a station of one single submodel and a call routes the job to the callee’s station rather than to a surrogate delay. It is the layering that lets two servers contend within one network, and the only one under which a routed call group (Section 6.5.2) can be expressed at all, since the group’s targets must share a submodel for a dispatch decision to mean anything. Squashing is refused rather than approximated for a replicated element, a cache task or a setup task, each of which carries state that only a submodel of its own can hold.
- `options.method=flat.ph`: the squashed layering with the activity graph composed into a **phase-type** service law, that is `flat.cs`’s single submodel carrying `srvn.ph`’s encoding. A caller task is one closed class and visits each server it uses once per invocation, served there by the composed law of the demand it places on that server, so the submodel carries one class per calling task instead of one per entry, activity and call. Squashing does not conflict with the composition: a task station’s service time is already the *inflated* entry time — host demand plus call counts times callee service and waiting — which the outer fixed point updates, as in `lqns --squashed-layering`. It refuses what `srvn.ph` refuses, and additionally the replicated elements and setup tasks that any squashing refuses, and the routed call groups that only the routing encoding can dispatch.
- `options.method=flat`: the alias for `flat.cs`. It resolves unconditionally rather than probing

`flat.ph`, because a model is squashed in order to express what only the routing encoding carries.

- `options.method=default`: a synonym of `srvn`, so a model solved without naming a method takes the better of the two SRVN encodings.
- `options.method=moment3`: solution by recursive 3-moment approximation of response time distributions. This method matches the first three moments (mean, variance, skewness) of response times at each layer by fitting acyclic phase-type distributions, improving accuracy for non-exponential service demands compared to the default mean-value method. It builds the class-switching layers and adds the distribution propagation on top of them.

The layering may equivalently be selected through `layering`, whose values are `'srvn'` (the default) and `'flat'` (`'squashed'` is accepted as a synonym). The two channels agree: a method that names a layering sets it, and a layering named in the configuration is reported back through `lnmethod` under its method name, so a model squashed through the configuration reads back as `flat.cs` even when no method named it.

The C++ port has no `options.config` for the layering: the method name is the only channel, so `opt.method = "flat"` is how the squashed layering is requested there.

Relaxation controls. The convergence behaviour of the layered iteration is governed by a family of `options.config` entries:

- `options.config.relax` (default `'fixed'`) selects the under-relaxation mode used when blending the new and previous iterates: `'none'` (no relaxation), `'fixed'` (constant factor), `'adaptive'` (factor adjusted from the recent error history), or `'auto'`.
- `options.config.relax_factor` (default 0.5) is the relaxation factor $\omega \in (0, 1]$ applied in the `'fixed'` mode. Smaller values yield more conservative updates and improve convergence on stiff models.
- `options.config.relax_min` (default 0.1) and `options.config.relax_history` (default 5): respectively the lower clamp on the relaxation factor and the error-history window used by the `'adaptive'` mode.
- `options.config.stochiter` (default `'auto'`) selects how the layered fixed point is driven when the layer solvers return noisy estimates, as they do when a layer is analysed by simulation. The value `'off'` keeps the plain Picard iteration, `'rm'` switches to Robbins-Monro stochastic approximation, in which successive iterates are averaged with a decaying step size so that the noise is filtered rather than propagated, `'crn'` additionally couples the layer solvers with common random numbers, and `'auto'` selects between them according to whether the model is classified as stochastic. The schedule is controlled by `options.config.stochiter_alpha` (default 0.6), the step-decay exponent, which must lie in $(0.5, 1]$ for the averaged iterate to converge, `options.config.stochiter_a0` (default 1.0), the initial step after burn-in, `options.config.stochiter_burnin` (default 5), the number of Picard iterations executed before the step starts decaying, and `options.config.stochiter_conseq` (default 3), the number of consecutive sub-tolerance iterations required before the iteration is declared converged. The last of these matters because a single sub-tolerance iterate is not evidence of convergence when the

residual is itself a random variable.

- `options.config.layer_init` (default `none`) selects the initialization of the layer throughput state. Setting it to `'bound'` seeds each element's initial throughput with the geometric mean $\sqrt{X^- X^+}$ of the Majumdar-Woodside robust box bounds (algorithm `lqn_boxbounds`). The initialization is correctness-preserving; for closed-chain layered models the outer iteration re-derives the throughputs from the first solve, so the converged solution is unchanged.
- `options.config.interlocking` (default `true`) applies the LQNS-style interlock correction to the residence time a layer attributes to a call, accounting for the traffic two tasks impute to each other along a shared downstream path ([61], following the static-analysis construction of LQNS V5). It is applied only when the layer's underlying MVA dispatch reaches an algorithm able to carry the correction matrix; a layer routed elsewhere (for instance to `qna`, `rqna`, `rqt` or the Smith Queue Decomposition path) computes without it, silently rather than by falling back to a different algorithm to force the correction in, because the algorithm swap itself was found to induce non-convergence on some models. See the internals manual, Chapter LN, for the mechanism and the reason it is a gate rather than a substitution.

Layer-wise sensitivities. `SolverLN` also reports the derivatives of the layer measures with respect to the service rates, by delegating to the layer solvers and concatenating their tables under a leading `Layer` column; these are exposed through the C++ sensitivity dispatch. In every edition, these are partial derivatives within a layer rather than total derivatives of the layered model.

Layered box bounds. As a fast, non-iterative alternative to the full decomposition, `SolverLN` exposes the throughput bound methods `options.method='mw.upper'` and `'mw.lower'` (algorithm `lqn_boxbounds`). These build a closed multiclass queueing network whose stations are the processors and whose classes are the reference-task call chains, and apply the Majumdar-Woodside robust box bounds [107] to return per-task upper and lower throughput bounds. This generalizes the classical LQN Type-1 throughput bound $X \leq N/(Z + D)$ (computed, e.g., by `lqns -b`) with a utilization-based upper bound and a contention-aware lower bound.

Conservation laws of a layered model. The algorithm `lqn_balance_equations` takes a `LayeredNetworkStruct` and enumerates the relations that any solution of that model must satisfy, one record per relation, together with the index sets to aggregate over and a printable form; it solves nothing, since the relation set is fixed by the model structure alone. Five families are emitted. `little` is Little's law [100] applied to a task's thread pool, which the threads traverse as a closed cycle of one delay stage (the surrogate think time the layered decomposition imputes to the task, plus the declared think time of a reference task) and one service stage: writing $B(t, k)$ for the mean number of threads of task t busy serving caller class k , that is $X(t)(Z(t) + z(t)) + \sum_k B(t, k) = N(t)$, equivalently $X(t)Z(t) = N(t)(1 - \sum_k U(t, k))$ for a queueing task whose per-class utilizations are normalized to $[0, 1]$. The caller classes are the calls targeting an entry of t together with any open-arrival stream it serves. The remaining four are the flow balances across a call, at an entry and at an activity, and the utilization law $\sum_a X(a)D(a) = m(h)U(h)$ at a processor. Because

each relation aggregates only over a node’s neighbours in the call graph, the set is also returned as sparse incidence matrices, which makes it usable as a consistency check on a solution or as a penalty in a learned surrogate of the layered model [137]. Supplying a solved `SolverLN` instantiates every relation and reports its residual, which vanishes to the solver’s own tolerance at the fixed point.

Transient analysis. `SolverLN` supports transient analysis of a layered model through `solver_ctmc_transient_analyzer`, which returns the time-varying mean queue lengths, utilizations and throughputs of every layer in the same block-diagonal layout (station by class, per layer) used by the steady-state results. Two schemes are available. The decoupled scheme freezes the inter-layer demands at the converged fixed point and then integrates each layer’s transient independently, which is inexpensive but ignores the fact that a layer’s demand on another layer itself varies over the transient. The coupled scheme reconciles the layers by waveform relaxation [97]: each layer’s fluid transient is driven by the time-varying demand trajectories taken from the other layers’ most recent transients, and the sweep is repeated until the trajectories stop changing in the sup-norm over time. The coupled channels are the task think times, which enter as a client delay, and the synchronous-call service demands, which enter at the caller’s client station; intra-layer host service is held at its equilibrium value. The time-varying demands are injected into each layer’s closing ODE through a per-event rate multiplier. The first coupled sweep uses the frozen equilibrium demands and therefore reproduces the decoupled result exactly, and at convergence every layer relaxes to its own fixed point, so the endpoint of the transient agrees with `getAvg`.

State transfer and solver switching The `LN` solver supports transferring its internal convergence state between solver instances, enabling hybrid solving schemes where a fast solver provides initial estimates that a more accurate solver then refines. This avoids restarting the iterative decomposition from scratch when switching solvers.

The C++ port has no state-transfer interface on the layered solver: a run is driven from its options, and a warm start is expressed by seeding the initial state of the model itself.

A typical use case is to obtain a quick initial solution with `MVA` and then refine it with a simulation-based solver: A C++ `SolverLN` is constructed around one layer solver and holds no exported iterate, so the state of a run cannot be handed to a second solver; the equivalent is to build a second `SolverLN` with a different `layer_solver`.

6.5.4 Conversion between Network and LayeredNetwork models

LINE provides two converters that bridge the `Network` (queueing network) and `LayeredNetwork` (layered queueing network) abstractions. These converters are useful when one wishes to apply a layered solver to an ordinary queueing network, or vice versa to analyse a layered model with a flat queueing network solver.

From Network to LayeredNetwork

The function `QN2LQN` maps a closed queueing network into a layered queueing network. Each closed chain is modelled by a reference task whose entries replicate the chain's visit pattern, and each queueing or delay station is mapped onto a processor hosting the corresponding task. Routing probabilities and class switching are reproduced through OR-fork activity precedences. Open classes (`Source/Sink`) are not currently supported by this converter and the input model must be closed. The conversion is invoked as `The queueing-to-layered transformation is not part of the C++ port.` The `QN2LQN` converter is invoked automatically by the `LQNS` solver when it is given a `that lacks a product-form solution and contains only closed classes`; in this case the model is converted on the fly and solved by `lqns` instead of `qnsolver` (§8.4.2.2).

From LayeredNetwork to Network

The reverse converter `LQN2QN` maps a `LayeredNetwork` into an extended queueing network in which synchronous calls are realised through reply signal classes (`SignalType.REPLY`). The caller class is blocked at the calling station until the reply class returns from the callee, and a class switch at the leaf node restores the caller class so that the request cycle can repeat. Phase-2 activities, when present, are mapped onto fork-join constructs so that the post-reply work is executed in parallel with the released caller. The conversion is invoked as `The layered-to-queueing transformation is not part of the C++ port either`; the layers `SolverLN` builds internally are `NetworkStruct` objects, and `line-cli -o layers` writes them out. Beyond synchronous calls, the converter maps the remaining call and task constructs of the layered model. Asynchronous calls become ordinary routing without a reply signal, since the caller does not block. Forwarding calls are translated into the caller-side pseudo rendezvous described in the forwarding section, so that the original client stays blocked until the last entry of the forwarding chain replies. Fork calls and their joins are mapped onto the corresponding `Fork` and `Join` nodes. Cache tasks are mapped onto a `Cache` node with the item entries as read classes and the hit and miss branches as the corresponding outgoing classes. The multiplicity of a task is enforced as a thread pool rather than as a plain multi-server station: a task with multiplicity m admits at most m concurrent requests across all of its entries, which is expressed as a finite capacity region over the task's stations, so that a task whose entries have different demands cannot exceed its declared concurrency. Open arrivals declared on an entry become a `Source` and `Sink` pair carrying an open chain into that entry, rather than a closed chain with a large population, so that the arrival rate is honoured exactly. The converter is available in all three codebases.

Figure 6.7 shows the resulting network for a single synchronous call.

This route is useful for validating an `LN` or `LQNS` solution against discrete-event simulation, since the resulting network can be analysed with any solver that supports signal classes (currently `LDES` and `JMT`).

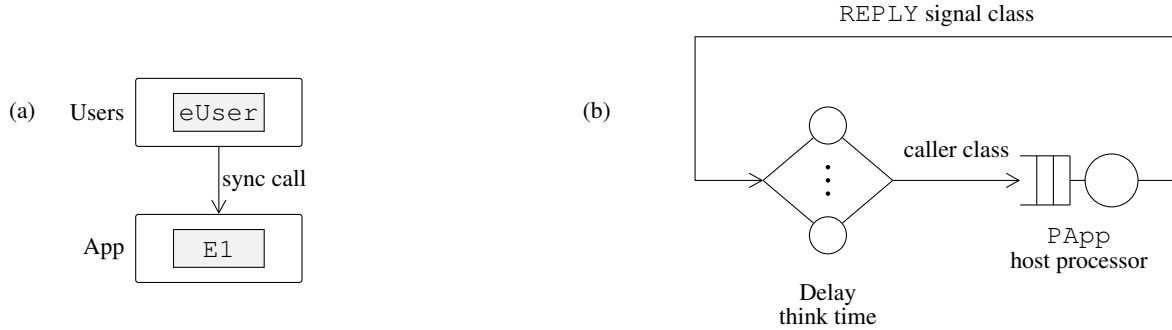


Figure 6.7: The output of LQN2QN. (a) A reference task issuing one synchronous call to an entry of a server task. (b) The flattened queueing network: the host processor of each task becomes a station, the reference task becomes a Delay holding its think time, and each step of the activity graph becomes a closed class. The caller is blocked while the callee runs, and is released by the REPLY signal class emitted by the replying activity.

6.5.5 Model import and export

A LayeredNetwork can be easily read from, or written to, a XML file based on the LQNS meta-model format¹. The read operation can be done using a static method of the LayeredNetwork class, i.e.,

```
const lqn::LqnStruct<double> lqn = lqn::read_lqnx("model.lqnx");
```

The reader accepts the same .lqnx interchange format as the other implementations, and is what line-cli -i lqnx uses. Conversely, the write operation is invoked directly on the model object

```
lqn::write_lqnx(model, "model.lqnx", "name");
```

The writer takes the builder's model rather than the finalized struct, and is what the LQNS wrapper hands to lqns. In both examples, filename is a string including both file name and its path.

Finally, we point out that it is possible to export a LQN to an XML file in the LQNX format by means of the function.

Ensemble merging

The function combines multiple independent models into a single Network containing disconnected subnetworks. Node and class names are prefixed with their model name to prevent conflicts, while all Source and Sink nodes are merged into single MergedSource and MergedSink nodes. Ensemble merging is not part of the C++ port.

¹<https://raw.githubusercontent.com/layeredqueueing/V5/master/xml/lqn.xsd>

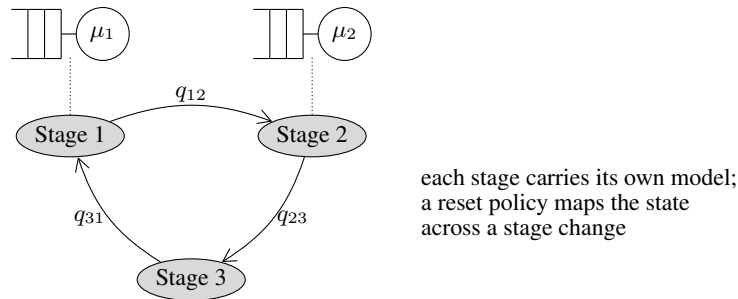


Figure 6.8: A random environment with three stages. The stage process is a continuous-time Markov chain or a semi-Markov process; each stage carries its own model, and a reset policy maps the state of the outgoing stage onto the incoming one.

6.6 Random environments

Random environments provide a framework for modeling systems whose input parameters, such as service rates, scheduling weights, or number of stations, vary according to the state of an external environment. We refer to these environmental states as *stages*. For instance, a system that alternates between normal-load and peak-load conditions can be represented by a two-stage environment, with each stage determining the number of servers available at a queueing station. Systems of this type arise in many settings, including servers subject to breakdown and repair, auto-scaling applications, and reliability models.

In LINE, an environment is described as a continuous-time Markov chain (CTMC) or as a semi-Markov process (SMP). Compared to CTMCs, SMPs relax the restriction of having exponential transitions between stages, but they are computationally harder to evaluate. As a first approximation, one may solve independent models for each stage and weight their solutions. However, depending on the frequency of transitions between stages, this can incur significant errors [34]. LINE automates the evaluation of such scenarios, applying the appropriate corrections based on the environment and system characteristics.

6.6.1 Environment definition

Figure 6.8 shows the structure of such a model.

Specifying the environment

To specify an environment, we first create an `Environment` object

```
const std::size_t E = 2;
env::Environment<double> envModel("UnreliableEnv", E);
```

The second argument is the number of stages, as in the other implementations; stages are indexed from zero in C++. We then add two stages

```
envModel.set_stage(0, "Online", "UP", network1.get_struct());
envModel.set_stage(1, "Offline", "DOWN", network2.get_struct());
```

A stage carries a `NetworkStruct`, and its index is zero-based. where the constructor specifies the stage name, an arbitrary string to classify the stage semantics, followed by the system model used while that environment stage is active.

We now describe that the transitions between stages are both exponential, with different rates

```
envModel.add_transition(0, 1, Exp(1.0));
envModel.add_transition(1, 0, Exp(2.0));
```

Self-loops are allowed. When multiple transitions are possible, the timer that elapses first determines the next stage.

The `Environment::stage` method summarizes the properties of an environment: There is no stage table in the C++ port: the stages are read back with `envModel.stage(e)` and the arcs with `envModel.arc(e, h)`, while the stage probabilities and holding times are part of what `env::solver_env` returns. In the table, the `Stage` column gives a numerical identifier for each stage, followed by its name, type classification, and a pointer to the sub-model associated with that stage.

Specifying system models for each stage

LINE places loose assumptions in the way the system should be described in each stage. It is just expected that the user supplies a model object, either a or a `LayeredNetwork`, in each stage, and that a transient analysis method is available in the chosen solver, a requirement fulfilled for example by `FLD`. When the stage models are `LayeredNetwork` objects, each stage is analysed by `LN` and the blending iteration is carried out over the layered ensembles, so that a layered application whose demands or multiplicities differ between environment stages can be studied without flattening it first. A worked two-stage layered example is provided in `renv_lqn_twostages` in the random-environment examples folder.

Specifying a reset policy

A reset policy specifies whether an environment transition instantaneously brings a redistribution of the jobs across the network. When the environment transitions, by default jobs in execution at a server are required all to restart execution at that server upon occurrence of a transition. This may not be appropriate in some models, for example when a station is removed from the model and its jobs can therefore no longer execute at that station. For such cases, one may define a custom reset policy, e.g.,

```
// move every job into station 0, without changing its class
const env::ResetMarginal resetRule = [] (const Matrix<double>& QExit) {
    Matrix<double> Q(QExit.rows(), QExit.cols(), 0.0);
    for (std::size_t c = 0; c < QExit.cols(); ++c) {
        double tot = 0.0;
        for (std::size_t s = 0; s < QExit.rows(); ++s) tot += QExit(s, c);
        Q(0, c) = tot;
    }
};
```

```

    }
    return Q;
};
envModel.add_transition(0, 1, Exp(1.0), resetRule);

```

$Q_{Exit}(i, r)$ is the queue length of class- r jobs at station i on leaving the source stage, and the rule must return a matrix of the same shape. The default reset keeps the marginal unchanged.

State-dependent environment rates In addition to the queue-length reset rule, the `add_transition` method accepts an optional sixth argument, a *reset rule for the outgoing environment-transition distribution*. This produces a state-dependent semi-Markov environment, in which the timing of stage transitions adapts to the system state at the moment of the previous stage change. The signature is

```

envModel.set_env_rate_reset(from, to,
    [(const Distrib<double>& original, const Matrix<double>& QExit,
      const Matrix<double>& UExit, const Matrix<double>& TExit) {
        return original; // the identity, which is the default
    }]);

```

The rate reset is installed separately from the transition, and is what `method = "statedep"` applies. When at least one transition supplies a non-trivial `resetEnvRatesFun`, ENV automatically switches to the state-dependent solution mode (see `options.method='statedep'` in the ENV configuration table below); this mode is incompatible with the semi-Markov mode `options.method='smp'`.

6.6.2 Solvers

The steady-state analysis of a system in a random environment is carried out in LINE using the blending method [34], which is an iterative algorithm leveraging the transient solution of the model. In essence, the model looks at the *average* state of the system at the instant of each stage transition, and upon restarting the system in the new stage re-initializes it from this average value. This algorithm is implemented in LINE by the `ENV` class, which is described next.

ENV

The `ENV` class (alias: `SolverENV`) applies the blending algorithm by iteratively carrying out a transient analysis of each system model in each environment stage, and probabilistically weighting the solution to extract the steady-state behavior of the system.

The solver offers two analyzer modes, selected through `options.method`. The default `'meanfield'` mode couples stages through the marginal mean queue length (the classic blending method); `'default'`, `'mean'`, `'blend'` and `'blending'` are accepted as names for it, and `'meancov'` is the same coupling carrying a covariance beside the mean, so that a stage restarts from a distribution rather than from a point mass. The alternative `'statevec'` mode carries the full per-stage state-probability vector across stage transitions instead of only its mean, yielding higher accuracy at the cost of an explicit state space; both are described

in the LINE internals manual (LINE-internals-cpp.pdf, Chapter ENV). The 'statevec' analyzer supports a CTMC backend for closed models and a MAM/LDQBD backend for open and load-dependent queues, and it accepts the additional option `options.sojourn=deterministic`, which models each stage sojourn as lasting exactly its mean (exact via uniformization) rather than as an exponential holding time.

As in the transient analysis of objects, LINE does not supply a method to obtain mean response times, since Little's law does not hold in the transient regime. To obtain the mean queue-length, utilization and throughput of the system one can call as usual the `avg` method on the ENV object, e.g.,

```
env::EnvOptions opt;
opt.stage_solver = "fluid";
const env::EnvAnalyzerSolution<double> res = env::solver_env(envModel, opt);
// res.QN, res.UN, res.TN, and the per-stage res.QExit / res.UExit / res.TExit
```

The stage solver is named rather than instantiated per stage; the fluid transient is the engine the C++ environment solver drives. Note that as model complexity grows, the number of iterations required by the blending algorithm to converge may grow large. In such cases, the `options.iter_max` option may be used to bound the maximum analysis time.

Per-stage results While the `avg` family of methods returns environment-averaged metrics, the underlying per-stage results computed by the blending algorithm remain accessible through the ENV solver. The full per-stage solver result for stage e is available at `res.stage[e]`, with the corresponding stage-local child solver retrievable via `res.stage[e].solver`. The steady-state probability of being in each environment stage is exposed as `res.probEnv`; the same values appear as the `Prob` column of the table returned by `Environment::stage`. The environment-averaged metrics returned by `getAvg` are the weighted combination $\sum_e \pi_e X_e$ where X_e is the per-stage queue-length, utilization or throughput and π_e is the corresponding entry of `probEnv`.

Stage holding-time inspection The semi-Markov holding time in each stage is precomputed when the environment is initialized. Stage e 's holding-time MMAP representation is stored as `res.stage[e].holdT` and is also reported in the `HoldT` column of `Environment::stage`, with the convenience alias `getStageT` producing the same table. When using the semi-Markov method (`options.method='smp'`), the per-stage sojourn-time CDF is exposed by the `computeSojournCdf(e, t)` method of ENV, which evaluates $\Pr\{S_e \leq t\}$ for the chosen stage e .

Decomposition/Aggregation Methods When the environment has many states, the ENV solver can use decomposition/aggregation methods to reduce computational complexity. The decomposition method can be configured using `options.config.da`. The available methods are based on nearly completely decomposable (NCD) Markov chain analysis:

- *Courtois decomposition* ('courtois'): The default method, based on [46]. It partitions the environment state space into macro-states and computes approximate steady-state probabilities.
- *Koury-McAllister-Stewart* ('kms'): An iterative aggregation-disaggregation method [91] that refines the Courtois solution through successive approximations.

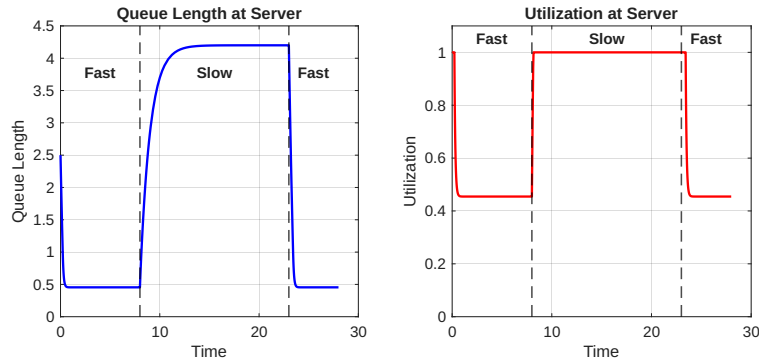


Figure 6.9: Transient queue length and utilization evolution along a sample path.

- *Takahashi's method* ('`takahashi`'): Another iterative aggregation-disaggregation approach [154] with different convergence properties than KMS.
- *Multigrid* ('`multi`'): A multi-level decomposition method [151] that applies hierarchical aggregation.

For iterative methods ('`kms`' and '`takahashi`'), the number of iterations can be controlled using `options.config.da_iter` (default: 10).

Table 6.4: EnvOptions fields (C++)

Field	Value	Description
method	"meanfield"	Mean-field coupling of the stages through the marginal mean queue length (default).
method	"statevec"	State-vector analyzer, propagating the per-stage state distribution across stage transitions.
method	"avg"	Fast-environment limit over one rate-averaged model.
method	"dec"	Slow-environment limit: each stage solved in isolation and recombined, cache ratios weighted by the arrival rate each stage sends into the cache.
method	"statedep"	State-dependent transition rates, applied when an arc carries a rate-reset function.
sojourn	"stochastic"	Exponential stage holding time (default).
sojourn	"deterministic"	Each stage lasts exactly its mean holding time.
stage_solver	"fluid"	Solver run in each stage; the fluid transient is the one available.
stage	FluidOptions	Options handed to that stage solver.
timespan_end	double	Transient horizon of the inner solve (default: 100).
tran_points	size_t	Points of the uniform transient grid the exit metrics are summed over (default: 1001).
iter_max	int	Cap on the outer fixed-point iterations (default: 100).
iter_tol	double	Convergence tolerance of that fixed point (default: 1e-4).

Environments derived from a Markovian arrival process

An environment does not always have to be written out by hand. When a model carries a non-renewal process, a MAP, MMPP2, MMAP or MPH, and the solver the user chose declares none of them, LINE does not refuse

the model outright. It freezes each modulating phase of the process into an environment stage, hands the resulting ensemble to `ENV` with the calling solver analysing every stage, and reports the recombined averages. The image is exact in its modulation when the process is an MMPP, whose D_1 blocks are diagonal, and intensity-matched otherwise.

The substitution is announced by a warning naming the number of stages and the kind of image, so an approximation is never returned silently in place of an exact answer. Three configuration fields govern it: `options.config.map_env` set to `off` restores the plain refusal, `options.config.map_env_method` pins the recombination method that `'auto'` would otherwise pick from the time scale of the environment relative to the model, and `options.config.map_env_maxstages` caps the number of stages the image may have, so that a high-order modulating process is refused rather than expanded into an unmanageable ensemble.

Results are written back onto the model the user built rather than onto the internal image. That distinction is invisible for queue lengths and throughputs, which travel in the average tables, but it is what makes a `Cache` work on this path: a cache reports its hit, delayed-hit and miss probabilities on the node itself, not in those tables. After the solve, `CacheMetrics` and `cache_metrics_of` describe the original `Cache`, and, since the hit and miss visit ratios are derived from that split, the `Hit` and `Miss` class throughputs follow it too instead of the even split that stands in before any cache has been analysed. Two columns of the cache table stay NaN on this path, `ResidT` and `ListCost`: neither quantity is recombined across stages, so it is reported as absent rather than as zero. The C++ port carries the transformation itself, as `map2renv` in `line/io/map2renv.h`, but not the automatic interception: a model with a non-renewal process has to be turned into an `Environment` explicitly before `env::solver_env` is called.

6.6.3 Examples

Example 1: Fast and slow service

This example demonstrates how to model a queueing system operating in a random environment, where system parameters (e.g., service rates) change according to an underlying environmental process.

The scenario models a server that alternates between “Fast” and “Slow” modes. In Fast mode, the service rate is 10.0 (low utilization). In Slow mode, the service rate is 0.8 (high utilization, near saturation). The environment switches from Fast→Slow at rate 0.5 (mean time in Fast mode = 2.0) and from Slow→Fast at rate 1.0 (mean time in Slow mode = 1.0).

We start by creating a base closed queueing network with a delay and a queue:

```
Network baseModel("BaseModel");
const std::size_t delay = baseModel.add_delay("ThinkTime");
const std::size_t queue = baseModel.add_queue("Fast/Slow Server", SchedStrategy::FCFS);

// Closed class with 5 jobs
const double N = 5;
const std::size_t jobclass = baseModel.add_closed_class("Jobs", N, delay);
baseModel.set_service(delay, jobclass, Exp(1.0)); // think time
baseModel.set_service(queue, jobclass, Exp(2.0)); // placeholder
```

```
// Connect the nodes in a cycle
Routing P;
P.set(delay, queue, 1.0);
P.set(queue, delay, 1.0);
baseModel.link(P);
```

Next, we create the random environment with two stages, each having a different service rate:

```
env::Environment<double> env("ServerModes", 2);

// Stage 0: fast mode (service rate 10.0, low utilization)
Network fastModel = baseModel; // a copy of the base model
fastModel.set_service(queue, jobclass, Exp(10.0));
env.set_stage(0, "Fast", "operational", fastModel.get_struct());

// Stage 1: slow mode (service rate 0.8, high utilization)
Network slowModel = baseModel;
slowModel.set_service(queue, jobclass, Exp(0.8));
env.set_stage(1, "Slow", "degraded", slowModel.get_struct());

// Transitions between the stages
env.add_transition(0, 1, Exp(0.5)); // Fast -> Slow
env.add_transition(1, 0, Exp(1.0)); // Slow -> Fast
```

A `qn::Network` is an ordinary value, so a stage variant is made by copying it and overwriting what changes. Finally, we solve the model using ENV with the Fluid solver (FLD) as the sub-solver for each stage:

```
env::EnvOptions opt;
opt.stage_solver = "fluid";
opt.timespan_end = 100.0;
opt.iter_tol = 0.01;
const env::EnvAnalyzerSolution<double> res = env::solver_env(env, opt);
// res.QN, res.UN, res.TN are the environment-averaged metrics
```

The same solve is available from the front end as `line-cli -f env.json -s env -a avg`. The results show that the server spends approximately 66.67% of time in Fast mode and 33.33% in Slow mode. The environment-averaged queue length at the server is approximately 2.0 jobs, which is a weighted combination of the queue lengths in the Fast and Slow stages.

Analyzing sample paths with `getSamplePathTable` In addition to computing steady-state metrics averaged over the environment distribution, ENV provides a method to analyze specific sample paths through the environment states. This is useful when you want to understand the transient behavior of the system as it transitions through a known sequence of environment states.

The `getSamplePathTable` method accepts a specification of a sample path—a list of (stage, duration) pairs—and computes the transient performance metrics for each segment. For each segment, the method:

1. Initializes the model from the previous segment's final queue lengths (or uniform distribution for the first segment)

2. Runs transient analysis for the specified duration using the Fluid solver
3. Extracts the initial and final values of queue length, utilization, and throughput

The output is a table with columns for segment index, stage name, duration, station, job class, and the initial/final values of each metric. An example of the transient behavior along a sample path is shown in Figure 6.9, which depicts a sample path transitioning through environment states Fast $\rightarrow$ Slow $\rightarrow$ Fast. In the Fast stage (service rate = 10.0), the server operates at low utilization ($\approx 45\%$), while in the Slow stage (service rate = 0.8), the server becomes saturated ($\approx 100\%$ utilization), causing queue buildup as arrivals accumulate faster than they can be processed.

Sample-path analysis of a prescribed stage sequence is not part of the C++ port; the environment solver reports the stationary blend and the per-stage exit metrics. The sample path can be specified using either stage names (strings) or stage indices (integers). Stages are 1-indexed in MATLAB and 0-indexed in Java/Python. The duration for each segment must be positive.

Example 2: Breakdown and repair

LINE provides convenience methods to simplify the common case of modeling node failures and repairs in random environments. These methods automatically create UP and DOWN stages for nodes that can break down, configure service rates appropriately, and set up breakdown and repair transitions.

The primary method is `addNodeFailureRepair`, which creates both breakdown and repair transitions in a single call: where `baseModel` is the network with normal (UP) service rates, 'Server1' is the node name, `Exp(0.1)` is the breakdown distribution (mean time to failure = 10 time units), `Exp(1.0)` is the repair distribution (mean time to repair = 1 time unit), and `Exp(0.5)` is the service distribution when the node is down.

```
env::Environment<double> env("ServerEnv", 2);
env.add_node_failure_repair(0, 1, baseModel.get_struct(), "Server1",
                           Exp(0.1), // breakdown time
                           Exp(1.0), // repair time
                           Exp(0.5)); // degraded service
```

The first two arguments name the stage slots that hold the UP and the DOWN model; the DOWN stage is created as `DOWN_Server1`, with the node's service replaced by the degraded one.

This method automatically:

- Creates an UP stage with the service rates of the base model
- Creates a DOWN stage with the specified reduced service rate for the failed node
- Adds a breakdown transition (UP $\rightarrow$ DOWN) with the given breakdown distribution
- Adds a repair transition (DOWN $\rightarrow$ UP) with the given repair distribution

Alternative methods for more control include `add_node_breakdown` and `add_node_repair`, which can be called separately. Their full signatures are where `nodeOrName` is either a `Node` object or a string node name, `breakdownDist` and `repairDist` are Markovian distributions controlling the timing of the

corresponding stage transitions, and `downServiceDist` is the service distribution applied to all classes at the failed node while it is in the DOWN stage. The first call to `addNodeBreakdown` also creates the UP stage from `baseModel` (if it does not yet exist), and each invocation creates a DOWN stage with auto-generated name `DOWN_<nodeName>`, so that several nodes can break down independently within the same environment. The optional `resetFun` argument follows the same convention as in the previous section and defaults to the identity, leaving queue lengths unchanged on the corresponding transition. Custom reset policies can also be supplied at the convenience entry point `add_node_failure_repair` as optional parameters:

```
env.add_node_breakdown(up, down, baseModel.get_struct(), node_name,
                      breakdown_dist, down_service_dist, breakdown_reset);
env.add_node_repair(node_name, repair_dist, repair_reset);
```

Reset policies can also be modified after environment creation using `setBreakdownResetPolicy` and `setRepairResetPolicy` methods.

Retrieving reliability metrics After solving an environment model with node breakdown and repair, reliability metrics can be retrieved using the method (aliases: `relT`, `getRelT`, `relTable`, `getRelTable`): Reliability metrics are not tabulated by the C++ port: the mean times to failure and to repair are the means of the breakdown and repair distributions the environment was given, and the availability is the stationary probability of the UP stage in the environment solution.

The reliability metrics returned are:

- **MTTF** (Mean Time To Failure): Expected time until breakdown from the UP state. For multiple failure modes, this uses the combined failure rate: $\lambda_{\text{total}} = \sum_i \lambda_i$.
- **MTTR** (Mean Time To Repair): Expected time to restore from the DOWN state. For multiple DOWN states, this is a weighted average based on steady-state probabilities.
- **MTBF** (Mean Time Between Failures): Sum of MTTF and MTTR.
- **Availability**: Steady-state probability of the system being in the UP state.

Example 3: Markov-modulated service

This example shows how to convert a queueing network with MMPP2 (Markov-Modulated Poisson Process) service into an equivalent random environment model using the `MAPQN2RENV` utility. MMPP2 is a 2-phase service process where the service rate switches between two values according to an underlying Markov chain. The `MAPQN2RENV` function automatically transforms such a model into a random environment with two stages, each having exponential service at the corresponding MMPP rate.

We create a closed queueing network with an MMPP2 service process: The conversion of a Markov-modulated queueing network into a random environment is not part of the C++ port; the modulating phases have to be written out as stages of a `qn::Environment`.

The resulting environment can then be analyzed using `ENV` as shown in Example 1.

Example 4: semi-Markov process environments

This example demonstrates the analysis of random environments with non-Markovian transition distributions using semi-Markov Processes. When environment transitions follow general distributions (e.g., Lognormal), the ENV solver should be run with the `'smp'` method.

Consider a system that alternates between two operational modes with log-normally distributed holding times in each stage.

Semi-Markov environments are not part of the C++ port: `env::solver_env` takes exponential stage transitions, and a non-exponential holding time has no analyzer behind it.

The ENV solver correctly handles this by using numerical integration to compute transition probabilities from the non-Markovian distribution functions, rather than assuming Markovian (memoryless) transitions.

Example 5: MMAP-modulated cache workload

A random environment can also drive a multiclass workload generated by a marked Markovian arrival process (MMAP). We feed a small Round-Robin cache with a marked MMPP2 whose two marks are bound, via `setMarkedArrival`, to two open read classes that share the modulating chain: phase 1 is bursty and emits mostly class-1 references at a high rate, phase 2 is calm and emits mostly class-2 references at a low rate. Viewing the modulating phase as the environment turns the MMAP into two phase-conditional Poisson streams (the diagonal D_1 blocks), so the cache can be solved by the ENV meta-solver. We use the state-vector method (`options.method='statevec'`), which carries the cache-state distribution across phase switches and averages each phase's sojourn-weighted distribution; for a Markovian environment it recovers the exact joint (cache $\times$ phase) solution, i.e., it matches CTMC on the true MMAP-fed cache. The C++ port builds this model: `set_marked_arrival` binds the two marks to the two read classes, so the true MMAP-fed cache is solved directly by LDES and by CTMC. The environment below is constructed explicitly, as it is above, since the C++ port does not derive one from the MMAP on its own. The per-read-class hit ratios returned by the state-vector ENV solver (`hitSTATEVEC`) match those of the exact CTMC on the true MMAP-fed cache (`hitCTMC`). The faster `'avg'` (fast-environment, rate-averaged) and `'dec'` (slow-environment, quasi-stationary decomposition) methods discard the within-phase temporal correlation of references and are therefore only approximate. A complete runnable version, including the LDES simulation and the two approximate methods, is in `examples/basic/cacheModel/cache_mmap_rr_env.m`.

Solving the marked model directly. Writing the environment out, as above, is the way to choose the recombination method and to compare the three of them. It is not, however, a precondition for solving the model: a solver that does not declare MMAP derives the same two-stage image on its own (Section 6.6.2) and writes the cache surface back onto the original `Cache` node, so the marked model can simply be handed to it. The two solvers reach different recombinations, NC the slow-environment limit and FLD the mean-field coupling, and neither limit is guaranteed to bound the exact value, since both discard the within-phase temporal correlation of references:

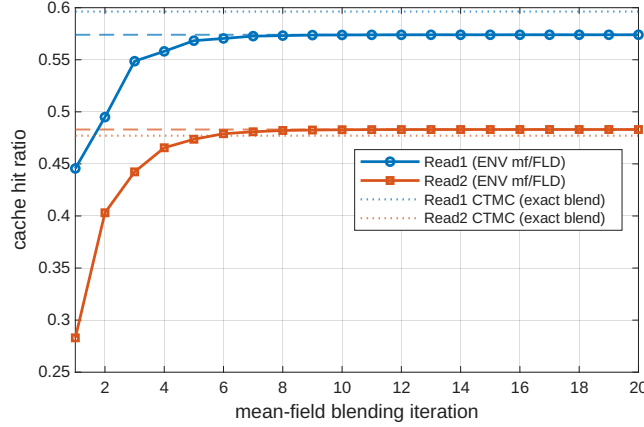


Figure 6.10: Mean-field ENV blending for the MMAP-modulated cache of Example 6.6.3. Starting from an empty cache, the per-read-class hit ratios produced by the default (mean-field) ENV analyzer with an FLD inner solver converge over the blending iteration to the environment-blended equilibrium (dashed), obtained by carrying each phase’s mean cache occupancy across phase switches. The exact CTMC blend (dotted) is approached up to the mean-field error of the small ($n=4$, $m=2$) cache.

	Read1	Read2
LDDES ($2 \cdot 10^5$ samples)	0.5970	0.4811
CTMC (exact)	0.5963	0.4771
NC (env.dec)	0.5991	0.5087
FLD (env.meanfield)	0.5739	0.4831

Each limit is near-exact on one class and off by two to three points on the other, in opposite directions: NC tracks the bursty class 1 that dominates the fast stage, FLD the calm class 2. For reference, the sampling error of the simulation at this sample count is 0.0007 and 0.0040.

Mean-field blending. When the per-phase cache state space is too large for the exact ‘statevec’ inner solver, the default (mean-field) ENV analyzer provides a fully fluid alternative driven by an FLD inner solver. Its refined mean-field (RMF) algorithm analyses each phase’s cache by integrating the mean-field occupancy drift over the phase sojourn; the mean cache occupancy is then carried across phase switches (the cache analog of the queue-length hand-off performed by `initFromMarginal`), and the per-read-class hit ratio is the `probEnv`-weighted, sojourn-averaged product of arrival rate and hit probability. This trades the exact joint (cache $\times$ phase) distribution of ‘statevec’ for a cheaper fluid approximation, and is available identically in the MATLAB, Java, and Python codebases. The C++ port drives the same mean-field coupling through `env::solver_env` with `method="meanfield"` and `stage_solver="fluid"`. Figure 6.10 shows the mean-field blending iteration converging, from an initially empty cache, to the fluid equilibrium `hitMF`; the residual gap to the exact CTMC blend is the mean-field error of the small ($n=4$, $m=2$) cache.

Chapter 7

Advanced analysis methods

The analyses collected here go beyond the steady-state averages of Chapter 4: they name a state of the model, follow it in time, or observe a single realization of it. They are requested through the same handles, but not every solver implements them; the feature tables of Section 1.2 state which one does.

7.1 Specifying states

In some analyses it is important to specify the state of the network, for example to assign the initial position of the jobs in a transient analysis. We thus discuss the support in LINE for state modeling.

7.1.1 Station states

We begin by explaining how to specify a state s_0 for a station. For example, it is not supported for shortest job first (`SchedStrategy.SJF`) scheduling, in which state must include the service time samples for the jobs and it is therefore a continuous quantity.

Suppose that the network has R classes and that service distributions are phase-type, i.e., that they inherit from `Markovian`. Let K_r be the number of phases for the service distribution in class r at a given station. Then, we define three types of state variables:

- c_j : class of the job waiting in position $j \leq b$ of the buffer, out of the b currently occupied positions. If $b = 0$, then the state vector is indicated with a single empty element $c_1 = 0$.
- k_j : service phase of the job waiting in position $j \leq b$ of the buffer, out of the b currently occupied positions.
- n_r : total number of jobs of class r in the station
- b_r : total number of jobs of class r in the station's buffer
- s_{rk} : total number of jobs of class r running in phase k in the server

Table 7.1: State descriptors for Markovian scheduling policies

Sched. strategy	Station state vector	State condition
EXT	$[Inf, s_{11}, \dots, s_{1K_1}, \dots, s_{R1}, \dots, s_{RK_R}]$	$\sum_k s_{rk} = 1, \forall r$
FCFS, FCFSPRIO, LCFS	$[c_b, \dots, c_1, s_{11}, \dots, s_{1K_1}, \dots, s_{R1}, \dots, s_{RK_R}]$	$\sum_r \sum_k s_{rk} = S$
LCFSPR, LCFSPI	$[c_b, k_b, \dots, c_1, k_1, s_{11}, \dots, s_{1K_1}, \dots, s_{R1}, \dots, s_{RK_R}]$	$\sum_r \sum_k s_{rk} = S$
SEPT, SIRO	$[b_1, \dots, b_R, s_{11}, \dots, s_{1K_1}, \dots, s_{R1}, \dots, s_{RK_R}]$	$\sum_r \sum_k s_{rk} = S$
PS, DPS, GPS, INF	$[s_{11}, \dots, s_{1K_1}, \dots, s_{R1}, \dots, s_{RK_R}]$	None

Here, by phase we mean the number of states of a distribution of class `Markovian`. If the distribution is not Markovian, then there is a single phase. With these definitions, the table below illustrates how to specify in `LINE` a valid state for a station depending on its scheduling strategy. There, S is the number of servers of the queueing station. All state variables are non-negative integers. The `SchedStrategy.EXT` policy is used for the `Source` node, which may be seen as a special station with an infinite pool of jobs sitting in the buffer and a dedicated server for each class $r = 1, \dots, R$.

States can be manually specified or enumerated automatically. `LINE` library functions for handling and generating states are as follows:

- `from_marginal`: enumerates all states that have the same marginal state $[n_1, n_2, \dots, n_R]$.
- `from_marginal_and_started`: restricts the output of `from_marginal` to states with given number of running jobs, all assumed to be in service phase $k = 1$.
- `from_marg_node`: the class-summed counterpart of `from_marginal`. It fixes only how many jobs the station holds in `TOTAL`, and returns the union of the per-class enumeration over every split of that total the station can hold. Classes disabled at the station are excluded from the split enumeration, so a disabled class cannot silently absorb a job.
- `from_marg_node_started`: the same for the pair (total number of jobs, total number of started jobs), taking the union of over every pair of per-class vectors consistent with both totals.
- `to_marginal`: extracts marginal statistics from a state, such as the total number of jobs in a given class that are running at the station in a certain phase.

Note that if a function call returns an empty state (`[]`), this should be interpreted as an indication that no valid state exists that meets the required criteria. Often, this is because the state supplied in input is invalid.

Example

We consider the example network in . We examine station 2 (the `Queue1` station), which is a three-server FCFS station. All four classes have exponential service times except class 2, which has Erlang-2 service times. We are interested in states with two running class-1 jobs and one running class-2 job, together with two buffered jobs belonging to classes 3 and 4, respectively. We can automatically generate this state space, which we store in the `space` variable, as: The C++ port enumerates a marginal with `qn : from_marginal (sn, ist, n, phases)`; the variants that also fix which jobs are running or started have no counterpart. Here,

each row of `space` corresponds to a valid state. The argument `qn` gives the number of jobs in the node for the 4 classes, while `phases` gives the number of running jobs in each class. This station has four valid states, differing on whether the class-2 job runs in the first or in the second phase of the Erlang-2 and on the relative position of the jobs of class 3 and 4 in the waiting buffer.

To obtain states where the jobs have just started running, we can instead use `from_marginal`. As above, only `from_marginal` is available. We see that the above state space restricted the one obtained with `from_marginal` to states where the job in class 1 is always in the first phase.

If we instead remove the specification of the running jobs, we can use `from_marginal` to generate all possible combinations of states depending on the class and phase of the running jobs. In the example, this returns a space of 20 possible states.

```
const std::vector<std::size_t> n{2, 1, 1, 1};
const std::vector<std::size_t> phases{1, 1, 1, 1};
const std::vector<std::vector<double>> space =
    qn::from_marginal(model.get_struct(), ist, n, phases);
```

`ist` is the 1-based station index, and each row of the result is one state of that station consistent with the marginal.

Assigning a prior to an initial state

It is possible to assign the initial state to a station using the `from_marginal` function on that station's object. LINE offers the possibility to specify a prior probability on the initial states, so that if multiple states have a non-zero prior, then the solver will need to solve an independent model using each one of those initial states, and then carry out a weighting of the results according to the prior probabilities. The default is to assign a probability 1.0 to the *first* specified state. The functions `get_prior` and `set_prior` can be used to check and change the prior probabilities for the initial states specified for a station or stateful node.

7.1.2 Network states

A collection of states that are valid for each station is not necessarily valid for the network as a whole. For example, if the sum of jobs of a closed class exceeds the population of the class, then the network state would be invalid. To identify these situations, LINE requires to specify the initial state of a network using functions supplied by the `Network` class. These functions are `from_marginal`, `from_network`, and `from_state`. They require a matrix `n` with elements (i, r) specifying the total number of resident class- r jobs at node i and the latter two require a matrix `s` with elements (i, r) with the number of running (or started) class- r jobs at node i . The user can also manually verify if the supplied network state is going to be valid using `is_valid`.

It is also possible to request LINE to automatically identify a valid initial state, which is done using the function `find_valid_initial_state` available in the `Network` class. This is going to select a state where:

- no jobs in open classes are present in the network;
- jobs in closed classes all start at their reference stations;

- the servers at each reference station are occupied by jobs of in class order, i.e., jobs in the firstly created class are assigned to the server, then spare server are allocated to jobs in the second class, and so forth;
- service or arrival processes are initialized in phase 1 for each job;
- if the scheduling strategy requires it, jobs are ordered in the buffer by class, with the firstly created class at the head and the lastly created class at the tail of the buffer.

The `method` is a wrapper for `from_marginal` to initialize the system as close as possible to the average steady-state distribution of the network. Since averages are typically not integer-valued, this function rounds the average values to the nearest integer and adjusts the result to ensure feasibility of the initialization.

7.1.3 Initialization of transient classes

Because of class-switching, it is possible that a class r with a non-empty population at time $t = 0$ becomes empty at some positive time $t' > t$ without ever being visited again by any job. LINE allows users to place jobs in transient classes and therefore it will not trigger an error in the presence of this situation. If a user wishes to prohibit the use of a class at a station, it is sufficient to specify that the corresponding service process uses the `Disabled` distribution.

Certain solvers may incur problems in identifying that a class is transient and in setting to zero its steady-state measures. For example, the JMT solver uses a heuristic whereby a class is considered transient if it has fewer samples than jobs initially placed in the corresponding chain the class belongs to. For such classes, JMT will set the values of steady-state performance indexes to zero.

7.1.4 State space generation

The state space of a model can be obtained by either invoking `get_state_space` or `get_states` on an instant of the CTMC solver, where the latter returns the state space cached during the solution of the CTMC.

LINE supports two state space generation methods, configurable using the option `options.config.state_space_gen` of the CTMC solver.

7.1.5 State probability analysis

Besides mean values, solvers that build an explicit state representation can return the stationary probability of the network states. Table 7.2 lists the available accessors, which differ in whether the state is global or local to a station, and in whether the phase information of non-exponential service is retained or aggregated away.

Reading these outputs requires the state encoding of Section 7.1.4: a state is a vector of the number of jobs of each class at each station, extended with the phase of each in-service job when the service process is non-exponential, and with the buffer order under FCFS scheduling, which is why FCFS state spaces are larger than the corresponding processor-sharing or infinite-server ones. The example files `statepr_aggr` and `statepr_allprobs_fcfs` show how to aggregate over classes and stations and how to enumerate all states of a FCFS network, from which quantities such as $P(\text{station empty})$ or $P(\text{total population} > k)$ follow by summation.

Table 7.2: State probability accessors

Method	Returns
	Probability of each global state of the network
<code>solver_mva_get_prob_sys_aggr</code>	The same, aggregated by class over all stations, so that phase and buffer-order detail is summed out
<code>solver_mva_get_prob_marg</code>	Probability mass function of the state of one station
<code>solver_mva_get_prob_aggr</code>	The same, aggregated by class: the joint per-class queue-length distribution at that station
<code>solver_mva_get_prob_marg</code>	Marginal queue-length distribution $P(n_{i,r} = k)$ of a single class r at station i , other classes marginalized out (MVA, MAM). Under NC the method takes only a station and returns the class-aggregated distribution at it
<code>solver_nc_getprob_sys_marg</code>	Joint law of the per-station TOTAL queue lengths, $P(n_1 = k_1, \dots, n_M = k_M)$, with the classes summed out (NC only). This is the joint counterpart of <code>getProbMarg</code> and the class-aggregated counterpart of <code>getProbSysAggr</code> ; see §7.1.5

For example, the marginal queue-length distribution of class `cclass1` at `queue1` can be obtained as follows:

```
mva::MvaOptions opt;
const mva::AvgResult<double> avg =
    mva::solver_mva_run_analyzer(model.get_struct(), opt, Matrix<double>());
const mva::MargResult<double> marg =
    mva::solver_mva_get_prob_marg(model.get_struct(), avg, opt, queue1, cclass1);
```

The marginal is computed from a solved model, so the average solve is passed in beside the query.

Joint law of the per-station total queue lengths

The methods above answer either a per-class question at one station or a per-class question about the whole network. The remaining question, the JOINT law of the per-station TOTALS with the classes summed out, is answered by `solver_nc_getprob_sys_marg` under the NC solver. The four state-probability methods answer questions on four different index sets:

Method	Random variable	States at 3 stations, 3 classes, populations 2, 2 and 1
<code>getProbAggr</code>	the per-class counts at one station	6 / 6 / 3
<code>getProbSysAggr</code>	the whole station-by-class table of counts	108
<code>getProbMarg</code>	the total count at one station	6
<code>getProbSysMarg</code>	the vector of per-station totals, classes summed out	21

Table 7.3: The four state-probability methods answer questions on different index sets. Each `getProbSysMarg` value is the sum of `getProbSysAggr` over every per-class table with the given row sums.

Each value returned by `getProbSysMarg` is therefore the sum of `getProbSysAggr` over the whole set of per-class tables with the prescribed row sums, and that set grows combinatorially, so summation is not a route to it. For a closed multiclass product-form network with load-independent single-server queues

plus infinite servers, the sum has instead a closed form. It is the permanent of a square demand matrix, built by repeating each class column as many times as that class has jobs and each station row as many times as that station holds, divided by the normalizing constant, by the factorial of each class population, and by one factorial per infinite-server station of the jobs it holds. The queueing stations need no such factorial of their own, the permanent supplying it. The permanent is evaluated with Ryser's inclusion-exclusion formula exploiting the column multiplicities [140], at a cost that grows with the product of the class populations rather than with the factorial of their sum.

The method takes an optional second argument selecting the estimator of the permanent: 'exact' (the default and the only exact one), 'bethe' [160], 'heur' [149], 'huberlaw' [82] and 'adapart' [94]. The four approximations require a demand matrix with full support and are refused, rather than applied to a floored matrix, whenever a class has zero demand at a station that holds jobs: Sinkhorn scaling does not converge without full support, and the Bethe estimate is a state-dependent lower bound whose gap does not cancel when the values are normalized against one another. The exact estimator has no such restriction, since a station holding no jobs contributes no row, a class with no jobs contributes no column, and a zero demand is an ordinary zero entry.

The closed form relies on one factorial per queueing station, which neither a multiserver nor a load-dependent station provides. Such models, as well as open and mixed ones, are refused by name rather than approximated. The following example sweeps the whole lattice of total states of a closed network, so that the reported probabilities sum to one:

```
nc::NcSolverOptions opt;
const std::vector<int> nvec{1, 1, 1};
const double p = nc::solver_nc_getprob_sys_marg(model.get_struct(), opt, nvec);
const double pb = nc::solver_nc_getprob_sys_marg(model.get_struct(), opt, nvec, "bethe");
```

The same query is available from the command line as `line-cli -s nc -a sysmarg`, with `--perm-engine` selecting the estimator.

Both the probability and its logarithm are returned, the latter being formed first so that a state whose probability underflows in double precision still reports a finite logarithm. A state whose total does not match the closed population is not an error but has probability zero, which is what allows the lattice sweep above to be written without a feasibility test. The example files `statepr_sys_marg` show the sweep in each language together with the check that the first moments of the law reproduce the mean queue lengths returned by the CTMC solver.

Transient state probabilities

For solvers that expose the underlying state space, LINE also provides transient counterparts of the steady-state probability methods, giving the distribution over network states at a finite time horizon (controlled by the `timespan` option). These are useful for analyzing warm-up behavior, ramp-up of arrival rates, or short-term reliability questions. Per-station transient probabilities are returned by `ctmc_get_tran_prob` (full state) and `ctmc_get_tran_prob_aggr` (aggregated by class), while system-level joint transient probabilities are returned by `ctmc_get_tran_prob_sys` and `ctmc_get_tran_prob_sys_aggr`. All

four methods are currently supported by the CTMC solver. JMT additionally provides the aggregated form (`ctmc_get_tran_prob_aggr`).

The following example computes the per-class transient probabilities at `queue1` from the default initial state up to time $t = 1$:

```
ctmc::CtmcOptions opt;
const auto res = ctmc::ctmc_get_tran_prob_aggr(model.get_struct(), opt, 0.0, 1.0, ind);
// one probability vector per point of the transient grid, with the
// aggregated marginal states it is indexed by
```

The horizon is passed explicitly: $\text{pi}(t)$ on an unstated horizon is not a quantity. The same query is `line-cli -s ctmc -a tranprob -tspan 0:1 -node <n>`.

7.2 Transient analysis

So far, we have seen how to compute steady-state average performance indexes, which are given by

$$E[n] = \lim_{t \rightarrow +\infty} E[n(t)]$$

where $n(t)$ is an arbitrary performance index, e.g., the queue-length of a given class at time t .

We now consider instead the computation of the quantity $E[n(t)|s_0]$, which is the *transient average* of the performance index, conditional on a given initial system state s_0 . Compared to $n(t)$, this quantity averages the system state at time t across all possible evolutions of the system from state s_0 during the t time units, weighted by their probability. In other words, we observe all possible stochastic evolutions of the system from state s_0 for t time units, recording the final values of $n(t)$ in each trajectory, and finally average the recorded values at time t to obtain $E[n(t)|s_0]$.

7.2.1 Computing transient averages

The computation of transient metrics proceeds similarly to the steady-state case. We first obtain the handles for transient averages: The C++ port has no transient handles: a transient solve is asked for by its horizon, and returns the trajectories directly. After solving the model, we will be able to retrieve *both* steady-state and transient averages as follows

```
ctmc::CtmcOptions opt;
const ctmc::CtmcTransient<double> tran =
    ctmc::solver_ctmc_transient_analyzer(model.get_struct(), opt, 0.0, 1.0);
// tran carries the time grid and the queue-length, utilization and
// throughput trajectories on it
```

The transient average queue-length at node i for class r is stored within `QNT` in row i and column r .

Note that the above code specifies a maximum time t for the output time series. This can be done using the `timespan` solver option. This applies also to average metrics. In the following example, the first model is solved at steady-state, while the second model reports averages at time $t = 1$ after initialization

```
// steady state
ctmc::CtmcOptions opt;
const mva::AvgResult<double> ss = ctmc::solver_ctmc_run_analyzer(model.get_struct(), opt);
// transient on [0, 1]
const ctmc::CtmcTransient<double> tr =
    ctmc::solver_ctmc_transient_analyzer(model.get_struct(), opt, 0.0, 1.0);
```

The front end prints the same two answers as `line-cli -s ctmc -a avg` and `-a tran -tspan 0:1`.

7.2.2 First passage times into stations

When the model is in a transient, the average state seen upon arrival to a station changes over time. That is, in a transient, successive visits by a job may experience different response time distributions. The function `solver_ctmc_cdf_respt`, implemented by JMT offers the possibility to obtain this distribution given the initial state specified for the model. As time passes, this distribution will converge to the steady-state one computed by solvers equipped with the function `solver_ctmc_cdf_respt`.

However, in some cases one prefers to replace the notion of response time distribution in transient by the one of *first passage time*, i.e., the distribution of the time to complete the *first visit* to the station under consideration. The function `solver_ctmc_cdf_respt` provides this distribution, assuming as initial state the one specified for the model, e.g., using `from_marginal` or `from_state`. This function is available in FLD and JMT and has a similar syntax as `solver_ctmc_cdf_respt`. A steady-state passage CDF is currently provided only by FLD (as `solver_ctmc_cdf_respt`); the generic dispatcher is reserved for future implementations and currently raises an error on all solvers.

7.3 Advanced metrics

This section collects analyses that go beyond the standard per-station and system-wide performance indexes. They include metrics specific to fork-join structures and finite capacity regions, the Age of Information for status-update systems, and user-defined reward models evaluated over the underlying Markov chain.

7.3.1 Fork-join metrics

Two metrics report a fork-join section as a whole rather than station by station. `FJQLen` is the mean number of jobs in the section, counted from dispatch at the fork to synchronization at the join, and `FJRespt` is the corresponding mean sojourn time, that is the time for all sibling tasks of a job to complete and be joined. Both therefore aggregate over every station lying between a fork and its join, which the ordinary per-station metrics cannot express. They are appended to the `solver_mva_run` output whenever the model contains a fork-join section, and are computed by the MVA, JMT and LDES solvers.

7.3.2 Finite capacity region queue length

When a station is part of a Finite Capacity Region (FCR) with `Queue` drop strategy, the reported queue length (`QLen`) includes both jobs currently inside the station and jobs that are blocked waiting to enter the FCR.

This semantic ensures consistency with blocking queueing network models where blocked jobs are logically associated with their destination queue. For example, if an FCR with capacity 2 contains a queue with 2 jobs, and 1 additional job is blocked waiting to enter, the reported queue length will be 3. This convention is consistent with JMT's handling of FCR blocking.

7.3.3 Age of Information analysis

FLD supports the computation of Age of Information (AoI) metrics for single-queue open systems with capacity constraints. AoI measures the freshness of information in status update systems, defined as the time elapsed since the last successfully received update was generated. When the model has a valid AoI topology (Source $\rightarrow$ Queue $\rightarrow$ Sink with one open class, one queue, one server, and queue capacity 1 or 2), the `mfq` method automatically switches to the AoI MFQ analysis and computes exact AoI and Peak AoI distributions. Age of Information is available in the C++ port at the API level only: the closed forms `aoi::aoi_fcfs_mm1`, `aoi_lcfspr_mgil` and their relatives take the arrival and service processes directly, and no solver reports AoI from a network model.

7.3.4 Reward models

LINE supports the definition of custom reward functions on network models, enabling the computation of application-specific performance metrics beyond the standard measures of queue length, utilization, and response time. Rewards associate a numerical value with each state of the underlying continuous-time Markov chain, allowing analysts to define metrics that capture domain-specific objectives such as energy consumption, revenue, or composite performance indicators. The reward framework supports both steady-state analysis, which computes the expected reward in equilibrium, and transient analysis, which tracks the evolution of reward values over time. This feature is available through the CTMC solver, which constructs the Markov chain representation of the network and evaluates reward functions over the state space.

Reward functions are defined using the `set_reward` method on the `Network` object. This method takes two arguments: a string name for the reward, which is used to identify it in the results, and a function that maps a state accessor object to a scalar reward value. The state accessor provides the method `at`, which takes a node and a job class as arguments and returns the number of jobs of that class at that node in the current state.

```
model.set_reward("QueueLength", [queue](const std::vector<double>& state) {
    return state[queue - 1];
});
```

A reward is a function of the state vector; it is evaluated on every state by the CTMC solver.

This example defines a reward equal to the number of jobs of class `oclass` at the `queue` node. More complex reward functions can be constructed by combining state information from multiple nodes and classes, or by applying arbitrary mathematical operations to state variables.

To simplify common use cases, a reward function that returns an indicator variable, taking the value 1 when the specified node is serving a job of the specified class and 0 otherwise. The `method` creates a reward

Table 7.4: Reward function templates in the `Reward` class (MATLAB method names shown, with the Python-native name in parentheses where it differs). In Java, the same metrics are written directly as `RewardFunction` lambdas.

Template	Arguments	Description
<code>queueLength(queue_length)</code>	<code>node, [jobclass]</code>	Number of jobs at <code>node</code> , totalled across classes or restricted to <code>jobclass</code> when supplied.
<code>utilization</code>	<code>node, [jobclass]</code>	Fraction of busy servers at <code>node</code> , computed as <code>min(jobs, nservers)</code> ; per-class when <code>jobclass</code> is supplied.
<code>blocking</code>	<code>node</code>	Indicator equal to 1 when <code>node</code> is at capacity (buffer full) and 0 otherwise.
<code>custom</code>	<code>fn</code>	Returns the user-defined reward function <code>fn</code> unchanged, documenting the intent that it is a custom reward.

function that returns an indicator for buffer-full conditions, taking the value 1 when the node’s buffer is at capacity and 0 otherwise.

There are no reward templates in the C++ port: a utilization or blocking reward is written as the function of the state it is.

The complete set of pre-built templates is summarized in Table 7.4. Each template is a factory that returns a reward function suitable for `set_reward`.

Once reward functions are defined on the model, they can be evaluated by creating a CTMC solver instance and calling the appropriate analysis methods. For steady-state analysis, the `solver_ctmc_avg_reward` method returns . For transient analysis, the `solver_ctmc_tran_reward` method returns, for each defined reward, the time-indexed trajectory of the expected reward $E[r(X(t))]$, together with the vector of time points and the reward names. Here $X(t)$ is the system state at time t and the expectation is taken over the transient distribution of the Markov chain, starting from the model’s initial state; a finite `timespan` must therefore be supplied to the solver.

```
ctmc::CtmcOptions opt;
const std::vector<double> R =
    ctmc::solver_ctmc_avg_reward(model.get_struct(), opt, rewards);
const std::vector<std::vector<double>> Rt =
    ctmc::solver_ctmc_tran_reward(model.get_struct(), opt, rewards, t0, t1);
```

The same two answers are `line-cli -s ctmc -a reward` with and without `--tspan`.

The transient trajectories can be plotted to visualize how each reward relaxes from its initial value towards the steady-state equilibrium. For example, the following snippet overlays the transient expected rewards on their steady-state values, producing Figure 7.1. The C++ port produces the two vectors above; plotting them is left to the caller.

The reward framework is particularly useful for defining objective functions in optimization problems, tracking custom performance indicators that combine multiple system states, or analyzing transient behavior of derived metrics. The examples `rewardModel_mmlk`, `rewardModel_templates`, and `rewardModel_aggregation` demonstrate various applications of reward models, including simple state-based rewards, the use of template functions, and the aggregation of reward values across multiple nodes or

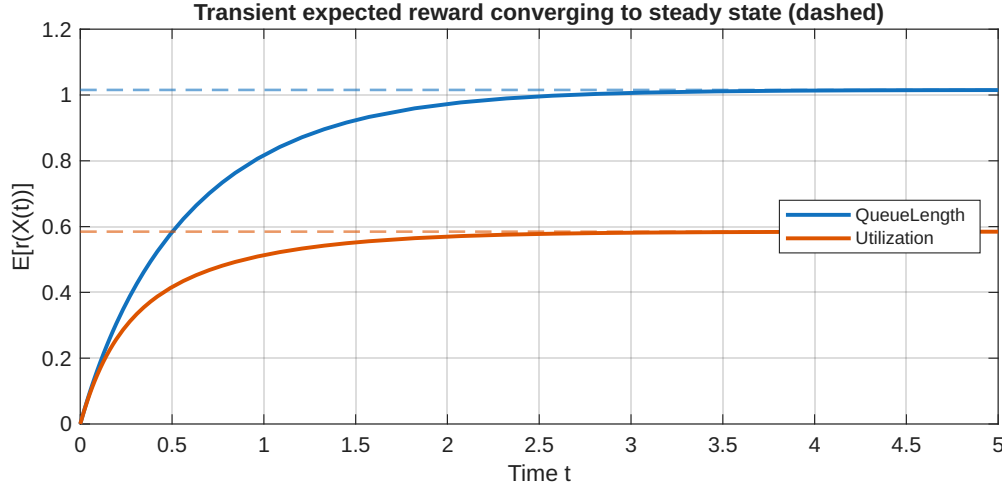


Figure 7.1: Transient expected reward $E[r(X(t))]$ for the queue length and utilization of an $M/M/1/3$ queue, converging over time to the steady-state values (dashed) computed by `solver_ctmc_avg_reward`.

classes.

7.4 Sample path analysis

Transient analysis returns the expected value $E[X(t)]$ of a performance index, whereas sample path analysis returns a single realization of the process $X(t)$, that is one trajectory produced by a specific sequence of random events. A different random seed yields a different sample path, and averaging many realizations at a fixed time recovers the transient average. Four methods extract such a trajectory from the stochastic process underlying the network:

- `solver_ctmc_sample`: the time-varying state of a given stateful node, labelled with the events that changed it.
- `solver_ctmc_sample_aggr`: the same, with the state aggregated to the number of jobs of each class at the node.
- `solver_ctmc_sample_sys`: the state of every stateful node in the model.
- `solver_ctmc_sample_sys_aggr`: the aggregated state of every stateful node.

The JMT solver supports only the aggregate variants, since the simulator does not export phase-level information; the SSA solver supports all four. For example, the following extracts ten samples for an $APH(2)/M/1$ queue, where the first column is the time at which the station enters the state shown next to it:

```
ctmc::CtmcOptions opt;
opt.samples = 10;
```

```
const ctmc::CtmcSamplePath<double> path =
    ctmc::solver_ctmc_sample_sys(model.get_struct(), opt, seed);
const Matrix<double> aggr =
    ctmc::solver_ctmc_sample_aggr(model.get_struct(), path, ind);
```

The sample path comes from the CTMC rather than from a simulator: the C++ port has no JMT bridge. The front end exposes it as `line-cli -s ctmc -a sample -node <n> -samples <k>`.

7.4.1 Confidence intervals for steady-state quantiles

A tail service-level agreement is written against a quantile rather than a mean, and that estimate needs an interval of its own. LINE provides two automated procedures for this purpose, (FQUEST), which works from a single sample path of arbitrary length [103], and (FIRQUEST), which works from a set of independent replications of equal length [104]. Both act on a sequence of observations rather than on a model, so they apply to any output series, and neither requires the user to choose a batch size, a batch count or a warm-up length. The two hypothesis tests they use internally are also exposed on their own as `and` and `.`

The output series must be continuous, that is a response, waiting or sojourn time: the method must not be applied to integer-valued output such as a queue length, whose marginal law has no density, nor to heavy-tailed service times. When the sample is too small to support the underlying asymptotics the procedure returns a conservative fallback interval and sets the `field` of the result, which should be inspected before an interval is quoted. As an illustration we estimate the 0.9-quantile of the waiting time in an $M/M/1$ queue with $\lambda = 0.8$ and $\mu = 1$:

```
// W is the sequence of waiting times observed in a simulation run
const sim::QuestResult<double> r = sim::sim_fquest(W, 0.9, 0.05);
// r.lower, r.estimate, r.upper
```

The quantile estimators are ported; the Lindley recursions that produce `W` are not, so the sequence has to come from elsewhere.

7.4.2 Conditional waiting times along a sample path

Useful for validating a simulated trajectory one customer at a time, LINE implements the conditional moments of Lindley's recursion [99, 124]. Given the waiting time of a customer, `returns` the exact conditional moments of the waiting time of the next customer in an $M/M/1$ queue, and `does` the same for hyperexponential interarrival and service times. Being conditional on the current state rather than stationary, they remain finite at any load, including a saturated queue.

For stations in tandem, `propagates` the recursion over an arbitrary number of stations and arbitrary interarrival and service times, reproducing a direct event-driven simulation of the tandem to machine precision, while `gives` the conditional mean waiting time at the downstream station of a two-station $M/M/1$ tandem.

7.4.3 Tail bounds for a tandem of two queues

End-to-end response times in a network are far harder to characterize than single-queue ones, since the waiting times a job accumulates along its path are neither independent nor identically distributed. For a tandem of two single-server FCFS stations fed by a renewal arrival process, LINE provides the polynomial-exponential upper bounds of [42] on the tails of the end-to-end waiting time and sojourn time, through `qsys_tandem_ub_ciucu`. Both stations serve the same hyperexponential law, a single phase giving exponential service, and the interarrival time enters only through its Laplace-Stieltjes transform, supplied as a function handle, so any light-tailed renewal input is admissible.

The bounds decay exponentially, at the rate fixed by the positive root of the transform equation that balances a service time against an interarrival time, multiplied by a polynomial of degree one; that polynomial factor is what lets them follow the concave bend of the tail a purely exponential bound cannot. They are attained on an $M/M/1$ tandem, where they return the exact tails of both quantities. The bound on the waiting time is available for exponential service only and is returned as `NaN` otherwise. As an illustration we bound the sojourn time of a $D/M/1$ tandem at three quarters utilization with unit service rate, at horizons 5, 10 and 20:

```
const std::function<double(const double&)> lst =
    [] (const double& s) { return std::exp(-4.0 * s / 3.0); };
const qsys::TandemUbResult<double> r = qsys::qsys_tandem_ub_ciucu<double>(
    {5.0, 10.0, 20.0}, lst, {1.0}, {1.0});
// r.theta, r.S, r.W
```

The derivative of the transform is optional in every language: when it is omitted, as above, it is obtained by a Richardson-extrapolated central difference. The bound is sharp where the input is not too variable. Against an exact reference for an $Erlang(2)/M/1$ tandem it exceeds the true sojourn-time tail by 2% where that tail is one in a hundred, and by 0.6% where it is five in ten billion, and it carries the correct asymptotic slope; under hyperexponential service with coefficient of variation between two and four it loosens to about a factor of two.

7.4.4 Multiserver queues with phase-type service

A queue with several servers is usually approximated by a single fast server, which reproduces the mean service time but neither the shape of the service law nor the correlation of the arrival stream. LINE instead solves two multiserver families exactly. `qsys_mapmc` solves the $MAP/M/c$ queue [68, 127], and `qsys_mapphc` solves $MAP/PH/c$, returning the queue length distribution, the per-server utilization, the probability of waiting, the moments of the waiting time and its complementary distribution function.

The state of the $MAP/PH/c$ chain is the *multiset* of the phases occupied by the busy servers: since the servers are identical, their identities carry no information, and a configuration $n = (n_1, \dots, n_{m_s})$ with $\sum_i n_i = k$ records how many of the k busy servers sit in phase i . There are $\binom{m_s+k-1}{k}$ such configurations rather than m_s^k [4]; for $m_s = 5$ and $c = 6$ that is 210 against 15625, which is what keeps the exact solution within reach. Levels $0, \dots, c-1$ form the boundary and levels c and above repeat, so the queue-length tail is

matrix-geometric.

The waiting time is phase type and is obtained without approximation. A customer that finds j others waiting must wait for exactly $j + 1$ service completions, so its delay is the $(j + 1)$ -st event time of the Markovian arrival process formed by the configuration chain; folding the matrix-geometric level distribution over j leaves a linear matrix differential equation, from which each moment follows by one solve of a generalized Sylvester equation. With a single service phase the whole construction collapses onto the Erlang-C formulae, which it reproduces to fourteen digits.

```
const qsys::MapPhcResult<double> r = qsys::qsys_mapphc(arrival, alpha, S, 3);
// r.meanQueueLength, r.meanWaitingTime, r.probWait, r.phaseCount
```

Both functions are also reached automatically by `SolverMAM` whenever a station in an open model matches them, so a multiserver queue fed by a correlated stream or serving a phase-type law is answered exactly rather than by the surrogate.

7.4.5 Per-class waiting times under general service

When customers of different classes need different service laws, and those laws are not phase type, the matrix-analytic machinery of the previous section no longer applies: no finite set of phases carries a deterministic or a uniform service time. `qsys_mmapgk1` solves this case through transforms, returning the waiting time moments, transform and distribution function of *each* class of a marked multiclass MMAP/G/1 FCFS queue [75]. The arrival process is marked, so the classes may be correlated with each other and with the arrival epochs.

Under FCFS a customer waits exactly the workload it finds on arrival, so the joint transform of workload and arrival phase satisfies a linear system whose coefficients are the arrival blocks and the service transforms. The vector of idle probabilities it needs is recovered without searching for the roots of a determinant, from the generator of the arrival phase observed when the queue empties by one customer. Each class then sees that workload biased by its own arrival block, so classes with identical service laws still differ in delay whenever their arrival epochs do.

```
const qsys::MmapGk1Result<double> r = qsys::qsys_mmapgk1(mmap, svc);
// r.meanWaitingTime[k], r.waitMoments[k][j], r.waitCDF[k][i]
```

With one class, Poisson arrivals and exponential or deterministic service the results reduce to the classical M/M/1 and Pollaczek-Khinchine formulae, which they reproduce exactly.

Chapter 8

Solvers and solution methods

This chapter documents how each solver is used: the class of models it accepts, the method names its `options.method` recognises, its configuration options, and the interfaces that exist only for that solver. The algorithms behind those method names, the internal architecture of each solver and the bibliographic sources are documented in the LINE internals manual (`LINE-internals-cpp.pdf`), which devotes one chapter to each solver and is not summarized here.

8.1 Choosing a solver

8.1.1 Loading a solver by name

We now describe the solution methods available within the solvers. Table 8.1 provides a global summary, while Figure 8.1 organizes the same methods by class of stochastic model. Some listed methods (e.g., `mg1`) are not associated with a specific solver because they do not fall within one of the reference formalisms. A solver that runs these methods can be instantiated as follows: There is no solver registry in the C++ port: each solver is a function, called by name. Note that the `LINE.load` notation can also be used to instantiate a custom solver pre-configured with the specified method. For example The CTMC solver is `ctmc::solver_ctmc_run`, or `line-cli -s ctmc`. runs the CTMC solver with default options. Solver-specific methods can be specified by appending their name to the method option, e.g. this command creates the MVA solver with the `amva` method enabled: The equivalent is `line-cli -s mva -method amva`.

8.1.2 Technique taxonomy and methods table

Figure 8.1 organizes the methods by class of stochastic model, and Table 8.1 lists, for each solver, the method names it accepts.

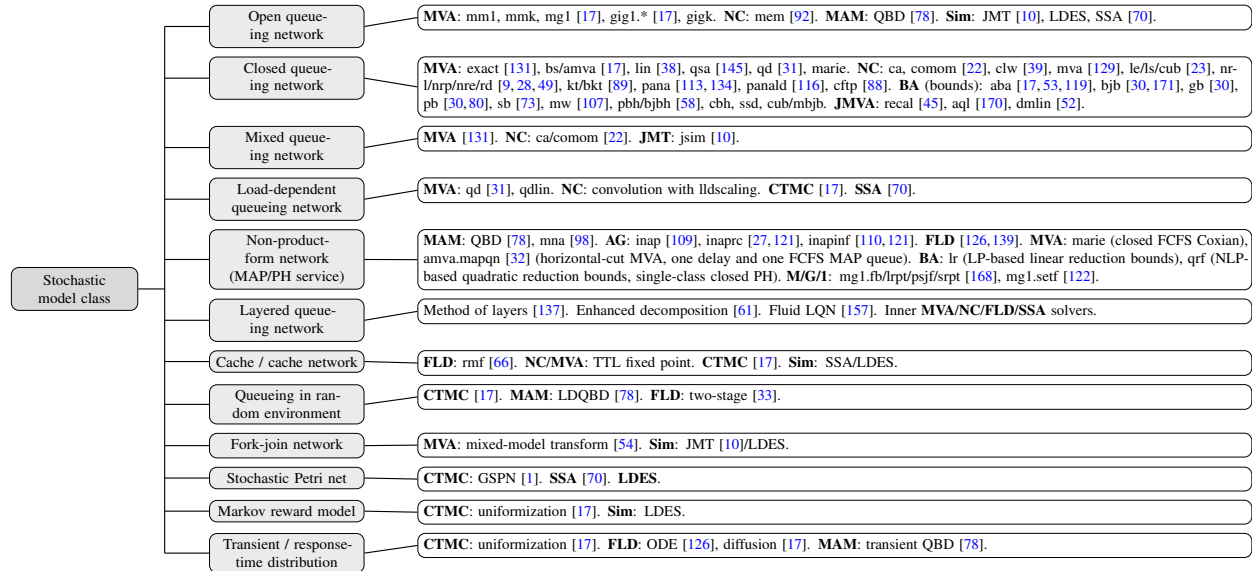


Figure 8.1: Taxonomy of solution techniques available in LINE for each class of stochastic model. Each leaf lists the short method name (as accepted by `options.method`) with its bibliographic reference; entries without a reference denote simulation backends or aliases. See Table 8.1 for the solver exposing each method.

Table 8.1: Solution methods for *Network* solvers. **Since** gives the LINE release in which the method first became available, under its current name or an earlier one (3.0.8 denotes the current development version).

Solver	Method	Description	Refs.	Since
CTMC	default	Solution based on global balance	[17, § 2.1.2]	2.0.0
CTMC	exact	Explicit alias of <code>default</code> : it pins the state-space path at the call site so that a later change of what <code>default</code> selects cannot re-baseline the result	–	3.0.7
CTMC	mdd	Multi-valued decision-diagram aggregation of a closed single-class chain: K coupled level-CTMCs instead of the $ S $ -state generator, with an exactness certificate reported per solve	[115]	3.0.7
LDES	default	Discrete-event simulator using the SSJ library	–	3.0.4
LDES	parallel	Independent replications of the <code>default</code> run, averaged (<code>config.replications</code> , else 8); not a second engine	–	3.0.7
LDES	para	Alias for the <code>parallel</code> method	–	3.0.7
FLD	default	ODE-based mean field approximations	[126, 139]	2.0.0
FLD	matrix	Alias for the <code>default</code> method	[126, 139]	2.0.21
FLD	closing	Fluid with closing method for open classes	[17, p. 507]	2.0.21
FLD	statedep	Kurtz’s mean field ODEs for closed models	[126]	2.0.0
FLD	pnorm	ODE approximation using p -norm smoothing for the processor-share constraint	[139]	3.0.2
FLD	softmin	Smoothed <code>statedep</code> with <code>softmin</code> replacing <code>min</code> functions	–	2.0.24

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Table 8.1 – Solution methods for Network solvers. *Continued from previous page*

Solver	Method	Description	Refs.	Since
FLD	diffusion	Diffusion approximation via Euler-Maruyama SDEs	[17, § 10.1.1]	3.0.2
FLD	mfq	Markovian fluid queue for single-queue systems	[77]	3.0.2
FLD	rmf	Refined mean field approximation for multi-list cache analysis	[66]	3.0.5
FLD	tbi	Cell-decomposed solution of the closing ODEs for large closed models	[148]	3.0.6
FLD	minnormal	Second-order (min-normal) moment closure, whose error stays bounded as $\rho \rightarrow 1$; the only fluid method serving GPS and the one reporting queue-length variances	[71]	3.0.7
FLD	refined, rmf	Model-aware aliases: min-normal closure plus the Gast-van Houdt $O(1/N)$ correction, or cache-queueing decomposition	[66, 67, 71]	3.0.7, 3.0.5
FLD	kp	Fluid and diffusion limits of the open time-inhomogeneous $(MAP_t/Ph_t/\infty)^N$ network, the mean and the covariance integrated jointly	[90]	3.0.7
FLD	dae	The min-normal closure stated and solved as one differential-algebraic system: in steady state the drift residual, the population conservation and the closure consistency are solved simultaneously by damped Newton, and over a finite horizon the index-1 DAE is integrated by RODAS, so the variance is the one the trajectory actually had. Finite capacity regions and station buffers are carried as algebraic constraints	[71, 72]	3.0.7
JMT	default	Alias for the jsim method	–	2.0.0
JMT	jmva	Alias for the jmva.mva method	–	2.0.0
JMT	jmva.mva	Exact MVA in JMVA	[131]	2.0.0
JMT	jmva.recal	Exact RECAL algorithm in JMVA	[45]	2.0.0
JMT	jmva.comom	Exact CoMoM algorithm in JMVA	[22]	2.0.0
JMT	jmva.amva	Approximate MVA, alias for jmva.bs.	–	2.0.0
JMT	jmva.aql	AQL algorithm in JMVA	[170]	2.0.0
JMT	jmva.bs	Bard-Schweitzer algorithm in JMVA	[17, § 9.1.1]	2.0.0
JMT	jmva.chow	Chow algorithm in JMVA	[40]	2.0.0
JMT	jmva.dmlin	De Souza-Muntz Linearizer in JMVA	[52]	2.0.0
JMT	jmva.lin	Linearizer algorithm in JMVA	[38]	2.0.0
JMT	jmva.ls	Logistic sampling in JMVA	[23]	2.0.0
JMT	jsim	Exact discrete-event simulation in JSIM	[10]	2.0.0
JMT	default	jsim; on a finite timespan, the mean of config.replications (default 10) independent transient runs	–	3.0.2
AG	inap	Iterative Numerical Approximation Procedure via RCAT	[109]	3.0.2
AG	inaprc	Improved INAP with weighted rates (no normalization)	[27, 121]	3.0.2
AG	inapinf	INAP with matrix-geometric infinite-state components	[110, 121]	3.0.6
MAM	default	Matrix-analytic solution of structured QBDs	[78]	2.0.0
MAM	dec.source	Decomposition with arrivals as from the source	–	2.0.2
MAM	dec.poisson	Decomposition based on Poisson arrival flows	–	2.0.2
MAM	dec.mmap	ETAQA departure decomposition: the departure process of each queue is approximated by a MAP and fed to the next	[133]	2.0.2
MAM	dec.source.mmap	dec.source with the per-class (marked) flows retained; selected automatically on general fork-join topologies	–	3.0.5
MAM	ldqbd	Level-dependent QBD: the queue length is solved from the rate matrices R_n of a level-dependent quasi-birth-death process, which admits load-dependent rates and, for a multi-server station with phase-type service, the exact multiset-of-phases chain	[4, 128]	3.0.2
MAM	mna	Decomposition based on MNA method	[98]	2.0.38

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Table 8.1 – Solution methods for Network solvers. *Continued from previous page*

Solver	Method	Description	Refs.	Since
MAM	bgchain	Mixed and closed networks: the closed classes are solved exactly as a background modulating chain and each open station as a level-dependent QBD driven by it; with no open class the chain alone answers, exactly	[36]	3.0.7
MVA	default	Approximate MVA, same as qd option	–	2.0.0
MVA	amva	Approximate MVA, same as qd option	–	2.0.0
MVA	aql	Aggregate queue length approximate MVA, single-server product-form only	[170]	2.0.17
MVA	bs	Bard-Schweitzer approximate MVA	[17, § 9.1.1]	2.0.16
MVA	lin	Linearizer approximate MVA	[38]	2.0.17
MVA	dmlin	de Souza e Silva-Muntz improved Linearizer, same estimator at a reduced cost	[52]	3.0.7
MVA	lcp	Bard large-customer-population approximate MVA, the arrival-instant queue length estimated by the time-averaged one	[8]	3.0.7
MVA	chow	JMT-compatible Chow full-population aggregate-queue estimate	[41]	3.0.7
MVA	pamb, pami, pamt	Hsieh-Lam noniterative proportional approximation methods: a proportional seed of the queue lengths followed by a fixed number of unrolled MVA steps	[81]	3.0.7
MVA	clust	de Souza e Silva-Lavenberg-Muntz clustering approximation for models with a large number of chains	[50]	3.0.7
MVA	qli, fli	Wang-Sevcik queue-line and fraction-line arrival-instant estimators	[163]	2.0.16
MVA	ab, schmidt-ext	Akyildiz-Bolch and Schmidt multiserver corrections of approximate MVA	[152]	3.0.2
MVA	tay	Tay-Suri arrival-instant approximate MVA, single-server stations only	[155]	3.0.7
MVA	sqni	Single-queue numerical iteration, closed form for one queueing station with a delay; offered only when the model has that shape	–	2.0.27
MVA	marie	Marie aggregation-decomposition, closed FCFS non-exponential (Coxian) service	–	3.0.6
MVA	amva.mapqn	Horizontal-cut mean value analysis, closed model of one exponential delay and one FCFS queue with a MAP service per class	[32]	3.0.7
MVA	qd	Queue-dependent approximate MVA	[31]	2.0.16
MVA	qdlin	Queue-dependent Linearizer approximate MVA	–	2.0.17
MVA	qsa	Queue-shift approximate MVA, single-server product-form only	[145]	3.0.7
MVA	exact	Exact solution, method depends on model	–	2.0.0
MVA	mva	Alias for the mva.amva method	[131], [20]	2.0.0
MVA	oi	Exact order-independent analyzer for closed networks with order-independent (OI) or pass-and-swap stations; auto-selected under default/exact and reported as the actual method	–	3.0.7
MVA	sqd	Smith Queue Decomposition, single-chain closed blocking-after-service (BAS) networks; offered only when the model is BAS	–	3.0.6
MVA	qna	Queueing Network Analyzer (two-moment decomposition) - open	[165]	2.0.22
MVA	rqna	Robust Queueing Network Analyzer (indices of dispersion) - open	[166]	3.0.6

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Table 8.1 – Solution methods for Network solvers. *Continued from previous page*

Solver	Method	Description	Refs.	Since
MVA	rqt	Robust Queueing Theory, worst-case system time bound - single-class open	[7]	3.0.7
MVA	scat	Neuse-Chandy SCAT approximate MVA, single-server product-form only	[120]	3.0.7
MVA	sum	Summation method (SUM/ESUM); closing method for open and mixed models	[17, § 9.2, § 10.1.4.4, § 10.1.5]	3.0.6
MVA	esum	Alias for the sum method	[17, § 10.1.4.4]	3.0.6
BA	auto.upper	Tightest feasible noniterative upper bound, selected automatically	–	3.0.7
BA	auto.lower	Tightest feasible noniterative lower bound, selected automatically	–	3.0.7
BA	aba.upper	Asymptotic bound analysis (upper bounds)	[17, § 9.4], [53, 119]	2.0.3
BA	aba.lower	Asymptotic bound analysis (lower bounds)	[17, § 9.4], [53, 119]	2.0.3
BA	mw.upper	Majumdar-Woodside robust box bounds, multiclass NBUE (upper bounds)	[107]	3.0.6
BA	mw.lower	Majumdar-Woodside robust box bounds, multiclass NBUE (lower bounds)	[107]	3.0.6
BA	bjb.upper	Balanced job bounds (upper bounds)	[30, Table 3], [171]	2.0.3
BA	bjb.lower	Balanced job bounds (lower bounds)	[30, Table 3], [171]	2.0.3
BA	gb.upper	Geometric square-root bounds (upper bounds)	[30]	2.0.3
BA	gb.lower	Geometric square-root bounds (lower bounds)	[30]	2.0.3
BA	pb.upper	Proportional bounds (upper bounds)	[30, Table3], [80]	2.0.3
BA	pb.lower	Proportional bounds (lower bounds)	[30, Table3], [80]	2.0.3
BA	sb.upper	Simple bounds (upper bounds, Thm. 3.2, $n = \min(3, N)$)	[73, Table3]	2.0.17
BA	sb.lower	Simple bounds (lower bounds, Eq. 1.6)	[73, Table3]	2.0.17
BA	harel.upper/lower	Harel-Namn-Sturm bounds extrapolated from the exact throughput at small populations, distinct from the power-sum sb family of the same paper	[73]	3.0.7
BA	looping.upper/lower	Eager multiclass looping bound	[56]	3.0.7
BA	pbh.upper/lower	Eager-Sevcik hierarchy initialized from the PB bound	[58]	3.0.6
BA	bjbh.upper/lower	Eager-Sevcik hierarchy initialized from the BJB bound	[58]	3.0.6
BA	cbh.upper/lower	Convolutional bound hierarchy, level-parameterized (Dowdy et al.)	–	3.0.6
BA	ssd.upper/lower	Multiserver disaggregation bounds (Suri-Dallery)	–	3.0.6
BA	cub.upper	Composite upper bound, multiclass, upper-only (Kerola)	–	3.0.6
BA	mbjb.lower	Multiclass balanced job bounds lower (Kerola eq. 10; seeds cub)	–	3.0.6
BA	sib.upper/lower	Successively-improving bounds, level-parameterized (Srinivasan; delay-free)	–	3.0.6
BA	ldac.lower	Load-dependent BCMP closed-open equivalence lower bound (Anselmi-Cremonesi)	–	3.0.6
BA	lr.upper/lower	Linear Reduction bounds, pure LP (linprog/HiGHS/simplex), single-class closed; bare lr means lr.upper	–	3.0.6
BA	qr/grf.mmi	Quadratic Reduction Framework, NLP-based (mutual-information objective), single-class closed PH service; single-server stations only, delay stations rejected	–	3.0.6, 3.0.4
BA	qrf.mmi.linear	QRF with an explicit linear constraint representation; the objective is still the nonlinear MEM entropy, so this is an NLP and is <i>not</i> the same as lr	–	3.0.4
BA	qrf.mem.mmi.ld	Further QRF reduction sub-methods (maximum entropy, load-dependent); NLP-based	–	3.0.4
BA	qrf.bas.*/rsrd	QRF blocking-after-service and RS-RD reduction bounds; LP-based	–	3.0.4

Continued on next page

Table 8.1 – Solution methods for Network solvers. *Continued from previous page*

Solver	Method	Description	Refs.	Since
BA	qrf.bethe	QRF with the tree-reweighted (Bethe) free entropy at $\lambda = 1/M$ in place of the mutual-information objective	[25], [162]	3.0.7
BA	qrf.bas.bethe	The same Bethe free entropy maximized over the blocking-after-service polytope	[25], [162]	3.0.7
BA	mapamva.upper/lower	MAP-AMVA bounds: a linear program over the exact mean-value balances of a closed MAP queueing network in the per-phase variables, the only BA family that consumes the CORRELATION between successive services rather than the service mean alone. Single-class closed, single-server, no delay, FCFS at the MAP station	[32]	3.0.7
BA	scb.upper/lower	Single-class bounds bracketing the aggregated multiclass system; <code>scb.lower</code> is the exact single-class throughput	[55]	3.0.7
BA	bgt.upper	Piecewise-linear-Lyapunov queue-length upper bound, open multiclass Markovian networks	[12]	3.0.7
BA	bpt.lower	Achievable-region sojourn-time lower bound, open multiclass Markovian networks	[13]	3.0.7
BA	snc.upper	Stochastic network calculus: MGF envelope bound on the response-time and queue-length TAIL, feed-forward open networks, any work-conserving policy	[59]	
BA	3.0.7 spnlp2.upper/lower	Petri-net moment relaxation: the uniformized evolution equation written for the first two marking moments and their covariances, plus the behavioural, probabilistic and Little's-law families, gives a polytope over which each measure is minimized and maximized. Exponential firing times; offered only on a model holding <code>Transition</code> nodes, and the only BA family offered there	[102]	3.0.7
BA	spnlp1.upper/lower	Variant dropping the second-moment, covariance and Little's law constraints	[102]	3.0.7
MVA	gigl.allen	Allen-Cunneen formula - GI/G/1	[17, § 6.3.4]	2.0.0
MVA	gigl.heyman	Heyman formula - GI/G/1	–	2.0.0
MVA	gigl.kingman	Kingman upper bound - GI/G/1	[17, § 6.3.6]	2.0.0
MVA	gigl.klb	Kramer-Langenbach-Belz formula - GI/G/1	[17, § 6.3.4]	2.0.0
MVA	gigl.kobayashi	Kobayashi diffusion approximation - GI/G/1	[17, § 10.1.1]	2.0.0
MVA	gigl.marchal	Marchal formula - GI/G/1	[17, § 10.1.3]	2.0.0
MVA	gigk	Kingman approximation - GI/G/k		2.0.1
MVA	mg1	Pollaczek–Khinchine formula - M/G/1	[17, § 3.3.1]	2.0.0
MVA	mg1.fb	Feedback/LAS scheduling - M/G/1/FB	[168]	3.0.6
MVA	mg1.lrrpt	Longest remaining PT - M/G/1/LRPT	[168]	3.0.6
MVA	mg1.psrf	Preemptive SJF - M/G/1/PSJF	[168]	3.0.6
MVA	mg1.srpt	Shortest remaining PT - M/G/1/SRPT	[168]	3.0.6
MVA	mg1.setf	Shortest elapsed time first - M/G/1/SETF	[122]	3.0.6
MVA	mm1	Exact formula - M/M/1	[17, § 6.2.1]	2.0.0
MVA	mmk	Exact formula - M/M/k (Erlang-C)		2.0.1
NC	default	Automated choice of solution method	–	2.0.0
NC	exact	Automated choice of exact solution method.	–	2.0.0
NC	ca	Multiclass convolution algorithm (exact): Buzen's recursion generalized to multiple closed chains	[21], [130]	2.0.0
NC	clw	Choudhury-Leung-Whitt generating function inversion (exact); limited load-dependent stations handled via Bertozzi-McKenna transforms	[39], [11]	3.0.6
NC	comom	Class-oriented method of moments for homogeneous models (exact)	[22]	2.0.8

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Table 8.1 – Solution methods for Network solvers. *Continued from previous page*

Solver	Method	Description	Refs.	Since
NC	comomld	Class-oriented method of moments for load-dependent models (exact), via factorization of the load-dependent normalizing constant	[28]	2.0.27
NC	cub	Grundmann-Moeller cubature rules (alias gm)	[23]	2.0.23
NC	ger	Exact normalizing constant in closed form by iterated residues of the rational generating function, one class per elimination	[69]	3.0.7
NC	gleint	Gauss-Legendre evaluation of the normalizing-constant integral	–	2.0.27
NC	mva	Product of throughputs on MVA lattice (exact)	[129, Eq. (47)]	2.0.0
NC	imci	Improved Monte carlo integration sampler	[164]	2.0.0
NC	is	Importance sampling over job orderings; closed models only, load-independent, load-dependent, order-independent and pass-and-swap	[43]	3.0.6
NC	kt	Knessl-Tier asymptotic expansion	[89]	2.0.8
NC	bkt	Knessl-Tier asymptotic expansion minus the exact Stirling remainder $\log N_r! - (N_r \log N_r - N_r + \frac{1}{2} \log(2\pi N_r))$ that steepest descent drops in each Laplace class direction (BKT, not part of the reference); with a think time it coincides with ble	[89]	3.0.7
NC	lekt	The estimator ble and bkt both compute, evaluated on the cheaper side (R -dimensional saddle point when $R \leq M$, M -dimensional logistic mode otherwise), with the $Z = 0$ constant $M(1 - \log(2\pi)/2) - r(N + M)$ so that both sides agree there too (not part of the references)	[23, 89]	3.0.7
NC	le	Logistic asymptotic expansion	[23]	2.0.0
NC	ble	Logistic asymptotic expansion with an empirical $(M - 1)(1 - \log(2\pi)/2)$ correction of the $\epsilon \rightarrow 0$ evaluation (BLE, not part of the reference); default for normally-loaded closed models	[23]	3.0.7
NC	aghq	Adaptive Gauss-Hermite quadrature of the simplex factor of the McKenna-Mitra integral, q nodes per simplex direction (<code>options.config.aghq_nodes</code> , default 3) at cost q^{M-1} ; $q = 1$ reproduces le exactly	[101]	3.0.7
NC	ls	Logistic sampling	[23]	2.0.0
NC	bk	Birman-Kogan saddle-point generating-function evaluation, exact pole treatment for bottlenecked dedicated chains	[15]	3.0.7
NC	bkue	Birman-Kogan single-chain uniform expansion, pole and saddle combined via erfc	[15]	3.0.7
NC	lc	Birman-Kogan load concealment, Gauss-Seidel over single-chain solves	[15]	3.0.7
NC	lc.ue	lc with the ue inner single-chain solver	[15]	3.0.7
NC	oi	Exact balanced-fairness normalizing constant for closed order-independent (OI) networks, with the OI functional-server identity for the mean queue lengths; auto-selected under <code>default/exact</code> and reported as the actual method	–	3.0.7
NC	nrl	Norlund-Rice integral with logit transformation	[28]	2.0.12
NC	nrp	Norlund-Rice integral with probit transformation	[28]	2.0.12
NC	nre	Norlund-Rice integral on a saddle-tilted contour, with a second-order Edgeworth correction	[28], [49], [9]	3.0.7
NC	sdr	Product form of a closed multiclass network with state-dependent routing, evaluated exactly by state enumeration (eq. 16); any branch topology	[93]	3.0.7

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Table 8.1 – Solution methods for Network solvers. *Continued from previous page*

Solver	Method	Description	Refs.	Since
NC	sdr.mva	Same model solved by the Section 4 mean value analysis and convolution instead of by enumeration; single-center branches only	[93]	3.0.7
NC	morrison	Heavy-usage asymptotic expansion of the generating function of a closed think+DPS network: one infinite-server station and one single-server discriminatory-processor-sharing station, exponential service, populations $K_j = Nb_j$ and usage $\rho = 1 - a/\sqrt{N}$. Non-product-form, so no normalizing constant is returned; the default (and only admissible method) on that shape	[118]	3.0.7
NC	pana	Panacea normal-usage asymptotic expansion of the normalizing constant	[112], [134]	2.0.0
NC	panald	Panacea asymptotic expansion, load-dependent	[116]	3.0.7
NC	rd	Reduction heuristic	[28]	2.0.12
NC	sampling	Automated selection of sampling method	–	2.0.8
NC	cftp	Perfect sampling of the closed single-class product-form stationary distribution by monotone coupling from the past; iid states are drawn from the balance function, so neither the state space nor the normalizing constant is computed	[88]	3.0.7
NC	cftp.approx	cftp with a deterministic cost per sample in place of the unbounded worst case, at the price of a residual bias	[88]	3.0.7
NC	mem	Maximum Entropy Method for open ($GE/GE/1$, $GE/GE/c$, $GE/GE/\infty$ building blocks), closed (pseudo-open decomposition plus convolution) and mixed queueing networks; auto-selected by default for non-Markovian open models	[92]	3.0.2
NC	lossn.exact	Open loss network in a DROP finite capacity region: exact normalizing constant of the product form over the integer state space $\mathbf{A}\mathbf{n} \leq \mathbf{C}$ by contour integration and residues, with the region's job, memory and linear admission rows as constraints; the loss-network default when $\mathbf{A}$, $\mathbf{C}$ are integer	[108]	3.0.7
NC	lossn.erlangfp	Same models, Erlang fixed-point (reduced-load) approximation; the loss-network default for fractional class sizes or capacities	[86, 87, 135]	3.0.2
NC	lossn.mci	Same models, importance-sampling Monte Carlo summation of the normalizing constant with confidence intervals; samples and seed options	[138]	3.0.7
NC	mem.blocking	Finite station buffers in a single-class open network: censored $GE/GE/c/0$; N stations under the loss rule, plus the holding-node expansion of transfer blocking (BAS); explicit request only	[92, 153]	3.0.6
SSA	default	Alias for nrm if the model supports it, otherwise serial.	–	2.0.0
SSA	nrm	Next reaction method for population models (e.g., PS/INF scheduling).	[2]	2.0.40
SSA	serial	CTMC stochastic simulation on a single core	[70]	2.0.0
SSA	para	Parallel simulations (independent replicas)	–	2.0.0

8.1.3 Warm start

`SolverOptions::set_init_sol` is available on every network solver and is not specific to simulation: an auxiliary solver supplies a steady-state job placement that becomes the initial state of the model, stored in `options.init_sol` as the station-major vector $[n_{1,1}, \dots, n_{1,K}, \dots, n_{M,K}]$. FLD starts the ODE integration

from it, `SSA` and `LDES` start the simulated trajectory from it, `JMT` preloads the stations with it, and the `AMVA` submethods of `MVA` use it as the starting iterate in place of the uniform N/M guess. Because the placement is written into the model object, it is visible to any other solver instance constructed on the same model. The same mechanism is used by `SolverLN`, which forwards each layer's previous-iteration solution, and is also reachable by setting `options.init_sol` by hand.

With `CTMC` as the auxiliary solver the initial state is the mode of the exact stationary distribution over the aggregate state space; with any other solver the steady-state mean queue lengths are rounded to an integer placement that conserves each closed-class population. The C++ port has no solver-to-solver transfer call: the caller computes the placement and hands it over as the same station-major vector, which the engine honours identically. Under `LDES` the simulation then starts approximately in steady state, so warmup removal is disabled (`tranfilter` set to `fixed` with `warmupfrac = 0`) and every sample contributes to the estimators.

8.1.4 Where the algorithms are documented

Each solver of this chapter has a chapter of the same name in the internals manual, which gives its solution paradigm, its analyzers and handlers, a sequence diagram of a solution request, and the algorithm and reference behind every method name. The solvers documented in other chapters of this manual, namely `AUTO` (§1.3), `ENV` and `LN`, have internals chapters as well. When this manual and the internals manual disagree, the user-facing statements (option names, supported features) are authoritative here, and the algorithmic ones there.

8.2 Native solvers

The solvers described in this section are implemented inside `LINE` itself. Each one analyzes the model with its own algorithms, in the codebase the model was written in, and requires no external tool: what they need is the runtime of that codebase and nothing else. They are presented in alphabetical order.

8.2.1 AG: agent-based solver

`AG` is `LINE`'s *agent-based* solver. It views the network as a collection of interacting agents rather than as a single stochastic process: every (station, class) pair becomes an isolated *agent*, solved exactly on its own state space, and the agents are coupled only through a small vector of scalars, one per synchronizing action, over which the analysis iterates to a fixed point. That vector is the entire interface between agents, so an agent is analyzed without ever forming the joint state space, and the cost of the analysis grows with the size of the largest agent rather than with the size of the network.

The decomposition is of the *model* rather than of the traffic: where `MAM` fits an arrival stream to each station and solves the station against it, `AG` solves each agent exactly and lets the agents meet only in those scalars. `AG` is also the only solver in `LINE` that reads the G-network signal marking, so a network with signals (§6.4.1) must be solved with it.

8.2.1.1 Methods and configuration options

The methods available in this release all realise the agent decomposition through the Reversed Compound Agent Theorem (RCAT), under which the coupling scalars are the reversed rates of the synchronizing actions: agent k carries the generator

$$Q_k(\mathbf{x}) = L_k + \sum_{c: \text{passive}(c)=k} x_c P_c^{(b)},$$

and publishes, for every action it is active on, one such rate read off its own stationary vector. Other agent decompositions may be added in later releases; the interface described here – the option names, the truncation control and the execution backends below – is that of the solver and not of any one method.

- `options.method=inap` (INAP, the default): Iterative Numerical Approximation Procedure [109]. Each agent is a quasi-birth-death process [121] whose level is the queue length and whose phase is the pair (arrival phase, service phase), so a phase-type arrival or service law is represented exactly rather than by its mean rate.
- `options.method=inaprc` (Improved INAP) [27]: the reversed rate is estimated by rate conservation instead of by the mean of the state-wise ratios; recommended when `inap` converges slowly.
- `options.method=inapinf` (Matrix-Geometric INAP) [110]: solves each open agent on its infinite state space through its geometric tail, avoiding the truncation of `maxStates`. Preferred for open signal networks with birth-death or catastrophe structure.

`options.config.maxStates` (default 100) truncates the queue-length dimension of an *open* agent; a closed class uses its own population instead, and `inapinf` ignores the setting entirely.

Execution backends. Because the coupling between agents is one scalar per action, the sweep parallelises exactly rather than approximately. `options.config.exec` selects who evaluates an agent, which never changes what the agent evaluates to:

- `serial` (default): one agent after another, on the caller's thread.
- `threads`: the agents of a sweep are fanned out over a local pool of `options.config.nworkers` workers (0 selects one per processor). The sweep is a Jacobi iteration, so the agent order is immaterial and this backend is *bit-identical* to `serial`.
- `cluster`: the agents are partitioned over `ag-worker` processes listed in `options.config.endpoints` as `'host:port'`, which may be on other machines. Each worker receives the static half of its agents once and then exchanges only the reversed rates, one double per action, per sweep.

A worker is started with

```
java -cp jline.jar jline.cli.AgWorker -p 5870
```

A worker that is unreachable, slow or lost mid-run is not fatal: its agents are solved on the coordinator instead, because any agent can be solved anywhere given the reversed rates, so a distributed run degrades to a slower run and never to a wrong one. `options.config.worker_timeout` (default 30 s) bounds the wait on one.

8.2.2 BA: bound analysis solver

The BA class (alias: `SolverBA`) is the *bound analysis* solver: it computes performance *bounds* for queueing networks. It differs from the other solvers in what it returns: rather than a point estimate, each method returns one guaranteed side of an interval containing the exact solution, so that the `.lower` and `.upper` members of a family bracket the true value. This is the property that makes bounds useful inside optimization loops and admission tests, where an approximation of unknown sign cannot be used safely. Because every bound is a function of the service demands, think times and populations alone, and never of the service distribution beyond its mean, BA is inexpensive. Most families require a closed product-form-parameterized model; the `bgt/bpt` and `snc` families instead cover open networks (below), so the feature set actually admitted depends on which family is requested. One family, `snc.upper`, returns a *tail quantile* rather than a bound on a mean, and is described at the end of this section.

8.2.2.1 Methods and configuration options

The available families are listed in Table 8.1: the asymptotic families, the bound hierarchies parameterized by `options.level`, the reduction bounds `lr` and `qr/qr.*`, the single-class bounds `scb.*`, the achievable-region bounds `bgt.upper/bpt.lower`, the stochastic network calculus bound `snc.upper`, the MAP bounds `mapamva.*`, and the Petri-net moment-relaxation bounds `spnlp.*`. Their algorithms and bibliographic sources are given in the internals manual, Chapter BA.

Three methods are one-sided by construction: `cub`, `bgt.upper` and `snc.upper` yield an upper bound only, while `mbjb`, `ldac` and `bpt.lower` yield lower bounds only. Several families carry structural restrictions beyond the feature set and reject a model rather than return a value that would not be a bound: `sb`, `sib` and `scb.*` do not admit infinite-server stations, `lr` and `scb.*` require a single-class closed model with single-server stations, and the whole `qr.*` family requires a closed single-class model without delay stations. `mapamva.*` asks for that same shape with a MAP service law at one station, and `spnlp.*` is offered only on a model holding `Transition` nodes, where it is in turn the only family offered. The blocking variants `qr.bas.mmi` and `qr.bas.mem` are unavailable in native Python and raise an explicit error there. The linear programs of the reduction bounds are solved to about eleven digits in MATLAB and native Python but only about five in Java, so the three should not be compared digit for digit.

The `bgt.upper/bpt.lower` family, together with `snc.upper`, admits an *open* model: it requires a Source station, a single server per queueing station (no delay, no multiserver), and deterministic, non-merging per-class routing; on an open model `list_valid_methods` narrows to exactly these three method names, and `linprog/HiGHS` is required to solve the linear program underlying the first two. Both bounds are deliberately loose (by one to several orders of magnitude on small networks): the sharp content is the stability certificate and the tail-decay rate they establish, not the numerical constant, so they should not be used where

a tight bracket is needed.

The stochastic network calculus family `snc.upper` answers a different question from every other family in this solver: its native output is a *tail quantile* with a certified violation probability, not a bound on a mean. Given a target violation probability, `solver_ba_snc_perc` returns, per station and class, the smallest delay whose exceedance probability is certified to stay below it, and `solver_ba_snc_perc` the corresponding queue-length quantile in jobs; `solver_ba_snc_perc` reports both in the layout of `getAvgTable`. The guarantee holds for *any* work-conserving scheduling policy at every station. The method describes each flow by a moment-generating-function envelope and each server by the counting process of its exponential service, composes them across the network with the min-plus operations (leftover service under blind multiplexing, concatenation of a tandem, departure envelope of a hop), and minimizes the resulting Chernoff bound over the envelope parameter. The underlying algebra is exposed as the `snc` API family in all four codebases.

The mean columns returned by `getAvgTable` under `snc.upper` are the integral of that tail bound, hence valid upper bounds on the mean response time, with the queue length following by Little's law from the exact open-network throughput. They are deliberately loose: on an $M/M/1$ they exceed the exact mean response time by a factor of about 3.5 at 30% utilization and 8.8 at 90%, because an integral over the whole axis is dominated by the prefactor of the bound rather than by its decay rate. The decay rate itself is asymptotically exact, so the ratio of the bounded quantile to the exact one falls towards one as the violation probability is tightened. Use the quantiles, not the mean columns, when the tail is what matters. Beyond the open-model requirements above, `snc.upper` needs a feed-forward station graph, deterministic routing downstream of the Source (a probabilistic split is admitted only at a Poisson Source, where thinning is exact), exponential service, and equal service rates among the classes sharing a station; each is refused by name rather than approximated. Worked illustrations are `snc_delay_quantile` and `snc_tandem_multiclass`.

The single-class bounds `scb.upper/scb.lower` answer a different question from every other family: given a single-class demand vector and population, they do not bracket that model's own solution but the unknown multiclass system it aggregates. `scb.lower` is in fact the *exact* single-class throughput, not an approximation; `scb.upper` inflates it by a gap that shrinks with the multiplicity of the dominant demand and is capped by the one-server-per-station capacity constraint. Because its lower side already equals the exact answer, `scb` is excluded from the `auto.*` candidate set below, which would otherwise collapse onto it. The companion model-free functions `pfqn:pfqn_scbgap`, `pfqn:pfqn_usumbound` and `pfqn:pfqn_minclasses` bound, respectively, the relative throughput error from merging classes, the sum of utilizations across stations, and the minimum number of classes consistent with a measured utilization sum; none requires a solved model. A worked illustration of all four, including the paper's own numeric examples, is given in `tut15_bound_analysis`.

The family `mapamva.upper/mapamva.lower` [32] is the only one in this solver that consumes the *correlation* between successive services rather than the service mean alone. Its linear program is written over the exact mean-value balances of a closed MAP queueing network, in the per-phase queue-length and utilization variables, so a workload whose burstiness moves the bottleneck between stations is bounded rather than averaged into a renewal process. It admits a single-class closed model with single-server stations and no delay, FCFS at the MAP station, and phase structure at one station only. A MAP or MMPP2 service law reaches this family, the `qrf.*chain` and `snc.upper`, and no other.

The family `spnlp.*` [102] bounds a stochastic Petri net, and is the only BA family offered on one: every other family is parameterized by demands and a population, which a marking is not. It relaxes the stationary chain to a *moment polytope*, the uniformized evolution equation written for the mean marking, its second moments and the pairwise covariances, together with the behavioural, probabilistic and Little's-law families, and then minimizes and maximizes each reported measure over it. Every stationary point of the true chain satisfies every row, so the two optima bracket the exact value. `spnlp2.upper` and `spnlp2.lower` require exponential firing times. The operational pair `spnlp1.upper/spnlp1.lower` drops the second-moment, covariance and Little's-law families and rests on flow balance, the place invariants and the state equation alone, so it needs only a mean firing time and admits any phase-type law, at the price of a much looser bracket. A worked illustration is `spn_lpbounds`.

The composite family `auto.upper/auto.lower` evaluates every noniterative family that admits the model and keeps the tightest side, so its bracket is never wider than that of any single family; it requires a single-class closed model and reports an error when no candidate is feasible. The level-parameterized hierarchies are deliberately excluded from the candidates, their accuracy being set by `options.level` rather than being intrinsic. On the Eager-Sevcik hierarchy `pbh` the bracket width falls monotonically as `options.level` rises and reaches zero once the level equals the population; a worked illustration is given in `tut15_bound_analysis`.

8.2.2.2 Reading bounds

The normal way to read a bound is `ba_bounds`, the counterpart of `AvgResult`. It evaluates both sides of the family of the selected method and reports the bracket per station and class, with columns `Qlower`, `Qupper`, `Tlower` and `Tupper`. It follows the labelling of `AvgResult` and likewise omits disabled station-class pairs.

```
ba::BaOptions opt;
opt.method = "gb.upper";
const ba::BaBounds<double> b = ba::ba_bounds(model.get_struct(), opt);
// b.Qlower, b.Qupper, b.Tlower, b.Tupper, each (station x class)
```

A side the chosen family does not produce is returned as NaN rather than as a fabricated number. For a one-sided family the absent side is reported as NaN rather than as zero, so that the absence of a bound stays distinguishable from a bound whose value happens to be zero. Requesting `cub`, which is upper-only, on the same model yields NaN throughout the `Qlower` and `Tlower` columns.

The programmatic accessor is `ba_bounds`, which returns the same information as a raw bracket rather than a formatted table, and is the form to use inside a loop. The relationship between the two mirrors that between `solver_mva_run` and `AvgResult`.

```
ba::BaOptions opt;
opt.method = "bjb.upper";
const ba::BaBounds<double> b = ba::ba_bounds(model.get_struct(), opt);
```

Both accessors inherit the option set of the solver they are called on, so `options.level` applies to the bracket as a whole.

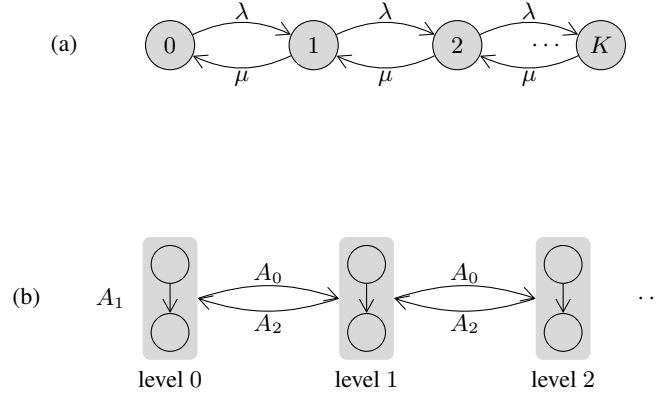


Figure 8.2: (a) The birth-death chain of an M/M/1/K queue, the kind of state space that CTMC enumerates. (b) The level structure of a QBD process, on which the matrix-analytic methods of MAM operate: A_0 , A_1 and A_2 collect the transitions that raise, preserve and lower the level.

8.2.3 CTMC: continuous-time Markov chain solver

The CTMC class solves the model by first generating the infinitesimal generator of the and then calling an appropriate solver. Steady-state analysis is carried out by solving the global balance equations defined by the infinitesimal generator. If the `keep` option is set to , the solver will save the infinitesimal generator in a temporary file and its location will be shown to the user.

Figure 8.2 shows the two state-space structures involved.

The solver refuses by default to solve models whose state space is predicted not to fit in memory, comparing the predicted peak footprint against a fraction (by default 0.6) of the memory available on the host; the `force` option bypasses this control. In models with infinite states, such as networks with open classes, the `cutoff` option reduces the CTMC to a finite process: a scalar is the maximum number of jobs that a class can place at an arbitrary station, while a matrix gives in row i and column r the maximum number of jobs of class r that can be placed at station i . Transient analysis integrates Kolmogorov's forward equations over the interval given by `timespan`, with the ODE solver choice of FLD, or by uniformization with Fox-Glynn weights, exposed through the Markov-chain API as `ctmc_foxglynn`.

8.2.3.1 Methods and configuration options

The following methods are available for the CTMC solver:

- `options.method=default`: direct solution of the global balance equations, which also covers a reducible generator.
- `options.method=mdd`: decision-diagram aggregation of a closed single-class chain, whose memory cost is governed by the diagram size rather than by the state space. Every solve reports an exactness certificate, and phase-type service under LCFSPR or FCFSPR is rejected. Transient analysis, reward

functionals and the symbolic export are unavailable under this method.

- `options.method=qrf.*`: approximations from the Quadratic Reduction Framework, namely `qrf.mmi`, `qrf.mem`, `qrf.bethe`, `qrf.mmi.ld`, `qrf.mmi.linear`, `qrf.bas.mmi`, `qrf.bas.mem`, `qrf.bas.bethe`, `qrf.bas` and `qrf.rsrd`, available only for a single-class model with a finite total population. The two `bethe` variants replace the mutual-information objective by the tree-reweighted (Bethe) free entropy at $\lambda = 1/M$ [162], on the plain and on the blocking-after-service polytope respectively. The blocking variants require the blocking topology in `options.config.qrf_params` and the load-dependent ones read `options.config.qrf_alpha`; a missing mandatory field aborts the solver.

The two configuration entries read by the QRF dispatcher are:

- `options.config.qrf_params`: optional struct describing the blocking configuration of `qrf.bas.*` variants. Fields: `f` (blocking-event index), `MR` (number of resources), `BB` (per-station buffer capacities), `F` (capacity vector), `MM`, `MM1`, `ZZ`, `ZM` (auxiliary index sets needed by the BAS engine). Defaults to an empty struct.
- `options.config.qrf_alpha`: optional $M \times N$ matrix of load-dependent scaling factors used by `qrf.mmi.ld` and `qrf.mmi.linear`. Row i , column n scales the service rate of station i when n jobs are present. Defaults to a matrix of ones.

Solving an aggregated model. The state space of a multiclass model grows with the per-class populations, and the two model adaptations of §5.6 are the standard way of making an otherwise intractable model solvable. Both are reachable from the CTMC solver itself, as opt-in configuration entries, so that the reduction and the recovery of the original metrics happen inside a single solve:

- `options.config.chain_aggregation` (default false): collapse every chain onto a single class with `aggregateChains` (§5.6.1), solve the collapsed model, and split the answer back per class. It is applied only when the model has fewer chains than classes, and is exact on a product-form model and an approximation otherwise, because one aggregate service law replaces the per-class ones weighted by α .
- `options.config.fes_stations` (default unset): the indices of the stations to collapse into one load-dependent station with `aggregateFES` (§5.6.2). Indices are 1-based, must be distinct, must name at least two stations, and may not name every station. The surviving stations are answered by the reduced chain directly; a collapsed station is recovered by conditioning its queue length on the population held by the flow-equivalent server and averaging over that population [36], the isolated subnetwork’s per-population queue lengths and utilizations being supplied by the API function `Fes_compute_metrics`, and the reduction leaves throughput unchanged. The decomposition is exact when the collapsed subnetwork is product-form.

When both entries are set, the flow-equivalent reduction is applied first. The method name reported back by records which path ran, as `default/chainaggr` or `default/fes`.

Immediate transitions introduced by `Router` nodes, for instance in a cache topology with state-dependent routing (§6.2.3), are eliminated automatically by stochastic complementation [114] before the chain is solved, so that per-class arrival and throughput metrics at downstream stations remain correct. The lower-level

primitives are exposed as the API functions `Ctmc_stochcomp` and `Dtmc_stochcomp` and may be used directly on custom chains. The remaining configuration options are listed in the next table.

Table 8.2: CtmcOptions fields (C++)

Field	Value	Description
method	"default"	Direct solution of the generator.
method	"mdd"	Level-iteration on a matrix-diagram decomposition, with <code>mdd_tol</code> and <code>mdd_maxiter</code> .
keep_filtration	bool	Keep the per-synchronization event filtration beside the generator; needed by the response-time distribution (default: false).
cutoff	double	Open jobs per class admitted into the state space; a negative value selects the reference's own $\lceil 6000^{1/(MK)} \rceil$.
cutoff_vec	vector	Per-class override of the cutoff, or empty.
state_max	size_t	Refuse a state space larger than this (default: 3000000).

8.2.3.2 Generator and state space access

The CTMC solver also provides access to the underlying infinitesimal generator and its decomposition into per-event filtration matrices via the `ctmc_get_generator` method:

```
ctmc::CtmcOptions opt;
const ctmc::CtmcGenerator<double> gen =
    ctmc::ctmc_get_generator(model.get_struct(), opt);
// gen.Q is the infinitesimal generator; the per-event filtration is kept
// only when opt.keep_filtration is set
```

Also available as `line-cli -s ctmc -a gen`. Here `infGen` is the infinitesimal generator matrix and `eventFilt` is a list of sparse matrices, one per event type, encoding the transitions associated with each event (arrival, departure, phase change, etc.). The generator can also be built symbolically, with each event filtration normalized by its smallest positive rate and scaled by a symbol $x_1, \dots, x_E$, one per event:

```
const ctmc::CtmcSymbolicGenerator<double> sym =
    ctmc::ctmc_symbolic_generator(model.get_struct(), opt);
```

Reached from the front end as `line-cli -s ctmc -a gen -symbolic`. No computer algebra is needed to assemble it, since the generator is linear in the symbols; substituting each symbol with its event's smallest positive rate recovers the numerical generator exactly. Solving with the symbolic generator does require a computer algebra system: that solve is `symbolic_solution`, and the symbolic backend it delegates to is described in §9.1.

A chain supplied directly by the user, rather than compiled from a network, is a model class of its own and is described in §6.1; the CTMC solver solves it as it stands, without generating a state space.

8.2.3.3 Reward analysis

The CTMC solver supports reward-based analysis through custom reward functions defined on network states. Users can define state-dependent reward functions that map a state vector and the network structure to a

scalar reward value. Multiple rewards can be defined by calling `setReward` multiple times with different names. The solver computes rewards using value iteration with uniformization. Results are retrieved via `solver.getAvgReward()`, which returns steady-state expected rewards, or `solver.getTranReward()`, which returns the transient expected-reward trajectories $E[r(X(t))]$ over time. Example applications include computing expected queue lengths, utilization, blocking probabilities, or custom cost functions.

A reward is registered with `model.set_reward(name, fn)`, where `fn` maps the state vector to a scalar; it is evaluated by the CTMC solver on every state.

8.2.4 FLD: fluid and mean-field solver

This solver implements fluid/mean-field approximation methods, based on the system of fluid ordinary differential equations for INF-PS queueing networks presented in [126]. The latter is based on Kurtz’s mean-field approximation theory. The C++ fluid solver integrates its drift with a port of LSODA (`line/util/lsoda.h`), which switches between Adams and BDF integration by itself; `FluidOptions::tol` is handed to it as both the absolute and the relative tolerance.

ODE variables corresponding to an infinite number of jobs, as in the job pool of a source station, or to jobs in a disabled class are not included in the solution vector. These rules apply also to the `options.init_sol` vector.

Nodes with zero or near-zero service time (e.g., `Router`, `ClassSwitch`) are treated as immediate nodes and are eliminated from the fluid equations before integration; FCFS stations are mapped into equivalent PS stations with a mixture service process. Both transformations are described in the LINE internals manual (`LINE-internals-cpp.pdf`, Chapter FLD).

8.2.4.1 Methods and configuration options

The following methods are available for the FLD solver:

- `options.method=default`: resolves to a concrete method from the model: `rmf` if the model contains `Cache` nodes, otherwise `minnormal` wherever that method applies, and failing that `closing` for models with `DPS` stations or `matrix` for the rest.
- `options.method=matrix`: matrix formulation of the fluid ODEs [139], for closed and open networks.
- `options.method=closing` or `statedep`: state-dependent fluid solver with closing approximations, useful for models with `DPS` scheduling.
- `options.method=pnorm`: p -norm smoothing of the processor-share constraint [139].
- `options.method=diffusion`: diffusion approximation, restricted to product-form closed networks.
- `options.method=mfg`: exact steady-state analysis of a `Source` $\rightarrow$ `Queue` $\rightarrow$ `Sink` model with phase-type arrivals and service [78]; closed networks and multiple queues are not supported.
- `options.method=refined` or `rmf`: model-aware aliases. Without a cache they apply the min-normal closure and the Gast–van Houdt $O(1/N)$ correction about the mean-field fixed point [67, 71]; with `Cache` nodes they select the refined mean-field cache–queueing decomposition [66].

- `options.method=tbi`: cell-decomposed solution of the closing ODEs for large closed models [148], the cells being reconciled by waveform relaxation [97]. Configured by `options.config.tbi_tol` (default 10^{-3}), `tbi_iter_max` (default 50), `tbi_sweep` ('gauss-seidel' or 'jacobi'), `tbi_parallel` (MATLAB only), and by the cell partition, given as `tbi_cells` or built automatically with `tbi_cellsize` stations per cell (default 5). Caching stations and immediate-node elimination are not supported.
- `options.method=minnormal`: second-order moment closure [71, 150, 159], which keeps the error bounded as $\rho \rightarrow 1$ where the first-order closures do not. It is the only LINE fluid method supporting GPS. Open and mixed models are admitted when the arrival process is Poisson, a non-exponential inter-arrival time being refused; throughputs are then exact and the error falls on the mean queue lengths. The state-level covariance and the queue-length variances are reported by , and are reliable for closed classes only.
- `options.method=kp`: fluid and diffusion limits of the open $(MAP_t/Ph_t/\infty)^N$ network of [90], i.e. of a network whose arrival and service processes are the time-inhomogeneous MAP_t/Ph_t distributions. It integrates the mean and the covariance jointly, the second moment being exposed through `getTranAvgVar`, which is available under this method only. Closed classes are rejected.
- `options.method=dae`: the same min-normal closure as `minnormal` – the same drift, the same rate factors, the same Lyapunov equation – stated as a single differential-algebraic system and solved as one, instead of by the outer substitution loop that alternates an ODE integration with a Lyapunov solve. See §8.2.4.2 below.

The algorithms behind these methods are documented in the internals manual, Chapter FLD.

FLD method selection. The default FLD method resolves from the model as described above. The `matrix` method formulates ordinary differential equations for the mean-field dynamics, while `statedep` uses state-dependent rate scaling and `softmin` smooths the capacity minimum. The `mfq` method solves Markovian fluid queues, and `diffusion` applies Euler–Maruyama to fluctuations around the mean field. The `refined/rmf` aliases select either the generic $O(1/N)$ correction or the cache decomposition.

8.2.4.2 The differential-algebraic formulation

The `minnormal` closure is already a differential system (the mean) coupled to an algebraic one (the covariance), and it is discharged by *successive substitution*: the mean ODE is integrated to its fixed point at a held variance, the Lyapunov equation is solved there, the variance is extracted and the cycle repeats. The `dae` method states the same closure as one system and solves it as one. Nothing about the closure changes – the drift, the rate factors and the Lyapunov equation are the same functions – only the way the coupled equations are discharged. The horizon decides which of two genuinely different problems is posed.

Steady state (an infinite timespan). One algebraic system collects three families of rows – the drift residual, one row per state coordinate; population conservation, one row per closed chain; and closure

consistency, one row per closable station – and is solved simultaneously by a damped Newton iteration. Convergence is quadratic near the root instead of the linear rate of substitution, and the answer is converged to `options.tol` rather than to the coarse tolerance of the outer loop. The fixed point is *not* found by integrating to it: a seed trajectory is integrated once, cheaply and at the first-order closure, because Newton needs a starting point inside the basin, and everything after that is algebraic. This matters most on stiff models, where the cost of integrating to a standstill is set by the slowest mode of the model rather than by the accuracy wanted. Writing conservation as a constraint rather than leaving it a consequence of the drift is also what makes the Newton system solvable: the drift Jacobian is singular along exactly the conserved directions, and the constraint rows supply the missing rank. The covariance itself is not a Newton unknown – it is linear in itself for a held mean, so it is eliminated by one Lyapunov solve per residual evaluation and only the per-station variances join the mean in the unknown vector, which keeps the work cubic rather than quartic.

Transient (a finite timespan). The index-1 system that evolves the mean under the drift, holds the conservation constraint as an algebraic row and evolves the covariance by its Lyapunov equation, with a singular mass matrix, is integrated by RODAS, the Rosenbrock method of order (3)4 for differential-algebraic systems [72]. Together with `kp` this is the only fluid method producing a *time-varying* second moment: `minnormal` evaluates the whole transient at the single stationary variance, whereas here the variance is the one the trajectory actually had at each instant. It is reported alongside the transient means.

Capacity limits as algebraic constraints. A finite capacity region (§5.5) and a station buffer are both a linear inequality on the fluid state, which this formulation can carry beside the drift and no ODE method can carry at all. All the declared forms reduce to the same family of weighted-sum inequalities on the fluid state: a region-global job cap is the row with unit weights, a per-class cap weights one class, a memory budget weights each class by its size, and an explicit linear constraint supplies the weights directly; a station cap is the same row restricted to one station. What differs between them is not the row but where a blocked job goes. Each capped *region* therefore gains one waiting-room coordinate per class, and every admission into it is split into an unthrottled leg from the upstream station into the room and a throttled leg from the room into the region, so that a job blocked by the cap has completed its upstream service and waits somewhere else rather than piling up at a station it has already left. A *station* buffer gets no room, matching the reference semantics in which the upstream departure is disabled and the blocked job is still counted where it was. Regions are accepted under the waiting-queue rule only; station buffers under either the waiting-queue or the drop rule. Blocking-after-service, blocking-before-service, RS-RD and the retrial rules each change the event set itself rather than constrain this drift, and are refused by name, as are per-class admission weights, which decide *which* job is admitted rather than how many.

Availability and limits. The method is present in all four codebases. It admits limited load dependence, like the rest of the closing family, and finite capacity regions, unlike every other fluid method; it rejects SIRO, LCFS and LCFSPR, which have no branch in the closing drift, and DPS and GPS, whose closures need covariance blocks between the class coordinates of a station that the DAE form carries no unknown for.

Two size gates guard the cost: `options.config.dae_maxstate` (default 100) caps the phase-resolved state of the simultaneous solve and `options.config.dae_maxcov` (default 25) the transient covariance. From the front end, `line-cli -f model.json -s fluid -method dae`; the stationary covariance is reached with `-a var`, which accepts `minnormal`, `refined` and `dae` for the stationary second moment and `kp` for the covariance along the trajectory.

8.2.4.3 Exporting the ODE system

The system of ODEs integrated by the mean-field methods (`default/matrix`, `pnorm`, `closing`, `statedep`, `softmin`) can be exported in symbolic form as a standalone L^AT_EX document using the `exportODEs` method. The export is both human readable, since the compiled document shows the equations together with a legend of the state variables, and machine readable, since the document header carries the state-variable mapping and event rates as structured comment lines (`% STATE` and `% EVENT`). Each state variable x_s represents the mean number of jobs of a class in a service phase at a station. Two notations are available:

- `scalar` (default): one expanded ODE per state variable, written in terms of station-level auxiliary functions such as the total station mass $n_i(x)$ and the capacity scaling $g_i(x) = \min(n_i(x), S_i)/n_i(x)$;
- `matrix`: the compact matrix form, $d\mathbf{x}/dt = W^\top \theta(\mathbf{x}) + \lambda$ for the `matrix/pnorm` methods (per Ruuskanen et al. [139], with θ the smoothed service vector and λ the exogenous arrival rates), and $d\mathbf{x}/dt = J r(\mathbf{x})$ for the `closing/statedep/softmin` methods, where J is the stoichiometry (jump) matrix and $r(\mathbf{x})$ the vector of event rate functions.

The exported document also reports the initial condition $\mathbf{x}(0)$ and remarks on the numerical regularizations applied by the solver. The exported system reflects the method set in the solver options; methods that are not ODE-based (`diffusion`, `mfq`, `rmf`) are not supported by this export. The ODE export is a front-end analysis in the C++ port: `line-cli -f model.json -s fluid -a odes` writes the system, and `--notation` selects the scalar or the matrix form.

8.2.5 LDES: discrete-event simulation engine

The LDES solver is LINE's own native discrete-event simulation engine, built on the SSJ library and shipped inside the canonical JAR. All three codebases drive the same engine, so a given model and seed produce identical sample paths and hence identical estimates.

The model features admitted by the solver are those declared by its feature set `()`, summarized in the tables of Appendix B. Finite-buffer stations are honoured on arrival according to the per-station `DropStrategy`, as are network-level `FiniteCapacityRegion` constraints. A BMAP assigned as a service process is interpreted as a batch service process (§5.3.6). The accuracy of the estimates is governed by the simulation length and by the transient filter and confidence-interval estimators of Table 8.3; the solver is validated against JMT by a two-sample t-test at the 1% significance level. Its initial state can be seeded from another solver as described in §8.1.3.

8.2.5.1 Methods and configuration options

Table 8.3 lists the configuration options. Defaults are those of the engine and are shared by all codebases.

Table 8.3: LDES configuration options

Option	Default	Description
<code>options.samples</code>	$2 \cdot 10^5$	Service-completion events to simulate (engine flag <code>-s</code>).
<code>options.seed</code>		Random number generator seed (<code>--seed</code>).
<code>options.method</code>		Simulation method; only default is defined.
	<code>mser5</code>	Warmup detection: <code>mser5</code> (MSER-5), <code>fixed</code> (fixed fraction), <code>none</code> .
	5	MSER batch size; used only when <code>tranfilter</code> is <code>mser5</code> .
	0.2	Fraction discarded; used only when <code>tranfilter</code> is <code>fixed</code> .
	<code>obm</code>	Estimator: <code>obm</code> (overlapping batch means), <code>bm</code> (batch means), <code>spectral</code> (Heidelberger-Welch), <code>none</code> .
	0.5	Overlap fraction of <code>obm</code> ; 0 reduces it to <code>bm</code> .
	10	Minimum batch size; the batch size used is $\max(\text{ciminbatch}, \sqrt{n})$ on n observations.
	100	Minimum post-warmup observations below which no interval is reported.
	0.25	Fraction of lowest frequencies in the log-periodogram regression; used only when <code>cimethod</code> is <code>spectral</code> .
	<code>false</code>	Stop as soon as the relative precision of every metric falls below <code>cnvgtol</code> .
	0.05	Target relative precision (CI half-width over mean).
	20	Minimum batches before the first convergence check.
	0 (auto)	Events between checks; 0 selects <code>samples/50</code> .
	<code>false</code>	Advance time on a discrete lattice instead of continuously.
	1.0	Slot length in model time units; meaningful only when <code>slotted</code> is set.
	1	Independent replications, each with its own RNG stream; intervals then use the cross-replication variance.
<code>options.init_sol</code>	<code>empty</code>	Warm-start placement, station-major vector (<code>--initsol</code>); set by <code>initFromSolver</code> .
	0	Highest busy period order measured (<code>--busyperiod</code>); 0 disables the measurement.
	<code>empty</code>	Station sets whose joint busy period is measured (<code>--busyperiod-subnet</code>).
	<code>empty</code>	Base URL of an LDES REST server; when set the model is POSTed to <code><url>/api/v1/solve</code> instead of being solved locally.

Setting `slotted` switches the engine from its default continuous time scale to a discrete (slotted) one, in which time advances in integer multiples of `slotlength`. Every sampled interarrival and service time must then fall on the slot lattice, a sample that does not lie on it being reported as an error rather than rounded; within a slot, simultaneous events are ordered by a fixed phase (service completions, internal movement, external arrivals, bookkeeping); and non-homogeneous Poisson arrivals and the processor-sharing family are rejected. The canonical discrete-time model is the Geo/Geo/1 queue, a `Source` with `Geometric` interarrivals feeding an `FCFS Queue` with `Geometric` service and a slot length of one, whose analytical counterpart is `qsys_geogeol`; with a batch source (§5.3.6) the stream is `GeoX` and the references are `qsys_geoxgeol`.

and `qsys_geoxgeol_moments`.

8.2.5.2 Busy period measurement

Setting `to K` makes the engine measure, along the simulated sample path, the mean busy periods of orders $1, \dots, K$ of every station, of every station-class pair, and of every station set declared through `.`. A busy period of order n runs from the instant an arrival raises the jobs held by the target to n up to the instant the target falls back below n ; the per-class variant counts only the jobs of one class. Periods still in progress when the warmup ends are discarded, their start not being observable, and a period of zero length is treated as the artifact of two simultaneous events rather than as an observation. The measurement is off by default and costs nothing when off.

```
ldes::LdesOptions opt;
opt.busy_period_orders = 3;
opt.busy_period_subnets.push_back({0, 1});
const ldes::LdesResult r = ldes::solver_ldes(model.get_struct(), opt);
```

The exact counterpart on product-form models is §8.2.8.2, against which the measurement agrees to within the simulation error.

8.2.5.3 Command-line interface

The LDES engine is also distributed as a standalone executable, either as the native binary `common/ldes` or as `common/ldes.jar`. The MATLAB and Python wrappers are thin JSON-mediated clients of this executable: they serialize the model to `model.json`, invoke `solve model.json -o result.json`, and parse the returned metrics. The same interface can be driven directly. The principal flags are summarized below.

Table 8.4: LDES command-line flags

Flag	Description
<code>--samples</code>	Number of simulation events to execute.
<code>--maxevents</code>	Hard cap on the number of simulated events.
<code>--maxtime</code>	Hard cap on simulated time.
<code>--seed</code>	Random number generator seed.
<code>--method</code>	Simulation algorithm selector.
<code>--cnvgon, --cnvgtol</code>	Enable convergence-based stopping and set its relative tolerance.
<code>--tranfilter</code>	Transient (warmup) filter: <code>mser5</code> , <code>fixed</code> or <code>none</code> .
<code>--warmupfrac</code>	Warmup fraction discarded under the <code>fixed</code> filter.
<code>--cimethod</code>	Confidence interval estimator.
<code>--replications</code>	Number of independent replications; drives the parallel analyzer.
<code>--numthreads</code>	Number of worker threads used for replications.
<code>--slotted</code>	Run on a discrete (slotted) time scale.
<code>--slotlength</code>	Slot length in model time units; implies <code>--slotted</code> .
<code>--timespan</code>	Time span for transient analysis.
<code>--initsol</code>	Warm-start initial state, as a station-major population vector.
<code>--trajectory</code>	Export the simulated sample path.
<code>--export-histogram</code>	Export the state histogram, which backs reward, <code>getProb</code> and sampling queries.

Continued on next page

Table 8.4 – Continued from previous page

Flag	Description
<code>--busyperiod</code>	Measure the busy periods of orders $1..K$ for every station and every station-class pair.
<code>--busyperiod-subnet</code>	Also measure the joint busy period of the given station set; repeatable.

8.2.6 MAM: matrix-analytic methods solver

This is a basic solver for some Markovian open queueing systems that can be analyzed using matrix analytic methods. The core solver is based on the BU tools library for matrix-analytic methods [78]. The solution of open queueing networks is based on traffic decomposition methods that compute the arrival process at each queue resulting from the superposition of multiple source streams.

The `getCdfRespT` method computes the response time distribution using matrix-analytic techniques. For processor-sharing (PS) queues with MAP arrivals, it uses the algorithm of [111]. The number of CDF points computed can be controlled via `options.config.num_cdf_pts` (default: 200).

8.2.6.1 Methods and configuration options

The following methods are available for the MAM solver:

- `options.method=default` or `dec.source` (Arrival Stream Decomposition): the default decomposition, applicable to open, closed, and mixed queueing networks.
- `options.method=mna` (Matrix Network Analyzer) [98]: more accurate than `dec.source`; recommended for networks with Markovian arrival processes, and restricted to FCFS and HOL scheduling. A departure stream split deterministically by round-robin scheduling, either at the station itself or at a downstream Router it feeds exclusively, is recognised as such and has its variance propagated by the exact round-robin formula rather than by the Bernoulli-thinning formula used for a probabilistic split.
- `options.method=dec.poisson` (Poisson Arrival Approximation): the fastest but least accurate decomposition, suitable for exploratory analysis or as a baseline.
- `options.method=dec.mmap` (ETAQA Departure Decomposition) [133]: computes MAP approximations of the departure processes; when the required state space exceeds `space_max` or the queue is unstable, the solver falls back to a scaled service process.
- `options.method=bgchain` (Background Modulating Chain): for *mixed* and *closed* networks, see below.

The algorithms behind these methods are documented in the internals manual, Chapter MAM.

The agent-based methods (`inap`, `inaprc`, `inapinf`) are *not* MAM methods: they belong to the AG solver of §8.2.1. MAM refuses them by name and says so.

The `dec.mmap` method exposes additional configuration parameters via `options.config`:

- `etaqa_trunc` (default: 8): QBD truncation level for the ETAQA computation; higher values yield more accurate departure processes at increased computational cost.

- `space_max` (default: 128): maximum state-space size $(\text{etaqa_trunc} + 1) \times n_a \times n_s$ beyond which ETAQA is skipped and a scaled service process is used instead.
- `merge` (default: 'super'): strategy for merging multiple marked MAP streams via superposition.
- `compress` (default: 'mixture.order1'): compression method to reduce the order of superposed MMAPs before downstream analysis.

For models containing fork-join sections, `options.config.fj_accuracy` sets the numeric accuracy parameter of the fork-join approximation (default 100) and `options.config.fj_tmode` selects how the fork-join T matrix is computed, either 'NARE' (default) or 'Sylvester'.

When an FCFS station declares a finite capacity K (§5.1.1), MAM analyses it as a loss system and reports the blocking probability as a reduction of the class throughputs, in preference to the infinite-buffer closed forms for PH/M/1, D/M/ c and MAP/D/ c stations, which are defined for $K = \infty$ only. An M/M/ c/K station is solved exactly; ; in the remaining cases the QBD of the infinite-buffer station is truncated at level K and renormalized, which yields one blocking probability shared by all classes.

MAM method selection. The MAM solver offers several decomposition approaches for networks with non-exponential service. The `dec.mna` method uses the MNA decomposition, which is generally the most accurate for networks with Markovian arrival processes. The `dec.source` method approximates arrivals at each station using the distribution observed at the source, while `dec.poisson` replaces all arrival flows with Poisson processes. The `inaprc` method provides an improved iterative approximation that avoids the normalization step of the standard `inap` method, and is recommended for models where `inap` converges slowly. The `inapinf` method [110] solves each isolated open component directly on its infinite state space by a matrix-geometric decomposition, yielding the exact marginal without state-space truncation – the scalar geometric with a single phase per level, and Neuts' rate matrix R [121] above it; it is preferred over `inap` for open G-networks with birth-death or catastrophe structure, and for heavy-tailed phase-type service, where truncation would otherwise bias the queue-length estimates. These methods accept any process with a Markovian (D_0, D_1) representation, and refuse the laws that have none (matrix-exponential and rational arrival processes), those that are time-inhomogeneous, discrete-time, or batch; they remain single-server only.

8.2.6.2 Background modulating chain for mixed and closed networks

The `bgchain` method applies to *mixed* and *closed* networks: the closed population vector of such a network is the sole part of the model whose state space is bounded, so it is a finite continuous-time Markov chain in its own right. The method solves that chain exactly, given the mean open occupancy, and hands each open station a station-local Markovian environment read off it, so the station becomes a level-dependent QBD whose phase carries the number of closed jobs competing for its server. The two halves meet at a fixed point on the mean open occupancy: the background chain is built from it, and the QBDs recompute it. A purely open model is refused by name, having no closed population vector to build the chain from, as are class priorities, which the per-station phase-type mixture cannot express, and fork-join sections, which do not conserve the closed population per station; `dec.source` is the fallback in all three cases.

A purely closed network is the degenerate case of the same construction rather than a separate algorithm. With no open class the fixed point has nothing to iterate: the capacity share the closed jobs hold never leaves $\min(e, c)$, which is exactly what they hold when no open work competes for the server, so the background chain alone answers, and it answers with the exact closed continuous-time Markov chain at chain granularity. On a product-form closed network it therefore reproduces exact MVA to machine precision, which is why it is what `default` resolves to on a closed model, ahead of `mna`. Two conditions gate that choice. The chain must fit within `bgstates_max`, since it enumerates the closed population vector over the stations the closed classes visit and so grows as $\binom{N+M-1}{M-1}$ per class; and the background chain reads only the *mean* service time, which is exact at a processor-sharing or infinite-server station by insensitivity and under any discipline when the law is exponential, but a first-moment surrogate otherwise. A non-exponential law at an FCFS station therefore defaults to `mna`, which carries the phase-type representation instead. Requesting `bgchain` by name overrides neither gate: the model is answered with the approximation stated.

With $R > 1$ closed chains the background chain would be exponential in R , so the method keeps one chain free at a time: the tagged chain is carried exactly and the remaining $R - 1$ are replaced by flow-equivalent aggregate classes whose population is their total and whose service times and routing are their throughput-weighted means [36]. Every chain takes its turn as the tagged one, reads its own metrics off the chain in which it is exact, and the open-class results are averaged over the passes. Which chains share an aggregate is decided by the similarity of their per-station service demands, so the aggregation stays where it is harmless.

At a processor-sharing station the open service law is replaced by the exponential of the same mean before the QBD is built. This is not a shortcut but the insensitivity property of PS: the QBD tracks one service phase for the whole station, so carrying a phase-type law there would make the open queue length inherit the coefficient-of-variation sensitivity of an M/PH/1 FCFS queue, which was measured 21% high on a HyperExp of squared coefficient of variation 4 where the exact answer is the exponential one to five digits. Measured against CTMC and exact MVA on mixed models of two to four stations, PS and INF stations agree to four or five significant digits under any service law, any number of servers and Poisson or MAP arrivals, where `dec.source` is 10 to 24% out; FCFS stations with class-independent rates agree to five digits, while class-dependent FCFS rates keep the closed queue lengths within about 1% and read the open queue length 14 to 20% low.

The method is controlled through `options.config`:

- `bgaggr` (default 1): the number G of aggregate classes, so that the background chain carries $1 + G$ classes. $G = 1$ is the classic tagged/aggregate pair and the cheapest option; $1 < G < R - 1$ splits the untagged chains into G groups; $G \geq R - 1$ aggregates nothing, which makes the background chain exact in the closed classes and lets a single pass answer for all of them.
- `bgstates_max` (default 20000): the cap on the number of background chain states, which is the product over the $1 + G$ classes of $\binom{N_b+M_c-1}{M_c-1}$. Exceeding it is an error naming both remedies, a smaller `bgaggr` or a larger cap.
- `qbdphases_max` (default 500): the cap on the number of environment phases carried by a station QBD.
- `bgenv` (default `lump`, MATLAB only): the environment axis handed to a station. The default lumps the

background chain onto the closed occupancy of that station; `full` passes the whole chain instead, with no lumping, so its phase count grows as the chain does.

8.2.6.3 Level-dependent QBDs

The MAM solver embeds an exact engine for Level-Dependent Quasi-Birth-Death processes [128], whose truncation level is chosen automatically from the traffic intensity and whose convergence is controlled by `options.tol` and `options.iter_max`. It is invoked either explicitly,

```
mam::MamOptions opt;
opt.method = "ldqbd";
const mva::AvgResult<double> res = mam::solver_mam_run_analyzer(model.get_struct(), opt);
```

or automatically. The explicit form is restricted to single-class networks with a single FCFS queue, in two regimes: a closed one, with an infinite-server delay station and a finite population, and an open one, with a `Source` feeding the queue and a level index truncated at `options.cutoff`. Multi-server and load-dependent stations are supported, and phase-type service is handled through matrix blocks. Multi-server phase-type service is solved *exactly*: the coordinate carried inside a level is then the multiset of the phases occupied by the $\min(n, c)$ busy servers, the same configuration space as the $MAP/PH/c$ queue of §7.4.4 and of order $\binom{\min(n, c) + m_s - 1}{m_s - 1}$ rather than $m_s^{\min(n, c)}$ [4], so level sizes grow over the boundary levels and repeat above them. A load-dependent factor scales the station's total service rate, shared over the busy servers. The automatic invocation happens on retrial topologies (§5.3.5), where the orbit retrial rate makes the generator level-dependent, and no method selector is required; the metrics then include the mean orbit length, the throughput, the server utilisation and the orbit blocking probability. A transient counterpart integrates Kolmogorov's forward equation on the level-truncated generator and is invoked automatically when `solver_ctmc_transient_analyzer` is called on such a model.

8.2.7 MVA: mean value analysis solver

The solver offers approximate mean value analysis (AMVA) (`options.method=`), but also exact MVA algorithms (`options.method=`). The default AMVA solver is based on Linearizer [38], unless there are two or less jobs in total within closed classes, in which case the solver runs the Bard-Schweitzer algorithm [144].

8.2.7.1 Methods and configuration options

Beyond the `default/exact` entry points, the solver exposes a catalogue of AMVA variants reported by `list_valid_methods`; the `amva.`-prefixed names are aliases of the corresponding short names. The most commonly used variants are:

- `amva.lin` (alias `lin`): Linearizer [38], the default AMVA for closed multi-class product-form models.
- `amva.bs` (alias `bs`): Bard-Schweitzer approximate MVA [144], automatically selected when the closed populations are small.

- `amva.qd` (alias `qd`) and `amva.qdlin` (alias `qdlin`): queue-dependent AMVA [31] and queue-dependent Linearizer, for limited load-dependent stations; `qd` is the engine behind `default` when load dependence is detected.
- `amva.qli` and `amva.fli`: Wang-Sevcik queue-line and fraction-line approximations [163].
- `amva.aql` (alias `aql`): aggregate queue-length approximation [170], which carries $R + 1$ population points and a correction term that removes the Schweitzer proportionality error. Restricted to strict product-form, load-independent, single-server models; it is listed only when the model qualifies.
- `amva.qsa` (alias `qsa`): the queue-shift approximation [145]. Where Linearizer extrapolates the *fractional* deviation of a per-class queue length, QSA extrapolates the *absolute* shift of the aggregate queue length seen when one customer is removed, so the unknowns are one per station rather than one per station-class. The core equation is imposed at the full population and at every population with one or two customers removed, the two-removal case through an affine extrapolation of the one-removal shifts, and the resulting system is solved as a whole by a damped Newton iteration. On the three stress networks published with the method it is between two and five times more accurate than Linearizer, and its advantage grows with the population. Because the shift is carried per station rather than per station-class, the gain holds when the service time at a queueing centre is the same for all classes, which is the regime the method was designed for and the one FCFS requires anyway; with strongly class-dependent demands Linearizer's per-class deviations can be the more accurate choice. Restricted, like `aql`, to strict product-form, load-independent, single-server models.
- `priomva` (alias `amva.priomva`): the preemptive-resume priority arm, offered only when a station uses `SchedStrategy.FCFSPRPRIO`. It applies the Chandy-Lakshmi approximation [37] where that paper derives it – a job in service is preempted by a higher-priority arrival – so unlike the non-preemptive `HOL` arm it carries no residual for the preempted job and scales the tagged job's own service by $1/(1 - \sigma_{k-1})$ as well as the queued work. `PRIOMVA` is Bolch's name for this method in PEPSY-QNS. Single-server stations only; multiserver PRS is refused by name. Accuracy degrades with population at high utilization, comparably to the `HOL` arm; see [57].
- `amva.scats` (alias `scats`): the Neuse-Chandy SCAT approximation [120], which shares the Bard-Schweitzer correction term $\Delta_{rit} = Q_{ir}(\mathbf{N} - \mathbf{e}_s)/(N - \mathbf{e}_s)_r - Q_{ir}(\mathbf{N})/N_r$ with Linearizer but refreshes it once per fixed-point pass rather than Linearizer's three, at roughly a third of the cost and with accuracy between plain Bard-Schweitzer and Linearizer. Restricted, like `aql` and `qsa`, to strict product-form, load-independent, single-server models.
- `amva.ab` (alias `ab`), `amva.schmidt` and `amva.schmidt-ext`: Akyildiz-Bolch and Schmidt approximations [152] for multi-server stations.
- `amva.tay` (alias `tay`): Tay-Suri arrival-instant approximate MVA [155], restricted to single-server stations.
- `egflin`, `gflin` and `sqni`: feedback Linearizer variants and single-queue numerical iteration, used internally for specialised topologies.

- `qna` and `rqna`: Whitt's Queueing Network Analyzer [17, § 6.7] for open networks with general distributions, and its robust counterpart [166] for single-class open networks with bursty non-renewal arrivals. Under default, `rqna` is auto-selected for multi-station bursty open networks, while a single Source-Queue-Sink model is routed to the queueing-system formulas; in either case the arrival autocorrelation is not silently discarded.
- `rqt`: Robust Queueing Theory [7], restricted to single-class open networks with a Source station. Rather than an expected value, it reports a worst-case system time over polyhedral uncertainty sets that replace the stochastic primitives, composing the robust-Burke queue-passage, superposition and thinning theorems of the reference along the visit-ratio path and closing with a heavy-traffic-regressed worst-case bound (`options.config.rqt_regime`, one of 'independent', 'normal' or 'pareto'; `rqt_exact` evaluates the exact worst case instead of the closed-form bound). Tail coefficients default to $\alpha = 2$ (finite variance) and are set per-stream by `rqt_alpha_a/rqt_alpha_s` in $(1, 2]$ to declare heavier tails, which LINE's distributions cannot express directly since they carry finite moments. Accuracy is heavy-traffic-biased: on M/M/1 the error is about 8% at $\rho = 0.9$ but exceeds 50% at $\rho = 0.5$, so `qna` or an exact method should be preferred at low utilization.
- `oi`, `sqd` and `sjn.mva/sjn.amva`: conditional MVA for order-independent and P&S stations, Smith Queue Decomposition for closed single-chain blocking-after-service models (§5.3.2), and shortest-job-next analysis of closed models with `SchedStrategy.SJF` stations [85]. None of the three is requested by the user: a model declaring the corresponding feature is routed to them automatically and the method name is reported back as the actual method used. The SJN route is configured by `options.config.sjn_lattice_max` (default 10^5 , above which the Schweitzer fixed point replaces the population lattice), `sjn_ns` (quadrature panels, default 32), `sjn_lfactor` (grid length, default 8) and `sjn_umax` (utilization cap, default 0.999); every other station must then be single-server FCFS, PS, LCFS-PR, SIRO or a delay, and open models are rejected.

For single-class closed models the solver additionally exposes the bound families `aba.*`, `bjb.*`, `gb.*`, `pb.*` and `sb.*`, and for multiclass closed models the Majumdar-Woodside robust box bounds `mw.upper/mw.lower` [107], also exposed for layered networks by `SolverLN`. For two-station open single-class networks it dispatches the `mm1/mmk/mg1/gig1.*`/`gigk.*` formulas; see Table 8.1 for the complete list. Queues with `POLLING` scheduling are routed to a dedicated polling analyzer implementing the `stationtime` method for cyclic polling with exhaustive, gated and limited service. The algorithms behind all these methods, and the policy by which `default` selects among them, are documented in the internals manual, Chapter MVA.

Extended features are handled through the configuration options of: `options.config.highvar` selects the correction applied to non-exponential service at FCFS stations, which is ignored by default and available in the single-server case only; `multiserver` selects the multi-server approximation; `np_priority` the non-preemptive priority approximation; and `fork_join` the transformation applied to fork-join sections, which are assumed to form a directed acyclic graph between each fork and its join. DPS stations, limited load dependence and class dependence are handled by built-in correction factors and need no option. The algorithms behind these settings are described in the internals manual, Chapter MVA.

Table 8.5: MvaOptions fields (C++)

Field	Value	Description
multiserver	"default"	"softmin" at PS queues and "seidmann" at FCFS queues.
multiserver	"seidmann"	Seidmann's decomposition [146].
multiserver	"softmin"	QD-AMVA's softmin approximation [31].
multiserver	"suri"	Suri et al.'s multiserver approximation [152].
np_priority	"default"	Non-preemptive priority handling; equals "cl".
np_priority	"cl"	Chandy-Lakshmi [37], in the arrival-instant utilization form of Eager-Lipscomb [57].
np_priority	"shadow"	Sevcik's shadow server [147].
highvar	"default"	No high-variance correction.
highvar	"interp"	Diffusion-M/G/k interpolation from [24].
highvar	"hvmva"	High-variance MVA as in [19].
tol, iter_tol, iter_max	numeric	Convergence tolerances and iteration cap of the fixed point.
rqt_regime	"independent"	Default Table 1 adaptation regime of rqt [7]; "normal"/"pareto" select the alternatives.
rqt_exact	false	rqt evaluates the exact worst case instead of the closed-form bound when true.
rqt_alpha_a, rqt_alpha_s	2.0	Tail coefficients of the rqt arrival/service streams, in (1, 2]; scalar only.

8.2.8 NC: normalizing constant solver

The NC class implements a family of solution algorithms based on the normalizing constant of state probability of product-form queueing networks. Contrary to the other solvers, this method typically maps the problem to certain multidimensional integrals, allowing the use of numerical methods such as Monte-Carlo sampling and asymptotic expansions in their approximation. The NC solver is the recommended back-end for the analysis of load-dependent stations specified through `set_load_dependence` (see §5.2.1); the `nrl`, `nrp`, `nre`, and `rd` methods listed below are designed for this case and remain compatible with the probability metrics `solver_mva_get_prob_marg`, `solver_mva_get_prob_aggr`, and `solver_mva_get_prob_norm_const_aggr`.

The NC solver is also the only one to answer `solver_nc_getprob_sys_marg`, the joint law of the per-station total queue lengths described in §7.1.5. That metric is not a method and is not requested through `options.method`: it is a permanent of the demand matrix replicated once per job, normalized by the same constant the solver already computes, and its optional second argument selects only the estimator of that permanent. Being a state probability it inherits the scope of the product form, so a multiserver station, a load-dependent station and an open or mixed model are refused by name rather than approximated.

When the model declares order-independent (`SchedStrategy.OI`) or pass-and-swap (P&S, `SchedStrategy.PAS`) stations, the NC solver dispatches automatically to a dedicated normalizing-constant analyzer, which is reported back as the actual method `oi`. It evaluates the balance functions of the OI stations by a convolution recursion instead of the product-form load terms, and it verifies the permutation-invariance condition that the declared station must satisfy. As with MVA, `oi` is not a method name the user requests: it is selected under `default` and `exact`. The importance-sampling method `is` provides a Monte Carlo alternative for OI and P&S networks whose state space makes the exact recursion impractical.

8.2.8.1 Methods and configuration options

The exact algorithms are `ca` (multiclass convolution [17, § 9.1]), `comom` (class-oriented method of moments [22], polynomial in the number of classes), `rgf` (recursion by generating functions [47], single-class closed models only) and `clw` (Choudhury-Leung-Whitt numerical inversion [39], exact up to a controllable aliasing error and efficient when a few chain populations are large). The method name `exact` selects among them automatically, dispatching load-dependent closed models to the load-dependent back-end.

The sampling estimators are `imci` [164], `ls` [23] and `is`, the last sampling job orderings rather than the integral representation, which is what makes it applicable to order-independent and P&S stations [43]; `samples` and `seed` control it. The selector `sampling` maps to `is` on such models and otherwise chooses among `imci`, `ls` and the McKenna-Mitra sampler.

The perfect samplers `cftp` and `cftp.approx` draw iid states from the stationary distribution of a closed single-class product-form network by monotone coupling from the past [88], walking the Gordon-Newell balance function one station pair at a time. The normalizing constant cancels in the conditional that each step inverts, so neither the state space nor $G(\mathbf{N})$ is computed and no normalizing constant is reported. The metrics are Monte Carlo estimates whose error decreases as `options.samples`^{-1/2} (default 10⁵), the exact variant having an unbounded worst-case cost per sample and the approximate one a deterministic cost with a residual bias. The model must be closed and single-class, built of `Queue`, `Delay` and `Router` nodes only, with exponential service under `INF`, `PS`, `FCFS`, `SIRO` or `LCFS` scheduling, infinite buffers and Markovian routing; anything else is refused rather than approximated. These two methods belonged to the CTMC solver until 2026-09-27.

The asymptotic and integral methods are `le` and `kt` [23, 89], `pana` and `panald` [113, 116, 134] for normally-loaded closed models, `cub`, `gm` and `gleint` (cubature and quadrature of the normalizing-constant integral [23]), `mmint2` (McKenna-Mitra integral [113], two-station closed models), `aghq` (quadrature of the simplex factor of that integral [101]), `nrl`, `nrp`, `nre` and `rd` (saddle-point and reduction approximations for load-dependent models [28]), and `propfair`, a fast proportionally-fair throughput heuristic. `ble` is a bias-corrected variant of `le` that adds the empirical term $(M - 1)(1 - \log(2\pi)/2)$ before taking the $\epsilon \rightarrow 0$ limit of the logistic expansion, which the reference itself requires to stay bounded away from zero; `le` evaluates the limit directly and keeps the resulting $O(1)$ bias, so it is retained unmodified as the published form while `ble` is what `default` now routes normally-loaded closed models through, having measured a lower median error across the whole example suite. The correction cancels in throughput ratios $G(\mathbf{N} - \mathbf{e}_r)/G(\mathbf{N})$, so it changes only the reported normalizing constant, not the mean-performance metrics derived from it. `bkt` plays the same role for `kt`: steepest descent on the generating function carries Stirling’s approximation of $\log N_r!$ in place of $\log N_r!$ in every class direction, and `bkt` subtracts that remainder, $\log N_r! - (N_r \log N_r - N_r + \frac{1}{2} \log 2\pi N_r)$, exactly, for each class the saddle point expands. With a think time `bkt` and `ble` coincide to the accuracy of their saddle-point solvers, being the same estimator computed in R and in M dimensions, so the cheaper of the two is whichever of R and M is smaller; without a think time they differ by a constant depending only on the total population. `lekt` exposes that identity as a method: it computes the common estimator on whichever side is cheaper, the Knessl–Tier saddle point when $R \leq M$ (or when a class self-loops, which that side extracts exactly) and the logistic mode otherwise, and without a think time it carries the constant

$M(1 - \log(2\pi)/2) - r(N + M)$ on the logistic side, r the Stirling remainder, so that the two sides agree there as well.

The method `aghq` acts on the simplex factor of the McKenna-Mitra integral rather than on the logistic transform of the whole integrand, and reads the mode and curvature that `le` already computes as a quadrature rule instead of as a closure. It evaluates that factor by adaptive Gauss-Hermite quadrature [101] with q nodes per simplex direction, set by `options.config.aghq_nodes` (default 3) at a cost of q^{M-1} integrand evaluations. The factor is where the error of `le` sits: without think times the integrand is homogeneous, so the radial direction separates and integrates exactly, leaving the simplex integral as the only approximated quantity. At $q = 1$ the single node is the mode and the method reproduces `le` exactly, so raising q buys accuracy in integrand evaluations rather than in a different approximation.

The `nre` method evaluates the same Norlund-Rice integral as `nrl` and `nrp`, but by steepest descent rather than by a Gaussian fit on the contour of unit radii. Two properties of the integrand are exploited. First, the integrand is invariant under a common shift of the class variables, since it is homogeneous of degree $\sum_r N_r$ and that degree cancels against the coefficient-extraction kernel; the redundant direction is therefore quotiented out and the integral is $(R - 1)$ -dimensional, whereas `nrl` and `nrp` integrate over R dimensions and let the logit or probit Jacobian supply a curvature term along the null direction that belongs to the change of variables rather than to the model. Next, the contour radii are tilted per class to the saddle point, i.e., to the $\mathbf{X}$ solving $X_r \partial \log h / \partial X_r = N_r$, obtained by a damped Newton iteration on the convex cumulant generating function of the tilted distribution; on the untilted contour the origin is not a stationary point of the phase, which is the leading source of bias of the two earlier methods. A second-order Edgeworth correction built from the third and fourth cumulants at the saddle [9, 49] is then applied, giving a relative error of order $O(1/(\sum_r N_r)^2)$. Every evaluation of the integrand sits at real positive demands, so, unlike `nrl` and `nrp`, the method requires no complex arithmetic and remains in the log domain throughout; its cost is $O(IR^2 + R^4)$ evaluations of a single-class load-dependent normalizing constant, hence polynomial in the number of classes.

The Birman-Kogan family [15] covers `bk` (saddle-point evaluation of the generating function, with an exact pole treatment for chains whose dedicated station is a bottleneck), `bkue` (the single-chain uniform-expansion variant that keeps the pole and the saddle together through an `erfc` term when they are close, falling back to `bk` on a multiclass model) and `lc/lc.ue` (Algorithm 2 of the reference, a Gauss-Seidel sweep that solves each chain against the residual capacity left by the others, with `mva` or `ue` as the inner single-chain solver).

The remaining method is `mem`, the Maximum Entropy Method [92] for open, closed and mixed networks with general (GE-type) arrival and service processes, exact for $M/G/1$, $M/M/c$ and infinite-server stations. It admits Source, Queue, Delay and Sink nodes with FCFS, PS, SIRO, LCFS, LCFS-PR and INF scheduling, rejecting priority disciplines, class switching and reducible routing; multiserver stations are supported in open models, and mixed models are restricted to single-server and infinite-server stations. It is used only on explicit request: `default` never selects it. Convergence is controlled by `options.config.mem_tol` (default 10^{-6}), `mem_maxiter` (default 1000) and `mem_verbose`. It is also the only NC method aware of a finite station buffer, every other method being product-form, so that a model carrying a binding `set_capacity` is rejected outright rather than solved as if the buffer were unbounded. That path is reported as `mem.blocking` and is restricted to single-class open models with FCFS capacitated stations whose arrival and service

processes have a squared coefficient of variation of at least one.

For cache models, the importance sampling method `sampling` estimates hit probabilities and miss rates by Monte Carlo, which is useful for large cache configurations where the exact methods become expensive:

```
nc::NcSolverOptions opt;
opt.method = "sampling";
opt.samples = 100000;
const mva::AvgResult<double> res = nc::solver_nc_run_analyzer(model.get_struct(), opt);
```

The algorithms behind these methods are documented in the internals manual, Chapter NC.

NC method selection. The normalizing constant solver provides both exact and approximate methods. For exact computation, `ca` (convolution algorithm) and `comom` (class-oriented method of moments) are available, with `comom` being generally faster for models with many classes. Among the approximate methods, `nr.logit` and `nr.probit` use Norlund-Rice integrals with different variable transformations and are suitable for large populations, while `nre` evaluates the same integral on a saddle-tilted contour with a second-order Edgeworth correction, which is markedly more accurate on load-dependent models at the same asymptotic cost. The `rd` method applies a reduction heuristic that recursively simplifies the network. The `cub` method employs cubature rules for numerical integration of the normalizing constant. The `mem.blocking` variant extends the same building block to finite station buffers, which no other NC method can represent: a capacitated station becomes a censored $GE/GE/c/0; N$ queue, exact for $M/M/1/N$ and $M/M/c/N$, and blocking after service is handled by the holding-node transformation of [153] that restores work conservation. For open networks, the `mem` method applies the Maximum Entropy Method [92] to estimate the joint queue-length distribution via a queue-by-queue decomposition into $GE/GE/1$, $GE/GE/c$ and $GE/GE/\infty$ building blocks; it is exact for $M/G/1$, $M/M/c$ and infinite-server stations and captures the propagation of non-exponential variability (including immediate feedback) through the network. The `default` method automatically selects `mem` through a benchmark-calibrated heuristic: open networks within its feature set whenever any arrival or service process has a non-unit squared coefficient of variation (except bursty external arrivals at low load, where the GE bound is loose), and closed networks that are non-product-form due to hypoexponential services or class-dependent FCFS rates; purely exponential models and hyperexponential closed FCFS models are instead solved by the native normalizing-constant path. For closed networks, `mem` implements the two-stage pseudo-open decomposition and convolution algorithm [92] on $G/G/1$ and $G/G/\infty$ building blocks; it conserves class populations exactly and reduces to the exact BCMP solution on Markovian single-class networks, but it is selected only on explicit request since the default path is exact for product-form closed models. Mixed open/closed models compose the two algorithms with product-form-style conditioning (capacity reduction of the closed subnetwork by the open load, inflation of the open queue lengths by the closed occupancy), exactly recovering mixed MVA in the product-form limit. The `oi` method analyzes closed order-independent (OI) networks, in which one or more stations serve jobs at a total rate that depends only on the multiset of classes present (the balanced-fairness rank rate [18]), possibly mixed with ordinary BCMP product-form stations: infinite-server (delay), processor sharing, LCFS-PR, SIRO and class-independent-rate FCFS, single- or multi-server. The normalizing constant is assembled

exactly over the population lattice, convolving the order-independent balanced-fairness recursion for the OI stations with the load-dependent BCMP weight tables of the remaining stations. The per-class mean queue lengths are then obtained without recourse to Little's law, from the order-independent functional-server identity $E[n_{i,r}] = G_{i,r}^+/G - 1$, the OI generalization of the load-dependent functional server, and the per-class throughputs from ratios of normalizing constants. The method is selected automatically, in place of the maximum-entropy heuristic above, whenever the closed model contains at least one order-independent station and every other station is BCMP product-form; a permutation-invariance check on the rate function guards the order-independence assumption at model-linking time. The `morrison` method is the one NC route that computes no normalizing constant at all, and it is the clearest illustration of the solver's wider remit. It analyzes the closed two-station network of [118]: one infinite-server (think) station and one single-server discriminatory-processor-sharing station, exponential service, each class alternating between the two. Unequal DPS weights destroy the product form, so there is no constant to expand; instead the substitution $P(\mathbf{n}) = \langle \mathbf{w}, \mathbf{n} \rangle f(\mathbf{n})$ clears the $1/\langle \mathbf{w}, \mathbf{n} \rangle$ denominators of the balance equations, whose generating function then satisfies a linear partial differential equation with affine coefficients. Rescaling about $z = 1$ and expanding in inverse powers of $\sqrt{N}$, in the moderately-heavy regime $\rho = 1 - a/\sqrt{N}$ with populations $K_j = Nb_j$, leaves a degenerate leading operator whose kernel is spanned by the functions of a single similarity variable; the solvability condition along its characteristic reduces the p -dimensional problem to one ordinary differential equation, whose bounded solution is a family of parabolic-cylinder integrals. The solver returns the resulting two-term approximations to the mean queue lengths and, by Little's law so that the reported table is self-consistent, to the mean sojourn times. Accuracy improves as the populations grow with the usage held near saturation, which is precisely the regime where a Markov chain is intractable and simulation converges slowly; the expansion also remains valid above saturation, where its leading term recovers the fluid approximation. It is selected automatically on that shape and is the only method admitted there, since every other NC route would silently drop the weights and answer with the egalitarian processor-sharing network; DPS models of any other shape are refused rather than approximated, and `logNormConstAggr` is returned as NaN.

8.2.8.2 Busy periods of a subnetwork

The busy period of a given order of a set of stations is the time from the instant an arriving job raises the jobs held by the set to that order, up to the next instant when the set falls back below it. At order one it is the ordinary busy period of the subnetwork, and at a high order it is a heavy-traffic period, so the family describes how long congestion of a given depth persists. Comparing idle and busy periods of a station yields its utilisation, and the busy periods of a high-priority class are the blocking times of the lower-priority ones.

The routine `pfqn_busyp`, and `solver_nc_busyp` which reads its inputs off a `NetworkStruct`, evaluates it exactly by the mean value analysis of [48]. The closed and the open forms are both a ratio: the numerator sums the normalizing constant of the subnetwork over every population at or above the order, weighted in the closed case by the normalizing constant of the complement at the matching residual population; the denominator takes the same constants one population lower and multiplies them by the total rate at which jobs enter the set from outside, which in the open case also collects the external arrivals. Both are

evaluated in the logarithmic domain, which keeps the individual normalizing constants from overflowing. The result is exact on single-chain product-form models and, being a function of the stationary distribution and of an ergodic flow rate alone, it is insensitive to the service time distributions beyond their means. A closed subnetwork must be a proper subset of the stations, since a busy period is started from outside it; an open subnetwork may be the whole network.

```
const std::vector<double> b = nc::solver_nc_busyp(model.get_struct(), {0, 1}, {1, 2, 3});
```

The same quantity can be measured by simulation, which is the recommended cross-check on models outside the product-form class: see §8.2.5.2.

8.2.9 SSA: stochastic simulation algorithm solver

The `SSA` class is a stochastic simulator for continuous-time Markov chains. It reuses parts of the `CTMC` machinery to generate network states and then samples the state dynamics by selecting among enabled events according to their rates. The sampling algorithms and state representation are described in the `LINE` internals manual (`LINE-internals-cpp.pdf`, Chapter `SSA`).

The state space is not generated upfront, but typically stored during the simulation, starting from the initial state. If the initialization of a station generates multiple possible initial states, `SSA` initializes the model using the first state found. The list of initial states for each station can be obtained using the `ctmc_get_state_space` function of the `Network` class.

8.2.9.1 Methods and configuration options

The `SSA` solver offers two general-purpose methods: `'serial'` and `'para'`. The serial method runs on a single core, while the parallel methods run on multicore. The `'default'` method resolves to `'nrm'` on models that support it (see below) and to `'serial'` otherwise.

`SSA` solver also implements the much faster next reaction method (`'nrm'`, see [2]). This is available for closed queueing models with exponential service under the `INF` and `PS` disciplines, as well as buffered `FCFS` and `LCFS` stations through integer (inverse-CDF) routing that preserves the per-buffer job order. The `'nrm'` method is selected by default on such models. Moreover, by setting `options.config.state_space_gen` to `it` it is possible to explicitly generate during the simulation the simulated state space and its steady-state probability.

`SSA` accepts a warmup-discard option `options.config.warmupfrac` (default `0.0`, disabled) that drops the leading fraction of collected samples before computing per-station means, mitigating transient bias on open networks; values of `0.1–0.3` typically bring open-network averages within $\sim 1\%$ of `LDES` (which uses `MSER-5` truncation). `SSA` also supports the full set of state-dependent dispatching strategies via a routing callback invoked at each event firing — `RAND`, `PROB`, `RROBIN`, `WRROBIN`, `JSQ`, `SQ`, with the `SQ` marginals matching the with-replacement, first-occurrence tie-break used by `LDES` and cached per $(d, ndest, n[\cdot])$ to amortise enumeration cost.

SSA method selection. The `serial` method implements Gillespie’s stochastic simulation algorithm on a single core, while `para` runs independent replicas in parallel for faster convergence. The `nrm` (next reaction method) offers significantly faster simulation for population models with processor-sharing or infinite-server scheduling, as it avoids recomputing all transition rates after each event.

8.3 Metasolvers

A metasolver does not analyze a model directly. It transforms the model, or decomposes it into a family of simpler ones, and delegates each resulting subproblem to one of the solvers of §8.2 or §8.4, which it may call many times; the features it admits and the accuracy it attains are therefore largely those of the solver it is given. Four are available in LINE, and only the last is documented here:

- `AUTO` chooses the back-end solver itself from structural features of the model, and falls back along a list of feasible solvers when the one it picked does not implement the requested operation (§1.3).
- `LN` solves a `LayeredNetwork` by decomposing it into layers and iterating a fixed point over their artificial delays, each layer being solved by a solver of the caller’s choosing (§6.5).
- `ENV` solves a model embedded in a random environment by analyzing one sub-model per environment stage and coupling the stages through their transition process (§6.6).
- `UQ` propagates uncertainty in the model parameters, solving one realization of the model per sample or quadrature node.

8.3.1 UQ: uncertainty quantification meta-solver

The `UQ` (uncertainty quantification) solver is a meta-solver that extends LINE’s capabilities to handle models with parameter uncertainty. In many practical scenarios, model parameters such as service rates or routing probabilities are not known precisely, but are instead characterized by a set of plausible alternatives with associated likelihoods. Rather than requiring the analyst to manually explore each alternative, the `UQ` solver automates this process by accepting parameters specified as probability distributions over discrete alternatives.

When a model contains parameters defined using the `Prior` distribution class, which encodes a discrete set of alternative parameter values along with their prior probabilities, the `UQ` solver detects these uncertain parameters and expands the model into a family of concrete instances. Each instance corresponds to one combination of parameter values from the alternatives. The solver then applies a user-specified inner solver to each instance, obtaining performance metrics for all alternatives. Finally, it aggregates the results using the prior probabilities as weights, producing expected performance metrics that account for the parameter uncertainty.

8.3.1.1 Methods and configuration options

The constructor for the `UQ` solver takes two arguments: the model containing uncertain parameters, and a solver factory that specifies which analysis method should be applied to each concrete instance. The solver

factory can be any of LINE's standard solvers such as MVA, CTMC, or JMT.

```
// the solver run at each design point
const uq::UqStageSolver<double> stage =
    [] (const qn::NetworkStruct<double>& sn) {
        mva::MvaOptions o;
        return mva::solver_mva_run_analyzer(sn, o, Matrix<double>());
    };
uq::UqOptions opt;
uq::SolverUq<double> post(model, stage, opt);
```

The per-design-point solver is a callable from a struct to an `AvgResult`, which is what a solver factory amounts to here.

The `UQ` solver provides several methods for retrieving results. The `solver_mva_run` method returns the prior-weighted expected values of all performance metrics, effectively averaging over the parameter uncertainty. The `method` returns the complete set of results for each alternative, allowing the analyst to examine how performance varies across different parameter values. Additionally, the `method` takes a metric type, node, and class as arguments and returns the posterior distribution of that metric as an `EmpiricalCDF` object, which can be used to compute quantiles or probabilities.

8.3.1.2 Priors

To specify parameter uncertainty, the analyst creates a `Prior` distribution by providing a list of alternative distributions and their corresponding probabilities. This prior distribution can then be used in place of a standard distribution when configuring model parameters such as service times.

```
const std::vector<Distrib<double>> alternatives{
    Exp(1.0),
    Exp(2.0),
    Exp(3.0)};
const std::vector<double> weights{0.3, 0.5, 0.2};
queue.set_service(jobClass, prior_discrete(alternatives, weights));
```

A `Prior` is a distribution like any other as far as the model is concerned; every solver except `SolverUq` refuses it by name.

In this example, the service time at the queue is uncertain, with three possible mean service rates having probabilities 0.3, 0.5, and 0.2 respectively. The `UQ` solver will analyze all three scenarios and provide both individual results and weighted averages. This capability is particularly useful for sensitivity analysis, robust design under uncertainty, and understanding the range of possible performance outcomes when input parameters cannot be measured precisely.

8.3.1.3 Interval parameters

Assigning probabilities to the alternatives is not always possible. An analyst may be able to bound a service demand between a best and a worst case, without being willing to say how the values in between are distributed. For this situation the `UQ` solver offers the `method`, which discards the prior weights and keeps

only the endpoints of the support, and its tabular form . Each metric is then returned as a pair of values, a lower and an upper endpoint, instead of a single expectation.

Two regimes are possible and the returned `exact` flag distinguishes them. If the model is a closed single-class network of load-independent single-server queues and delay stations, and the uncertainty affects service demands only, then the interval is computed by the algorithm of Lüthi and Haring [106], exposed at the API level as `pfqn_mva_interval`. Mean value analysis of such a network is monotone in each of its inputs: the throughput decreases in every demand and in the think time and increases in the population, while the queue length and the residence time of a station increase in the demand of that station and decrease in the demands of the others. The range of each output over the input box is therefore attained at a corner of the box, and the interval is obtained by a small number of ordinary MVA solutions, without expanding the model into a family of instances at all. The reported interval is exact: every value inside it is realized by some admissible choice of the parameters.

If the model does not meet these conditions, for instance because it has several job classes, the solver falls back on the range of the results across the instances it did solve, and reports `exact` as false together with the condition that was violated. This fallback covers the whole support when the uncertain parameters take finitely many values, but it is only an inner approximation when a parameter varies continuously, since the sampled values are interior points of its range rather than its endpoints.

Such an interval is the range of the model's own predictions over the set of admissible parameters, conditional on the true parameters lying inside the declared intervals. It says nothing about the accuracy of mean value analysis itself, and is therefore a different object from the bounds returned by the `BA` solver, which bracket the exact solution of a model whose parameters are known. The two must not be combined.

8.4 External solvers (wrappers)

The solvers in this section do not implement an analysis of their own: each drives a third-party tool as an external process. `LINE` transforms the model into the input format of that tool, runs it, and parses its output back into the `LINE` result tables, so that a wrapper is used exactly like a native solver and returns the same objects. Two consequences follow. First, the tool must be installed and reachable – on the system `PATH`, or at a location given through the options – and the solver is unavailable otherwise; the environment check of §2.1.2 reports which of them can be found. Second, the accuracy, the supported model features and the failure modes are those of the back end rather than of `LINE`.

8.4.1 JMT: Java Modelling Tools wrapper

The class is a wrapper for the `JMT` and consists of a model-to-model transformation from the `data` structure into the `JMT`'s input XML formats (either `.jsimg` or `.jmva`) and a corresponding parser for `JMT`'s results. Upon first invocation, the `JMT` JAR archive will be searched in the `MATLAB` path and if unavailable automatically downloaded.

8.4.1.1 Methods and configuration options

This solver offers several methods. The default method is the JSIM solver (`method`), which runs JMT's discrete-event simulator. For parallel simulation, the solver supports both serial and parallel execution methods. The alternative method is the JMVA analytical solver (`method`), which is applicable only to queueing network models that admit a product-form solution. This can be verified calling `isProductForm` prior to running the JMVA solver.

For transient analysis, the `'default'` method runs independent simulation replications and interpolates the resulting sample paths to compute transient averages whenever the `timespan` option is set to a finite interval (e.g., $[0, T]$ for some finite T). The number of replications is the configuration option `nsamples`, 10 by default, the same knob `SolverLDES` uses; replication k is seeded `seed+k-1`. An explicit `'jsim'` always performs a single run.

In the transformation to JSIM, artificial nodes will be automatically added to the routing table to represent class-switching nodes used in the simulator to specify the switching rules. One such class-switching node is defined for every ordered pair of stations (i, j) such that jobs change class in transit from i to j .

8.4.2 LQNS: layered queueing network solver wrapper

The `LQNS` class wraps the `lqns` analytical solver and the `lqsim` simulator of the LQNS suite [62]. On a `LayeredNetwork` model, the model is exported to the LQNS XML interchange format (`.lqnx`), the back-end command is run as an external process, and the resulting `.lqxo` file is parsed back into the layered result tables. The same class also accepts a flat `Model`, which it solves with the `qnsolver` tool of the same suite (§8.4.2.2), in the same way as JMT hosts both JSIM and JMVA. It is, together with the native metasolver `LN`, one of the two ways of solving a layered model in LINE, and the two are directly comparable on the same model, as shown in §6.5.

8.4.2.1 Methods and configuration options

- `options.method=default` or `lqns`: the `lqns` analytical solver with its default settings. The method name `lqns.default` is an alias retained for backward compatibility.
- `options.method=exactmva`: `lqns` with exact MVA in each layer instead of the default approximate one.
- `options.method=srvn` and `srvn.exactmva`: the same two, with SRVN layering.
- `options.method=sim` or `lqsim`: the `lqsim` simulator, run for `options.samples` events (passed as `-A`).

`options.config.multiserver` forwards a `-Pmultiserver=...` pragma that selects the LQNS multi-server approximation (`conway` [44], `reiser` [131], `rolia` [64, 136], `zhou` [172, 173], `suri` [152] or `schmidt` [143]; default maps to `rolia`), and is passed verbatim, so any pragma the installed LQNS understands is accepted. `options.config.remote` and `options.config.remote_url` dispatch the run to an `lqns-rest` service instead of a local binary, which is useful where the tools cannot be installed.

Table 8.6: LQNS solution methods on a (qnsolver)

Method	Flag	Description
default, qns	-mrolia or none	Sets <code>config.multiserver</code> to <code>rolia</code> , so a model with multi-server FCFS stations is solved by <code>-mrolia</code> ; without such stations no flag is emitted and <code>qnsolver</code> runs its built-in product-form solution.
qns.conway	-mconway	Conway’s multi-server MVA approximation [44], built on the multinomial all-servers-busy probability of de Souza e Silva and Muntz [51].
qns.reiser	-mreiser	Reiser-Lavenberg load-dependent MVA, exact for a product-form multi-server station [129, 131].
qns.rolia	-mrolia	Rolia’s multi-server approximation [136, 137], in the per-class <i>Rolia-Franks</i> form of [64].
qns.zhou	-mzhou	Zhou’s arrival-theorem binomial (AB) multi-server approximation [172, 173].
qns.suri	none	Suri-Sahu-Vernon approximation [152]. Sets <code>config.multiserver</code> to <code>suri</code> for compatibility with the homonymous option of MVA. <code>qnsolver</code> has no flag for it: without multi-server stations the back end runs its built-in solution, while a model with multi-server stations is refused on the <code>qnsolver</code> path.
qns.schmidt	none	Schmidt’s approximation [143]. Sets <code>config.multiserver</code> to <code>schmidt</code> for compatibility with the homonymous option of MVA. As for <code>qns.suri</code> , <code>qnsolver</code> has no flag for it, so a model with multi-server stations is refused on the <code>qnsolver</code> path.

LINE ships no LQNS binary, names no container image holding one, and never pulls one: LQNS is distributed by its authors under an evaluation agreement that forbids providing the software to third parties, so the binary must be one the user has obtained and is licensed for. It is available from <http://www.sce.carleton.ca/rads/lqns/>, and `lqns` and `lqsim` must be on the system `PATH`. The wrapper, its result mapping and remote execution are covered in full in §6.5.

8.4.2.2 qnsolver methods on a Network

When the model given to LQNS is a `rather` than a `LayeredNetwork`, the solver drives the standalone `qnsolver` command-line tool distributed with the LQNS suite. It performs a model-to-model transformation from the `data` structure into a JMVA-format input file (`.jmva`), invokes `qnsolver` as an external process, and parses the resulting per-station chain throughputs, queue-lengths, response times, and utilisations back into the LINE result tables. This path is recommended for product-form closed networks with multi-server FCFS stations, where it offers several well-known multi-server approximations not implemented natively by MVA. It replaces the former standalone QNS solver, which has been removed.

Product-form or open models are sent to `qnsolver`. A closed model without a product-form solution is instead converted automatically to a layered queueing network via `QN2LQN` and solved by `lqns`. The `qnsolver` executable is located on the system `PATH`; its presence can be checked with `,` and if it is missing the call aborts.

The methods available on a `are` listed in Table 8.6. Each one maps to a multi-server approximation of `qnsolver`, selected through the flag shown in the second column; for instance `.` The `qns*` methods are refused on a `LayeredNetwork`, and the layered methods listed above are not offered on a `.` The result tables report the method as `qns.conway`, `qns.rolia`, and so on, or as `default/qns.rolia` when default was requested.

The selected method is reflected in the actual back-end command line, e.g. `qnsolver -l model.jmva`

`-mconway -o result.jmva`, so users can replicate these runs by hand if needed.

Chapter 9

Parameter inference and optimization

The estimator that fits a model to measurement data, and the optimization layer that searches over model parameters, are model-level facilities: both drive a solver in a loop and rewrite the model between iterations. Model-level parameter inference is not yet part of the C++ port. Optimization is available through the public `line::opt` headers: `OptimizationProblem`, flat and layered decision variables, differential-evolution and bisection solvers, projected-gradient modes, Pareto sweeps, decomposition workflows, scenarios, and analytic sensitivity dispatch.

What the port does carry is the numerical layer underneath the estimator: `infer::infer_gibbs` (Gibbs sampling of queue-length observations), `infer::infer_qmle` and `infer::infer_rps` (maximum-likelihood and response-time estimators on a given demand structure), and, for layered models, `infer_lqn_ekf`, `infer_lqn_getobs`, `infer_lqn_jacobian` and `infer_lqn_findbyname`. The estimators that build an augmented `Network` and solve it inside a search, such as `infer_mlps` and `infer_minps`, have no C++ counterpart.

The executable studies in `cpp/examples/opt/opt.cpp` show the complete C++ workflow, including layered host-demand tuning and frozen layers.

9.1 Symbolic analysis backend

Some analyses are exact only in symbolic form, such as the stationary distribution of a Markov chain as a rational function of its transition rates. The backend is selected by `options.symbolic`, written by `SolverOptions::set_symbolic`, whose accepted values are `'auto'` (the default: the native engine, and an external service only where there is none), `'sage'` (always the service) and `'native'`. A service is located from an explicit URL, from the `LINE_SAGE_URL` environment variable, from a service already listening on port 8085 or 8080, or from a container started by `LINE` on a free port. Expressions are exchanged as plain infix strings with exact rational coefficients, so the exchange introduces no rounding.

Beyond those three values the field is read as an instruction: `'none'` (equivalently `'off'`) keeps every computation local and makes a method that needs algebra refuse rather than dial out, an `http://` URL names

a service to use directly, and anything else is taken as the name of a Docker image to start. The C++ HTTP client carries no TLS, so an `https://` URL is refused by name rather than attempted; terminate TLS in front of the service, or reach it over `http://`. Each request is bounded by `options.symbolic_timeout`, in seconds, which defaults to 300. That is a real knob and not a formality: a symbolic solve grows superpolynomially in the number of states, and the equilibria of a smoothed `softmin` drift are transcendental in e^{-p^*x} and outrun the default on models that are otherwise unremarkable.

Exact sensitivities and the fluid Jacobian

Exact parametric sensitivities follow from the symbolic solution by the chain rule, which leaves only the map from the parameter to the affected rates to be differenced; a parameter that reshapes an event rather than scaling it invalidates the chain rule and is refused with an explanatory error rather than approximated silently. The mean-field drift can likewise be exported as expressions and differentiated to give the Jacobian, provided the drift is differentiable: the methods that scale service rates by $\min(n_i, S_i)$ are not, and a smoothed variant is selected through `options.config.pstar`. The Jacobian is what tells a fixed point apart from a limit cycle and gives the local convergence rate of the fluid approximation, neither of which a numerical integration reports. Equilibria, the solutions of $f(x) = 0$, are returned only when asked for, and an empty answer records what the engine could solve in closed form rather than asserting that none exist.

```
const fluid::FluidSymSystem sys =
    fluid::fluid_symodes(model.get_struct(), "pnorm", 8.0, std::vector<double>());
const fluid::FluidSymbolicDrift d = fluid::fluid_symbolic_drift(sys);

fluid::FluidSymbolicOptions symopt;
symopt.backend = "sage";
symopt.equilibria = true;
const fluid::FluidJacobian J = fluid::fluid_jacobian(sys, symopt);
// J.J[i][j] is d f_i / d x_j; J.engine names what produced it
```

Symbolic stationary solution

The symbolic generator of §8.2.3.2 is assembled without any computer algebra, being linear in the symbols $x_1, \dots, x_E$; solving $\pi Q = 0$ with $\sum_s \pi_s = 1$ over the field of rational functions in those symbols is what needs a backend, and that is what `symbolic_solution` returns. The distribution comes back both entry by entry and split over one common denominator, the latter being the form to use when the entries are to be compared or combined, since it spares the engine a repeated `together`.

```
ctmc::CtmcSymbolicOptions symopt;
symopt.backend = "sage";
const ctmc::CtmcSymbolicSolution<double> s =
    ctmc::ctmc_symbolic_solution(model.get_struct(), ctmc::CtmcOptions(), symopt);
// s.pi over s.space, s.num over s.den, s.engine naming the backend that answered
```

The same result is reached from the solver facade as `SolverCTMC::symbolic_solution`, which reads the backend from the solver's own options.

A symbolic result is not portable as text. Symbol numbering follows event enumeration order, and the printed normal form depends on the engine and its version, so two codebases can return the same function under different spellings. Substitute rates and compare numbers instead: substituting each symbol with its event's smallest positive rate, the value the symbol was introduced to divide out, recovers the numerical stationary distribution exactly. Without a backend the getter raises, rather than quietly answering numerically under a symbolic name. The C++ port carries no computer algebra system of its own, so the service is the only route: `--arith exact` is rational arithmetic, not symbolic algebra.

Parametric sensitivity of a Markov chain

The sensitivity of a steady-state reward to a scalar model parameter θ follows Trivedi and Bobbio [158], Sec. 9.7, and is computed by `ctmc::solver_ctmc_sensitivity`. The parameter is described by a name, its nominal value and a setter that writes it onto the model, which always receives a private copy; the reward is a rate per state, given either as a vector over the state space or as a function of it. Both the unscaled sensitivity $dE[r]/d\theta$ of Eq. (9.79) and the scaled, dimensionless $(\theta/E[r]) dE[r]/d\theta$ of Eq. (9.80) are reported, together with the derivative of the distribution itself. Omitting the reward returns that derivative alone. Note that $dr/d\theta$ is taken to be zero: a reward whose own rates depend on θ needs the second term of Eq. (9.83) and is not covered.

Two methods are available, and they differ in *where* the differencing happens.

- `'fd'` (the default) central-differences the generator with the state space held fixed, then solves the differentiated balance equations once. The state space is unaffected by the perturbation because it depends on the topology and the cutoff and not on rate values, so the scheme is exact to $O(\text{step}^2)$ and needs no backend at all.
- `'symbolic'` solves π as a rational function of the rate symbols, differentiates that exactly with respect to each symbol, and closes with the chain rule $d\pi/d\theta = \sum_e (\partial\pi/\partial x_e)(dx_e/d\theta)$. Only the rate map $x_e(\theta)$ is still differenced, and that map is affine in θ in the common cases (a rate set to θ , or scaled by it), where a central difference reproduces it exactly. The whole $O(\text{step}^2)$ error of `'fd'` comes from differencing through the solve, which this avoids. A parameter whose perturbation reshapes an event's filtration instead of merely scaling it is refused, since the chain rule no longer holds.

The step defaults to $\max(|\theta|, 1) \cdot 10^{-6}$ and can be set on the parameter. Agreement between the two methods is evidence about both, since they share nothing but the model.

```
ctmc::CtmcSensParam<double> mu;
mu.name = "mu";
mu.value = 1.0;
mu.set = [] (qn::NetworkStruct<double>& x, double v) {
    x.service[1][0] = lang::Distrib<double>::exp_rate(v);
    x.refresh_rates();
};
ctmc::CtmcSymbolicOptions symopt;
symopt.backend = "sage";
const ctmc::CtmcSens<double> r = ctmc::solver_ctmc_sensitivity(
    model.get_struct(), ctmc::CtmcOptions(), mu, reward, "symbolic", symopt);
```

```
// r.S, r.SS (valid only if r.scaled_valid), r.dpi, r.pi
```

Measures and first passage times

Two further operations, `sym::SymEngine::ctmcMeasures` and `sym::SymEngine::ctmcPassage`, are exposed on the backend itself rather than on a solver, because they take a generator written in symbols and not a model. Every mean the CTMC solver reports is a *numeric linear functional* of π , so a caller can build one weight vector per measure with no algebra at all and recover each $w \cdot \pi$ as a rational function in a single round trip: a throughput is $\pi \cdot \text{depRates}$, a queue length $\pi \cdot \text{stateSpaceAggr}$, a marginal probability $\pi \cdot \text{indicator}$ and a mean reward $\pi \cdot r$. Weights are parsed in the same field as the generator, so an expression is admissible and not only a number. Ratios of two measures are requested by index, which is how a response time is obtained from a queue length over a throughput; a ratio whose denominator is identically zero comes back marked `undefined` with a reason, never as 0.

The passage operation gives the first passage time into a target set. Writing A for the complement of the target, $S = Q(A, A)$ for the sub-generator, $s_0 = -S\mathbf{1}$ for the exit vector and α for the initial law restricted to A , it follows Harrison and Knottenbelt [74],

$$L(s) = \alpha (sI - S)^{-1} s_0 + \text{atom}, \quad (-S) M^{(n)} = n M^{(n-1)}, \quad M^{(0)} = \mathbf{1},$$

where *atom* is the initial mass already inside the target set. The moments come from the recursion and not from differentiating *L*: the recursion carries no transform symbol, stays inside the same fraction field and costs `nmax` right solves against one matrix, whereas differentiating leaves the field and its evaluation at $s = 0$ can hit 0/0 wherever a factor of s failed to cancel. The transform symbol is adjoined to the field only when a transform is asked for and must differ from every rate symbol. Requests name what they want among `1st`, `1stall`, `moments` and `momall`, the `all` forms reporting one result per starting state instead of one for the initial law; states that cannot reach the target at all are reported separately rather than silently given an infinite time.

```

const std::shared_ptr<sym::SymEngine> engine = sym::sym_resolve("sage");
const std::vector<std::vector<std::string> > Q = {{ {"-x1", "x1", "0", "0"},
                                                    {"x2", "-x1-x2", "x1", "0"},
                                                    {"0", "x2", "-x1-x2", "x1"}},
                                                    {"0", "0", "x2", "-x2"}}};
const std::vector<std::vector<std::string> > W = {{ {"0", "1", "2", "3"},
                                                    {"0", "x1", "x1", "x1"}}};
const sym::CtmcMeasures m =
    engine->ctmcMeasures(Q, {"x1", "x2"}, W, {"QN", "TPUT"}, {});

const std::vector<std::vector<std::string> > S = {
    {"-x1", "x1", "0"}, {"x2", "-x1-x2", "x1"}, {"0", "x2", "-x1-x2"}}};
const sym::Passage p = engine->ctmcPassage(S, {"", "", "x1"}, {"1", "0", "0"},
    {"0", {"x1", "x2"}, "s"}, {"1st", "moments"}, 2);

```

On this chain the mean queue length and the throughput evaluate at $x_1 = 1$, $x_2 = 2$ to $11/15$ and $7/15$, and the first passage from empty to full has mean 11 and second moment 228. Evaluating is the way to check a symbolic result: the engines substitute values and compare numbers for exactly the reason given above, that printed normal forms are not comparable across engines.

Conclusion

LINE provides a common framework for specifying, solving, and validating queueing networks and related stochastic models across four language editions. The methods in this manual range from exact product-form and state-space algorithms to matrix-analytic approximations, fluid models, simulation, inference, and optimization. Their value depends on making solver assumptions and language support explicit, reporting reproducible numerical evidence, and selecting an analysis whose cost is commensurate with the research question. Continued development should therefore preserve cross-language parity while strengthening reproducibility, scalable state representations, and integration with empirical data. These objectives determine the long-term utility of LINE as both a research platform and an engineering tool for the performance analysis of networked and service-based systems.

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Appendix A

Examples

Table A.1: Examples

Example	Problem
ag_closed_network	Closed network solved through the agent-based interface
ag_gnetwork	G-network with negative customers and signals
ag_jackson_network	Jackson network solved through the agent-based interface
ag_multiclass_closed	Multiclass closed network solved through the agent-based interface
ag_tandem_open	Open tandem queue solved through the agent-based interface
cache_replc_rr	A small cache model with an open arrival process and random replacement
cache_replc_fifo	A small cache model with a closed job population and FIFO replacement
cache_replc_lru	A cache model with least-recently-used replacement
cache_replc_hlru	h-LRU ($\text{LRU}(m)$) replacement over h lists of capacities $m_1, \dots, m_h$
cache_replc_qlru	q-LRU replacement, where a missed item is admitted with probability q
cache_replc_climb	CLIMB (transposition) replacement, where a hit moves an item up one position
cache_compare_replc	Comparison of replacement policies across the CTMC, MVA and NC solvers
cache_replc_routing	A caching model with state-dependent output routing
cache_mmap_rr_env	MMap-fed cache with two correlated classes in a random environment
cache_rmfm_transient	Transient cache analysis by refined mean-field approximation
retrieval_simple	Cache with a single-queue retrieval system (delayed hits)
retrieval_ps	Retrieval system served by a single processor-sharing station
retrieval_chain	Cache with a chained retrieval system (delayed hits)
retrieval_routing	Cache with probabilistic per-item routing through a retrieval system
retrieval_default	Retrieval system whose per-item miss routing and service are taken by default
cdf_respt_closed	Station response time distribution in a single-class single-job closed network
cdf_respt_closed_threeclasses	Station response time distribution in a multi-chain closed network
cdf_respt_open_twoclasses	Station response time distribution in a multi-chain open network
cdf_respt_distrib	Simulation-based station response time distribution analysis
cdf_respt_populations	Station response time distribution under increasing job populations
cl_basic	Open cluster with four processor-sharing servers and random dispatching
cl_closed	Closed cluster with a think stage and three processor-sharing servers
cl_multiclass	Two-class open cluster, e.g. interactive against batch traffic
cl_compare	Comparison of random and round-robin dispatching on the same cluster
cl_sweep	Arrival-rate sweep on a two-server cluster
dispatch_open	Open cluster built through the <code>Cluster</code> builder
dispatch_closed	Closed cluster built through the <code>Cluster</code> builder
cqn_repairmen	Solving a single-class exponential closed queueing network

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Table A.1 – Examples. *Continued from previous page*

Example	Problem
cqn_repairmen_multi	Multi-server closed queueing network with repairmen
cqn_twoclass_hyperl	Solving a closed queueing network with a multi-class FCFS station
cqn_threeclass_hyperl	Solving exactly a multi-chain product-form closed queueing network
cqn_twoclass_erl	Closed network with round robin scheduling
cqn_multiserver	Local state space generation for a station in a closed network
cqn_multiserver_nc	Closed delay and multi-server FCFS queue solved by normalizing constants
cqn_twoqueues_multi	Closed queueing network with two multi-server queues
cqn_online	1-line exact MVA solution of a cyclic network of PS and INF stations
cqn_bcmp_theorem	Comparison of different scheduling policies that preserve the product-form solution
cqn_multichain_cs	Closed network with three chains, five classes and class switching
cqn_lcfs_multiclass	Multiclass closed network with LCFS and LCFS-PR scheduling
cqn_bas_blocking	Closed network with blocking-after-service (BAS) finite-buffer blocking
cqn_scheduling_dps	Parameterization of a discriminatory processor sharing (DPS) station
cqn_mmpp2_service	Automatic detection of solvers that cannot analyze the model
cs_implicit	Class switching with implicit routing
cs_single_diamond	Class switching controlled by a reducible Markov chain, single diamond pattern
cs_multi_diamond	Class switching controlled by a reducible Markov chain, multiple diamond patterns
cs_transient_class	Class switching with transient classes
ctmc_tandem_mml	Tandem of two M/M/1 stations solved by the CTMC solver
ctmc_spn_mml	M/M/1 queue modelled as a stochastic Petri net and solved by CTMC
ctmc_spn_vs_nc_closed_qn	Closed network as a stochastic Petri net: CTMC against NC comparison
est_ubr_open	Utilization-based regression (UBR) demand estimation on an open network
est_ubr_closed	UBR demand estimation on a closed network
est_ubr_changepoint	UBR sliding-window change-point detection on an open network
est_ubo_open	Utilization-based optimization (UBO) demand estimation on an open network
est_ubo_closed	UBO demand estimation on a closed network
est_ubo_changepoint	UBO sliding-window change-point detection on an open network
est_ubo_variants_closed	Comparison of UBO and maximum-likelihood estimators on a closed network
est_erps_open	Extended regression for processor sharing (ERPS) estimation, open network
est_erps_closed	ERPS demand estimation on a closed network
est_mle_open	Maximum-likelihood demand estimation on an open network
est_qmle_closed	Queue-length maximum-likelihood estimation on a closed network
est_mcmc_closed	Markov chain Monte Carlo estimation on a closed network
est_ekf_open	Extended Kalman filter estimation on an open network
est_ekf_closed	Extended Kalman filter estimation with automatic method selection
est_ekf_changepoint	Extended Kalman filter sliding-window change-point detection
est_trace_closed	Comparison of trace-based estimators (MLPS, FMLPS, Gibbs) on a PS station
fcr_lossn	Loss network analysis using finite capacity regions
fcr_constraints	Constraint types available for finite capacity regions
fcr_lincon	Linear admission constraints for finite capacity regions
fcr_mmlwaitq	Finite capacity region with blocking compared against an M/M/1 queue
fcr_mmlkdrop	Finite capacity region with dropping compared against an M/M/1/K queue
fcr_oqnwaitq	Finite capacity region with blocking in a multiclass open network
fcr_oqndrop	Finite capacity region with dropping in a multiclass open network
fes_single_class	Single-class flow-equivalent server (FES) aggregation
fes_aggregation	Flow-equivalent server aggregation of a subnetwork
ld_fes_singleclass	Single-class FES aggregation with verification of Norton's theorem
ld_fes_multiclass	Multiclass FES aggregation with verification of Norton's theorem
ld_multiserver_fcfs	Solving a single-class load-dependent closed model
ld_multiserver_ps_twoclasses	Solving a two-node multiclass load-dependent closed model
ld_multiserver_ps	Solving a three-node multiclass load-dependent closed model
ld_class_dependence	Load-dependent model with class dependence
fj_basic_open	A simple single class open fork-join network
fj_basic_closed	A simple single class closed fork-join network

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Table A.1 – Examples. *Continued from previous page*

Example	Problem
fj_tiny_closed	Minimal closed fork-join model solved exactly by the native CTMC solver
fj_twoclasses_forked	A multiclass open fork-join network
fj_basic_nesting	A closed model with nested forks and joins
fj_deep_nesting	Fork-join network with deep nesting
fj_nojoin	An open model with a fork but without a join
fj_serialfjs_open	Two open fork-join subsystems in tandem
fj_serialfjs_closed	Closed serial fork-join network
fj_complex_serial	Complex serial fork-join network
fj_cs_prefork	Fork-join network with class switching before the fork
fj_cs_postfork	Two-class fork-join with a class that switches into the other after the fork
fj_cs_multi_visits	Two fork-join loops within the same chain
fj_route_overlap	A model with overlapping routes in a fork-join network
fj_asymm	Asymmetric fork-join network
fj_delays	Fork-join network with delays
fj_threebranches	Fork-join network with three branches
init_state_fcfs_exp	Specifying an initial state and prior in a single class model
init_state_fcfs_nonexp	Specifying an initial state and prior in a multiclass model
init_state_ps	Specifying an initial state and prior in a model with class-switching
ldes_warmstart	Warm start of the LDES simulator from an auxiliary solver
lcq_singlehost	Single-host layered cache queueing network
lcq_threehosts	Three-host layered network with a multi-level cache
lqn_basic	Basic layered network example
lqn_serial	Analyze a layered network specified in a LQNS XML file
lqn_twotasks	Layered network with two tasks
lqn_multi_solvers	Specifying and solving a basic layered network with several solvers
lqn_setup	Layered network with a setup/delay-off task
lqn_server_pools	Layered network whose processor servers are not interchangeable, declared with a class-compatibility graph
lqn_workflows	Workflow modeling in layered networks
lqn_bpmn	BPMN to layered network transformation
lqn_moment3	Third-moment (higher-order) analysis of a layered network
lqn_transient	Transient (time-dependent) analysis of a layered network
lqn_paramident	Identification of hidden LQN parameters from measured performance
lqn_sockshop	Layered model of the Sock Shop microservice benchmark
lqn_ofbiz	Layered model of the Apache OFBiz enterprise application
largescale_tbi	Large-scale transient fluid analysis with trajectory-based iteration (TBI)
mam_transient_mapmapl	Transient analysis of a MAP/MAP/1 queue by matrix-analytic methods
mqn_basic	Solving a queueing network model with both closed and open classes
mqn_singleserver_ps	Mixed model with single-server PS stations
mqn_singleserver_fcfs	Mixed model with single-server FCFS stations
mqn_multiserver_ps	A difficult mixed model with sparse routing among multi-server PS nodes
mqn_multiserver_fcfs	Mixed model with multi-server FCFS nodes
mqn_multichain_cs	Mixed network with multiple chains and class switching
oqn_basic	Solving a queueing network model with open classes, scalar cutoff options
oqn_online	1-line solution of a tandem network of PS and INF stations
oqn_cs_routing	Solving a queueing network model with open classes, matrix cutoff options
oqn_multichain_cs	Open network with three chains, five classes and class switching
oqn_trace_driven	Trace-driven simulation of an M/M/1 queue
oqn_nhpp	Open network with a non-homogeneous Poisson (cyclic) arrival process
oqn_vsinks	A model illustrating the emulation of multiple sinks
oqn_fourqueues	A large multiclass example with PS and FCFS
opt_server_sizing	Minimum number of servers of an M/M/c queue under a utilization bound
opt_bisection_sizing	Exact server sizing by bisection, using $O(\log n)$ solver evaluations
opt_population_sizing	Largest closed-class population sustaining a service-level target

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Table A.1 – Examples. *Continued from previous page*

Example	Problem
opt_service_rate opt_load_balancing opt_decomposition opt_robust_sizing opt_pareto_frontier	Cheapest continuous service rate meeting a response-time target Routing split from a source to a fast and a slow PS station Two-variable problem (server count and service rate) solved by decomposition Worst-case server sizing across average-load and peak-load scenarios Cost against utilization-bound Pareto frontier by parametric sweep
pas_mmk pas_compatibility_5class pas_selfloop pas_saturation pas_closed_tandem_fig6 pas_cyclic_ctmc_vs_bruteforce pas_cyclic_normconst_bruteforce pas_nc_sampling	Pass-and-swap (P&S) station as a multiserver order-independent M/M/K queue Pass-and-swap queue with a five-class compatibility (swapping) graph Pass-and-swap queue whose swapping graph includes a self-loop Pass-and-swap saturation handling for large or unbounded buffers Closed tandem of two pass-and-swap queues Validation of the CTMC solver on a closed cyclic P&S network Brute-force normalizing constant of a cyclic P&S network Importance-sampling normalizing-constant analysis of a closed P&S network
polling_exhaustive_exp polling_exhaustive_det polling_gated polling_klimited switchover_basic	Exhaustive polling with exponential switchover times Exhaustive polling with deterministic switchover times Gated polling K-limited polling Basic switchover times model
prio_hol_open prio_hol_closed prio_psprio prio_identical	A multiclass open example with PS, SIRO, FCFS and head-of-line priority A high-load multiclass example with PS, SIRO, FCFS and head-of-line priority A repairmen model with PS priority scheduling Priority model with identical classes
renv_basic renv_twostages_repairmen renv_threestages_repairmen renv_fourstages_repairmen renv_node_breakdown renv_genqn renv_lqn_twostages renv_container_terminal renv_rotterdam_blending example_mapqn2renv	Introductory random-environment model Solving a model in a 2-stage random environment with exponential rates Solving a model in a 3-stage random environment with Erlang rates Solving a model in a 4-stage random environment with Coxian rates Node failure and repair through the random-environment breakdown API Random environment over an automatically generated queueing network Layered network operating in a two-stage random environment Daily-cycle container terminal in a random environment Semi-open queueing network with environment blending Transformation of an MMPP-service queue into a random-environment model
rewardModel_mmk rewardModel_multiclass rewardModel_aggregation rewardModel_templates	Reward-based CTMC analysis using custom reward functions (setReward, getReward, getAvgReward) Multi-class reward analysis Aggregation operations on rewards Predefined reward templates
sdroute_closed sdroute_twoclasses_closed sdroute_open sdroute_jsq	A model with round-robin routing A model with round-robin routing after multi-class PH and MAP service A load-balancer modeled as a router Join-the-shortest-queue state-dependent routing
solver_custom	Template for a user-defined solver plugged into the solver interface
statepr_aggr statepr_aggr_large statepr_sys_aggr statepr_sys_aggr_large statepr_allprobs_ps statepr_allprobs_fcfs	Computing marginal state probabilities for a node Computing marginal state probabilities for a node under class-switching Computing joint state probabilities for a system with two nodes under class-switching Computing joint state probabilities under class-switching and with delay nodes Computing probabilities under PS class-switching and with delay nodes Computing probabilities under PS and FCFS class-switching and with delay nodes
spn_basic_open spn_basic_closed spn_open_sevenplaces spn_closed_twoplaces spn_closed_fourplaces spn_twomodes	Simulation of a simple stochastic Petri net model Basic closed stochastic Petri net Simulation of a complex stochastic Petri net model Closed stochastic Petri net with two places Closed stochastic Petri net with four places Stochastic Petri net with two modes

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Table A.1 – Examples. *Continued from previous page*

Example	Problem
spn_fourmodes	Stochastic Petri net with four modes
spn_inhibiting	Stochastic Petri net with inhibiting arcs
spn_pareto_service	Stochastic Petri net with Pareto-distributed firing times
spn_queueing_place	Closed queueing Petri net (QPN) with two queueing places
tut01_mml_basics	M/M/1 queue basics
tut02_mgl_multiclass_solvers	Multiclass M/G/1 queue across solvers
tut03_repairmen	Machine interference (repairmen) problem
tut04_lb_routing	Round-robin load balancing
tut05_completes_flag	Use of the completes flag on a class
tut06_cache_lru_zipf	Queueing network with an LRU cache under Zipf-distributed references
tut07_delayed_hit_cache	Delayed-hit cache with a retrieval queue
tut08_respt_cdf	Response time distribution and percentiles
tut09_opt_load_balancing	Optimal load balancing
tut10_dep_process_analysis	Departure process analysis
tut11_lqn_basics	Layered queueing network basics
tut12_random_env	Random environments and the ENV solver
tut13_posterior_analysis	Posterior analysis with uncertain parameters
tut14_cluster	Open cluster model
tut15_loss_network	Finite-capacity loss network
tut15_bound_analysis	Bound analysis with the BA solver
tut16_queueing_petri_net	Closed queueing Petri net
tut17_colored_gspn	Colored generalized stochastic Petri net
wf_serial	Serial workflow model
wf_parallel	Parallel workflow model (AND-fork/join)
wf_branch	Probabilistic branching workflow model (OR-fork/join)
wf_loop	Workflow model with a loop
wf_erlang	Workflow with Erlang-distributed activity durations
wf_aph	Workflow with acyclic phase-type activity durations
wf_complex	Complex workflow combining sequence, branching and parallelism

Appendix B

Supported language features

B.1 Solver features

Once a model is specified, it is possible to use the `features` function to obtain a list of the features of a model. For example, the following conditional statement checks if the model contains a FCFS node. The C++ port has no feature-usage query on the model: each solver checks the struct against its own feature set and refuses what it cannot analyse, by name, when it is called. Every LINE solver implements the `supports` to check if it supports all language features used in a certain model. There is no `supports` predicate either: a solver that cannot analyse a model throws `UnsupportedError` naming the feature.

B.1.1 Which solvers can solve this model

The question a model raises first is not which features it uses but which solvers and solver methods can consume them, and `auto_find_solver` answers it directly from the model. `and` and `or` are aliases: the same question is asked both as “which solver do I use” and as “which method do I pass”, and the answer serves both.

The result carries one row per pair, with columns `Solver` (the method family), `Method` (the method name to pass as a solver method), `Runnable`, `Class`, `Metrics` and `Reason`. Because `Method` is a method name the solver constructor accepts, a row can be acted on without translation:

`Class` states what kind of answer the method returns: `exact`, `approx`, `bound` or `simulation`. Exactness is claimed of the *model* and not of the algorithm in the abstract, since the exactness of a normalizing constant or of mean value analysis is a property of the product-form model it is computed on: `mva.exact` reports `exact` on a product-form network and `approx` on one without a product-form solution, and the matrix-analytic methods report `exact` only on the single-queue shape the QBD is stated for. `Metrics` lists the measure groups the family can report – `avg`, `tran`, `cdf`, `prob`, `tranprob`, `sample`, `cache`, `loss`, `orbit`, `moment` and `sens` – and is what the optional first argument filters on; a group may equally be named by any accessor that returns it. `Reason` is empty on a runnable row and otherwise names the offending features, so a refusal says what to change rather than that something is unsupported.

The command line offers the same report, `line-cli -f model.json --find-solver`, optionally narrowed by a measure and optionally including the refused pairs (`--find-solver-all`).

The gate `findSolver` applies is the one solver selection already applies before delegating, asked of every candidate rather than of the first feasible one, and `list_valid_methods` is the `Method` column of its runnable rows. The two therefore cannot disagree.

A supported feature set is a necessary but not always sufficient condition for a method to run. The BA solver is the clearest example: its feature set is deliberately narrow, admitting closed product-form-parameterized models only, because bounds are computed from the service demands and populations alone. Several of its methods then impose further structural restrictions that the feature set cannot express, such as the absence of infinite-server stations for `sb` and `sib`, or a single-class single-server model for `lr`. Those methods raise an error on an inadmissible model rather than returning a value that would not be a valid bound. Consequently `may` accept a model for which a particular BA method is still unavailable, and `list_valid_methods` reports the method names rather than their per-model admissibility.

B.2 State-dependent routing and service

A model is state-dependent when its routing decisions or its service rates are functions of the current state. On the routing side the strategies are those of Section B.6: `RROBIN` cycles through the destinations in a fixed order, `WRRROBIN` does so through a list in which each destination appears as many times as its integer weight, and `JSQ` sends each job to the destination holding the fewest jobs. On the service side, `set_load_dependence` scales the station rate with its occupancy, which is how congestion effects, speed scaling and batch service are represented.

Support differs by solver. `CTMC` handles both families exactly, by carrying the dispatcher state and the occupancy-dependent rates in the state descriptor; the simulators `JMT`, `LDES` and `SSA` evaluate them on the sample path; `MVA` and `NC` handle load-dependent rates through their queue-dependent algorithms but relax state-dependent routing to its probabilistic counterpart. The example below dispatches jobs round-robin from a `Router` to three parallel queues.

```
Router router(model, "Dispatcher");
Queue queue1(model, "Queue1", SchedStrategy::FCFS);
Queue queue2(model, "Queue2", SchedStrategy::FCFS);
Queue queue3(model, "Queue3", SchedStrategy::FCFS);
model.set_routing(router, jobclass, RoutingStrategy::RROBIN);
Routing P;
P.set(router, queue1, 1.0);
P.set(router, queue2, 1.0);
P.set(router, queue3, 1.0);
```

The destination list is the set of arcs leaving the router: round robin expands uniformly over the nodes it is connected to. Examples demonstrating state-dependent routing can be found in the examples directory, including models that use `RROBIN` for load distribution and `JSQ` for dynamic load balancing.

Table B.1: Solver support for average performance metrics

Function	Regime	Network Solver							
		CTMC	LDES	FLD	JMT	MAM	MVA	NC	SSA
avg	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
avg_table	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
sn_deaggregate_chain_results	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
sn_deaggregate_chain_results	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::CN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::CN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::AN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::QN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::QN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::RN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::RN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::CN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::TN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::TN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::XN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::UN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
AvgResult::UN	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
	Steady-state	✓	✓	✓	✓	✓	✓	✓	✓
solver_ctmc_transient_analyzer	Transient	✓	✓	✓	✓	✓			

B.3 Class functions

The table below lists the steady-state and transient analysis functions implemented by the solvers. Since the features of the LINE solver are the union of the features of the other solvers, in what follows it will be omitted from the description.

The functions listed above with the `_table` suffix (e.g., `avg_table`) provide results in tabular format corresponding to the corresponding core function (e.g., `avg`). The features of the core functions are as follows:

- `avg`: returns the mean queue-length, utilization, mean response time (for one visit), and throughput for each station and class.
- `sn_deaggregate_chain_results`: returns the mean queue-length, utilization, mean response time (for one visit), and throughput for every station and chain.

Table B.2: Solver support for advanced metrics

Function	Regime	Network Solver							
		CTMC	LDES	FLD	JMT	MAM	MVA	NC	SSA
solver_ctmc_cdf_respt	Steady-state	✓	✓	✓	✓	✓			
	Steady-state			✓					
	Steady-state			✓					
solver_mva_get_prob_marg	Steady-state	✓	✓						
solver_mva_get_prob_aggr	Steady-state	✓	✓	✓	✓		✓	✓	
solver_mva_get_prob_marg	Steady-state					✓	✓	✓	
	Steady-state	✓	✓						✓
solver_mva_get_prob_sys_aggr	Steady-state	✓	✓		✓			✓	
solver_mva_get_prob_norm_const_aggr	Steady-state				✓		✓	✓	
	Steady-state			✓		✓	✓		
solver_ctmc_avg_reward	Steady-state	✓	✓						
ctmc_get_tran_prob	Transient		✓	✓					
	Transient		✓		✓				
	Transient	✓	✓						
	Transient	✓	✓						
	Transient	✓	✓						
	Transient	✓	✓						
	Transient	✓	✓						
solver_ctmc_tran_reward	Transient	✓	✓						
solver_ctmc_sample	Transient		✓						✓
solver_ctmc_sample_aggr	Transient		✓		✓				✓
solver_ctmc_sample_sys	Transient		✓						✓
solver_ctmc_sample_sys_aggr	Transient		✓		✓				✓

- **AvgResult : :CN:** returns the system response time and system throughput, as seen as the reference node, by chain.
- **solver_ctmc_cdf_respt:** returns the distribution of response times (for one visit) for the stations at steady-state.
- **solver_mam_passage_time:** returns the distribution of *passage* (first-visit) times at steady-state for the stations of interest, complementing the transient counterpart .
- **solver_mva_get_prob_marg:** returns the marginal queue-length distribution $P(n_{i,r} = k)$ for a single class r at a station i , as opposed to the joint per-class distribution returned by **solver_mva_get_prob_aggr**.
- **fluid_aoi:** returns the mean and variance of Age of Information (AoI) and Peak AoI for single-queue open systems with capacity constraints.
- **aoi_cdf:** returns the cumulative distribution function of AoI and Peak AoI.
- **node_metrics:** behaves similarly to **avg**, but returns performance metrics for each node and class. For example, throughputs at the sinks can be obtained with this method.
- **solver_mva_get_prob_marg:** returns state probabilities at equilibrium at a given station.

- `solver_mva_get_prob_aggr`: returns marginal state probabilities for jobs of different classes at a given station.
- `solver_mva_get_prob_sys_aggr`: returns joint probabilities for jobs of different classes at all stations.
- `solver_mva_get_prob_norm_const_aggr`: returns the normalizing constant of the state probabilities for the model.
- `solver_ctmc_transient_analyzer`: returns transient mean queue length, utilization and throughput for every station and chain from a given initial state.
- `fluid_tran_passage_time`: returns the distribution of first passage times in transient regime.
- `ldes_cdf_respt`: returns the distribution of response times in transient regime.
- `solver_ctmc_sample`: returns the transient marginal state for a station from a given initial state.
- `solver_ctmc_sample_aggr`: returns the transient marginal state for jobs of different classes at a given station from a given initial state.
- `solver_ctmc_sample_sys`: returns the transient marginal system state for a station from a given initial state.
- `solver_ctmc_sample_sys_aggr`: returns the transient marginal system state for jobs of different classes at a given station from a given initial state.
- `solver_ctmc_avg_reward`: returns the steady-state expected value $E[r] = \sum_s \pi(s) r(s)$ of each Markov reward function r defined via `set_reward`. The CTMC solver computes it exactly from the stationary distribution, while the LDES solver estimates it by time-integrating the reward over the exact joint state along the simulated sample path, so it is correct also for nonlinear rewards.
- `solver_ctmc_tran_reward`: returns the transient expected reward $E[r(X(t))]$ over time for each reward function. The CTMC solver uses the transient distribution from uniformization, while the LDES solver returns the reward trajectory sampled along the simulated path.

B.4 Node types

The table below shows the node types supported by the different solvers. It should be noted that the FLD solver is capable of handling `and` nodes, but due to low accuracy when run on open models this feature is disabled in the current release.

B.5 Scheduling strategies

The table below shows the supported scheduling strategies within LINE queueing stations. Each strategy belongs to a policy class:

- preemptive resume (`SchedStrategyType.PR`)

Table B.3: Solver support for nodes

Strategy	Network Solver							
	CTMC	LDES	FLD	JMT	MAM	MVA	NC	SSA
Cache	✓	✓	✓	✓		✓	✓	✓
ClassSwitch	✓	✓	✓	✓	✓	✓	✓	✓
Delay	✓	✓	✓	✓	✓	✓	✓	✓
Fork	✓	✓		✓	✓	✓		✓
Join	✓	✓		✓	✓	✓		✓
Queue	✓	✓	✓	✓	✓	✓	✓	✓
Sink	✓	✓	✓	✓	✓	✓	✓	✓
Source	✓	✓	✓	✓	✓	✓	✓	✓

- preemptive non-resume, i.e. restart with an independent resample (`SchedStrategyType.PNR`)
- non-preemptive (`SchedStrategyType.NP`)
- non-preemptive priority (`SchedStrategyType.NPPrio`)
- preemptive-resume priority (`SchedStrategyType.PRPRio`).

The `SchedStrategyType.PR` policy class includes scheduling strategies like `LCFSPR`, where a pre-empted job resumes service from the point it was interrupted. In contrast, the `SchedStrategyType.PNR` policy class includes strategies like `LCFSPI` (Last-Come-First-Served Preemptive-restart Independent), where a preempted job restarts from scratch with a new service time sample that is independent of the past.

The table primarily refers to invocation of the `avg` methods. Specialized methods, such as transient or response time distribution analysis, may be available only for a subset of the scheduling strategies supported by a solver.

The table lists both classical and specialized disciplines; the definitions of the latter are recalled in Table 3.2. One entry is worth isolating because it is not a scheduling discipline at all: `SQ` is a *routing* strategy (Section B.6), dispatching each job to the shortest of d randomly sampled queues. The specialized disciplines are supported mainly by `CTMC`, `LDES`, `JMT` and `SSA`, as the feature table below records. Worked models are in the `examples` folder, notably the exhaustive and gated polling examples and the `DPS` examples with class-dependent weights.

B.6 Routing strategies

The table below summarises which routing strategies are honoured by each network solver. Probabilistic strategies (`RAND`, `PROB`) reduce to the entries of the routing matrix, so every solver supports them. State-dependent strategies (`RROBIN`, `WRROBIN`, `JSQ`, `SQ`) require either an explicit dispatcher in the simulator (`LDES`, `JMT`) or a per-event marginal-probability callback (`SSA`). Analytical solvers (`MVA`, `NC`, `MAM`, `FLD`) approximate non-product-form policies by their probabilistic relaxation and are not appropriate for dispatching-sensitive analyses; `CTMC` handles `RROBIN`, `WRROBIN`, `JSQ` and `SQ` exactly, by augmenting the state descriptor with the dispatcher state, at the cost of an enlarged state space.

Table B.4: Solver support for scheduling strategies

Strategy	Class	Network Solver							
		CTMC	LDES	FLD	JMT	MAM	MVA	NC	SSA
FCFS	NP	✓	✓	✓	✓	✓	✓	✓	✓
INF	NP	✓	✓	✓	✓	✓	✓	✓	✓
SIRO	NP	✓	✓		✓		✓	✓	✓
SEPT	NP	✓	✓		✓				✓
SRPT	NP		✓		✓				
FSP	NP		✓						
SJF	NP		✓		✓				
FCFSPRIO	NPPrio	✓	✓		✓		✓		✓
EDD	NPPrio		✓		✓				
EDF	PRPrio		✓		✓				
PS	PR	✓	✓	✓	✓	✓	✓	✓	✓
DPS	PR	✓	✓	✓	✓		✓		✓
GPS	PR	✓	✓		✓				✓
PSPRIO	PRPrio	✓	✓		✓				✓
DPSPRIO	PRPrio	✓	✓		✓				✓
GPSPRIO	PRPrio	✓	✓		✓				✓
HOL	NPPrio	✓	✓	✓	✓	✓	✓		✓
LCFS	NP	✓	✓	✓	✓		✓	✓	✓
LCFSPR	PR	✓	✓	✓	✓		✓	✓	✓
LPS	PR	✓	✓		✓				✓
POLLING	NP	✓	✓		✓		✓		✓
PAS	NP	✓	✓				✓	✓	✓
OI	NP	✓	✓				✓	✓	✓

B.7 Statistical distributions

The table below summarizes the current level of support for arrival and service distributions within each solver. `Replayer` represents an empirical trace read from a file, which will be either replayed as-is by the JMT solver, or fitted automatically to a `Cox` by the other solvers. Note that JMT requires that the last row of the trace must be a number, *not* an empty row.

Table B.5: Solver support for routing strategies

Strategy	Network Solver							
	CTMC	LDES	FLD	JMT	MAM	MVA	NC	SSA
RAND	✓	✓	✓	✓	✓	✓	✓	✓
PROB	✓	✓	✓	✓	✓	✓	✓	✓
RROBIN	✓	✓		✓				✓
WRROBIN	✓	✓		✓				✓
JSQ	✓	✓		✓				✓
SQ	✓	✓		✓				✓
DISABLED	✓	✓	✓	✓	✓	✓	✓	✓

Table B.6: Solver support for statistical distributions

Distribution	Network Solver							
	CTMC	LDES	FLD	JMT	MAM	MVA	NC	SSA
APH	✓	✓		✓	✓	✓	✓	✓
BMAP					✓			
Coxian	✓	✓	✓	✓	✓	✓	✓	✓
Cox2		✓	✓	✓				
Exp	✓	✓	✓	✓	✓	✓	✓	✓
Erlang	✓	✓	✓	✓	✓	✓	✓	✓
HyperExp	✓	✓	✓	✓	✓	✓	✓	✓
MAP	✓	✓		✓	✓			✓
ME		✓		✓	✓			
MMPP2	✓	✓		✓	✓			✓
PH	✓	✓		✓				✓
RAP		✓		✓	✓			
Disabled	✓	✓	✓	✓	✓	✓	✓	✓
Det		✓		✓				
Gamma		✓		✓				
Lognormal		✓		✓				
Pareto		✓		✓				
Replayer		✓		✓				
Uniform		✓		✓				
Weibull		✓		✓				

Appendix C

Solver options

Solver options are encoded in LINE in a structure array that is internally passed to the solution algorithms.

This can be specified as an argument to the constructor of the solver. For example, the following two constructor invocations are identical

```
SolverMVA default_solver(model);  
SolverOptions options;           // shared defaults  
SolverMVA explicit_solver(model, options);
```

`SolverOptions` is the shared public option surface. Solver-specific option structures remain available below the facade for specialized analyzers. Modifiers to the default options can either be specified directly in the `options` data structure, or alternatively be specified as argument pairs to the constructor, i.e., the following two invocations are equivalent

```
SolverOptions options;  
options.set_samples(1000000);  
SolverSSA solver(model, options);  
AvgTable result = solver.avg_table();
```

Available solver options are as follows:

- `C++ SolverOptions` does not expose the object-level `cache` option used by the other editions; a facade solver caches its first result until `reset()` is called.
- `config(struct)` this is data structure to pass solver-specific configuration options to customize the execution of particular methods.
- `cutoff(integer  $\geq 1$ )` requires to ignore states where stations have more than the specified number of jobs. A finite cutoff is required to analyze open classes using the CTMC solver; it defaults to 10 and is auto-adjusted for open and mixed models.
- `force(logical)` requires the solver to proceed with analyzing the model. This bypasses checks and therefore can result in the solver either failing or requiring an excessive amount of resources from the system.

- `iter_max` (integer ≥ 1) controls the maximum number of iterations that a solver can use, where applicable. If `iter_max` = n , this option forces the `FLD` solver to compute the ODEs over the timespan $t \in [0, 10n/\mu^{\min}]$, where $\mu^{\min}$ is the slowest service rate in the model. For the `MVA` solver this option instead regulates the number of successive substitutions allowed in the fixed-point iteration.
- `iter_tol` (double) controls the numerical tolerance used for convergence of iterative methods. In the `FLD` solver this option regulates both the absolute and relative tolerance of the ODE solver.
- `level` (integer ≥ 1 , BA only) sets the depth of the bound hierarchies `pbh`, `bjbh`, `cbh` and `sib`, and defaults to 2. Raising it spends more computation and tightens the bracket monotonically toward the exact solution, which is reached at a finite depth: at `level` equal to the closed population for `pbh` and `bjbh`, and at `level` equal to the number of stations for `cbh`. The option has no effect on the noniterative families or on the reduction bounds. See Section 8.2.2.
- `init_sol` (solver dependent) re-initializes iterative solvers with the given configuration of the solution variables. In the case of `MVA`, this is a matrix where element (i, j) is the mean queue-length at station i in class j . In the case of `FLD`, this is a model-dependent vector with the values of all the variables used within the ODE system that underpins the fluid approximation. In the case of `LDES`, this is a matrix of initial queue lengths used for transient analysis, where jobs are distributed according to this matrix at simulation start instead of placing all closed class jobs at reference stations.
- `confint` (logical or real $\in (0, 1)$) enables confidence interval reporting for simulation-based solvers (`JMT`, `SSA`, `LDES`). When set to `true`, a 95% confidence interval is used. When set to a numeric value between 0 and 1 (e.g., 0.99 for 99% confidence), that confidence level is used. When enabled, the `getAvgTable` output displays metrics in the format `mean  $\pm$  half-width`. Default is `false` (disabled).
- `cimethod` (string, `LDES` only) specifies the confidence interval computation method. Valid values are `'obm'` (overlapping batch means, default), `'bm'` (non-overlapping batch means), `'spectral'` (Heidelberger–Welch spectral analysis), or `'none'` (disable CI computation). The `'spectral'` method estimates the spectral density at frequency zero via quadratic log-periodogram regression at low frequencies, accounting for autocorrelation between batches and producing wider but more statistically honest confidence intervals than batch means methods.
- `ciminbatch` (integer ≥ 2 , `LDES` only) minimum batch size for confidence interval computation. Actual batch size is $\max(\text{ciminbatch}, \sqrt{n})$ where n is the sample count. Default is 10.
- `ciminobs` (integer ≥ 10 , `LDES` only) minimum number of post-warmup observations required before computing confidence intervals. Default is 100.
- `config.variates` (string, `LDES` only) specifies the variance reduction method for simulation. Valid values are `'none'` (no variance reduction, default), `'antithetic'` (antithetic variates using synchronized 1-U method to generate paired samples with negative correlation), `'control'` (control variates using mean-based correction that applies post-hoc corrections based on deviation of sampled means from known theoretical means), or `'both'` (combined antithetic and control variates). Default is `'none'`.
- `events` (integer ≥ 1 , `SSA` and `LDES`) event budget of the discrete-event simulation solvers: the number

of reaction firings simulated by SSA, and the number of service completion events simulated by LDES. When set, it overrides `samples`, which remains accepted as a deprecated alias for the event budget; when unset (default), the solvers fall back to `samples` (10^4 events for SSA, $2 \cdot 10^5$ for LDES).

- `keep` (logical) determines if the model-to-model transformations store on file their intermediate outputs. In particular, if `verbose` ≥ 1 then the location of the `.jsimg` models sent to JMT will be printed on screen.
- `method` (string) configures the internal algorithm used to solve the model.
- `samples` (integer ≥ 1) statistical sample budget of the simulation- and sampling-based solvers; its unit is solver-dependent. For JMT it is the number of samples collected *for each* performance index, with a required minimum of $5 \cdot 10^3$. For NC, it is the number of Monte Carlo draws used by the sampling-based normalizing-constant methods. For SSA and LDES, it is a deprecated alias of `events` (see above) and is used only when `events` is not set.
- `seed` (integer ≥ 1) controls the seed used by the pseudo-random number generators. For example, simulation-based solvers will give identical results across invocations only if called with the same `seed`.
- `stiff` (logical) requires the solver to use a stiff ODE solver.
- `timeout` (double, seconds, default `Inf`) sets a cooperative wall-clock budget measured from solver launch. The budget is polled at checkpoints inside the long-running phases of the analysis, in particular during state-space generation and inside iterative analyzers. When the budget is exhausted the solver raises an error rather than returning a partially converged or partially enumerated result, so that a timed-out run cannot be mistaken for a completed one. The default `Inf` disables the check.
- `timespan` (real interval) requires the transient solver to produce a solution in the specified temporal range. If the value is set to `the` solver will only return a steady-state solution. For the FLD solver and in simulation, this setting has the same computational cost of `the` therefore the latter is used as default for this solver.
- `timestep` (double) controls the fixed time interval for transient analysis in the CTMC solver. When specified, the solver generates equally-spaced time points instead of using adaptive time stepping. If not specified or set to empty, the solver uses adaptive ODE time stepping. This option only affects transient analysis and is ignored for steady-state computations.
- `tol` default numerical tolerance for all uses other than the ones where `iter_tol` is used.
- `tranfilter` (string, LDES only) specifies the transient (warmup) detection method. Valid values are `'mser5'` (MSER-5 automatic truncation, default), `'fixed'` (fixed fraction warmup removal), or `'none'` (no transient filtering). The SSA solver does not expose adaptive transient detection but supports a configurable warmup discard via `warmupfrac` (see below).
- `mserbatch` (integer ≥ 1 , LDES only) batch size for the MSER transient detection algorithm. Only used when `tranfilter` = `'mser5'`. Default is 5 (standard MSER-5).
- `warmupfrac` (real $\in [0, 1)$, SSA and LDES) fraction of collected samples discarded as warmup before computing steady-state averages. For LDES this is used only when `tranfilter` = `'fixed'` (default 0.2);

for SSA it controls the initial-sample discard directly (default 0.0, i.e. disabled). Values of 0.1–0.3 are typical on open networks.

- `obmoverlap` (real $\in [0, 1]$, LDES only) overlap fraction for overlapping batch means confidence intervals. A value of 0.5 means 50% overlap (standard OBM). Only used when `cimethod='obm'`. Default is 0.5.
- `spectralLowFreqFrac` (real $\in (0, 1]$, LDES only) fraction of the lowest periodogram frequencies used for the log-periodogram regression in the Heidelberg–Welch spectral method. A smaller value concentrates the regression near frequency zero, giving a more local estimate of the spectral density; a larger value uses more frequencies for a more stable but potentially biased estimate. Only used when `cimethod='spectral'`. Default is 0.25.
- `cnvgon` (logical, LDES only) enables convergence-based stopping. When enabled, simulation stops when the confidence interval half-width relative to the mean falls below the convergence tolerance for all metrics. Default is `false`.
- `cnvgtol` (real $\in (0, 1)$, LDES only) convergence tolerance threshold. Simulation stops when $(\text{CI half-width}/\text{mean}) < \text{cnvgtol}$ for all metrics. A value of 0.05 means 5% relative precision. Default is 0.05.
- `cnvgbatch` (integer ≥ 1 , LDES only) minimum number of batches required before checking for convergence. More batches provide more reliable confidence interval estimates. Default is 20.
- `cnvgchk` (integer ≥ 0 , LDES only) number of events between convergence checks. A value of 0 means auto-calculate as `samples/50`. Default is 0 (auto).
- `verbose` controls the verbosity level of the solver. Supported levels are 0 for silent, 1 for standard verbosity, 2 for debugging.

Table C.1: Solver option structs of the C++ port and their defaults

Struct	Fields	Defaults
<code>mva::MvaOptions</code>	<code>method, tol, iter_tol, iter_max, multiserver, highvar, np_priority</code>	"default", 1e-4, 1e-6, 1000, "default", "default", "default"
<code>nc::NcSolverOptions</code>	<code>method, tol, iter_tol, iter_max, highvar, samples</code>	"default", 1e-4, 1e-4, 1000, "interp", 100000
<code>ctmc::CtmcOptions</code>	<code>method, keep_filtration, cutoff, state_max</code>	"default", false, unset, 3000000
<code>mam::MamOptions</code>	<code>method, tol, iter_max, space_max</code>	"default", 1e-4, 100, 128
<code>fluid::FluidOptions</code>	<code>method, tol, iter_tol, iter_max, timespan_end, timestep, seed</code>	"default", 1e-4, 1e-4, 200, ∞ , 0.01, 23000
<code>ssa::SsaOptions</code>	<code>method, samples, seed, verbose</code>	"default", 10000, 23000, false
<code>ba::BaOptions</code>	<code>method, level</code>	"default", 2
<code>env::EnvOptions</code>	<code>method, sojourn, stage_solver, timespan_end, tran_points, iter_max, iter_tol</code>	"meanfield", "stochastic", "fluid", 100, 1001, 100, 1e-4
<code>ln::LnOptions</code>	<code>method, layer_solver, iter_max, iter_tol, tol, interlocking, relax</code>	"default", "mva", 200, 5e-3, 1e-4, true, "fixed"

Appendix D

Interoperability

This chapter collects the facilities that LINE offers for moving models and results in and out of the tool. We first describe the scripts that import external models and export LINE models to reusable scripts, then the portable JSON serialization format used for model interchange across the MATLAB, Java, and Python codebases, and finally the command-line interface (`line-cli`) that drives the solvers from the terminal.

D.1 Model import and export

LINE offers a number of scripts to import external models into `Network` object instances that can be analyzed through its solvers. The available scripts are as follows:

- `JMT2LINE` imports a JMT simulation model (`.jsimg` or `.jsimw` file) instance.

This script requires in input the filename and desired model name, and returns a single output, e.g., The C++ port reads no JMT model: its input formats are the LINE JSON model and the `.lqnx` layered interchange. where `sn` is an instance of the `class`.

JSON model serialization. LINE models can be saved to and loaded from a portable JSON format conforming to the `line-model.schema.json` specification. This enables model exchange across codebases and integration with external tools. Supported model types include `Network`, `LayeredNetwork`, `Workflow`, and `Environment`.

```
const Network model = io::read_network_json<double>("mymodel.json");
```

The C++ port reads the JSON model format the other implementations write, and does not write it: a model built in C++ is solved in place.

D.1.1 Supported JMT features

Table D.1.1 lists the JSIMgraph/JSIMwiz model features supported by the JMT2LINE transformation. We indicate as “Fully” supported a feature that is supported in the import and such that the resulting model can be solved in LINE using at least JMT. A feature with “Partial” support implies that some core aspects of this feature available in JSIM are not available in LINE.

A few notes are needed to clarify the entries with partial support:

- `Fork` and `Join` are supported with their default policies. Advanced policies, such as partial joins or setting a distribution for the forked tasks on each output link, are not supported yet.
- a single `Sink` and a single `Source` can be instantiated in a LINE model, whereas there is no such constraint in JMT.
- `Finite Capacity Region` is supported on export (LINE to JMT), but the JMT2LINE import does not yet parse JSIM blocking regions.

Table D.1: Supported JSIM features for automated model import and analysis

JMT Feature	Support	Notes
Distributions	Full	Phase-Type, Burst (MAP), Burst (MMPP2), Deterministic, Disabled, Exponential, Erlang, Gamma, Hyperexponential, Coxian, Lognormal, Pareto, Uniform, Zero Service Time, Replayer, Trace, Weibull
Classes	Full	Open class, Closed class, Class priorities
Metrics	Full	Number of customers, Residence Time, Throughput, Response Time, Throughput per sink, Utilization, Arrival Rate
Nodes	Full	ClassSwitch, Place, Delay, Queue, Router, Transition
Routing	Full	Random, Probabilities, Round Robin, Weighted Round Robin, Power of k, Join the Shortest Queue
Mechanisms	Full	Polling, Setup times, Delay-off times, Switchover times, Impatience, Load Dependence, Retrial, Soft deadlines, Heterogeneous servers, Server Compatibilities
Scheduling	Full	FCFS, FCFSPRIO, LCFS, LCFS-PR, SIRO (Random), SJF, SEPT, LJF, LEPT, PS, DPS, GPS, PS Priority, DPS Priority, GPS Priority, LPS, EDD, EDF, SRPT, FSP
Nodes	Partial	Fork, Join, Source, Sink, Finite Capacity Region
Distributions	No	Burst (General), Normal
Nodes	No	Scaler, Semaphore
Routing	No	Shortest Response Time, Least Utilization, Fastest Service, Load Dependent, Class Switch Routing
Metrics	No	Drop rate, Response time per sink, Power
Scheduling	No	TBS, QBPS
Mechanisms	No	Parallelism

D.2 JSON model format

LINE defines a portable JSON format for saving and loading queueing network models. The format is supported across the MATLAB, Java and Python-native codebases and enables model interchange without

loss of information. A formal JSON Schema specification is available at `doc/line-model.schema.json` in the LINE distribution.

The reference for this format is the implementation, that is the four bridges

- `matlab/src/io/linemodel_save.m`
- `matlab/src/io/linemodel_load.m`
- `python/line_solver/io/linemodel_io.py`
- `jar/src/main/java/jline/io/LineModelIO.java`

Where a reader accepts more than one spelling of a key, the spelling emitted by the writers is described below as *canonical* and the remaining ones as *legacy aliases*.

D.2.1 API

Models are saved and loaded using each codebase's JSON I/O functions:

```
const Network model = io::read_network_json<double>("mymodel.json");
// an Environment model is read by io::read_environment_json<double>(path)
```

D.2.2 Top-level structure

Every LINE JSON file has the following envelope:

```
{
  "format": "line-model",
  "version": "1.0",
  "model": { ... }
}
```

The model object must contain a `type` field set to one of `Network`, `LayeredNetwork`, `Workflow`, or `Environment`. Any other value is rejected by the loaders.

Models held in a foreign format (`.lqnx`, `.jmva`, `.jsimg`) are not referenced from within a LINE JSON file; they are passed directly to the tool that reads them, which dispatches on the file extension. See Section [D.3](#) for the `line-cli` behaviour.

D.2.3 Network models

A `Network` model describes an extended queueing network. Its fields are:

Table D.2: Network model fields

Field	Type	Description
<code>type</code>	<code>"Network"</code>	Discriminator of the model kind. Required.
<code>name</code>	<code>string</code>	Model name. Node and class names must be unique within it. Required.

Field	Type	Description
nodes	array of Node	Node definitions, in the order that fixes the node indices used by the routing matrix. Required.
classes	array of JobClasses	Job class definitions, in the order that fixes the class indices. Required.
routing	RoutingMatrix	Routing probability matrix, given per class pair over the nodes. Required.
routingStrategies	{key: {key: RoutingStrategy}}	Per-node, per-class routing strategy: {nodeName: {className: strategy}}. Written at model level; the readers also accept it nested inside the 'routing' object and prefer the nested copy. Only strategies other than PROB and RAND are emitted, since the routing matrix already captures those.
routingWeights	{key: {key: {key: number}}}	Per-node, per-class, per-destination routing weights for WRROBIN: {nodeName: {className: {destNode: weight}}}. Written at model level; the readers also accept it nested inside the 'routing' object and prefer the nested copy.
routingParams	{key: {key: SQParams}}	Per-node, per-class parameters of the parameterized routing strategies: {nodeName: {className: params}}. Absent params make a SQ node fall back to d=2, which warns.
finiteCapacityRegions	array of FiniteCapacityRegion	Finite capacity regions constraining groups of stations jointly.
globalDependence	GlobalDependence	Global (Whittle) rate scaling $\phi(n)$ over the whole network state, declared through setGlobalDependence. Model level rather than per node, since its argument is not station-local.
rewards	array of Reward	Reward definitions evaluated on the state space by the CTMC solver.

The `routingStrategies`, `routingWeights` and `routingParams` objects are written at model level. The readers also accept them nested inside the `routing` object, and prefer the nested copy when both are present.

Nodes

Each node is an object with at least `name` and `type` fields. Table D.3 lists the node types emitted by the writers. Additional fields depend on the node type.

Table D.3: Node types

Type	Description
Source	Generates arrivals for open classes
Sink	Absorbs departing jobs
Queue	Queueing station with one or more servers
Delay	Infinite-server station
Fork	AND-fork for parallel branching
Join	AND-join for synchronization; references a paired Fork via <code>forkNode</code>
Router	Probabilistic routing node (no queueing)
Cache	Cache node with hit/miss class transformation

Type	Description
ClassSwitch	Deterministic class-switching node
Place	Place node for stochastic Petri nets
Transition	Transition node for stochastic Petri nets
Logger	Logging/monitoring node

A node of any other class is serialized under its own class name, obtained by reflection; such a node is not restored by the readers.

Table D.4 lists the fields available on station nodes (Queue, Delay, Source, Place).

Table D.4: Station node fields

Field	Type	Description
name	string	Node name, unique within the model. Routing and per-class maps key on it. Required.
type	Source, Sink, Queue, Delay, Fork, Join, ...	Node type. A node of any other class is serialized under its own class name, obtained by reflection, and is not restored by the readers. Required.
scheduling	SchedStrategy	Scheduling strategy of the station. Omitted for nodes that do not serve jobs.
servers	integer	Number of servers (default 1). There is no infinite sentinel: use a Delay node for an infinite-server station.
buffer	integer	Total buffer capacity; omit for infinite.
classCap	{key: integer}	Per-class buffer capacity: {className: capacity}.
dropRule	{key: DropRule}	Per-class drop rule: {className: rule}.
service	{key: Distribution}	Per-class service/arrival distribution: {className: Distribution}.
markedClasses	array of string	Source only: marked (MMAP) arrival binding; class names ordered by mark index, so mark k emits jobs of the k-th listed class. All listed classes share one MMAP arrival entry in 'service'.
schedParams	{key: number}	Per-class scheduling parameters (DPS weights, GPS shares, priorities): {className: value}.
serverTypes	array of ServerType	Heterogeneous server types for this node.
heteroSchedPolicy	ORDER, ALIS, ALFS, FAIRNESS, FSF, RAIS	Heterogeneous server scheduling policy.
switchoverTimes	array of object	Class-to-class switchover times: [{from, to, distribution}]. A per-class (rather than class-pair) switchover omits 'to'.
pollingType	GATED, EXHAUSTIVE, KLIMITED	Polling strategy for POLLING scheduling. Crosses as the uppercase enum name.
pollingPar	integer	K limit for KLIMITED polling.
loadDependence	LoadDependence	Load-dependent scaling factor f(n) at this station.
classDependence	ClassDependence	Class-dependent scaling $\beta_{i,r}(n)$ at this station. A sibling of loadDependence, not one of its modes.
jointDependence	JointDependence	Joint-dependent (non-product-form) scaling $\eta_i(n)$ at this station. A sibling of classDependence; the handle may read the joint per-class vector arbitrarily, so the demands are not product form.

Field	Type	Description
oiServiceRate	{key: {key: number}}	Order-independent per-state service rate table: {className: {state: rate}}.
oiCutoffs	array of integer	Per-class cutoffs of the oiServiceRate table.
swapGraph	array of array of number	Pass-and-swap graph, read only for a station scheduled PAS.
departureDiscipline	{key: N, FIFO}	Place only: per-class departure discipline {className: rule}.
balking	{key: object}	Per-class balking configuration. Keys are class names.
retrial	{key: object}	Per-class retrial configuration. Keys are class names.
patience	{key: object}	Per-class patience (reneging) configuration. Keys are class names. A node-scoped patience overrides the class-scoped one declared on JobClass.
orbitImpatience	{key: Distribution}	Per-class orbit impatience distribution: {className: Distribution}.
immediateFeedback	{key: boolean}	Per-class immediate feedback flag: {className: bool}.
setupTime	{key: Distribution}	Per-class server setup time: {className: Distribution}.
delayOffTime	{key: Distribution}	Per-class server delay-off time: {className: Distribution}.
initialState	number or array of number	Initial state vector of a stateful node: a Place's token counts, a Cache's contents, a closed pass-and-swap (PAS) queue's ordered job placement, and the state row of any other stateful node. Written for every stateful node that has one, because a reader calls the model initialized only when all of them do.
stateSpace	array of array of number	Explicit state space of a stateful node. Emitted together with statePrior.
statePrior	array of number	Prior probability over the rows of stateSpace.

Table D.5 lists the fields of the remaining node types.

Table D.5: Fields of non-station nodes

Field	Type	Description
classSwitchMatrix	{key: {key: number}}	For ClassSwitch nodes: explicit switching probability matrix. {fromClass: {toClass: probability}}.
csMatrix	array of array of number	For ClassSwitch nodes: legacy 2D array form of classSwitchMatrix, indexed by class order.
tasksPerLink	integer	For Fork nodes: number of tasks generated per outgoing link.
forkNode	string	For Join nodes: name of the paired Fork node.
joinStrategy	STD, PARTIAL, Quorum, QUORUM, Guard	For Join nodes: join strategy. The writers emit PARTIAL for a quorum join; the readers accept QUORUM as an alias of Quorum. An unrecognized value falls back to STD.
joinQuorum	integer	Quorum count for a quorum join.
itemSizes	integer or array of integer	Cache nodes: per-item storage cost (size), positive integers, one per item. Also readable as cache.itemSizes.

Field	Type	Description
costCaps	integer or array of integer	Cache nodes: cap on the total storage cost of the items resident in each list; a single value declares one cache-wide cap. Requires itemSizes. Also readable as cache.costCaps.
modes	array of TransitionMode	For Transition (SPN) nodes: firing modes.

Cache node fields are given in Section [D.2.3](#).

Table D.6: Scheduling strategies

Strategy	Description
INF	Infinite servers (Delay)
FCFS	First Come First Served
FCFSPR	FCFS with preemptive resume
FCFSPI	FCFS with preemptive restart
LCFS	Last Come First Served
LCFSPR	LCFS with preemptive resume
LCFSPI	LCFS with preemptive restart
SIRO	Service In Random Order
SJF	Shortest Job First
LJF	Longest Job First
PS	Processor Sharing
DPS	Discriminatory Processor Sharing (uses schedParams for weights)
GPS	Generalized Processor Sharing
SEPT	Shortest Expected Processing Time
LEPT	Longest Expected Processing Time
SRPT	Shortest Remaining Processing Time
SRTPRIO	SRPT with priorities
HOL	Head-Of-Line priority
FCFSPRIO	FCFS with priorities
FORK	Fork node pseudo-strategy
EXT	External world pseudo-strategy
REF	Reference station pseudo-strategy
POLLING	Polling discipline (uses pollingType, pollingPar)
PSPRIO	PS with priorities
DPSRIO	DPS with priorities
GPSRIO	GPS with priorities
LCFSRIO	LCFS with priorities
LCFSPRPRIOR	LCFS preemptive resume with priorities
LCFSPIPRIOR	LCFS preemptive restart with priorities
FCFSPRPRIOR	FCFS preemptive resume with priorities
FCFSPIPRIOR	FCFS preemptive restart with priorities
EDD	Earliest Due Date
EDF	Earliest Deadline First
LPS	Limited Processor Sharing
PSJF	Preemptive Shortest Job First
FB	Foreground-Background
LRPT	Longest Remaining Processing Time
SETF	Shortest Elapsed Time First
PAS	Pass and swap; the swap relation travels in swapGraph

Strategy names cross as uppercase enum names.

Cache nodes

Cache configuration is written as flat node-level keys. The readers accept both these keys and a nested `cache` object using the alternative names given in the last column of Table D.7, and prefer the nested object when both are present.

Table D.7: Cache node fields

Field	Type	Description
<code>numItems</code>	integer	Cache nodes: total number of cacheable items. Also readable as <code>cache.items</code> .
<code>itemLevelCap</code>	integer or array of integer	Cache nodes: capacity per list level. Also readable as <code>cache.capacity</code> .
<code>replacementStrategy</code>	ReplacementStrategy	Cache nodes: replacement policy. Also readable as <code>cache.replacement</code> .
<code>admissionProb</code>	number	Cache nodes: probability of admitting a missed item.
<code>hitClass</code>	{key: string}	Cache nodes: hit class mapping {inputClassName: outputClassName}.
<code>missClass</code>	{key: string}	Cache nodes: miss class mapping {inputClassName: outputClassName}.
<code>popularity</code>	{key: Distribution}	Cache nodes: per-class item popularity distributions {className: Distribution}.
<code>accessGraph</code>	array of array of number	Cache nodes: item access graph.
<code>accessProb</code>	array	Cache nodes: explicit per-class item access probabilities.
<code>retrievalSystem</code>	object	Cache nodes: delayed-hit retrieval configuration.
<code>cache</code>	CacheConfig	Nested alternative to the flat cache keys above. The writers emit the flat keys; the readers accept both and prefer this nested object when present.

Table D.8: Cache replacement policies

Policy	Description
RR	Random replacement
FIFO	First-in first-out
SFIFO	Strict first-in first-out
LRU	Least recently used
HLRU	h -LRU
CLIMB	CLIMB
QLRU	q -LRU

Load and class dependence

The `loadDependence` object specifies scaling of the service rate in the total station population n :

```
"loadDependence": { "type": "loadDependent", "scaling": [1.0, 1.8, 2.4] }
```

where scaling holds $f(n)$ for $n = 1, 2, \dots$

The `classDependence` object is a sibling key of `loadDependence`, not one of its modes. It specifies the class-dependent scaling $\beta_{i,r}(n)$ in the per-class population vector $\mathbf{n}$ at the station. The underlying object is a function handle, which cannot be serialized, so the writer materializes it over the per-class box lattice $0 \leq n_r \leq c_r$, where c_r is the r -th entry of `cutoffs`:

Table D.9: Class dependence fields

Field	Type	Description
<code>type</code>	<code>classDependent</code>	Shape of the class-dependent rate scaling.
<code>cutoffs</code>	array of integer	Per-class cutoffs, ordered by class index. A closed class is bounded by its population; an open class uses a saturation cutoff beyond which beta is constant.
<code>scaling</code>	{key: array of number}	Lattice table keyed by the comma-joined 0-based per-class counts ("n1,n2,...,nR"). Each value is the length-R vector [beta_1(n), ..., beta_R(n)]; a scalar-valued handle is broadcast to R equal entries by the writer.
<code>peak</code>	array of number	Declared peak (maximum) rate scaling per class [p_1,...,p_R], used to normalize utilization as $U_r = T_r * S_r / p_r$ (the same $T*S/c$ convention as multiserver stations). A scalar peak is broadcast to R entries by the writer. When absent (legacy models) the reader derives it as the lattice peak of scaling.

A reader rebuilds a handle that clamps the population to the cutoffs, so β saturates beyond the tabulated range. When `peak` is absent (legacy models) the reader derives it as the lattice peak of the scaling.

The `jointDependence` object is the non-product-form twin of `classDependence`, storing the joint-dependent scaling $\eta_i(\mathbf{n})$ (see §5.2.4). It is materialized over the same per-class box lattice and rebuilt with clamping; unlike `classDependence`, the handle may read foreign class marginals, so the demands are not product form.

Table D.10: Joint dependence fields

Field	Type	Description
<code>type</code>	<code>jointDependent</code>	Shape of the joint-dependent rate scaling.
<code>cutoffs</code>	array of integer	Per-class cutoffs, ordered by class index. A closed class is bounded by its population; an open class uses a saturation cutoff beyond which eta is constant.
<code>scaling</code>	{key: array of number}	Lattice table keyed by the comma-joined 0-based per-class counts ("n1,n2,...,nR"). Each value is the length-R vector [eta_1(n), ..., eta_R(n)]; a scalar-valued handle is broadcast to R equal entries by the writer.

Field	Type	Description
peak	array of number	Declared peak (maximum) rate scaling per class $[p_1, \dots, p_R]$, used to normalize utilization as $U_r = T_r * S_r / p_r$ (the same $T * S / c$ convention as multiserver stations). A scalar peak is broadcast to R entries by the writer. When absent the reader derives it as the lattice peak of scaling.

The `globalDependence` object is the model-level twin of the two above. It stores the global (Whittle) scaling $\phi(\mathbf{n})$ declared through `setGlobalDependence` (field `gdscaling`), whose argument is the whole $(M \times R)$ network state rather than one station's slice, so it is materialized over a lattice of that state rather than over a per-station box. Only the (station, class) `slots` a class can occupy carry a coordinate: a source holds no jobs, and a class with zero per-class capacity at a station never appears there, so those entries are pinned to zero. The restriction is lossless, since no service event ever fires at such a slot, and it is what keeps the lattice finite. The `slots` array names each coordinate by station and class, so a reordering on the writing side cannot shift one silently; `cutoffs` bounds each coordinate, a closed class at its own population and an open one at `cutoff`, the third argument of the setter. Each key is the comma-joined slot counts and each value the full $(M \times R)$ scaling flattened row-major, with `peak` flattened the same way. A reader rebuilds a handle that clamps the population to `cutoffs`, exactly as the two objects above do. A lattice above 2×10^5 points is refused by name rather than truncated: a silently shortened table would leave a delegated backend solving the same model at unscaled rates outside the tabulated range, disagreeing with the native solver without any diagnostic.

Table D.11: Global dependence fields

Field	Type	Description
type	globalDependent	Shape of the global rate scaling.
stations	array of string	Station names in station-index order, fixing the row order of the flattened $(M \times R)$ values below.
classes	array of string	Class names in class-index order, fixing the column order of the flattened $(M \times R)$ values below.
slots	array of object	The varying coordinates of the lattice, each named by station and class so a reordering on the writing side cannot shift one silently. Every (station, class) pair outside this list is pinned to 0.
cutoffs	array of integer	Per-slot cutoffs, in slot order. A closed class is bounded by its population, an open class by 'cutoff' below; the reader clamps to these, so ϕ saturates beyond the tabulated range.
cutoff	integer	The open-class truncation the writer used, i.e. the third argument of <code>setGlobalDependence</code> . It plays no part in solving and is carried so a reader restores the same declaration.

Field	Type	Description
scaling	{key: array of number}	Lattice table keyed by the comma-joined 0-based slot counts ("c1,c2,...,cP"), or "0" when no slot varies. Each value is the full (M x R) scaling flattened row-major, so the reader restores the matrix without re-deriving which return form the handle used.
peak	array of number	Declared peak (maximum) rate scaling, the (M x R) matrix flattened row-major, used to normalize utilization as $U = T \cdot S / \text{peak}$ (the same $T \cdot S / c$ convention as multiserver stations).

Impatience, balking and retrials

Three node-level objects, each keyed by class name, describe customer impatience:

```

"balking": { "Class1": { "strategy": "QUEUE_LENGTH",
                           "thresholds": [ { "minJobs": 5, "maxJobs": -1,
                                              "probability": 0.5 } ] } },
"retrial": { "Class1": { "delay": { "type": "Exp", "params": { "lambda": 2.0 } },
                           "maxAttempts": -1 } },
"patience": { "Class1": { "distribution": { "type": "Exp",
                                                "params": { "lambda": 0.1 } },
                           "impatienceType": "reneging" } }

```

The **balking** strategy is one of `QUEUE_LENGTH`, `EXPECTED_WAIT`, or `COMBINED`; a **maxJobs** of `-1` denotes infinity. The **maxAttempts** field is `-1` when the number of retrials is unlimited. The **impatienceType** is one of `reneging`, `balking`, or `retrial`.

A patience distribution may also be attached to a job class rather than to a station; see Section D.2.3. A node-scoped patience overrides the class-scoped one.

Petri net nodes

A **Place node** is a station: it carries scheduling, servers, service, departureDiscipline and initialState. A **Transition node** carries a modes array:

Table D.12: Transition mode fields

Field	Type	Description
name	string	Mode name, unique within its transition. Required.
distribution	Distribution	Firing time distribution.
timingStrategy	TIMED, IMMEDIATE	Whether the mode fires after a delay or immediately.
numServers	integer or "Infinity"	Number of firing servers; the string "Infinity" denotes an infinite-server mode.
firingPriority	integer	Integer priority breaking ties among simultaneously enabled modes; higher fires first.
firingWeight	number	Weight randomizing the choice among enabled modes of equal highest priority.

Field	Type	Description
enablingConditions	array of Arc	Token multiplicity required on each input place for the mode to be enabled.
inhibitingConditions	array of Arc	Token multiplicity on each input place above which the mode is inhibited.
firingOutcomes	array of Arc	Tokens deposited on each output place when the mode fires.

Every arc object has the same three fields: `node` (name of the Place), `class` (name of the job class), and `count` (number of tokens).

```
{ "name": "T1", "type": "Transition",
  "modes": [
    { "name": "model", "timingStrategy": "TIMED", "numServers": 1,
      "distribution": { "type": "Exp", "params": { "lambda": 1.0 } },
      "enablingConditions": [ { "node": "P1", "class": "Class1", "count": 1 } ],
      "firingOutcomes": [ { "node": "P2", "class": "Class1", "count": 1 } ] }
  ] }
```

Job classes

Each class is an object with at least `name` and `type` fields.

Table D.13: Job class fields

Field	Type	Description
name	string	Class name, unique within the model. Per-class maps key on it. Required.
type	Open, Closed, SelfLooping, Signal	Class type. SelfLooping specializes Closed, so the writers test for it first. Required.
population	integer	For Closed and SelfLooping classes: number of jobs.
refNode	string	Reference node name (Source for Open, home station for Closed/SelfLooping/ClosedSignal).
priority	integer	Class priority (0=highest).
deadline	number	Class deadline (Inf = no deadline).
isReferenceClass	boolean	True if the class is the reference class of its chain.
patience	Distribution	Class-scoped patience distribution. A node-scoped patience overrides it.
impatienceType	ImpatienceType	Qualifies the class-scoped patience.
replySignalClass	string	Name of the class carrying the reply of a reply signal. Without it, REPLY signals are inert after a round trip.
signalType	negative, reply, catastrophe	For Signal classes: signal type.
openOrClosed	Open, Closed	For Signal classes: whether the signal is open or closed.
targetClass	string	For Signal classes: name of the associated job class.
removalDistribution	Distribution	For Signal classes: distribution for batch removal count.
removalPolicy	random, fcfs, lcf s	For Signal classes: policy for selecting jobs to remove.

Since `SelfLooping` classes specialize `Closed` ones, the writers test for them first; a reader that ignores the `SelfLooping` value would silently obtain an ordinary closed class.

Routing

Routing is specified by a probability matrix. The `routing` object has its `type` field set to `matrix` and its `matrix` object uses composite keys of the form `"fromClass,toClass"`, each mapping to a nested dictionary `{fromNode: {toNode: probability}}`.

```
"routing": {
  "type": "matrix",
  "matrix": {
    "Class1,Class1": {
      "Source": { "Queue1": 0.5, "Queue2": 0.5 },
      "Queue1": { "Sink": 1.0 },
      "Queue2": { "Sink": 1.0 }
    },
    "Class1,Class2": {
      "Queue1": { "Queue2": 0.3 }
    }
  }
}
```

Only non-zero probabilities need to be specified. Same-class routing (e.g., `"Class1,Class1"`) describes routing without class switching; cross-class entries (e.g., `"Class1,Class2"`) describe class-switching transitions.

Routing strategies other than `PROB` and `RAND`, which the matrix already captures, are carried by the model-level `routingStrategies` object:

```
"routingStrategies": { "Router1": { "Class1": "RROBIN" } }
```

Table D.14: Routing strategies

Strategy	Description
RAND	Random selection with equal probability
PROB	Probabilistic routing, given by the routing matrix
RROBIN	Round-robin over the destinations
WRROBIN	Weighted round-robin; uses <code>routingWeights</code>
JSQ	Join the shortest queue
FIRING	Firing routing, used by Petri net transitions
SQ	Shortest queue of d , $SQ(d)$; uses <code>routingParams</code>
DISABLED	Class may not depart the node

`DISABLED` is emitted explicitly because the routing matrix records only positive probabilities and therefore cannot express it.

Weights for `WRROBIN` are given per node, class and destination:

```
"routingWeights": { "Router1": { "Class1": { "Queue1": 2.0, "Queue2": 1.0 } } }
```

The `routingParams` object mirrors that shape and carries the parameters of the parameterized strategy SQ, whose entry is `{ "d": int }`.

Omitting `routingParams` for a SQ node makes the reader warn and fall back to $d = 2$.

Finite capacity regions

A finite capacity region groups stations under a shared capacity constraint.

Table D.15: Finite capacity region fields

Field	Type	Description
name	string	Region name, unique within the model. Required.
stations	array of object	Names of the stations the region spans; the constraint applies to their joint population. Required.
globalMaxJobs	integer	Maximum total jobs in the region.
globalMaxMemory	integer	Maximum total memory in the region.
classMaxJobs	{key: integer}	Per-class maximum jobs: {className: maxJobs}.
classMaxMemory	{key: integer}	Per-class maximum memory: {className: maxMemory}.
constraintA	array of array of number	Linear constraint matrix A of $A*x \leq b$.
constraintB	array of number	Linear constraint capacity vector b of $A*x \leq b$.
dropRule	{key: DropRule}	Per-class drop rule: {className: rule}.

Rewards

The `rewards` array carries reward definitions for CTMC analysis in a declarative form:

```
"rewards": [ { "name": "q1", "type": "QLen", "node": "Queue1", "class": "Class1" } ]
```

The `type` field is one of `QLen`, `Util`, or `Blocking`, matching the `Reward.queueLength`, `Reward.utilization` and `Reward.blocking` templates. The `class` field is optional; when omitted the reward aggregates over all classes.

Only a reward created through a `Reward.*` template carries the structural metadata needed to reproduce it. A reward defined from a bare function handle, or through `Reward.custom`, is not reproducible from JSON: the writer warns and omits it rather than emit a reward that would be wrong on reload.

D.2.4 Distributions

Distributions are used throughout the format for service times, arrival processes, think times, and host demands. A distribution object always contains a `type` field and one of several parameterization modes.

Direct parameterization

The `params` object provides distribution-specific parameters.

Table D.16: Distribution types and parameters

Type	Parameters	Description
Exp	lambda	Exponential of rate λ . <code>rate</code> is accepted as a read-side alias
Det	value	Deterministic
Erlang	lambda, k	Erlang (per-phase rate λ , k phases)
HyperExp	p: [], lambda: []	Hyperexponential (weights and rates, n phases)
Coxian	mu: [], phi: []	Coxian (phase rates μ and completion probabilities ϕ)
Gamma	alpha, beta	Gamma (alpha shape, beta scale)
Lognormal	mu, sigma	Lognormal
Normal	mu, sigma	Normal
Uniform	a, b	Uniform over $[a, b]$
Pareto	alpha, scale	Pareto (alpha shape, scale scale)
Weibull	alpha, beta	Weibull. Note the roles: alpha is the <i>scale</i> and beta the <i>shape</i>
Zipf	s, n	Zipf (exponent, items)
Geometric	p	Geometric
Binomial	p, n	Binomial
Poisson	lambda	Poisson
Bernoulli	p	Bernoulli
Discrete Uniform	min, max	Discrete uniform
Discrete Sampler	p: [], x: []	Empirical discrete (probs and values)
EmpiricalCDF	x: [], F: []	Empirical CDF (support and cdf values)
NHPP	breakpoints: [], rates: [], cyclic	Non-homogeneous Poisson process, piecewise-constant intensity (cyclic when <code>cyclic</code> is true)
MAPt	breakpoints: [], D0: [[[]]], D1: [[[]]], cyclic	Time-inhomogeneous MAP: one (D_0, D_1) pair per segment, segment j in force on $[b_j, b_{j+1})$
PHt	breakpoints: [], alpha: [[[]]], S: [[[]]], cyclic	Time-inhomogeneous phase-type: one (α, S) pair per segment
MPH	alpha: [], S: [[[]]], exit: [[[]]]	Marked phase-type: K exit vectors partitioning $-Se$; the renewal special case of MMAP
MMAPt	breakpoints: [], D0: [[[]]], D1k: [[[[]]]], cyclic	Marked time-inhomogeneous MAP: D1k nests mark then segment
MPHt	breakpoints: [], alpha: [[[]]], S: [[[]]], exit: [[[]]], cyclic	Marked time-inhomogeneous phase-type: exit nests mark then segment

Type	Parameters	Description
BMMAPt	breakpoints: [], D0: [[[]]], D1kb: [[[[[]]]]], cyclic	Batch marked time-inhomogeneous MAP: D1kb nests mark, then batch size, then segment, and block (c, b, j) releases b jobs all of mark c in segment j . The batch axis is dense in $b = 1, \dots, B$; the reader derives the batch-aggregated D1k and the aggregate D1, so neither travels
MMPP2	lambda0, lambda1, sigma0, sigma1	Two-state MMPP
MMDP2	r0, r1, sigma0, sigma1	Two-state MMDP: deterministic rates in the two states, and the two switching rates
ME	alpha: [], A: [[[]]]	Matrix-exponential
RAP	H0: [[[]]], H1: [[[]]]	Rational arrival process
DMAP	D0: [[[]]], D1: [[[]]]	Discrete MAP
BMAP	D: [[[[]]]]	Batch MAP; $D[0] = D_0, D[1] = D_1, \dots$
MarkedMMPP	D: [[[[]]]], K	Marked MMPP (M3PP) with K marking types
Replayer	fileName, mean	Trace replay; see Section D.2.4
Immediate	(none)	Zero service time
Disabled	(none)	No service (class disabled at this node)

Distributions carrying no `params` block, namely PH, APH, MAP, MMAP and Expolynomial, are described in Sections D.2.4 and D.2.4.

Parameterization of MMDP2. The two-state Markov-modulated deterministic process is parameterized by deterministic *rates* in every codebase: `r0` and `r1` are the rates in the two states, and `sigma0`, `sigma1` the switching rates. The stationary mean rate is $(r_0\sigma_1 + r_1\sigma_0)/(\sigma_0 + \sigma_1)$ and the reported mean is its reciprocal. The Python-native class formerly took inter-arrival *times* `d0`, `d1`; that spelling is no longer written or read.

Fit specification

Instead of direct parameters, a distribution can be constructed by fitting from summary statistics via the `fit` object. This is the preferred mode when the user specifies a distribution by its moments rather than its native parameters.

Table D.17: Fit methods

Field	Type	Description
method	fitMean, fitMeanAndSCV, fitMeanAndOrder, fitCentral, fitRawMoments	Fitting method: which moments the parameters are derived from. Required.
mean	number	Target mean, matched exactly by every method.
scv	number	Squared coefficient of variation.
order	integer	Number of phases/stages.

Field	Type	Description
moments	array of number	The three moments for fitCentral (central) or fitRawMoments (raw). Must have length 3. Alternatively give the named fields: mean/var/skew for fitCentral, m1/m2/m3 for fitRawMoments. Both methods resolve to Cox2 when type is Coxian or Cox2, and to APH otherwise.
var	number	Variance, for fitCentral given by named fields.
skew	number	Skewness, for fitCentral given by named fields.
m1	number	First raw moment, for fitRawMoments given by named fields.
m2	number	Second raw moment, for fitRawMoments given by named fields.
m3	number	Third raw moment, for fitRawMoments given by named fields.

The `moments` array must have length 3 for `fitCentral` and `fitRawMoments`; a shorter array is rejected. Both methods resolve to `Cox2` when the `type` is `Coxian` or `Cox2`, and to `APH` otherwise.

Example:

```
{ "type": "Erlang", "fit": { "method": "fitMeanAndOrder", "mean": 2.0, "order": 3 } }
```

Phase-type and MAP representations

Phase-type distributions (PH, APH, Coxian, Cox2) can be specified via a `ph` object containing:

- `alpha`: initial probability row vector α
- `T`: sub-generator matrix T (negative diagonal, non-negative off-diagonal)

Example:

```
{ "type": "PH", "ph": { "alpha": [1.0, 0.0], "T": [[-2.0, 2.0], [0.0, -1.0]] } }
```

Markovian Arrival Processes are specified via a `map` object containing `D0`, the background transition matrix D_0 , and `D1`, the arrival transition matrix D_1 .

Marked MAPs (MMAP) instead use an `mmap` object containing `D0` and `D1k`, an array holding one arrival matrix per mark. The aggregate $D_1 = \sum_k D_{1k}$ is rebuilt on load. A 1×1 matrix may be encoded as a bare scalar.

Special distributions

- `Replayer`: replays inter-arrival times from a trace file, given by `params.fileName`. The writer also emits `params.mean` and, when it can fit one, a `ph` block; the reader uses the file when it resolves and otherwise falls back to the fitted phase-type. A trace built from in-memory samples lives in a generated temporary file whose path does not survive a round trip, and the writer warns in that case.

- **Expolynomial:** specified via an `expolynomial` object with `density` (a density expression in expolynomial (GEN) format, such as `"0.5 * x^2 * Exp[-0.3 x]"`), `eft` (earliest firing time) and `lft` (latest firing time, with the string `"Inf"` allowed for an unbounded support).
- **Prior:** a weighted mixture of alternative distributions for posterior or uncertainty analysis, specified by `distributions` (an array of distribution objects) and `probabilities` (an array of weights).
- **Workflow:** an activity graph that is compiled to a phase-type distribution, specified by a `workflow` object holding `activities` (each with a name and a nested distribution) and `precedences` between them. The compiled phase-type law is what the solvers see; the graph is retained so that a round trip does not lose the structure.

A distribution for which no branch matches is written as its own type name carrying `params.mean` and `params.scv`, with a warning. On reading, an unrecognized type carrying a mean and an SCV is reconstructed as an APH matching those two moments, again with a warning.

D.2.5 LayeredNetwork models

A `LayeredNetwork` model describes a layered queueing network (LQN). It is organized hierarchically into hosts (processors), tasks, entries, and activities.

Table D.18: LayeredNetwork model fields

Field	Type	Description
<code>type</code>	<code>"LayeredNetwork"</code>	Discriminator of the model kind. Required.
<code>name</code>	<code>string</code>	Model name. Required.
<code>hosts</code>	<code>array of LHost</code>	Host (processor) definitions. Canonical key; the MATLAB and Python readers accept 'processors' as a legacy alias, the Java reader does not. Required.
<code>processors</code>	<code>array of LHost</code>	Legacy alias of 'hosts'. Read by MATLAB and Python only.
<code>tasks</code>	<code>array of LTask</code>	Task definitions; each names the host it runs on. Required.
<code>entries</code>	<code>array of LEntry</code>	Entry definitions; each names the task that offers it. Required.
<code>activities</code>	<code>array of LActivity</code>	Activity definitions; each names the task it belongs to. Required.
<code>precedences</code>	<code>array of LPrecedence</code>	Precedence relations over the activities of a task.

The canonical key for the host array is `hosts`; the MATLAB and Python readers accept `processors` as a legacy alias, the Java reader does not.

Hosts

Table D.19: Host fields

Field	Type	Description
name	string	Host (processor) name, unique within the model. Required.
multiplicity	integer	Number of processors (default 1).
scheduling	FCFS, PS, INF, HOL, RAND	Host scheduling (default INF).
quantum	number	Scheduling quantum (time slice) for PS.
speedFactor	number	Processor speed factor (default 1.0).
replication	integer	Number of replicas.

Tasks

Table D.20: Task fields

Field	Type	Description
name	string	Task name, unique within the model. Required.
host	string	Name of the host. Canonical key; the MATLAB and Python readers accept 'processor' as a legacy alias.
processor	string	Legacy alias of 'host'. Read by MATLAB and Python only.
multiplicity	integer	Number of task threads/copies (default 1).
scheduling	REF, FCFS, INF, HOL, POLL	Task scheduling strategy.
priority	integer	Task priority at its host, 0 being the highest.
thinkTime	Distribution	Think time distribution (for REF tasks). Alternative to the thinkTimeMean/thinkTimeSCV pair.
thinkTimeMean	number	Mean think time (for REF tasks).
thinkTimeSCV	number	Think time squared coefficient of variation.
replication	integer	Number of replicas.
fanIn	object	Fan-in from the named task.
fanOut	object	Fan-out to the named task.
taskType	SetupTask, FunctionTask, CacheTask	Specialized task kind. 'FunctionTask' is the legacy name of 'SetupTask' and is still read; setup and delay-off times are accepted on any task.
setupTime	Distribution	Server activation (setup) time distribution. Alternative to the setupTimeMean/setupTimeSCV pair.
setupTimeMean	number	Mean setup time paid when the task resumes from an idle period.
setupTimeSCV	number	Squared coefficient of variation of the setup time.
delayOffTime	Distribution	Delay-off time distribution, i.e. the idle period before a server powers off. Alternative to the delayOffTimeMean/delayOffTimeSCV pair.
delayOffTimeMean	number	Mean idle time the task waits before powering off.
delayOffTimeSCV	number	Squared coefficient of variation of the delay-off time.
totalItems	integer	Number of cacheable items (CacheTask).
cacheCapacity	integer or array of integer	Cache capacity per list level (CacheTask).
replacementStrategy	ReplacementStrategy	Replacement policy (CacheTask).

A think time may also be given as a `thinkTime` distribution object instead of the `thinkTimeMean/thinkTimeSCV` pair; likewise `setupTime` and `delayOffTime` accept distribution objects.

Entries

Table D.21: Entry fields

Field	Type	Description
<code>name</code>	<code>string</code>	Entry name, unique within the model. Required.
<code>task</code>	<code>string</code>	Name of parent task. Required.
<code>arrival</code>	<code>Distribution</code>	Open arrival distribution for this entry.
<code>forwarding</code>	<code>array of object</code>	Forwarding to other entries.
<code>entryType</code>	<code>ItemEntry</code>	Specialized entry kind.
<code>totalItems</code>	<code>integer</code>	Number of items (<code>ItemEntry</code>).
<code>accessProb</code>	<code>Distribution</code>	Item popularity distribution (<code>ItemEntry</code>).

Activities

Table D.22: Activity fields

Field	Type	Description
<code>name</code>	<code>string</code>	Activity name, unique within its task. Required.
<code>task</code>	<code>string</code>	Name of parent task. Required.
<code>hostDemand</code>	<code>Distribution</code>	Host demand distribution.
<code>boundToEntry</code>	<code>string</code>	Entry name this activity is bound to. Canonical key; the MATLAB and Python readers accept 'boundTo' as a legacy alias.
<code>boundTo</code>	<code>string</code>	Legacy alias of 'boundToEntry'. Read by MATLAB and Python only.
<code>repliesTo</code>	<code>string</code>	Entry name this activity replies to.
<code>callOrder</code>	<code>string</code>	Call order for this activity's calls.
<code>synchCalls</code>	<code>array of LCall</code>	Synchronous (blocking) calls to entries.
<code>asynchCalls</code>	<code>array of LCall</code>	Asynchronous (non-blocking) calls to entries.

Each call object has an `entry` field (target entry name) and an optional `mean` field (mean number of calls, default 1).

Activity precedences

Precedence relations define control flow within a task. Two schemas are accepted. The first is the *typed* form, written by the MATLAB and Python bridges, in which the pre/post split is implied by the type: a fork names its predecessor first, a join names its successor last.

Table D.23: Typed precedence fields

Field	Type	Description
task	string	Parent task name.
type	Serial, AndFork, AndJoin, OrFork, OrJoin, Loop, ...	Kind of precedence relation between the pre and post activity sets. Required.
activities	array of string	Ordered list of activity names.
probabilities	array of number	For OrFork: branch probabilities. Defaults to uniform.
loopCount	number	For Loop: mean number of iterations.
preActivity	string	For Loop: the loop trigger activity; the loop body is then carried whole in 'activities'.

The second is the *explicit* form, written by the Java bridge, which carries the precedence types verbatim and so needs no shape inference. A precedence object carrying `preActs` or `postActs` is read in this form.

Table D.24: Explicit precedence fields

Field	Type	Description
task	string	Parent task name. Required.
preActs	array of string	Predecessor activity names.
postActs	array of string	Successor activity names.
preType	pre, pre-AND, pre-OR	Pre-condition type (default 'pre').
postType	post, post-AND, post-OR, post-LOOP, post-CACHE	Post-condition type (default 'post').
preParams	array of number	Parameters of the pre-condition, such as an AND-join quorum. Defaults to all predecessors required.
postParams	array of number	Parameters of the post-condition, such as OR-fork probabilities or a loop count.

D.2.6 Workflow models

A `Workflow` model describes an activity graph that compiles to a phase-type distribution. Its `type` field is set to "Workflow" and it carries a name, an `activities` array of {name, distribution} objects, and a `precedences` array using the typed precedence schema of Table D.23.

A workflow may equally appear as a distribution, through a `workflow` object with the same `activities` and `precedences` fields.

D.2.7 Environment models

An `Environment` model describes a queueing network operating in a random environment. Each environment stage is associated with a `Network` sub-model.

Table D.25: Environment model fields

Field	Type	Description
type	"Environment"	Discriminator of the model kind. Required.
name	string	Model name. Required.
numStages	integer	Number of environment stages.
stages	array of EnvStage	Environment stages, in the order that fixes the 0-based stage indices used by the transitions. Required.
transitions	array of object	Transitions between stages. 'from' and 'to' are 0-based stage indexes, not names. There is no scalar rate field: the stage holding time is a full Distribution object.

A transition object holds `from` and `to` as 0-based *stage indexes*, not stage names, and a distribution object (Section D.2.4) giving the stage holding time. There is no scalar rate field: an exponential holding time of rate λ is written `{"type": "Exp", "params": {"lambda":  $\lambda$ }}`.

Environment state not carried by the format

The reset policy of an environment transition (Section 6.6.1) and its state-dependent rate rule are function-valued and are not serialized. The consequences are described in Section D.2.9.

D.2.8 Examples

M/M/1 queue

```
{
  "format": "line-model",
  "version": "1.0",
  "model": {
    "type": "Network",
    "name": "M/M/1",
    "nodes": [
      { "name": "Source", "type": "Source",
        "service": {
          "Class1": { "type": "Exp",
            "fit": { "method": "fitMean", "mean": 1.0 } } } },
      { "name": "Queue", "type": "Queue", "scheduling": "FCFS",
        "service": {
          "Class1": { "type": "Exp",
            "fit": { "method": "fitMean", "mean": 0.5 } } } },
      { "name": "Sink", "type": "Sink" }
    ],
    "classes": [
      { "name": "Class1", "type": "Open" }
    ],
    "routing": {
      "type": "matrix",
      "matrix": {
        "Class1,Class1": {
          "Source": { "Queue": 1.0 },

```

```

        "Queue": { "Sink": 1.0 }
    }
}
}
}

```

Two-task layered queueing network

```

{
  "format": "line-model",
  "version": "1.0",
  "model": {
    "type": "LayeredNetwork",
    "name": "lqn_twotasks",
    "hosts": [
      { "name": "P1", "scheduling": "PS" },
      { "name": "P2", "scheduling": "PS" }
    ],
    "tasks": [
      { "name": "T1", "host": "P1", "multiplicity": 100,
        "scheduling": "REF",
        "thinkTime": { "type": "Erlang",
          "fit": { "method": "fitMeanAndOrder",
            "mean": 10, "order": 1 } } },
      { "name": "T2", "host": "P2", "multiplicity": 1,
        "scheduling": "INF" }
    ],
    "entries": [
      { "name": "E1", "task": "T1" },
      { "name": "E2", "task": "T2" }
    ],
    "activities": [
      { "name": "A1", "task": "T1",
        "hostDemand": { "type": "Exp", "params": { "lambda": 1.0 } },
        "boundToEntry": "E1",
        "synchCalls": [ { "entry": "E2" } ] },
      { "name": "A2", "task": "T2",
        "hostDemand": { "type": "Exp", "params": { "lambda": 1.0 } },
        "boundToEntry": "E2", "repliesTo": "E2" }
    ]
  }
}

```

Random environment

The model below alternates between an Up stage and a Down stage in which the queue serves at half its nominal rate. Each stage carries a complete Network sub-model, and the two stages are connected by exponential transitions.

```

{
  "format": "line-model",
  "version": "1.0",
  "model": {
    "type": "Environment",
    "name": "server_breakdown",
    "numStages": 2,
    "stages": [
      { "name": "Up",
        "model": {
          "type": "Network", "name": "up",
          "nodes": [
            { "name": "Delay", "type": "Delay",
              "service": { "Class1": { "type": "Exp",
                "params": { "lambda": 0.2 } } } },
            { "name": "Queue1", "type": "Queue", "scheduling": "FCFS",
              "service": { "Class1": { "type": "Exp",
                "params": { "lambda": 1.0 } } } } }
          ],
          "classes": [
            { "name": "Class1", "type": "Closed",
              "population": 3, "refNode": "Delay" }
          ],
          "routing": {
            "type": "matrix",
            "matrix": {
              "Class1,Class1": {
                "Delay": { "Queue1": 1.0 },
                "Queue1": { "Delay": 1.0 }
              }
            }
          }
        }
      },
      { "name": "Down",
        "model": {
          "type": "Network", "name": "down",
          "nodes": [
            { "name": "Delay", "type": "Delay",
              "service": { "Class1": { "type": "Exp",
                "params": { "lambda": 0.2 } } } },
            { "name": "Queue1", "type": "Queue", "scheduling": "FCFS",
              "service": { "Class1": { "type": "Exp",
                "params": { "lambda": 0.5 } } } } }
          ],
          "classes": [
            { "name": "Class1", "type": "Closed",
              "population": 3, "refNode": "Delay" }
          ],
          "routing": {
            "type": "matrix",
            "matrix": {
              "Class1,Class1": {
                "Delay": { "Queue1": 1.0 },
                "Queue1": { "Delay": 1.0 }
              }
            }
          }
        }
      }
    ]
  }
}

```

```

    }
  }
}
],
"transitions": [
  { "from": 0, "to": 1,
    "distribution": { "type": "Exp", "params": { "lambda": 0.1 } } },
  { "from": 1, "to": 0,
    "distribution": { "type": "Exp", "params": { "lambda": 1.0 } } }
]
}
}

```

D.2.9 State not carried by the format

Some model state is function-valued and therefore has no JSON representation. A round trip through the format silently drops it, and the reloaded model reverts to the corresponding default. Users must reapply such state programmatically after loading.

Table D.26: Model state not carried by the JSON format

State	Consequence of a round trip
Environment reset policy (Section 6.6.1)	Reverts to the identity rule, so jobs keep their positions across a stage transition instead of being redistributed
Environment state-dependent rate rule (Section 6.6.1)	Reverts to the identity rule. ENV then no longer selects <code>options.method='statedep'</code> , so the reloaded model is solved as a plain random environment
State-dependent routing functions	Reverts to the routing matrix
Custom rewards (Section D.2.3)	Omitted with a warning at save time

The reset policy is a per-transition function, supplied as the fourth argument of `add_transition`, and may differ on each transition. It is not a model-level enumeration and there is no field in the JSON format that selects between clearing and keeping the network state.

D.2.10 Serialization conventions

The following conventions apply to the JSON serialization:

- Matrices are stored as nested JSON arrays, e.g., `[[1, 2], [3, 4]]` for a 2×2 matrix. A 1×1 matrix may be encoded as a bare scalar.
- Enumerations cross as names, never as ordinal numbers, since the numeric values of the enumerations differ across codebases.

- NaN values are represented as JSON `null`.
- Infinity values are clamped to $\pm 10^{308}$, except where a key documents the string "Infinity".
- All identifiers (node names, class names) are strings and must be unique within their scope.
- ClassSwitch nodes that are auto-generated by `link()` are excluded from serialization.
- An absent key means the corresponding feature is not configured.
- Optional fields may be omitted; the loader applies default values as documented in the schema.

D.3 Command-line interface (line-cli)

The LINE command-line interface (`line-cli`) provides a standalone tool for solving queueing network models directly from the terminal. This interface is useful for batch processing, integration with external tools, scripting, and environments where interactive use of MATLAB, Python, or Java is not desirable.

Two distinct programs carry this name, and this section documents the first of them:

- `line-cli.py`, the Python script at the root of the distribution, which drives `common/jline.jar` and therefore needs a Java runtime and Python. It is the subject of the rest of this section.
- `cpp/build/line-cli`, the native executable built from `cpp/` in Section 2.1, which needs neither. It reads the same model formats and takes the same `-f`, `-s` and `-a` options, so the usage below transfers, but it is a separate program with its own option set: run it with `--help` for the flags it actually accepts, and note that it takes them directly rather than after a `solve` subcommand (`line-cli -f example.json -s mva -a avg`).

D.3.1 Installation

The CLI tool requires:

- Java 8+: The Java Runtime Environment (JRE) version 8 or later.
- Python 3.11+: Python 3.11 or later.
- `jline.jar`: The LINE solver JAR file, typically located in the `common/` directory.

The CLI is included in the standard LINE distribution. After extracting the release archive, verify the installation by running:

```
python line-cli.py info
```

If the JAR file is not automatically detected, set the `LINE_JAR_PATH` environment variable:

```
export LINE_JAR_PATH=/path/to/jline.jar
```

D.3.2 Basic Usage

The CLI supports several commands for solving models and querying information.

Solving a Model. To solve a queueing network model, use the `solve` command with the optional `-s` (or `--solver`) option to specify the solver algorithm (when omitted, the solver defaults to `auto`):

```
python line-cli.py solve model.jsimg -s mva
```

If you are unsure of which solver to use, the `auto` solver provides an automatic choice. To list all available solvers:

```
python line-cli.py list solvers
```

The following table summarizes the solvers presently interfaced with the cli tool:

Solver	Name	Formats	Notes
auto	Automatic Selection	All	Selects best solver for model
ag	Agent-based (RCAT/INAP)	json/jsim	Reversed-rate product form
ba	Bound Analysis	json/jsim	Closed-form bounds
ctmc	CTMC	json/jsim	Exact, small models only
env	Random Environment	json	Environment models only
fld	Fluid ODE	json/jsim	Fast, approximate
jmt	Java Modelling Tools	json/jsim	External simulation
ldes	LDES	json/jsim/pnml/lqnx/xml	Simulation-based (alias des)
ln	Layered Network	json/lqnx/xml	For LQN models (MVA layers)
ln.mva, ln.nc, ln.comom	Layered Network	json/lqnx/xml	Layer engine named
lqns	LQN Solver	json/jsim/lqnx/xml	External LQNS tool; qns.* methods
mam	Matrix Analytic	json/jsim	Supports Fork-Join percentiles
mva	Mean Value Analysis	json/jsim	Fast, general purpose
nc	Normalizing Constant	json/jsim	Exact analysis
ssa	Stochastic Simulation	json/jsim	State-space sampling
uq	Uncertainty Quantification	json/jsim	Needs <code>--uq-solver</code>

The input format is automatically detected from the file extension. Supported formats include:

- `.json`: LINE's own portable model format, carrying a , a or an
- `.jsimg`, `.jsim`, `.jsimw`: JMT simulation model formats
- `.lqnx`, `.xml`: Layered Queueing Network Solver XML format
- `.pnml`: place/transition net (ISO/IEC 15909-2)

Models can also be piped via stdin when the input format is specified:

```
cat model.jsimg | python line-cli.py solve -i jsimg -s mva
```

The CLI also supports multiple output formats via the `-o` or `--output-format` option:

```
python line-cli.py solve model.jsimg -s mva -o table # Human-readable table
python line-cli.py solve model.jsimg -s mva -o json  # JSON format
python line-cli.py solve model.jsimg -s mva -o csv   # Comma-separated values
```

```
python line-cli.py solve model.jsimg -s mva -o raw # Raw solver output
```

If unspecified, the table output format is used by default.

To save results to a file, further add the `-O` option:

```
python line-cli.py solve model.jsimg -s mva -o json -O results.json
```

Analysis Types. The `-a` or `--analysis` option specifies which metrics to compute:

```
python line-cli.py solve model.jsimg -s mva -a all # Average and system ...
metrics (default)
python line-cli.py solve model.jsimg -s mva -a avg # Average metrics only
python line-cli.py solve model.jsimg -s mva -a sys # System metrics only
```

Advanced analysis types include:

- stage, chain, node, nodechain: stage-, chain-, node-, and node-chain-level averages
- cdf-respt, cdf-passt: Response/passag time CDFs
- perct-respt: Response time percentiles (MAM solver)
- tran-avg, tran-cdf-respt, tran-cdf-passt: transient averages and transient CDFs
- prob, prob-aggr, prob-marg: State probabilities (CTMC/SSA)
- prob-sys, prob-sys-aggr, prob-sys-marg: System state probabilities
- sample, sample-aggr, sample-sys, sample-sys-aggr: State trajectory sampling (SSA)
- reward, reward-steady, reward-value: Reward metrics (CTMC)
- cache, item: cache hit/miss metrics, and the per-item table
- orbit, loss, region-loss, deadline: retrial orbit, class-level and finite-capacity-region loss, and deadline-miss metrics
- normconst: the log normalizing constant $\log G(N)$ (nc, mva)
- busyperiod: mean busy period of a subnetwork, with `--busyperiod-subnet` naming it and `--busyperiod` the orders (nc, ldes)
- sens: sensitivity of the mean metrics to the service demands
- generator, statevec, moments, interval: solver-internal reports (CTMC generator, fluid state vector, moment closure, UQ envelope)

To list all analysis types:

```
python line-cli.py list analysis
```

D.3.3 Server Modes

The CLI can run as a server to accept solve requests over the network.

WebSocket Server. Start a WebSocket server for real-time communication:

```
python line-cli.py server -p 5863
```

This starts a WebSocket server on port 5863 (default). Clients can connect to `ws://localhost:5863` to submit models and receive results.

REST API Server. The `rest` subcommand is at present an alias of `server`: it execs the JAR with `-p`, so it starts the same WebSocket server on the requested port (8080 by default) and does not answer HTTP requests.

```
python line-cli.py rest -p 8080
```

The HTTP interface is a separate Maven module, `io/rest-api/`, which is not part of the root build and has to be packaged and launched on its own:

```
cd io/rest-api
mvn clean package
java -cp target/line-rest.jar:common/jline.jar jline.rest.LineRestServer --port ...
8080
```

It serves JSON under the base path `/api/v1`, for instance:

- POST `/api/v1/models/solve`: Submit a model for solving
- POST `/api/v1/models/solve/async`: Submit a model as a job, tracked under `/api/v1/jobs`
- GET `/api/v1/health`: Health check endpoint
- GET `/api/v1/solvers`: Solver catalogue and per-solver capabilities

The complete contract, including the analysis and metrics routes, is given by `io/rest-api/openapi.yaml`.

D.3.4 Advanced Options

Stochastic Solver Options. For stochastic solvers (`ldes`, `ssa`, `jmt`), a random seed can be specified for reproducibility:

```
python line-cli.py solve model.jsimg -s ssa -d 12345
```

Probability and Sampling Options. For probability and sampling analysis:

```
# State probability at node 0
python line-cli.py solve model.jsimg -s ctmc -a prob -n 0

# Sample 5000 events from node 1
python line-cli.py solve model.jsimg -s ssa -a sample -n 1 --events 5000

# Compute specific percentiles
python line-cli.py solve model.jsimg -s mam -a perct-respt --percentiles ...
50,90,95,99
```

Output Metrics. The CLI outputs performance metrics organized in two tables.

Average Metrics (per station and job class):

QLen	Average queue length (number of jobs)
Util	Server utilization (0–1)
RespT	Response time (seconds)
ResidT	Residence time (seconds)
Tput	Throughput (jobs/second)
ArrvR	Arrival rate (jobs/second)

System Metrics (per chain):

SysRespT	End-to-end response time
SysTput	System throughput
SysQLen	Total jobs in system

D.3.5 Command Reference

Command/Option	Description
<code>solve <file></code>	Solve a queueing model
<code>info</code>	Display system information
<code>list solvers</code>	List available solvers
<code>list formats</code>	List supported formats
<code>list analysis</code>	List analysis types
<code>server</code>	Start WebSocket server
<code>rest</code>	Start WebSocket server on port 8080 (alias of <code>server</code>)
<code>-s, -solver</code>	Solver algorithm
<code>-i, -input-format</code>	Input format (auto-detected)
<code>-o, -output-format</code>	Output format (table/json/csv/raw)
<code>-a, -analysis</code>	Analysis type(s)
<code>-d, -seed</code>	Random seed
<code>--events</code>	Maximum simulation events (<code>ldes</code> , <code>ssa</code>)
<code>--samples</code>	Simulation samples / Monte Carlo draws
<code>--cutoff</code>	State-space cutoff per open class (<code>ctmc</code> , <code>ssa</code>)
<code>--timespan</code>	Horizon of the transient analyses, <code>T1</code> , <code>T0</code> , <code>T1</code> or <code>T0:T1</code>
<code>--method</code>	Algorithm within the chosen solver
<code>--uq-solver</code>	Engine <code>uq</code> runs at each design point
<code>--busyperiod-subnet</code>	Stations forming the <code>-a busyperiod</code> subnetwork
<code>-n, -node</code>	Node index (for <code>prob/sample</code>)
<code>-c, -class-idx</code>	Class index (for <code>prob-marg</code>)
<code>-v, -verbose</code>	Verbose output
<code>-q, -quiet</code>	Suppress status messages
<code>-O, -output-file</code>	Write results to file
<code>-p, -port</code>	Server port
<code>-H, -host</code>	Server host address

D.3.6 Examples

Sample Output. Running the CLI on a sample open queueing network model:

```
python line-cli.py solve example.json -s mva
```

produces the following output:

```
Solving example.json...

Average Metrics
=====
Station  Class  Metric  Value  Unit
-----
Source 1  Class A  QLen    0
Source 1  Class A  Util    0
Source 1  Class A  RespT   0
Source 1  Class A  ResidT  0
Source 1  Class A  ArvR    0
Source 1  Class A  Tput    2
Source 1  Class B  QLen    0
Source 1  Class B  Util    0
Source 1  Class B  RespT   0
Source 1  Class B  ResidT  0
Source 1  Class B  ArvR    0
```

```

Source 1  Class B  Tput    1
Queue 1   Class A  QLen    1.3333
Queue 1   Class A  Util    0.4
Queue 1   Class A  RespT    0.66667
Queue 1   Class A  ResidT    0.44444
Queue 1   Class A  ArvR     2
Queue 1   Class A  Tput     2
Queue 1   Class B  QLen     1
Queue 1   Class B  Util    0.3
Queue 1   Class B  RespT    1
Queue 1   Class B  ResidT    0.33333
Queue 1   Class B  ArvR     1
Queue 1   Class B  Tput     1
Queue 2   Class C  QLen    0.81818
Queue 2   Class C  Util    0.45
Queue 2   Class C  RespT    0.27273
Queue 2   Class C  ResidT    0.27273
Queue 2   Class C  ArvR     3
Queue 2   Class C  Tput     3

```

System Metrics

=====

Station	Class		Metric	Value	Unit
Chain1	Class A Class B Class C	SysRespT	0.77778		
Chain1	Class A Class B Class C	SysTput	3		

Execution time: 0.252s

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